

## Santa Ana Regional Water Quality Control Board

July 28, 2021

James Herberg, General Manager  
Orange County Sanitation District  
10844 Ellis Avenue  
Fountain Valley, CA 92708

**WASTE DISCHARGE REQUIREMENTS AND NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM PERMIT FOR ORANGE COUNTY SANITATION DISTRICT PUBLICLY OWNED TREATMENT WORKS (RECLAMATION PLANT NO. 1, TREATMENT PLANT NO. 2, COLLECTION SYSTEM AND OUTFALLS), ORDER NO. R8-2021-0010, NPDES NO. CA0110604.**

Dear Mr. Herberg:

Enclosed is an original copy of Order No. R8-2021-0010, NPDES Permit No. CA0110604, for your records. The Order includes waste discharge requirements and National Pollutant Discharge Elimination System permit for Orange County Sanitation District's Publicly Owned Treatment Works (Reclamation Plant No. 1, Treatment Plant No. 2, Collection System and Outfalls)..

This Order was adopted by the Santa Ana Water Board during their June 18, 2021 Board meeting. If you have any questions regarding this matter, please contact myself at (951) 320-2008 and/or at [Ryan.Harris@waterboards.ca.gov](mailto:Ryan.Harris@waterboards.ca.gov).

Sincerely,

**Ryan Harris**  
Digitally signed by  
Ryan Harris  
Date: 2021.07.28  
15:23:14 -07'00'

Ryan L. Harris, PE  
Water Resources Control Engineer

Enclosure: Original signed Order No. R8-2021-0010

CC: Dindo Carrillo, [DCARRILLO@OCSD.COM](mailto:DCARRILLO@OCSD.COM)  
Susan Beeson, [Susan.Beeson@waterboards.ca.gov](mailto:Susan.Beeson@waterboards.ca.gov)

CIWQS Place ID:  
Plant 1 – 758392  
Plant 2 – 259158

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
SANTA ANA REGION**

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**ORDER NO. R8-2021-0010  
NPDES NO. CA0110604**

**WASTE DISCHARGE REQUIREMENTS AND  
NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM PERMIT  
FOR ORANGE COUNTY SANITATION DISTRICT  
PUBLICLY OWNED TREATMENT WORKS  
(RECLAMATION PLANT NO. 1, TREATMENT PLANT NO. 2,  
COLLECTION SYSTEM, AND OUTFALLS)**

As set forth in this State Order/federal Permit, the following Discharger (Table 1) is authorized to discharge from the locations listed in Table 2 in accordance with both the State waste discharge requirements (WDRs) and the federal National Pollutant Discharge Elimination System (NPDES) permit requirements. The Order/Permit are consolidated under 40 CFR § 124.4(c)(2).

**Table 1. Discharger Information**

|                                          |                                                                           |
|------------------------------------------|---------------------------------------------------------------------------|
| <b>Discharger</b>                        | Orange County Sanitation District                                         |
| <b>Name of Facilities<br/>(and POTW)</b> | Reclamation Plant No. 1 (Fountain Valley), collection system and outfalls |
|                                          | Treatment Plant No. 2 (Huntington Beach), collection system and outfalls  |
| <b>Facility Addresses</b>                | 10844 Ellis Avenue, Fountain Valley, CA 92708-7018                        |
|                                          | 22212 Brookhurst Street, Huntington Beach, CA 92646-8406                  |
|                                          | Orange County                                                             |

**Table 2. Discharge Location**

| <b>Discharge Point</b>                  | <b>Effluent Description</b>                                | <b>Discharge Point Latitude (North)</b> | <b>Discharge Point Longitude (West)</b> | <b>Receiving Water</b>                |
|-----------------------------------------|------------------------------------------------------------|-----------------------------------------|-----------------------------------------|---------------------------------------|
| 001<br>(120" Outfall)                   | Secondary treated effluent and reverse osmosis concentrate | 33° 34' 36" N                           | 118° 00' 36" W                          | Pacific Ocean<br>(4.5 miles offshore) |
| 002<br>(78" Outfall)                    | Secondary treated effluent and reverse osmosis concentrate | 33° 36' 56" N                           | 117° 58' 13" W                          | Pacific Ocean<br>(1.5 miles offshore) |
| 003<br>(Santa Ana River Overflow Weirs) | Secondary treated effluent and reverse osmosis concentrate | 33° 38' 06" N                           | 117° 57' 20" W                          | Santa Ana River                       |

**Table 3. Administrative Information for State WDRs**

|                                                                                                                                                                                                           |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| This Order was adopted on:                                                                                                                                                                                | <b>June 18, 2021</b>    |
| This Order shall become effective on:                                                                                                                                                                     | <b>August 1, 2021</b>   |
| This Order shall expire on:                                                                                                                                                                               | <b>July 31, 2026</b>    |
| The Discharger shall file a ROWD as an application for reissuance of WDRs in accordance with title 23, California Code of Regulations, and an application for reissuance of a NPDES permit no later than: | <b>February 1, 2026</b> |
| The USEPA and California Regional Water Quality Control Board, Santa Ana Region have classified this discharge as follows:                                                                                | <b>Major</b>            |

I, Hope A. Smythe, Executive Officer, do hereby certify that this State Order with all attachments is a full, true, and correct copy of the Order adopted by the California Regional Water Quality Control Board, Santa Ana Region, on the date indicated above.

**Hope Smythe**  Digitally signed by Hope Smythe  
Date: 2021.06.21 11:36:38 -07'00'

Hope A. Smythe, Executive Officer

**Table 4. Administrative Information for Federal NPDES Permit**

|                                                                                                                                |                                |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| This Permit was issued on:                                                                                                     | <b>Date of signature below</b> |
| This Permit shall become effective on:                                                                                         | <b>August 1, 2021</b>          |
| This Permit shall expire on:                                                                                                   | <b>July 31, 2026</b>           |
| The Discharger shall file an application for reissuance of a NPDES permit in accordance with 40 CFR § 122.21(d) no later than: | <b>February 1, 2026</b>        |
| The USEPA has classified this discharge as follows:                                                                            | <b>Major</b>                   |

This federal NPDES Permit is signed and issued on the date indicated above, for the Regional Administrator.

**TOMAS**  
**TORRES**  
Digitally signed by  
TOMAS TORRES  
Date: 2021.06.23  
15:55:32 -07'00'  
Tomás Torres, Water Division Director

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## I. FACILITY INFORMATION

Information describing the Orange County Sanitation District (hereinafter “OC San”, “Permittee” or “Discharger”) is summarized in Tables 1 and 2, and in sections I and II of the Fact Sheet (Attachment F). Section I of the Fact Sheet also includes information regarding the Facility’s permit application.

## II. FINDINGS

The California Regional Water Quality Control Board, Santa Ana Region (hereinafter “Santa Ana Water Board”) and the United States Environmental Protection Agency, Region 9 (hereinafter “USEPA”) find:

**A. Legal Authorities.** This Order serves as waste discharge requirements (WDRs) pursuant to article 4, chapter 4, division 7 of the California Water Code (CWC; commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) (Public Law 92-500, as amended, 33 U.S.C. 1251 et seq.) and implementing regulations adopted by USEPA and chapter 5.5, division 7 of the CWC (commencing with section 13370). It shall serve as the State’s National Pollutant Discharge Elimination System (NPDES) permit authorizing the Discharger to discharge into waters of the United States at the discharge locations described in Table 2, subject to the WDRs in this Order (hereinafter “Order”). The Order/Permit are consolidated under 40 CFR § 124.4(c)(2).

Although Discharge Point 001 is beyond the limit of State-regulated ocean waters, effluent plume migration into State waters warrants joint regulation of the discharge by USEPA and the Santa Ana Water Board. Therefore, pursuant to 40 CFR § 124.4, this Order shall serve as a joint consolidated State and federal NPDES Permit authorizing the Discharger to discharge into waters of the United States at the discharge locations described in Table 2, subject to the limitations, conditions and all other requirements set forth herein (hereinafter “Order/Permit”). The Permit is both a federal permit and a state permit by operation of law.

**B. Background and Rationale for Requirements.** The Santa Ana Water Board and USEPA developed the requirements in this Order/Permit based on information submitted as part of the application, through monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for the requirements in this Order/Permit, is hereby incorporated into and constitutes Findings for this Order/Permit. Attachments A through E and G through H are also incorporated into this Order/Permit.

**C. Permit Renewal Contingency.** The Discharger’s federal NPDES Permit renewal is contingent upon:

1. Determination by the U.S. Fish and Wildlife Service (USFWS) and the National Oceanic and Atmospheric Administration (NOAA) National Marine Fisheries Service (NMFS) that the discharge is consistent with the federal Endangered Species Act (ESA) of 1973, as amended (16 U.S. Code (U.S.C.) section 1531 et seq.);



2. Determination by the NOAA NMFS that the proposed discharge is consistent with the Magnuson-Stevens Fishery Conservation and Management Act (MSA) of 1976, as amended (16 U.S.C. section 1801 et seq.);
3. Determination by the California Coastal Commission (CCC) that the proposed discharge is consistent with the Coastal Zone Management Act (CZMA) of 1972, as amended (16 U.S.C. section 1451 et seq.); and
4. The Santa Ana Water Board's certification/concurrence that the discharge will comply with applicable State water quality standards.

**D. Notification of Interested Parties.** The Santa Ana Water Board and USEPA have notified the Discharger and interested agencies and persons of their intent to jointly issue consolidated WDRs and NPDES Permit requirements; and has provided an opportunity to submit written comments and recommendations. Details of the notification are provided in the Fact Sheet and the joint public notice for this Order/Permit.

**E. Consideration of Public Comment.** The USEPA considered all written comments pertaining to the discharge. The Santa Ana Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Santa Ana Water Board's Public Hearing are provided in the Fact Sheet (Attachment F).

THEREFORE, IT IS HEREBY ORDERED, that this Order/Permit supersedes Order No. R8-2012-0035 except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the Water Code (commencing with section 13000) and regulations adopted thereunder, and the provisions of the CWA and regulations and guidelines adopted thereunder, the Discharger shall comply with the requirements in this Order/Permit. This action in no way prevents the Santa Ana Water Board and/or USEPA from taking enforcement action for violations of the previous Order/Permit.

### III. DISCHARGE PROHIBITIONS

- A. The discharge of waste/pollutants in a manner or at locations that have not been specifically authorized by this Order/Permit is prohibited.
- B. The discharge of any radiological, chemical, or biological warfare agent or high-level radioactive waste into the ocean is prohibited.
- C. Waste shall not be discharged to designated Areas of Special Biological Significance, except as provided in Chapter III.E of the Ocean Plan (i.e., Implementation Provisions for Marine Managed Areas).
- D. Pipeline discharge of sludge to the ocean is prohibited by federal law; the discharge of municipal and industrial waste sludge directly to the ocean, or into a waste stream that discharges to the ocean, is prohibited by the California Ocean Plan. The discharge of sludge digester supernatant directly to the ocean, or to a waste stream that discharges to the ocean without further treatment, is prohibited.
- E. The bypassing of untreated wastes containing concentrations of pollutants in excess of those in Table 4 or Table 3 of the California Ocean Plan to the ocean is prohibited.

- F. The discharge of trash to surface waters of the State or the deposition of trash where it may be discharged into surface waters of the State is prohibited.
- G. Discharge at Discharge Point 001 (120" outfall) is prohibited when the discharge does not receive a minimum initial dilution of at least 181:1 ( $D_m = 180$ ), as modeled assuming no currents. Compliance shall be achieved by proper operation and maintenance of the discharge outfall to ensure that it (or its replacement, in whole or part) is in good working order and is consistent with, or can achieve better mixing than, 181:1. The Discharger shall notify the Santa Ana Water Board and USEPA if outfall ports will be retrofitted, as the 181:1 dilution assumes all outfall ports are operational.
- H. Unless regulated by appropriate waste discharge requirements, the discharge to surface or groundwaters of waste which contains the following substances is prohibited: toxic substances or materials; pesticides; PCBs (polychlorinated biphenyls); mercury or mercury compounds; radioactive substances or material in excess of levels allowed by the California Code of Regulations. This list is not necessarily all-inclusive.
- I. The discharge of untreated sewage to any surface water stream, natural or man-made, or to any drainage system intended to convey storm water runoff to surface water streams is prohibited.
- J. The discharge of treated sewage to streams, lakes or reservoirs, or to tributaries thereto, which are designated MUN and which are used as a domestic water supply is prohibited unless approved by the California Department of Public Health. The discharge of treated sewage to waterbodies which are excepted from MUN (Basin Plan Table 3-1) but which are tributary to waters designated MUN and are used as a domestic water supply is prohibited unless the discharge of treated sewage to the drinking water supply is precluded or approved by the California Department of Public Health.
- K. The discharge of wastewater to a water of the United States from any locations other than Discharge Point 001 (120" outfall) is prohibited, except during the following situations:
  - 1. Emergency discharge of disinfected secondary effluent and/or reverse osmosis (RO) concentrate to Discharge Point 002 (78" outfall) when the flow rate exceeds the hydraulic capacity of Discharge Point 001, provided that discharges through Discharge Point 001 (120" outfall) and discharges to Orange County Water District (OCWD)'s water recycling facilities are maximized to a functional capacity that is necessary to prevent infrastructure or equipment damage before wastewater is discharged through Discharge Point 002;
  - 2. Emergency discharge of disinfected secondary effluent and/or RO concentrate to Discharge Point 003 (Santa Ana River overflow weirs) in the event of an extreme emergency (e.g., tsunami, earthquake, flood, and acts of war or terrorism) that precludes discharging all wastewater to Discharge Points 001 and 002, provided that discharges through Discharge Points 001 (120" outfall) and 002 (78" outfall) are maximized before wastewater is discharged through Discharge Point 003; or

3. Discharge of disinfected secondary effluent and/or RO concentrate to Discharge Point 002 (78" outfall) during planned essential maintenance or capital improvement projects to assure efficient operation of Discharge Point 001 (120" outfall) when there is no other feasible alternative.

The Santa Ana Water Board Executive Officer and USEPA Water Division Director are notified of the pending discharge as soon as possible. During a planned diversion such as essential maintenance or capital improvement projects, this notification shall be submitted a minimum of 10 days prior to diverting flow to Discharge Point 002 (78" outfall), and include the rationale for the discharge, the expected time, date, and the duration of the discharge. Projects warranting such a diversion must be approved by the Santa Ana Water Board Executive Officer and USEPA Water Division Director prior to diverting flow to Discharge Point 002. The Discharger shall take all steps required by the Santa Ana Water Board Executive Officer or USEPA Water Division Director to minimize any harm resulting from the discharge.

#### **IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS**

##### **A. Discharge Specifications – Discharge Points 001 and 002**

1. Waste management systems that discharge to the ocean must be designed and operated in a manner that will maintain the indigenous marine life and a healthy and diverse marine community.
2. Waste discharged to the ocean must be essentially free of: Material that is floatable or will become floatable upon discharge; settleable material or substances that may form sediments which will degrade benthic communities or other aquatic life; substances which will accumulate to toxic levels in marine waters, sediments or biota; substances that significantly decrease the natural light to benthic communities and other marine life; and materials that result in aesthetically undesirable discoloration of the ocean surface.
3. Waste effluents shall be discharged in a manner which provides sufficient initial dilution ( $D_m = 180$  for Discharge Point 001 and  $D_m = 36$  for Discharge Point 002), with exception of radioactivity, to minimize the concentrations of substances not removed in the treatment.
4. Waste that contains pathogenic organisms or viruses should be discharged a sufficient distance from naturally occurring shellfishing and water contact sports are to maintain applicable bacterial standards without disinfection. Where conditions are such that an adequate distance cannot be attained, reliable disinfection in conjunction with a reasonable separation of the discharge point from the area of use must be provided. Disinfection procedures at Discharge Point 002 (78" outfall) that do not increase effluent toxicity and that constitute the least environmental and human hazard should be used.

##### **B. Effluent Limitations – Discharge Points 001 and 002**

**1. Final Effluent Limitations – Discharge Point 001 (120” outfall)**

- a. The Discharger shall maintain compliance with the following effluent limitations at Discharge Point 001, with compliance measured at Monitoring Location EFF-001 as described in the Monitoring and Reporting Program (MRP), Attachment E:

**Table 5. Effluent Limitations – Discharge Point 001 (120” outfall)**

| Parameter                                                                                | Units                    | Effluent Limitations <sup>[1],[2]</sup> |                               |                                  |                                          |                                          |                                 |
|------------------------------------------------------------------------------------------|--------------------------|-----------------------------------------|-------------------------------|----------------------------------|------------------------------------------|------------------------------------------|---------------------------------|
|                                                                                          |                          | Average Monthly <sup>[3]</sup>          | Average Weekly <sup>[3]</sup> | Maximum Daily <sup>[3],[4]</sup> | Instantaneous Minimum <sup>[3],[4]</sup> | Instantaneous Maximum <sup>[3],[4]</sup> | Six-Month Median <sup>[3]</sup> |
| Carbonaceous Biochemical Oxygen Demand, 5-day @ 20°C (CBOD <sub>5</sub> ) <sup>[5]</sup> | mg/L                     | 25                                      | 40                            | –                                | –                                        | –                                        | –                               |
|                                                                                          | lbs/day                  | 42,951                                  | 68,722                        | –                                | –                                        | –                                        | –                               |
|                                                                                          | % removal <sup>[6]</sup> | ≥ 85                                    | –                             | –                                | –                                        | –                                        | –                               |
| Total Suspended Solids (TSS)                                                             | mg/L                     | 30                                      | 45                            | –                                | –                                        | –                                        | –                               |
|                                                                                          | lbs/day                  | 51,541                                  | 77,312                        | –                                | –                                        | –                                        | –                               |
|                                                                                          | % removal <sup>[6]</sup> | ≥ 85                                    | –                             | –                                | –                                        | –                                        | –                               |
| pH                                                                                       | standard units           | –                                       | –                             | –                                | 6.0                                      | 9.0                                      | –                               |
| Oil and Grease                                                                           | mg/L                     | 25                                      | 40                            | –                                | –                                        | 75                                       | –                               |
|                                                                                          | lbs/day                  | 42,951                                  | 68,722                        | –                                | –                                        | 128,853                                  | –                               |
| Settleable Solids                                                                        | ml/L                     | 1.0                                     | 1.5                           | –                                | –                                        | 3.0                                      | –                               |
| Turbidity                                                                                | NTU                      | 75                                      | 100                           | –                                | –                                        | 225                                      | –                               |
| Total chlorine residual                                                                  | mg/L                     | –                                       | –                             | 1.45                             | –                                        | 10.86                                    | 0.36                            |
|                                                                                          | lbs/day                  | –                                       | –                             | 2,491                            | –                                        | 18,658                                   | 618                             |
| Acute toxicity <sup>[7]</sup>                                                            | Pass or Fail             | –                                       | –                             | Pass                             | –                                        | –                                        | –                               |
| Chronic toxicity <sup>[7]</sup>                                                          | Pass or Fail             | –                                       | –                             | Pass                             | –                                        | –                                        | –                               |
| Radioactivity                                                                            | pCi/L                    | <sup>[8]</sup>                          |                               |                                  |                                          |                                          |                                 |
| Benzidine                                                                                | µg/L                     | 0.0125                                  | –                             | –                                | –                                        | –                                        | –                               |
|                                                                                          | lbs/day                  | 0.0215                                  | –                             | –                                | –                                        | –                                        | –                               |

| Parameter                       | Units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Effluent Limitations <sup>[1],[2]</sup> |                               |                                  |                                          |                                          |                                 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------|----------------------------------|------------------------------------------|------------------------------------------|---------------------------------|
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Average Monthly <sup>[3]</sup>          | Average Weekly <sup>[3]</sup> | Maximum Daily <sup>[3],[4]</sup> | Instantaneous Minimum <sup>[3],[4]</sup> | Instantaneous Maximum <sup>[3],[4]</sup> | Six-Month Median <sup>[3]</sup> |
| Hexachloro-benzene              | µg/L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.0380                                  | —                             | —                                | —                                        | —                                        | —                               |
|                                 | lbs/day                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.0653                                  | —                             | —                                | —                                        | —                                        | —                               |
| Toxaphene                       | µg/L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.0380                                  | —                             | —                                | —                                        | —                                        | —                               |
|                                 | lbs/day                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.0653                                  | —                             | —                                | —                                        | —                                        | —                               |
| PCBs <sup>[3]</sup>             | µg/L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.0034                                  | —                             | —                                | —                                        | —                                        | —                               |
|                                 | lbs/day                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.0058                                  | —                             | —                                | —                                        | —                                        | —                               |
| TCDD Equivalents <sup>[3]</sup> | pg/L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.7059                                  | —                             | —                                | —                                        | —                                        | —                               |
|                                 | lbs/day                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.00000<br>12                           | —                             | —                                | —                                        | —                                        | —                               |
| Notes:                          | <p><sup>[1]</sup> Mass emission effluent limitations (in lbs/day) are based on the average daily influent flow of 206 MGD projected for 2025, taken from the Discharger's 2017 Master Plan.</p> <p><sup>[2]</sup> The minimum probable initial dilution used to calculate WQBELs for Ocean Plan Table 3 pollutants is 181:1 (<math>D_m=180</math>).</p> <p><sup>[3]</sup> See Attachment A – Definitions.</p> <p><sup>[4]</sup> The maximum daily effluent limitations shall apply to flow weighted 24-hour composite samples. The instantaneous maximum/minimum effluent limitations shall apply to grab samples.</p> <p><sup>[5]</sup> In lieu of the parameter BOD<sub>5</sub> and the BOD<sub>5</sub> levels specified for effluent quality in 40 CFR § 133.102(a)(1), (2), and (3), the parameter CBOD<sub>5</sub> and the CBOD<sub>5</sub> levels specified for effluent quality in 40 CFR § 133.102(a)(4) are substituted and reported by the Discharger.</p> <p><sup>[6]</sup> In each calendar month, the arithmetic mean of CBOD<sub>5</sub> and TSS, by mass, of effluent samples collected at Monitoring Location EFF-001 as described in the MRP, shall not exceed 15 percent of the arithmetic mean of CBOD<sub>5</sub> and TSS, by mass, of influent samples collected at Monitoring Locations INF-001 and INF-002 as described in the MRP, at approximately the same times during the same periods.</p> <p><sup>[7]</sup> The maximum daily effluent limitation shall be reported as “Pass” or “Fail”, and “% Effect”. See section IV.B.1.b-c below and section V of Attachment E.</p> <p><sup>[8]</sup> Not to exceed limits specified in Title 17, Division 1, Chapter 5, Subchapter 4, Group 3, Article 3, Section 30253 of the California Code of Regulations. Reference to Section 30253 is prospective, including future changes to any incorporated provisions of federal law, as the changes take effect.</p> |                                         |                               |                                  |                                          |                                          |                                 |

- b. **Acute Toxicity:** The acute toxicity WQBEL for Discharge Point 001 (120" outfall) is expressed as a null hypothesis ( $H_0$ ) and regulatory management decision ( $b$  value) of 0.80 for the acute toxicity methods in Attachment E of this Order/Permit. The null hypothesis for this discharge is:

$H_0$ : Mean response (5.56% effluent)  $\leq$  0.80 mean response (Control).

Results obtained from an acute toxicity test shall be analyzed using the Test of Significant Toxicity (TST) statistical approach (EPA 833-R-10-003, 2010; Appendix A). Compliance with this acute toxicity WQBEL is demonstrated by rejecting the null hypothesis and reporting "0" = "Pass".

Percent Effect" (or Effect, in %) = [(Control mean response – IWC mean response)  $\div$  Control mean response]  $\times$  100

- c. **Chronic Toxicity:** The chronic toxicity WQBEL for Discharge Point 001 is expressed as a null hypothesis ( $H_0$ ) and regulatory management decision ( $b$  value) of 0.75 for the chronic toxicity methods in Attachment E of this Order/Permit. The null hypothesis for this discharge is:

$H_0$ : Mean response (0.556% effluent)  $\leq$  0.75 mean response (Control).

Results obtained from a chronic toxicity test shall be analyzed using the TST statistical approach (EPA 833-R-10-003, 2010; Appendix A). Compliance with this acute toxicity WQBEL is demonstrated by rejecting the null hypothesis and reporting "0" = "Pass".

Percent Effect" (or Effect, in %) = [(Control mean response – IWC mean response)  $\div$  Control mean response]  $\times$  100

## 2. Final Effluent Limitations – Discharge Point 002 (78" outfall)

- a. The Discharger shall maintain compliance with the following effluent limitations at Discharge Point 002 (78" outfall), with compliance measured at Monitoring Location EFF-002 or EMG-001 as described in the Monitoring and Reporting Program, Attachment E:

**Table 6. Effluent Limitations – Discharge Point 002 (78" outfall)**

| Parameter                                                                               | Units                    | Effluent Limitations <sup>[1],[2]</sup> |                               |                                  |                                          |                                          |                                 |
|-----------------------------------------------------------------------------------------|--------------------------|-----------------------------------------|-------------------------------|----------------------------------|------------------------------------------|------------------------------------------|---------------------------------|
|                                                                                         |                          | Average Monthly <sup>[3]</sup>          | Average Weekly <sup>[3]</sup> | Maximum Daily <sup>[3],[4]</sup> | Instantaneous Minimum <sup>[3],[4]</sup> | Instantaneous Maximum <sup>[3],[4]</sup> | Six-Month Median <sup>[3]</sup> |
| Carbonaceous Biochemical Oxygen Demand 5-day @ 20°C (CBOD <sub>5</sub> ) <sup>[5]</sup> | mg/L                     | 25                                      | 40                            | –                                | –                                        | –                                        | –                               |
|                                                                                         | lbs/day                  | 47,955                                  | 76,728                        | –                                | –                                        | –                                        | –                               |
|                                                                                         | % removal <sup>[6]</sup> | $\geq$ 85                               | –                             | –                                | –                                        | –                                        | –                               |



| Parameter                       | Units                      | Effluent Limitations <sup>[1],[2]</sup> |                               |                                  |                                          |                                          |                                 |
|---------------------------------|----------------------------|-----------------------------------------|-------------------------------|----------------------------------|------------------------------------------|------------------------------------------|---------------------------------|
|                                 |                            | Average Monthly <sup>[3]</sup>          | Average Weekly <sup>[3]</sup> | Maximum Daily <sup>[3],[4]</sup> | Instantaneous Minimum <sup>[3],[4]</sup> | Instantaneous Maximum <sup>[3],[4]</sup> | Six-Month Median <sup>[3]</sup> |
| Total Suspended Solids (TSS)    | mg/L                       | 30                                      | 45                            | —                                | —                                        | —                                        | —                               |
|                                 | lbs/day                    | 57,546                                  | 86,319                        | —                                | —                                        | —                                        | —                               |
|                                 | % removal <sup>[6]</sup>   | ≥ 85                                    | —                             | —                                | —                                        | —                                        | —                               |
| pH                              | standard units             | —                                       | —                             | —                                | 6.0                                      | 9.0                                      | —                               |
| Oil and Grease                  | mg/L                       | 25                                      | 40                            | —                                | —                                        | 75                                       | —                               |
|                                 | lbs/day                    | 47,955                                  | 76,728                        | —                                | —                                        | 143,865                                  | —                               |
| Settleable Solids               | ml/L                       | 1.0                                     | 1.5                           | —                                | —                                        | 3.0                                      | —                               |
| Turbidity                       | NTU                        | 75                                      | 100                           | —                                | —                                        | 225                                      | —                               |
| Total chlorine residual         | mg/L                       | —                                       | —                             | 0.296                            | —                                        | 2.22                                     | 0.074                           |
|                                 | lbs/day                    | —                                       | —                             | 568                              | —                                        | 4,258                                    | 142                             |
| Ammonia as N                    | mg/L                       | —                                       | —                             | 88.8                             | —                                        | 222                                      | 22.2                            |
|                                 | lbs/day                    | —                                       | —                             | 170,336                          | —                                        | 425,840                                  | 42,584                          |
| Fecal coliform density          | MPN /100 mL <sup>[7]</sup> | [8]                                     |                               |                                  |                                          |                                          |                                 |
| Enterococcus density            | CFU /100 mL <sup>[7]</sup> | [8]                                     |                               |                                  |                                          |                                          |                                 |
| Chronic toxicity <sup>[9]</sup> | Pass or Fail               | —                                       | —                             | Pass                             | —                                        | —                                        | —                               |
| Radioactivity                   | pCi/L                      | [10]                                    |                               |                                  |                                          |                                          |                                 |
| TCDD Equivalents <sup>[3]</sup> | pg/L                       | 0.1443                                  | —                             | —                                | —                                        | —                                        | —                               |
|                                 | lbs/day                    | 0.00000 028                             | —                             | —                                | —                                        | —                                        | —                               |

| Parameter | Units | Effluent Limitations <sup>[1],[2]</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                               |                                  |                                          |                                          |                                 |
|-----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------|------------------------------------------|------------------------------------------|---------------------------------|
|           |       | Average Monthly <sup>[3]</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Average Weekly <sup>[3]</sup> | Maximum Daily <sup>[3],[4]</sup> | Instantaneous Minimum <sup>[3],[4]</sup> | Instantaneous Maximum <sup>[3],[4]</sup> | Six-Month Median <sup>[3]</sup> |
| Notes:    |       | <p><sup>[1]</sup> Mass emission effluent limitations (in lbs/day) are based on the Discharger's conservative estimate for hydraulic flow capacity of the outfall of 230 MGD.</p> <p><sup>[2]</sup> The minimum probable initial dilution used to calculate WQBELs for Ocean Plan Table 3 pollutants and bacteria is 37:1 (<math>D_m=36</math>).</p> <p><sup>[3]</sup> See Attachment A – Definitions.</p> <p><sup>[4]</sup> The maximum daily effluent limitations shall apply to flow weighted 24-hour composite samples. The instantaneous maximum/minimum effluent limitations shall apply to grab samples.</p> <p><sup>[5]</sup> In lieu of the parameter BOD<sub>5</sub> and the BOD<sub>5</sub> levels specified for effluent quality in 40 CFR § 133.102(a)(1), (2), and (3), the parameter CBOD<sub>5</sub> and the CBOD<sub>5</sub> levels specified for effluent quality in 40 CFR § 133.102(a)(4) are substituted and reported by the Discharger.</p> <p><sup>[6]</sup> In each calendar month, the arithmetic mean of CBOD<sub>5</sub> and TSS, by mass, of effluent samples collected at Monitoring Location EFF-002 or EMG-001 as described in the MRP, shall not exceed 15 percent of the arithmetic mean of CBOD<sub>5</sub> and TSS, by mass, of influent samples collected at Monitoring Locations INF-001 and INF-002 as described in the MRP, at approximately the same times during the same periods.</p> <p><sup>[7]</sup> Results may be reported as either Most Probable Number (MPN)/100 mL if the laboratory method used provides results in MPN/100 mL or Colony Forming Units (CFU)/100 mL if the laboratory method used provides results in CFU/100 mL.</p> <p><sup>[8]</sup> See section IV.B.2.b below for the effluent limitations for fecal coliform density and <i>enterococcus</i> density.</p> <p><sup>[9]</sup> The maximum daily effluent limitation shall be reported as "Pass" or "Fail", and "% Effect". See section IV.B.2.c below and section V of Attachment E.</p> <p><sup>[10]</sup> Not to exceed limits specified in Title 17, Division 1, Chapter 5, Subchapter 4, Group 3, Article 3, Section 30253 of the California Code of Regulations. Reference to Section 30253 is prospective, including future changes to any incorporated provisions of federal law, as the changes take effect.</p> |                               |                                  |                                          |                                          |                                 |

- b. **Bacteria Water Quality Based Effluent Limitations (WQBELs):** The discharge at Discharge Point 002 (78" outfall) shall meet the following effluent limitations for fecal coliform density and *enterococcus* density, with compliance measured at Monitoring Location EFF-002 or EMG-001, as described in the MRP:

**Fecal Coliform**



1. The 30-day geometric mean of fecal coliform density shall not exceed 7,400 MPN/100 mL, calculated based on the five most recent samples; and
2. The single sample maximum shall not exceed 14,800 MPN/100 mL.

**Enterococcus**

1. The six-week rolling geometric mean of enterococcus bacteria, calculated weekly, shall not exceed 1,110 CFU/100 mL or MPN/100 mL; and
2. No more than 10 percent of all enterococcus bacteria samples collected in a calendar month shall exceed a statistical threshold value of 4,070 CFU/100 mL or MPN/100 mL.

Compliance with these enterococcus limits shall be evaluated as follows:

- Six-week rolling geometric mean. Compliance with this limit shall be determined weekly by calculating the geometric mean of all enterococcus sample results from the past six weeks.
  - 10 percent of sample. Compliance with this limit shall be determined based on measured sample results. The Discharger shall not report interpolated results. If the Discharger has 9 or fewer sample results in a calendar month, compliance shall be based on the highest result. If the Discharger has 10 to 19 sample results, compliance shall be based on the second highest result, and so on.
- c. **Chronic Toxicity:** The chronic toxicity WQBEL for Discharge Point 001 is expressed as a null hypothesis ( $H_0$ ) and regulatory management decision ( $b$  value) of 0.75 for the chronic toxicity methods in Attachment E – Monitoring and Reporting Program. The null hypothesis for this discharge is:

$H_0$ : Mean response (2.78% effluent)  $\leq$  0.75 mean response (Control).

Results obtained from a chronic toxicity test shall be analyzed using the TST statistical approach (EPA 833-R-10-003, 2010; Appendix A). Compliance with this chronic toxicity WQBEL is demonstrated by rejecting the null hypothesis and reporting “0” = “Pass”.

Percent Effect” (or Effect, in %) = [(Control mean response – IWC mean response)  $\div$  Control mean response]  $\times$  100

**3. Interim Effluent Limitations – Not Applicable**

**C. Land Discharge Specifications – Not Applicable**

**D. Recycling Specifications – Not Applicable**

**V. PERFORMANCE GOALS AND MASS EMISSION BENCHMARKS**

**A. Performance goals – Discharge point 001**

The performance goals for Discharge Point 001 (120” outfall) are prescribed below in Table 7 in this Order/Permit. Performance goals are based upon last 5-year actual

performance data for the OC San's secondary treatment plants and are specified only as an indication of the treatment efficiency of the plants. The performance goals are not considered enforceable effluent limitations or standards for the regulation of discharge from the treatment facility.

The Discharger shall maintain existing treatment levels and the effluent quality at or below the performance goal concentrations. Any two consecutive exceedances of the performance goals shall trigger an investigation into the cause of the exceedance. If the exceedance persists in three successive monitoring periods, the Discharger shall submit a written report to the Santa Ana Water Board and USEPA on the nature of the exceedance, the results of the investigation including the cause of the exceedance.

The Santa Ana Water Board and USEPA may reopen the Order/Permit to modify any of the performance goals if the Discharger submits a request and demonstrates that the change is warranted, including results of completion of GWRS final expansion.

#### **B. Mass Emission Benchmarks – Discharge point 001**

The following 12-month average mass emission benchmarks for Discharge Point 001 (120" outfall) are prescribed below in Table 7. For each parameter with a mass emission benchmark, the Discharger shall report the annual mass emission and the effluent concentrations and flows used to calculate the annual mass emission in the annual pretreatment report and annual receiving water monitoring report (effluent chapter).

These mass emission benchmarks are not enforceable water quality-based effluent limitations. They may be re-evaluated and revised during the five-year permit term. For this Order/Permit, the mass emissions benchmarks (in metric tons per year; MT/yr) were determined based on 2015 through 2019 effluent mass emission and the Discharger's annual average influent flow of 206 MGD projected for 2025 (see section V of the fact sheet in Attachment F).

**Table 7. Performance Goals and Mass Emission Benchmarks – Discharge point 001**

| <b>Parameter</b>                     | <b>Average Monthly Performance Goals (µg/L)</b> | <b>12-month Average Mass Emission Benchmarks (MT/yr)</b> |
|--------------------------------------|-------------------------------------------------|----------------------------------------------------------|
| <b>Marine Aquatic Life Toxicants</b> |                                                 |                                                          |
| Arsenic, total recoverable           | 6.62                                            | 1.88                                                     |
| Cadmium, total recoverable           | 0.24                                            | 0.07                                                     |
| Chromium (VI) <sup>[1]</sup>         | 1.55                                            | 0.44                                                     |
| Copper, total recoverable            | 18.31                                           | 5.21                                                     |
| Lead, total recoverable              | 0.62                                            | 0.18                                                     |
| Mercury, total recoverable           | 0.0071                                          | 0.002                                                    |
| Nickel, total recoverable            | 23.50                                           | 6.69                                                     |
| Selenium, total recoverable          | 21.90                                           | 6.23                                                     |

| Parameter                                       | Average Monthly Performance Goals (µg/L) | 12-month Average Mass Emission Benchmarks (MT/yr) |
|-------------------------------------------------|------------------------------------------|---------------------------------------------------|
| Silver, total recoverable                       | 0.16                                     | 0.05                                              |
| Zinc, total recoverable                         | 46.01                                    | 13.09                                             |
| Cyanide, total recoverable <sup>[2]</sup>       | 5.88                                     | 1.67                                              |
| Ammonia as Nitrogen                             | 36,743                                   | 10,457                                            |
| Total Chlorine Residual <sup>[3]</sup>          | --                                       | 38.09                                             |
| Non-chlorinated Phenols <sup>[4]</sup>          | 1.56                                     | 0.44                                              |
| Chlorinated Phenols <sup>[4]</sup>              | 0.54                                     | 0.15                                              |
| Endosulfan <sup>[4]</sup>                       | 0.011                                    | 0.003                                             |
| Endrin                                          | 0.021                                    | 0.006                                             |
| Hexachlorocyclohexane (HCH) <sup>[4]</sup>      | 0.011                                    | 0.003                                             |
| Radioactivity <sup>[3]</sup>                    | --                                       | --                                                |
| <b>Human Health Toxicants – Non-Carcinogens</b> |                                          |                                                   |
| Acrolein                                        | 10.65                                    | 3.03                                              |
| Antimony                                        | 2.54                                     | 0.72                                              |
| Bis(2-chloroethoxy) methane                     | 10.65                                    | 3.03                                              |
| Bis(2-chloroiso-propyl)ether                    | 4.26                                     | 1.21                                              |
| Chlorobenzene                                   | 4.26                                     | 1.21                                              |
| Chromium (III) <sup>[1]</sup>                   | 1.55                                     | 0.44                                              |
| Di-n-butyl-phthalate                            | 1.80                                     | 0.51                                              |
| Dichlorobenzenes <sup>[4]</sup>                 | 2.13                                     | 0.61                                              |
| Diethyl phthalate                               | 0.76                                     | 0.22                                              |
| Dimethyl phthalate                              | 4.26                                     | 1.21                                              |
| 4,6-dinitro-2-methylphenol                      | 10.65                                    | 3.03                                              |
| 2,4-dinitrophenol                               | 10.65                                    | 3.03                                              |
| Ethylbenzene                                    | 4.26                                     | 1.21                                              |
| Fluoranthene                                    | 2.13                                     | 0.61                                              |
| Hexachlorocyclopentadiene                       | 10.65                                    | 3.03                                              |
| Nitrobenzene                                    | 0.38                                     | 0.11                                              |
| Thallium                                        | 0.20                                     | 0.06                                              |
| Toluene                                         | 0.19                                     | 0.05                                              |

| Parameter                                   | Average Monthly Performance Goals (µg/L) | 12-month Average Mass Emission Benchmarks (MT/yr) |
|---------------------------------------------|------------------------------------------|---------------------------------------------------|
| Tributyltin                                 | 0.25                                     | 0.07                                              |
| 1,1,1-Trichloroethane                       | 4.26                                     | 1.21                                              |
| <b>Human Health Toxicants – Carcinogens</b> |                                          |                                                   |
| Acrylonitrile                               | 4.26                                     | 1.21                                              |
| Aldrin                                      | 0.004                                    | 0.001                                             |
| Benzene                                     | 4.26                                     | 1.21                                              |
| Benzidine <sup>[3]</sup>                    | --                                       | 0.004                                             |
| Beryllium                                   | 1.07                                     | 0.30                                              |
| Bis(2-chloroethyl) ether                    | 2.13                                     | 0.61                                              |
| Bis(2-ethylhexyl) phthalate                 | 3.90                                     | 1.11                                              |
| Carbon tetrachloride                        | 4.26                                     | 1.21                                              |
| Chlordane <sup>[4,5]</sup>                  | 0.004                                    | 0.001                                             |
| Chlorodibromomethane                        | 4.25                                     | 1.21                                              |
| Chloroform                                  | 16.60                                    | 4.72                                              |
| DDT <sup>[4]</sup>                          | 0.011                                    | 0.003                                             |
| 1,4-dichloro-benzene                        | 0.43                                     | 0.12                                              |
| 3,3'-dichloro-benzidine                     | 1.47                                     | 0.42                                              |
| 1,2-dichloroethane                          | 4.26                                     | 1.21                                              |
| 1,1-dichloro-ethylene                       | 4.26                                     | 1.21                                              |
| Dichlorobromomethane                        | 8.98                                     | 2.56                                              |
| Dichloro-methane                            | 4.26                                     | 1.21                                              |
| 1,3-dichloropropene                         | 4.26                                     | 1.21                                              |
| Dieldrin                                    | 0.007                                    | 0.002                                             |
| 2,4-dinitrotoluene                          | 10.65                                    | 3.03                                              |
| 1,2-diphenyl-hydrazine                      | 2.13                                     | 0.61                                              |
| Halomethanes                                | 0.43                                     | 0.12                                              |
| Heptachlor                                  | 0.009                                    | 0.003                                             |
| Heptachlor epoxide                          | 0.004                                    | 0.001                                             |
| Hexachlorobenzene <sup>[3]</sup>            | --                                       | 0.01                                              |
| Hexachlorobutadiene                         | 2.13                                     | 0.61                                              |
| Hexachloroethane                            | 2.13                                     | 0.61                                              |

| Parameter                                               | Average Monthly Performance Goals (µg/L)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 12-month Average Mass Emission Benchmarks (MT/yr) |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Isophorone                                              | 2.13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.61                                              |
| N-Nitroso-dimethylamine                                 | 10.65                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3.03                                              |
| N-Nitrosodi-N-propylamine                               | 10.65                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3.03                                              |
| N-Nitroso-diphenylamine                                 | 2.13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.61                                              |
| Polycyclic Aromatic Hydrocarbons (PAHs) <sup>[4]</sup>  | 1.59                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.45                                              |
| Total Polychlorinated Biphenyls (PCBs) <sup>[3,4]</sup> | --                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.001                                             |
| TCDD Equivalents <sup>[3,4]</sup>                       | --                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2.01×10 <sup>-7</sup>                             |
| 1,1,2,2-Tetrachloroethane                               | 4.26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.21                                              |
| Tetrachloroethylene                                     | 1.58                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.45                                              |
| Toxaphene <sup>[3]</sup>                                | --                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.01                                              |
| Trichloroethylene                                       | 4.26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.21                                              |
| 1,1,2-Trichloroethane                                   | 4.26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.21                                              |
| 2,4,6-Trichlorophenol                                   | 0.54                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.15                                              |
| Vinyl chloride                                          | 4.26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.21                                              |
| Notes:                                                  | <p><sup>[1]</sup> The Discharger may at its option meet both the chromium III and the chromium VI performance goals or mass emission benchmarks by analyzing for total recoverable chromium.</p> <p><sup>[2]</sup> Cyanide: If the Discharger can demonstrate to the satisfaction of the Santa Ana Water Board (subject to USEPA approval) that an analytical method is available to reliably distinguish between strongly and weakly complexed cyanide, effluent limitations for cyanide may be met (or performance goals may be evaluated) by the combined measurement of free cyanide, simple alkali metal cyanides, and weakly complexed organometallic cyanide complexes. In order for the analytical method to be acceptable, the recovery of free cyanide from metal complexes must be comparable to that achieved by the approved method in 40 CFR § 136.</p> |                                                   |

| Parameter | Average Monthly Performance Goals (µg/L)                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 12-month Average Mass Emission Benchmarks (MT/yr) |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|           | <sup>[3]</sup> Enforceable effluent limitations for these parameters have been determined due to RP.<br><sup>[4]</sup> See Attachment A for definition of terms.<br><sup>[5]</sup> Chlordane: The Discharger may temporarily suspend the monitoring requirements for alpha- and gamma-chlordene, if analytical standards for these compounds are not available. However, the Discharger is required to resume detection and quantification practices as soon as standards become available. |                                                   |

## VI. RECEIVING WATER LIMITATIONS

### A. Surface Water Limitations

The discharge of waste by the Discharger shall not cause a violation of the California Ocean Plan water quality objectives and USEPA water quality criteria specified below. Compliance shall be determined from samples collected at stations representative of the area within the waste field where initial dilution is completed.

#### 1. Bacterial Characteristics

##### a. State Water Board Water-Contact Objectives

- i. Within a zone bounded by the shoreline and a distance of 1,000 feet from the shoreline or the 30-foot depth contour, whichever is further from the shoreline, and in areas outside this zone used for water contact sports, as determined by the Santa Ana Water Board or USEPA (i.e., waters designated as REC-1), but including all kelp beds, the following bacterial water quality objectives shall be maintained throughout the water column:

##### Fecal Coliform

A 30-day geometric mean of fecal coliform density shall not exceed 200/100 mL, calculated based on the geometric mean of the five most recent samples from each site, and a single sample maximum shall not exceed 400/100 mL.

##### Enterococci

A 6-week rolling geometric mean of enterococci, calculated weekly, shall not exceed 30 CFU or MPN per 100 mL. The geometric mean value shall be applied based on a statistically sufficient number of samples, which is generally not less than five samples distributed over a 6-week period.

A statistical threshold value of 110 CFU or MPN per 100 mL shall not be exceeded by more than 10 percent of all enterococci samples collected in a calendar month, calculated in a static manner.

- ii. The Initial Dilution Zone of wastewater outfalls shall be excluded from designation as kelp beds for purposes of bacterial standards. Adventitious assemblages of kelp on waste discharge structures (e.g., outfall pipes and multiport diffusers) do not constitute kelp beds for purposes of bacterial standards.

b. State Water Board Beach Notification Levels

Minimum protective bacteriological standards for coastal waters adjacent to public beaches and for public water-contact sports areas in ocean waters are established in the California Code of Regulations, Title 17 (beginning at div. 1, ch. 5, section 7958 et seq.), which are not water quality objectives. When a public beach or public water-contact sports area fails to meet these standards, the California Department of Public Health or the local public health officer may post with warning signs or otherwise restrict use of the public beach or public water-contact sports area until the standards are met. The regulations impose more frequent monitoring and more stringent posting and closure requirements on certain high-use public beaches that are located adjacent to a storm drain that flows in the summer.

c. State Water Board Shellfish Harvesting Standards

At all areas where shellfish may be harvested for human consumption, as determined by the Santa Ana Water Board and USEPA, the following bacterial objectives shall be maintained throughout the water column: The median total coliform density shall not exceed 70 per 100 mL, and not more than 10 percent of the samples shall exceed 230 per 100 mL.

d. USEPA Recreational Water Quality Criteria (RWQC)

Ocean waters beyond the outer limit of the territorial sea shall not exceed the following RWQC for *Enterococcus* density beyond the zone of initial dilution in areas where primary contact recreation occurs. The 2012 RWQC describes the criteria are designed to protect "primary contact recreation", including swimming, bathing, surfing, water skiing, tubing, water play by children, and similar water contact activities where a high degree of bodily contact with the water, immersion and ingestion are likely (Recreational Water Quality Criteria, EPA-820-F-12-058, 2012, p. 6.).

Estimated illness rate of 32 per 1,000 primary contact recreators:

A 30-day geometric mean shall not exceed 30 CFU or MPN per 100 mL, which is calculated based on a statistically sufficient number of samples (generally not less than five samples equally spaced over any 30-day period).



A statistical threshold value corresponding to the 90th percentile of the same water quality distribution shall not exceed 110 CFU or MPN per 100 mL in the same 30-day interval.

**2. Physical Characteristics**

- a. Floating particulates and grease and oil shall not be visible.
- b. The discharge of waste shall not cause aesthetically undesirable discoloration of the ocean surface.
- c. Natural light shall not be significantly reduced at any point outside the initial dilution zone as the result of the discharge of waste.
- d. The rate of deposition of inert solids and the characteristics of inert solids in ocean sediments shall not be changed such that benthic communities are degraded.
- e. Trash from the discharge shall not be present in ocean waters, along shorelines or adjacent areas in amounts that adversely affect beneficial uses or cause nuisance.

**3. Chemical Characteristics**

- a. The dissolved oxygen concentration shall not at any time be depressed more than 10 percent from that which occurs naturally, as the result of the discharge of oxygen demanding waste materials.
- b. The pH shall not be changed at any time more than 0.2 units from that which occurs naturally.
- c. The dissolved sulfide concentration of waters in and near sediments shall not be significantly increased above that present under natural conditions.
- d. The concentration of substances, set forth in Chapter II, Table 3 of the California Ocean Plan, in marine sediments shall not be increased to levels which would degrade indigenous biota.
- e. The concentration of organic materials in marine sediments shall not be increased to levels which would degrade marine life.
- f. Nutrient materials shall not cause objectionable aquatic growths or degrade indigenous biota.
- g. Numerical water quality objectives established in Table 3 of the California Ocean Plan shall not be exceeded as a result of discharges from the facility through Discharge Points 001 and 002 (as computed using an applicable dilution factor).

**4. Biological Characteristics**

- a. Marine communities, including vertebrate, invertebrate, and plant species, shall not be degraded.
- b. The natural taste, odor, and color of fish, shellfish, or other marine resources used for human consumption shall not be altered.

- c. The concentration of organic materials in fish, shellfish, or other marine resources used for human consumption shall not bioaccumulate to levels that are harmful to human health.

**5. Radioactivity**

- a. Discharge of radioactive waste, which meets the definition of “pollutant” at 40 CFR § 122.2, shall not degrade marine life.

**B. Groundwater Limitations – Not Applicable**

**VII. PROVISIONS**

**A. Standard Provisions**

1. The Discharger shall comply with all Standard Provisions included in Attachment D.
2. The Facility shall be protected to reduce infrastructure vulnerability to extreme wet weather events, flooding, storm surges, and projected sea level rise resulting from current and future impacts associated with climate change.
3. This Order/Permit expires on July 31, 2026, after which, the terms and conditions of this Order/Permit are automatically continued pending issuance of a new Order, provided that all requirements of USEPA's NPDES regulations at 40 CFR 122.6 and the State's regulations at CCR title 23, section 2235.4 regarding the continuation of expired permits and waste discharge requirements are met.
4. The Discharger shall comply with the following Santa Ana Water Board standard provisions and USEPA Region 9 standard provisions. In the event that there is any conflict, duplication, or overlap between provisions or requirements specified by this Order/Permit, the more stringent provision or requirement shall apply:
  - a. Consistent with the requirements in Attachment D. Standard Provisions – Reporting, section V.E. Twenty-Four Hour Reporting of this Order/Permit, in the event the Discharger does not comply or will be unable to comply for any reason, with any prohibition, effluent limitation, or receiving water limitation of this Order/Permit, the Discharger shall notify the Santa Ana Water Board by telephone (951) 782-4130 and USEPA by telephone (415) 947-4179 and by email R9NPDES@epa.gov within 24 hours of having knowledge of such noncompliance and shall confirm this notification in writing within five (5) working days, unless the Santa Ana Water Board and USEPA waives written confirmation. Other noncompliance requires written notification as above, at the time of the next self-monitoring report (SMR)/discharge monitoring report (DMR).
  - b. Neither the treatment nor the discharge of wastes shall cause, or threaten to cause, a pollution, contamination, or nuisance or as defined in California Water Code section 13050.
  - c. The Discharger shall take all reasonable steps to minimize and correct any adverse impact to receiving waters resulting from noncompliance with this Order/Permit, including accelerated or additional monitoring as may be necessary to determine the nature and impact of the noncomplying discharge.

- d. The Discharger shall file with the Santa Ana Water Board and USEPA a Report of Waste Discharge/application for permit modification at least 180 days before making any material change in the character, location, or volume of the discharge. A material change includes, but is not limited to, the following: Adding a major industrial waste discharge to a discharge of essentially domestic sewage, or adding a new process or product by an industrial facility resulting in a change in the character of the waste; significantly changing the disposal method or location, such as changing the disposal to another drainage area or waterbody; significantly changing the method of treatment; increasing the treatment plant design capacity beyond that specified in this Order/Permit.
- e. The Discharger shall maintain a full and complete copy of this Order/Permit at the site so that it is available to site operating personnel, the Santa Ana Water Board, the California State Water Resources Control Board (hereinafter "State Water Board"), and USEPA at all times. Key operating personnel shall be familiar with its content.
- f. Collected screenings, sludge, and other solids removed from liquid wastes shall be managed in accordance with federal, state, and local regulations (see Attachment G - Biosolids).
- g. Permit Transfer. In the event of any change in control or ownership of land or waste discharge facility presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of this Order/Permit by letter, a copy of which shall be sent to the Santa Ana Water Board and USEPA. As an alternative to transfers under 40 CFR § 122.61(a), this Permit/Order may be automatically transferred to a new permittee if: The minimum 30 day notice to the USEPA Water Division Director includes a written agreement between the Discharger and new permittee containing a specific date for transfer of permit responsibility, coverage, and liability between them; and the Director does not notify the Discharger and the proposed new permittee of his/her intent to modify or revoke and reissue the Permit. (A modification under this paragraph may also be a minor modification under 40 CFR § 122.63.) If this notice is not received, the transfer is effective on the date specified in the written agreement between the Discharger and the new permittee. (40 CFR § 122.61(b).)
- h. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
- i. The requirements prescribed herein do not authorize the commission of any act causing injury to the property of another, nor protect the Discharger from its liabilities under federal, state, or local laws, nor guarantee the Discharger a capacity right in the receiving waters.
- j. Termination of Permits. The causes for terminating a permit during its term, or for denying a permit renewal application, are found at 40 CFR § 122.64 and are incorporated into this Order/Permit by reference.

- k. Availability of Reports. Except for data determined to be confidential under 40 CFR § 2, all reports prepared in accordance with the terms of this Order/Permit shall be available for public inspection at the offices of the Santa Ana Water Board and USEPA. As required by the CWA, permit applications, permits, and effluent data shall not be considered confidential. (Pursuant to CWA section 308.)
- l. Severability. The provisions of this Order/Permit are severable, and if any provision of this Order/Permit, or the application of any provision of this Order/Permit to any circumstance, is held invalid, the application of such provision to other circumstances, and the remainder of this Order/Permit shall not be affected thereby. (Pursuant to CWA section 512.)
- m. Civil and Criminal Liability. Except as provided in standard conditions (Attachment D) on Bypass and Upset, nothing in this Order/Permit shall be construed to relieve the Discharger from civil or criminal penalties for noncompliance. (Pursuant to CWA section 309.)
- n. Oil and Hazardous Substances Liability. Nothing in this Order/Permit shall be construed to preclude the institution of any legal action or relieve the Discharger from any responsibilities, liabilities, or penalties to which the Discharger is or may be subject under CWA section 311.
- o. State, Tribe, or Territory Law. Nothing in this Order/Permit shall be construed to preclude the institution of any legal action or relieve the operator from any responsibilities, liabilities, or penalties established pursuant to any applicable State, Tribe, or Territory law or regulation under authority preserved by CWA section 510.

## **B. Monitoring and Reporting Program (MRP) Requirements**

The Discharger shall comply with the MRP, and future revisions thereto, in Attachment E of this Order/Permit.

## **C. Special Provisions**

### **1. Reopener Provisions**

This Order/Permit may be reopened for modification prior to its expiration date in accordance with the requirements set forth at 40 CFR § 122 and 124 to:

- a. Include an effluent limitation if monitoring establishes that the discharge causes, has the reasonable potential to cause, or contributes to an excursion above a California Ocean Plan Table 3 water quality objective.
- b. Revise or modify the Order/Permit if present or future investigations demonstrate that the discharges governed by this Order/Permit have or will have a reasonable potential to cause or contribute to, or will cease to have, adverse impacts on water quality or beneficial uses of the receiving waters.
- c. Include effluent limitations for pollutants determined to be present in the discharge.

- d. Address any changes in State or federal plans, policies, or regulations that would affect the quality requirements of the discharge.
- e. Include conditions or effluent or receiving water limitations based on newly available information (e.g., effluent toxicity, dilution, significant change in waste flow, strategic process study results, etc.).
- f. Include revised effluent limitations or conditions to address acute or chronic toxicity in the effluent or receiving water, as a result of the discharge; or to implement new, revised, or newly interpreted water quality standards applicable to acute or chronic toxicity.
- g. Revise performance goals or mass emission benchmarks contained in this Order/Permit (see Section V).
- h. Incorporate a promulgated CWA section 405(d) standard for sewage sludge use or disposal more stringent than any requirements for sludge use or disposal in this Order/Permit, or control a pollutant or practice not limited in this Order/Permit. (40 CFR § 122.44(c).)
- i. Incorporate any effluent standard or discharge prohibition (including any schedule of compliance specified in such effluent standard or prohibition) established under CWA section 307(a) for a pollutant which is present in the discharge, and such standard or prohibition is more stringent than any requirement for that pollutant in this Order/Permit.
- j. This Order/Permit may be modified, or revoked and reissued, based on the results of Magnuson-Stevens Fishery Conservation and Management Act and/or Endangered Species Act section 7 consultation(s) with the National Marine Fisheries Service and/or U.S. Fish and Wildlife Service.
- k. This Order/Permit may be reopened and modified if new or revised water quality objectives or total maximum daily loads (TMDLs) come into effect for contiguous water bodies (whether statewide, regional, or site-specific). In such cases, effluent limitations in this Order/Permit may be modified as necessary to reflect the updated water quality objectives and wasteload allocations in the TMDLs. Adoption of effluent limitations contained in this Order/Permit is not intended to restrict in any way future modifications based on legally adopted water quality objectives, TMDLs, or as otherwise permitted under federal regulations governing NPDES permit modifications.
- l. Or as otherwise authorized by law.

## **2. Southern California Bight Monitoring Exchange**

The MRP (Attachment E) may be modified by the Santa Ana Water Board and USEPA to enable the Discharger to participate in comprehensive regional monitoring activities conducted in the Southern California Bight during the term of this permit. The intent of regional monitoring activities is to maximize the efforts of all monitoring partners using a cost-effective monitoring design and to best utilize the pooled scientific resources of the region. During these coordinated monitoring efforts, the



Discharger's sampling and analytical effort may be reallocated to provide a regional assessment of the impact of wastewater discharges to the Southern California Bight; however, certain core elements (i.e., monthly water quality monitoring, quarterly REC-1 water quality monitoring, quarterly benthic monitoring, semi-annual trawl fish monitoring, and weekly Orange County Regional Shoreline REC-1 cooperative monitoring) shall remain unchanged. Anticipated modifications to the monitoring program will be coordinated so as to provide a comprehensive picture of the ecological and statistical significance of monitoring results and to determine cumulative impacts of various pollutant sources. If predictable relationships among the biological, water quality and effluent monitoring variables can be demonstrated, it may be appropriate to decrease the Discharger's monitoring effort. Conversely, the monitoring program may be intensified if it appears that the objectives cannot be achieved through the Discharger's existing monitoring program. These changes will improve the overall effectiveness of monitoring in the Southern California Bight. Minor changes may be made without further public notice.

### **3. Special Studies, Technical Reports and Additional Monitoring Requirements**

#### **a. Toxicity Reduction Requirements**

If the discharge exceeds an effluent limitation for toxicity specified in Table 5 and 6, the Discharger shall conduct a Toxicity Reduction Evaluation (TRE) defined in Attachment A, in accordance with an approved TRE Work Plan, and take actions to mitigate the impact of the discharge and prevent recurrence of toxicity. A TRE is not an effluent limitation for toxicity.

##### **i. TRE Work Plan**

Within 90 days of the Order/Permit effective date, the Discharger shall prepare and submit a copy of their Initial Investigation Toxicity Reduction Evaluation (TRE) Workplan to the Santa Ana Water Board and USEPA for review and approval by the Santa Ana Water Board Executive Officer and USEPA Water Division Director. The TRE Work Plan must be developed in accordance with USEPA guidance, EPA manual *Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants* (EPA/833/B-99/002, 1999), and be of adequate detail to allow the Discharger to immediately initiate a TRE as required in this Provision. This plan shall outline the procedures for identifying the source(s) of, and reducing or eliminating effluent toxicity and shall describe the steps the Discharger intends to follow if toxicity is measured above an acute or chronic WET permit limit. The TRE Work Plan should include, at minimum:

- (a) A description of the investigation and evaluation techniques that would be used to identify potential causes and sources of the exceedance, effluent variability, and/or efficiency of the treatment system in removing toxic substances. This shall include a description of an accelerated chronic toxicity testing program;
- (b) A description of the methods to be used for investigating and maximizing

in-house treatment system efficiency and good housekeeping practices, and a list of all chemicals used in operations at the facility;

- (c) A description of the evaluation process to be used to determine if implementation of a more detailed Toxicity Reduction Evaluation and Toxicity Identification Evaluation (TRE/TIE) is necessary;
- (c) If a TIE is necessary, an indication of who would conduct the TIEs (i.e., an in-house expert or outside contractor).

**ii. Accelerated Toxicity Monitoring Specifications and TRE/TIE Initiation**

When a WET permit limit is exceeded during regular toxicity monitoring, and the testing meets all test acceptability criteria, the Discharger shall initiate accelerated monitoring as required below. The Discharger shall initiate a TRE to address effluent toxicity if any WET testing results exceed the WET permit limit during accelerated monitoring.

- (a) If an acute or chronic WET permit limit is exceeded and the source(s) of toxicity is easily identified (e.g., a temporary plant upset), then the Discharger shall make necessary corrections to the facility and shall conduct one (1) additional toxicity test using the same species and test method. This test shall begin within 14 days of notification by the laboratory of the exceedance of the WET permit limit. If the additional toxicity test does not exceed the WET permit limit, then the Discharger may return to their regular testing frequency.
- (b) If an acute or chronic WET permit limit is exceeded and the source of toxicity is not known, then the Discharger shall conduct four (4) additional toxicity tests using the same species and test method, approximately every two weeks, over an eight (8) week period. This testing shall begin within 14 days of notification by the laboratory of the exceedance of the WET permit limit. The following protocol shall be used for accelerated monitoring and TRE initiation:
  - (1) If none of the additional four (4) consecutive toxicity tests exceed the WET permit limit, then the Discharger may cease accelerated monitoring and resume their regular toxicity monitoring. However, notwithstanding the accelerated monitoring results, if there is adequate evidence of a pattern of effluent toxicity, the Santa Ana Water Board Executive Officer and/or USEPA Water Division Director may require that the Discharger initiate a TRE.
  - (2) If one of the additional toxicity tests (in paragraphs ii(a) or ii(b) above) exceeds the WET permit limit, then, within 14 days of receipt of this test result, the Discharger shall cease accelerated monitoring and initiate a TRE using as guidance, based on the type of treatment facility, USEPA manual *Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants* (EPA/ 833/B-99/002, 1999). In conjunction, within 30 days of notification by the laboratory of any



test result exceeding the WET permit limits during accelerated monitoring, the Discharger shall develop and submit a TRE Action Plan to the Santa Ana Water Board and USEPA, which shall include specific actions undertaken by the Discharger to investigate, identify, and correct the cause(s) of toxicity, including a TRE WET monitoring schedule; specific actions the Discharger will take to mitigate the impact of the discharge and prevent the recurrence of toxicity; and a schedule for these actions.

- (c) The Discharger may initiate a TIE as part of a TRE to identify the causes of toxicity using the same species and test method and, as guidance, USEPA test method manuals: *Toxicity Identification Evaluation: Characterization of Chronically Toxic Effluents*, Phase I (EPA/600/6-91/005F, 1992); *Methods for Aquatic Toxicity Identification Evaluations, Phase II Toxicity Identification Procedures for Samples Exhibiting Acute and Chronic Toxicity* (EPA/600/R-92/080, 1993); *Methods for Aquatic Toxicity Identification Evaluations, Phase III Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity* (EPA/600/R-92/081, 1993); and *Marine Toxicity Identification Evaluation (TIE): Phase I Guidance Document* (EPA/600/R-96-054, 1996).

**b. Dilution Model Update**

No later than 4 years after the effective date of this Order/Permit, the Discharger shall submit a work plan for an updated dilution modeling that is reflective of current operations, including the increase in RO concentrate that may be discharged upon completion of the groundwater replenishment system (GWRS) final expansion project. The purpose of the study is to determine initial dilution under a reasonable worst-case scenario (e.g., strongest density stratification and zero current). The discharger shall include explanation of the conditions and data used, assessment of compliance with applicable water quality objectives, and determination of an appropriate dilution ratio for each outfall. Specifically, the work plan shall include:

- Model inputs and assumptions,
- Describe trapping levels and any boundary effects in the observed range of the ambient density stratification,
- Assume zero current, consistent with the California Ocean Plan,
- Account for a decrease in flow and an increase in pollutant concentrations,
- Include WET assessment using synthetic effluent to approximate future final conditions, and
- Include a sensitivity analysis for how different inputs affect the initial dilution, including effluent temperature and density.

**c. Ocean Outfall Condition Assessment and Scoping Study**

Within 180 days after completion of outfall condition assessment project for Discharge Point 001 (120" outfall) or Discharge Point 002 (78" outfall) if performed during this permit term, the Discharger shall submit a report of each

comprehensive ocean outfall condition assessment to the Santa Ana Water Board and USEPA, which shall include, but not limited to:

- Field work findings (e.g., Barnacle Collar removal and manhole cover removal/ replacement),
- Any environmental impacts caused during the field works,
- Videographic/photographic records of the interior of the outfall pipe and ballast and manhole cover conditions,
- 3-D mapping,
- Analysis of the impacts of low flows on diffuser hydraulics and plugging, and
- The potential rehabilitation projects recommendation.

#### **4. Best Management Practices and Pollution Prevention**

##### **a. Pollutant Minimization Program**

The Discharger shall develop and conduct a Pollutant Minimization Program (PMP) as described below when there is evidence (e.g., sample results reported as detected, but not quantified (DNQ) when the effluent limitation is less than the method detection limit (MDL), sample results from analytical methods more sensitive than those methods required by this Order/Permit, presence of whole effluent toxicity, health advisories for fish consumption, results of benthic or aquatic organism tissue sampling) that a pollutant is present in the effluent above an effluent limitation and either:

- i. The concentration of the pollutant is reported as DNQ and the effluent limitation is less than the reported minimum level (ML), using definitions described in Attachment A and reporting protocols described in MRP section XII.B.4.
- ii. The concentration of the pollutant is reported as not detected (ND) and the effluent limitation is less than the MDL, using definitions described in Attachment A and reporting protocols described in MRP section XII.B.4.

By March 1 of each year, the Discharger shall submit its annual PMP Report to the Santa Ana Water Board and USEPA, for the previous calendar year. The PMP report shall include, but not be limited to, the following actions and submittals acceptable to the Santa Ana Water Board and USEPA:

- i. An annual review and semi-annual monitoring of potential sources of the reportable pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling;
- ii. Quarterly monitoring for the reportable pollutant(s) in the influent to the wastewater treatment system;
- iii. Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable pollutant(s) in the effluent at or below the effluent limitation;

- iv. Implementation of appropriate cost-effective control measures for the pollutant, consistent with the control strategy; and
- v. An annual status report that shall be sent to the Santa Ana Water Board and USEPA including: All PMP monitoring results for the previous year; a list of potential sources of the reportable pollutant(s); a summary of all actions undertaken pursuant to the control strategy; and a description of actions to be taken the following year.

**b. Storm Water Management Plan**

Onsite storm water at this POTW shall be captured, treated, and discharged with the treated municipal wastewater regulated under this Order/Permit. The Discharger shall file with the Santa Ana Water Board and USEPA, within 180 days of the effective date of this Order/Permit, an updated Storm Water Management Plan for discharges of storm water associated with industrial activities excluding construction activities at its treatment/reclamation plants to prevent trash being present in the discharge. The Storm Water Management Plan must include the trash management to incorporate the 2015 trash amendments of the Ocean Plan.

**5. Construction, Operation and Maintenance Specifications**

- a. The Discharger's wastewater treatment plants shall be supervised and operated by persons possessing certificates of appropriate grade, pursuant to Title 23, Division 3, Chapter 26 of the California Code of Regulations. The Discharger shall report annually to the Santa Ana Water Board and USEPA a roster of such personnel, including job titles, duties, and level of State certification for each individual.
- b. The Discharger shall develop an Operation and Maintenance Manual (O&M Manual). If an O&M Manual has been developed, then the Discharger shall update it as necessary to conform to the most recent plant changes and requirements. The O&M Manual shall be readily available to operating personnel onsite. The O&M Manual shall include:
  - i. A detailed description of safe and effective operation and maintenance of treatment processes, process control instrumentation, and equipment.
  - ii. A description of the treatment plant organization showing the number of employees, duties and qualifications, plant attendance schedules (daily, weekends and holidays, part-time, etc.), and emergency contact information. The description should include documentation that the personnel are knowledgeable and qualified to operate the treatment facility so as to achieve the required level of treatment at all times.
  - iii. A description of laboratory and quality assurance procedures.
  - iv. Process and equipment inspection and maintenance schedules.

- v. A description of safeguards (e.g., standby or emergency power and/or storage capacity or other means) to assure that, should there be reduction, loss, or failure of electric power, the Discharger will be able to comply with the requirements of this Order/Permit.
- vi. A description of and reference to the most current spill preventive and contingency plan for controlling accidental discharges, and for minimizing the effect of such events (see the paragraph c below).
- c. **Spill Preventive and Contingency Plan (SPCP).** The Discharger shall file with the Santa Ana Water Board and USEPA, within 180 days after the effective date of this Order/Permit, the Discharger's spill preventive (fail-safe) and contingency plan (response and cleanup) in an up-to-date condition. The Discharger shall amend this plan whenever there is a change (e.g., in the design, construction, operation, or maintenance of the Facility) which materially affects the potential for spills and the response required for each potential spill. The Discharger shall review and amend the plan as appropriate after each spill from the POTW. At a minimum, this plan shall:
  - i. Identify the possible sources of accidental discharges, untreated or partially treated waste bypass, overflows, and contaminated drainage that reach water bodies including dry channels and beach sands. Loading and storage areas, power outage, waste treatment outage, and failure of process equipment, tanks, and collection system sewer pipes and pump stations should be considered.
  - ii. Evaluate the effectiveness of present facilities and procedures and when they become operational. Describe present facilities and procedures needed for effective preventive and contingency plans.
  - iii. Describe any new facilities and procedures needed. Predict the effectiveness of the proposed facilities and procedures and provide an implementation schedule containing interim and final dates when they will be constructed, implemented, or operational.
  - iv. Describe the activities and protocols to address clean-up of spills and containment measures, public notifications, monitoring, and the procedures to be carried out if floatable material is visible on the water surface near the discharge point or has been washed ashore.
  - v. Describe proposed and completed training programs and schedules to train and familiarize plant operating personnel with the Discharger's SPCP for controlling accidental discharges and for minimizing the effects of such events. (California Water Code sections 13267(b) and 13268.)

## **6. Special Provisions for Publicly-Owned Treatment Works (POTWs)**

- a. **Biosolids.** The Discharger shall manage its sludge and biosolids in accordance with federal regulations (40 CFR § 257, 258 and 503) and the requirements specified in Attachment G of this Order/Permit.

- b. Pretreatment.** The Discharger shall implement and enforce its approved pretreatment program in accordance with federal pretreatment regulations (40 CFR § 403); pretreatment standards promulgated under CWA sections 307(b), 307(c), 307(d), and 402(b); pretreatment requirements specified under 40 CFR § 122.44(j); and the requirements specified in Attachment H of this Order/Permit.
- c. Collection System.** The Discharger is subject to the requirements of and must comply with State Water Resources Control Board (State Water Board) Order 2006-0003-DWQ, Statewide General Waste Discharge Requirements for Sanitary Sewer Systems, including monitoring and reporting requirements as amended by State Water Board Order WQ 2013-0058-EXEC and any subsequent order. Regardless of the coverage obtained under the General Order, as amended, the Discharger's collection system is part of the POTW that is subject to this Order/Permit. As such, pursuant to federal regulations, the Discharger must properly operate and maintain its collection system (40 CFR § 122.41(e)), report any non-compliance (40 CFR § 122.41(l)(6) and (7)), and mitigate any discharge from the collection system in violation of this Order/Permit (40 CFR § 122.41(d)).
- d. Resource Recovery from Anaerobically Digestible Material.** If the Discharger will receive hauled-in anaerobically digestible material for injection into an anaerobic digester, the Discharger shall notify the Santa Ana Water Board and USEPA and develop and implement Standard Operating Procedures for this activity. The Standard Operating Procedures shall be developed prior to receiving hauled-in anaerobically digestible material. The Standard Operating Procedures shall address material handling, including unloading, screening, or other processing prior to anaerobic digestion; transportation; spill prevention; and spill response. In addition, the Standard Operating Procedures shall address avoidance of the introduction of materials that could cause interference, pass-through, or upset of the treatment processes; avoidance of prohibited material; vector control; odor control; operation and maintenance; and the disposition of any solid waste segregated from introduction to the digester. The Discharger shall train its staff on the Standard Operating Procedures and shall maintain records for a minimum of five years for each load received, describing the hauler, waste type, and quantity received. In addition, the Discharger shall maintain records for a minimum of five years for the disposition, location, and quantity of cumulative pre-digestion-segregated solid waste hauled offsite.
- e. Ensuring Adequate Treatment Capacity.** The Discharger shall submit a written report to the Santa Ana Water Board Executive Officer and USEPA Water Division Director within 90 days after the monthly average daily dry-weather influent flow rate equals or exceeds 75 percent of the daily dry-weather design flow of the treatment plants (i.e.,  $0.75 \times 332 \text{ MGD} = 249 \text{ MGD}$ ). The Discharger's senior administrative officer shall sign a letter in accordance with the Standard Provisions (Attachment D) which transmits the report and certifies that the policy-making body is adequately informed of the influent flow rate relative to the POTW design capacity. The report shall include the following: Daily average influent flow



for the calendar month, the date on which the maximum daily flow occurred, and the rate of that maximum flow; the Discharger's best estimate of when the daily average influent flow for a calendar month will equal or exceed the design capacity of the treatment plants (i.e., 332 MGD); and the Discharger's intended schedule for studies, design, and other steps needed to provide additional treatment for the wastewater from the collection system before the waste flow exceeds the capacity of the POTW.

- f. Asset Management.** The Discharger shall develop an asset management program to cover the POTW. The Discharger shall:
- i. Develop and utilize an asset management program within two years of the effective date of this Order/Permit. This program shall include a detailed inventory of critical assets; condition rating and/or likelihood of failure of said assets; rehabilitation and replacement planning, capacity assurance planning, and maintenance strategy to ensure the Discharger's system meets a desired level of service and plan for future needs and requirements; and funding source to support the planned asset maintenance, rehabilitation, and replacement activities. Critical assets may include, but are not limited to sewer lines, manholes, outfalls, pump stations, force mains, and wastewater treatment facility assets.
  - ii. Create and submit to the Santa Ana Water Board and USEPA an Asset Management Plan (AMP) within one year of the effective date of this Order/Permit. The AMP shall be updated and re-evaluated every five years. The AMP shall include the following components: A Rehabilitation and Replacement Plan identifying and prioritizing upcoming rehabilitation and replacement projects for critical assets and outlining a proposed schedule for completion of each project; a Maintenance Plan identifying major maintenance activities, frequency performed for critical assets, and estimates of ongoing and projected costs of maintenance activities; and a Sewer Collection System Map incorporating assets from the asset management inventory. Finally, the AMP shall include estimated costs for the Rehabilitation and Replacement Plan and the Maintenance Plan. Expenses may include operational, administrative, interest, or capital expenses. The cost estimate shall include a determination of whether the planned expenditures are capital or operational and the source of funds: user or connection fees, grants, bonds, or reserves.

## **7. Other Special Provisions**

- a. Monitoring Data Accessibility.** The Discharger shall make monitoring data accessible to the public via the Internet. Within 180 days of the effective date of this Order/Permit, the Discharger shall submit a report to the Santa Ana Water Board and USEPA that updates the Discharger's plans and activities making monitoring data accessible to the public via the Internet, including implementation schedules. The Santa Ana Water Board and USEPA shall be informed of any change, in writing, within 30 days of the change.

## **8. Compliance Schedules – Not Applicable**

### **VIII. COMPLIANCE DETERMINATION**

#### **A. Effluent Limitations**

Compliance with the effluent limitations contained in section IV of this Order/Permit shall be determined as specified below.

1. Sufficient sampling and analysis shall be required to determine compliance with the effluent limitation. If the analytical result of any single sample (daily discharge) monitored monthly, or less frequently, exceeds the AMEL (or 6-month median effluent limitation), then the Discharger shall increase the monitoring frequency to weekly until compliance with the effluent limitation is demonstrated.

2. Compliance with Effluent Limitations expressed as Single Constituents

Dischargers are out of compliance with the effluent limitation if the concentration of the pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reported Minimum Level (i.e., ML or RL).

3. Compliance with Effluent Limitations expressed as Sum of Several Constituents

Dischargers are out of compliance with an effluent limitation which applies to the sum of a group of chemicals (e.g., PCBs) if the sum of the individual pollutant concentrations is greater than the effluent limitation. Individual pollutants of the group will be considered to have a concentration of zero if the constituent is reported as ND or DNQ.

4. Multiple Sample Data

When determining compliance with a measure of central tendency (arithmetic mean, geometric mean, median, etc.) of multiple sample analyses and the data set contains one or more reported determinations of DNQ or ND, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:

- a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
- b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.

5. Average Monthly Effluent Limitation (AMEL)

If the average (or when applicable, the median determined by subsection 4 above for multiple sample data reduction) of daily discharges over a calendar month exceeds (is higher than) the AMEL for a given parameter, an alleged violation will be



flagged and the Discharger will be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month). However, an alleged violation of the AMEL will be considered one violation for the purpose of assessing State mandatory minimum penalties. If only a single sample is collected during the calendar month and the analytical result for that sample exceeds the AMEL, the Discharger may be considered out of compliance for that calendar month. For those average monthly effluent limitations that are based on the 6-month median water quality objectives in the 2019 Ocean Plan, the daily value used to calculate these average monthly values for intermittent discharges, shall be considered to equal zero for days on which no discharge occurred. The Discharger will only be considered out of compliance for days when the discharge occurs. If no sample (daily discharge) is taken over any one calendar month, no compliance determination can be made for that month with respect to effluent violation determination, but compliance determination can be made for that month with respect to reporting violation determination.

A month will begin on the first day of the calendar month and end on the last day of the calendar month, in order to calculate and report a consecutive (uninterrupted) average value for the AMEL for a calendar month.

#### 6. Average Weekly Effluent Limitation (AWEL)

If the average (or when applicable, the median determined by subsection 4 above for multiple sample data reduction) of daily discharges over a calendar week exceeds (is higher than) the AWEL for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for each day of that week for that parameter (e.g., resulting in seven days of non-compliance). However, an alleged violation of the AWEL will be considered one violation for the purpose of assessing State mandatory minimum penalties. The average of daily discharges over a calendar week that exceeds the AWEL for a parameter will be considered out of compliance for that week only. If only a single sample is collected during the calendar week and the analytical result for that sample exceeds the AWEL, the Discharger will be considered out of compliance for that calendar week. If no sample (daily discharge) is taken over a calendar week, no compliance determination can be made for that week with respect to effluent violation determination, but compliance determination can be made for that week with respect to reporting violation determination.

A calendar week will begin on Sunday and end on Saturday. Partial calendar weeks at the end of the calendar month will be carried forward to the next month in order to calculate and report a consecutive seven-day average value for the AWEL on Saturday.

#### 7. Maximum Daily Effluent Limitation (MDEL)

If a daily discharge on a calendar day exceeds (is higher than) the MDEL for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for that day for that parameter. If no sample (daily discharge) is taken over a calendar day, no compliance determination can be made for that day

with respect to effluent violation determination, but compliance determination can be made for that day with respect to reporting violation determination.

8. Instantaneous Minimum Effluent Limitation

If the analytical result of a single grab sample exceeds (is lower than) the instantaneous minimum effluent limitation for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for that single sample for that parameter. Non-compliance for each single grab sample will be considered separately (e.g., the analytical results of two grab samples taken over a calendar day that are lower than the instantaneous minimum effluent limitation would result in two instances of non-compliance with the instantaneous minimum effluent limitation).

9. Instantaneous Maximum Effluent Limitation

If the analytical result of a single grab sample exceeds (is higher than) the instantaneous maximum effluent limitation for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for that single sample for that parameter. Non-compliance for each single grab sample will be considered separately (e.g., the analytical results of two grab samples taken over a calendar day that both are higher than the instantaneous maximum effluent limitation would result in two instances of non-compliance with the instantaneous maximum effluent limitation).

10. Six-month Median Effluent Limitation (i.e., 180-day Median Effluent Limitation)

If the median of daily discharges over any 180-day period exceeds (is higher than) the 6-month median effluent limitation for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for each day of that 180-day period (e.g., resulting in 180 days of non-compliance in any 180-day period). The next assessment of compliance will occur when the next sample is taken. If only a single sample is collected during a given 180-day period and the analytical result for that sample exceeds the six-month median, the Discharger will be considered out of compliance for the 180-day period. If no sample (daily discharge) is taken over a 180-day period, no compliance determination can be made for that period with respect to effluent violation determination, but compliance determination can be made for that period with respect to reporting violation determination.

11. Mass and Concentration Limitations

Compliance with mass and concentration effluent limitations for the same parameter shall be determined separately with their respective limitations. When the concentration of a constituent in an effluent sample is determined to be ND or DNQ, the corresponding mass emission rate determined from that sample concentration shall also be reported as ND or DNQ.

12. Mass Emission Rate

The daily discharge mass emission rate for any calendar day is calculated using the following equations:

$$\text{Daily Discharge mass emission rate (lb/day)} = \frac{8.34}{N} \sum_{i=1}^N Q_i C_i$$

in which “N” is the number of samples taken over any calendar day. If grab samples are taken, “C<sub>i</sub>” is the constituent concentration (mg/L) and “Q<sub>i</sub>” is the flow rate (MGD) associated with each “N” grab sample. If composite samples are taken, “C<sub>i</sub>” is the constituent concentration (mg/L) in each composite sample and “Q<sub>i</sub>” is the average flow rate (MGD) during the period over which sample compositing occurs.

The daily discharge concentration of a constituent shall be determined from the flow-weighted average of the same constituent in the combined waste stream using the following equation:

$$\text{Daily discharge concentration} = \frac{1}{Q_t} \sum_{i=1}^N Q_i C_i$$

in which “N” is the number of component waste streams. “C<sub>i</sub>” is the constituent concentration (mg/L) and “Q<sub>i</sub>” is the flow rate (MGD) associated with each “N” component waste stream. “Q<sub>t</sub>” is the total flow rate of the combined waste stream.

### 13. Bacterial Standards and Analysis

- a. The geometric mean used for determining compliance with bacterial standards is calculated with the following equation:

$$\text{Geometric Mean} = (C_1 \times C_2 \times \dots \times C_n)^{1/n}$$

where n is the number of days samples were collected during the period and C is the concentration of bacteria (MPN/100 mL or CFU/100 mL) found on each day of sampling.

- b. For bacterial analyses, sample dilutions should be performed so the expected range of values is bracketed (for example, with multiple tube fermentation method or membrane filtration method, 2 to 16,000 per 100 mL for total and fecal coliform, at a minimum, and 1 to 1000 per 100 mL for *Enterococcus*). The detection methods used for each analysis shall be reported with the results of the analyses.
- c. Detection methods used for coliforms (total and fecal) and enterococcus shall be those presented in Table IA of 40 CFR § 136, unless alternate methods have been approved by USEPA pursuant to 40 CFR § 136, or improved methods have been determined by the Santa Ana Water Board and/or USEPA.

### 14. Sample Reporting Protocols

The Discharger must report with each sample result the reported Minimum Level, selected and used in accordance with Ocean Plan Chapters III.C.5 and 6, the laboratory’s current Method Detection Limit. In accordance with Ocean Plan Chapter III.C.7, the Discharger must also report the results of analytical determinations for

the presence of chemical constituents in a sample using the following reporting protocols:

- a. Sample results greater than or equal to the reported Minimum Level must be reported "as measured" by the laboratory (i.e., the measured chemical concentration in the sample).
- b. Sample results less than the reported Minimum Level, but greater than or equal to the laboratory's Method Detection Limit, must be reported as "Detected, but Not Quantified", or "DNQ". The laboratory must write the estimated chemical concentration of the sample next to DNQ as well as the words "Estimated Concentration" (may be shortened to "Est. Conc.").
- c. Sample results less than the laboratory's Method Detection Limit must be reported as "Not Detected", or "ND".

## ATTACHMENT A – DEFINITIONS

### **Arithmetic Mean ( $\mu$ )**

Also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

Arithmetic mean =  $\mu = \Sigma x / n$       where:  $\Sigma x$  is the sum of the measured ambient water concentrations, and  $n$  is the number of samples.

### **Areas of Special Biological Significance (ASBS)**

Those areas designated by the State Water Board as ocean areas requiring protection of species or biological communities to the extent that maintenance of natural water quality is assured. All Areas of Special Biological Significance are also classified as a subset of STATE WATER QUALITY PROTECTION AREAS (SWQPA). ASBS are also referred to as State Water Quality Protection Areas – Areas of Special Biological Significance (SWQPA-ASBS).

### **Average Monthly Effluent Limitation (AMEL)**

Means the highest allowable average of daily discharges over a calendar month, calculated as the sum of all “daily discharges” measured during a calendar month divided by the number of “daily discharges” measured during that month. (40 CFR § 122.2)

### **Average Weekly Effluent Limitation (AWEL)**

Means the highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of “daily discharges” measured during that week. (40 CFR § 122.2)

### **Best Management Practices (BMPs)**

Means schedules of activities, prohibitions of practices, maintenance procedures, and other management practices to prevent or reduce the pollution of “waters of the United States”. BMPs also include treatment requirements, operating procedures, and practices to control plant site runoff, spillages or leaks, sludge or waste disposal, or drainage from raw material storage. (40 CFR § 122.2)

### **Chlordane**

Shall mean the sum of chlordane-alpha, chlordane-gamma, chlordene-alpha, chlordene-gamma, nonachlor-alpha, nonachlor-gamma, and oxychlordane.

### **Chlorinated Phenols**

The sum of 2-chlorophenol, 2,4-dichlorophenol, 4-chloro-3-methylphenol, 2,4,6-trichlorophenol, and pentachlorophenol.

### **Coefficient of Variation (CV)**

CV is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

### **Composite Sample**

Means a time-proportioned mixture of not less than eight discrete aliquots obtained at equal time intervals (e.g., 24-hour composite means a minimum of eight samples collected every three hours). The volume of each aliquot shall be directly proportional to the discharge flow

rate at the time of sampling, but not less than 100 ml. The compositing period shall equal the specified sampling period, or 24 hours, if no period is specified.

For a composite sample, if the duration of the discharge is less than 24 hours but greater than 8 hours, at least eight flow-weighted individual sample portions shall be taken during the duration of the discharge and composited. For a discharge duration of 8 hours or less, eight individual "grab samples" may be substituted and composited.

### **Daily Discharge**

Daily Discharge is defined as either (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass; or (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day), or by the arithmetic mean of analytical results from one or more grab samples taken over the course of one day.

For composite sampling, if one day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

### **DDT**

Shall mean the sum of 4,4'DDT, 2,4'DDT, 4,4'DDE, 2,4'DDE, 4,4'DDD, 2,4'DDD, and 4,4'DDMU.

### **Degrade**

Degradation shall be determined by comparison of the waste field and reference site(s) for characteristic species diversity, population density, contamination, growth anomalies, debility, or supplanting of normal species by undesirable plant and animal species. Degradation occurs if there are significant differences in any of three major biotic groups, namely, demersal fish, benthic invertebrates, or attached algae. Other groups may be evaluated where benthic species are not affected, or are not the only ones affected.

### **Detected, but Not Quantified (DNQ)**

Sample results that are less than the reported Minimum Level (ML), but greater than or equal to the laboratory's Maximum Detection Limit (MDL). Sample results reported as DNQ are estimated concentrations.

### **Dichlorobenzenes**

Shall mean the sum of 1,2- and 1,3-dichlorobenzene.

### **Dilution Credit**

Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.



**Downstream Ocean Waters**

Shall mean waters downstream with respect to ocean currents.

**Dredged Material**

Any material excavated or dredged from the navigable waters of the United States, including material otherwise referred to as “spoil”.

**Eelgrass Beds**

Are aggregations of the aquatic plant species of the genus *Zostera*.

**Enclosed Bays**

Are indentations along the coast which enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. This definition includes but is not limited to: Humboldt Bay, Bodega Harbor, Tomales Bay, Drakes Estero, San Francisco Bay, Morro Bay, Los Angeles Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

**Endosulfan**

Shall mean the sum of endosulfan-alpha and -beta and endosulfan sulfate.

**Estuaries and Coastal Lagoons**

Are waters at the mouths of streams that serve as mixing zones for fresh and ocean waters during a major portion of the year. Mouths of streams that are temporarily separated from the ocean by sandbars shall be considered as estuaries. Estuarine waters will generally be considered to extend from a bay or the open ocean to the upstream limit of tidal action but may be considered to extend seaward if significant mixing of fresh and salt water occurs in the open coastal waters. The waters described by this definition include but are not limited to the Sacramento-San Joaquin Delta as defined by section 12220 of the California Water Code, Suisun Bay, Carquinez Strait downstream to Carquinez Bridge, and appropriate areas of the Smith, Klamath, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay Rivers. Estuaries do not include inland surface waters or ocean waters.

**Facility or Activity**

Means any NPDES “point source” or any other facility or activity (including land or appurtenances thereto) this is subject to regulation under the NPDES program. (40 CFR § 122.2)

**Geometric Mean (GM)**

Is a type of mean or average that indicates the central tendency or typical value of a set of numbers by using the product of their values (as opposed to the arithmetic mean which uses their sum). The geometric mean is defined as the  $n$ th root of the product of  $n$  numbers. The formula is expressed as:  $GM = (C_1 \times C_2 \times \dots \times C_n)^{1/n}$ , where  $C$  is the sample value and  $n$  is the number of samples taken.

**Grab Sample**

Is a single sample collected during a period of time, not to exceed 15 minutes, which represents the composition of the discharge only at a particular time and place. Grab samples

shall be collected during normal peak loading conditions for the parameter of interest, which may or may not occur during hydraulic peaks.

**Halomethanes**

Shall mean the sum of bromoform, bromomethane (methyl bromide) and chloromethane (methyl chloride).

**HCH**

Shall mean the sum of the alpha, beta, gamma (lindane) and delta isomers of hexachlorocyclohexane.

**Indicator Bacteria**

Includes total coliform bacteria, fecal coliform bacteria (or *E. coli*) and/or Enterococcus bacteria.

**Initial Dilution**

Is the process which results in the rapid and irreversible turbulent mixing of wastewater with ocean water around the point of discharge.

For a submerged buoyant discharge, characteristic of most municipal and industrial wastes that are released from the submarine outfalls, the momentum of the discharge and its initial buoyancy act together to produce turbulent mixing. Initial dilution in this case is completed when the diluting wastewater ceases to rise in the water column and first begins to spread horizontally.

For shallow water submerged discharges, surface discharges, and non-buoyant discharges, characteristic of cooling water wastes and some individual discharges, turbulent mixing results primarily from the momentum of discharge. Initial dilution, in these cases, is considered to be completed when the momentum induced velocity of the discharge ceases to produce significant mixing of the waste, or the diluting plume reaches a fixed distance from the discharge to be specified by the Santa Ana Water Board and/or USEPA, whichever results in the lower estimate for initial dilution.

**Instantaneous Maximum Effluent Limitation**

The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum effluent limitation).

**Instantaneous Minimum Effluent Limitation**

The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum effluent limitation).

**Kelp Beds**

Are aggregations of marine algae of the order *Laminariales*, including species in the genera *Macrocystis*, *Nereocystis*, and *Pelagophycus*. Kelp beds include the total foliage canopy throughout the water column.

**Mariculture**

The culture of algae, plants, and animals in marine waters independent of any pollution source.

**Marine Managed Areas**

Are named, discrete geographic marine or estuarine areas along the California coast designated by law or administrative action, and intended to protect, conserve, or otherwise

manage a variety of resources and their uses. According to the California Public Resources Code (§§ 36600 et seq.) there are six classifications of marine managed areas, including State Marine Reserves, State Marine Parks and State Marine Conservation Areas, State Marine Cultural Preservation Areas, State Marine Recreational Management Areas, and State Water Quality Protection Areas.

### **Material**

(a) In common usage: (1) the substance or substances of which a thing is made or composed (2) substantial; (b) For purposes of the Ocean Plan relating to waste disposal, dredging and the disposal of dredged material and fill, MATERIAL means matter of any kind or description which is subject to regulation as waste, or any material dredged from the navigable waters of the United States. See also, DREDGED MATERIAL.

### **Maximum Daily Effluent Limitation (MDEL)**

The highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period) (40 CFR § 122.2). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day. Also, Maximum Daily Discharge Limitation (MDDL).

### **Median**

The middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements (n) is odd, then the median =  $X_{(n+1)/2}$ . If n is even, then the median =  $(X_{n/2} + X_{(n/2)+1})/2$  (i.e., the midpoint between the  $n/2$  and  $n/2+1$ ).

### **Method Detection Limit (MDL)**

Is the minimum measured concentration of a substance that can be reported with 99 percent confidence that the measured concentration is distinguishable from method blank results, as defined in 40 CFR § 136, Appendix B.

### **Minimum Level (ML)**

Is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method-specified sample weights, volumes and processing steps have been followed.

### **Mixing Zone**

Mixing Zone is a limited volume of receiving water that is allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

### **Multiport Diffusers**

Are linear structures consisting of spaced ports or nozzles that are installed on submerged marine outfalls.

### **Municipal Separate Storm Sewer System (MS4)**

Has the same meaning as set forth in 40 CFR § 122.26(b)(8).

**Natural Light**

Reduction of natural light may be determined by the Santa Ana Water Board and USEPA by measurement of light transmissivity or total irradiance, or both, according to the monitoring needs of the Santa Ana Water Board and USEPA.

**No Discharge Zone (NDZ)**

Is an area in which both treated and untreated sewage discharges from vessels are prohibited. Within NDZ boundaries, vessel operators are required to retain their sewage discharges onboard for disposal at sea (beyond 3 nautical miles from shore) or onshore at a pump-out facility.

**Non-Chlorinated Phenols**

The sum of 2,4-dimethylphenol, 2-nitrophenol, 4-nitrophenol, 2,4-dinitrophenol, 4,6-dinitro-2-methylphenol, and phenol.

**Non-Storm Water Discharge**

Is any runoff that is not the result of a precipitation event. This is often referred to as “dry weather flow”.

**Not Detected (ND)**

Sample results which are less than the laboratory’s MDL.

**Ocean Waters**

Are the territorial marine waters of the State as defined by California law to the extent these waters are outside of enclosed bays, estuaries, and coastal lagoons. If a discharge outside the territorial waters of the State could affect the quality of the waters of the State, the discharge may be regulated to assure no violation of the Ocean Plan will occur in ocean waters.

**PAHs (polynuclear aromatic hydrocarbons)**

The sum of acenaphthylene, anthracene, 1,2-benzanthracene (benzo[a]anthracene), benzo[b]fluoranthene, benzo[k]fluoranthene, 1,12-benzoperylene (benzo[ghi]perylene), benzo[a]pyrene, chrysene, dibenzo[ah]anthracene, fluorene, indeno[1,2,3-cd]pyrene, phenanthrene and pyrene.

**PCBs (polychlorinated biphenyls) as Aroclors**

The sum of chlorinated biphenyls whose analytical characteristics resemble those of Aroclor-1016, Aroclor-1221, Aroclor-1232, Aroclor-1242, Aroclor-1248, Aroclor-1254 and Aroclor-1260.

**PCBs as congeners**

The sum of the following 41 individually quantified PCB congeners or mixtures of isomers of a single congener in a co-elution: PCB-18, 28, 37, 44, 49, 52, 66, 70, 74, 77, 81, 87, 99, 101, 105, 110, 114, 118, 119, 123, 126, 128, 138, 149, 151, 153, 156, 157, 158, 167, 168, 169, 170, 177, 180, 183, 187, 189, 194, 201, and 206.

**Permitting Authority**

Means the State Water Board or Regional Waterboard, whichever issues the permit, and USEPA which issues the permit.

**Pollutant Minimization Program (PMP)**

PMP means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of the 2019 Ocean Plan Table 3 pollutants through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Santa Ana Water Board and USEPA may consider cost effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to Water Code section 13263.3(d), shall be considered to fulfill the PMP requirements.

### **Pollution Prevention**

Any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in CWC section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State Water Resources Control Board (State Water Board), Santa Ana Water Board, or USEPA.

### **Publicly-Owned Treatment Works (POTWs)**

A treatment works as defined by section 212 of the CWA, which is owned by a State or municipality (as defined by section 502(4) of the Act). This definition includes any devices and systems used in the storage, treatment, recycling and reclamation of municipal sewage or industrial wastes of a liquid nature. It also includes sewers, pipes and other conveyances only if they convey wastewater to a POTW Treatment Plant. The term also means the municipality which has jurisdiction over the Indirect Discharges to and the discharges from such treatment works. (40 CFR § 403.3(q).)

### **Reported Minimum Level (RML)**

Is the ML (and its associated analytical method) chosen by the Discharger for reporting and compliance determination from the MLs included in their permit, which is also known as the Reporting Level (RL). The MLs included in this permit correspond to approved analytical methods for reporting a sample result that are selected by the Santa Ana Water Board in accordance with Ocean Plan Chapter III.C.5. The ML is based on the proper application of method-specific analytical procedures and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the reported ML. (See Chapter III.C.6 of the 2019 Ocean Plan.).

### **Salinity**

Is a measure of the dissolved salts in a volume of water. For the purposes of the Ocean Plan, salinity shall be measured using a standard method approved by the Santa Ana Water Board

and USEPA (e.g., Standard Method 2520 B, EPA Method 120.1, EPA Method 160.1) and reported in parts per thousand (ppt). For historical salinity data not recorded in parts per thousand, the Santa Ana Water Boards and USEPA may accept converted data at their discretion.

**Seawater**

Is salt water that is in or from the ocean.

**Sensitive Habitats**

For the purposes of the Ocean Plan, are kelp beds, rocky substrate, surfgrass beds, eelgrass beds, oyster beds, spawning grounds for State or federally managed species, market squid nurseries, or other habitats in need of special protection as determined by the Water Boards.

**Shellfish**

Are organisms identified by the California Department of Public Health as shellfish for public health purposes (i.e., mussels, clams and oysters).

**Significant Difference**

Is defined as a statistically significant difference in the means of two distributions of sampling results at the 95 percent confidence level, unless otherwise specified by the permitting authority.

**Single Sample Maximum (SSM)**

A maximum value not to be exceeded in any single sample.

**Six-Month Median Effluent Limitation (i.e., 180-Day Median Effluent Limitation)**

The highest allowable moving median of daily values for any 180-day period in which daily values represent flow weighted average concentrations within a 24-hour period. For intermittent discharges, the daily value shall be considered to equal zero for days on which no discharge occurred.

**Standard Deviation ( $\sigma$ )**

Standard Deviation is a measure of variability that is calculated as follows:

$$\sigma = (\sum[(x - \mu)^2]/(n - 1))^{0.5}$$

where:

x is the observed value;

$\mu$  is the arithmetic mean of the observed values; and

n is the number of samples.

**State Water Quality Protection Areas (SWQPAs)**

Non-terrestrial marine or estuarine areas designated to protect marine species or biological communities from an undesirable alteration in natural water quality. All AREAS OF SPECIAL BIOLOGICAL SIGNIFICANCE (ASBS) that were previously designated by the State Water Board in Resolutions 74-28, 74-32, and 75-61 are now also classified as a subset of State Water Quality Protection Areas and require special protections afforded by the Ocean Plan.

**State Water Quality Protection Areas – General Protection (SWQPA-GP)**

Designed by the State Water Board to protect marine species and biological communities from undesirable alteration in natural water quality within State Marine Parks and State Marine Conservation Areas.



### Statistical Threshold Value (STV)

Is defined for the bacteria water quality objectives as a set value that approximates the 90th percentile of the water quality distribution of a bacterial population. The STV for the enterococcus water quality objective is 110 CFU/100mL.

### Storm Water

Has the same meaning as set forth in 40 CFR § 122.26(b)(13).

### Surfgrass Beds

Are aggregations of marine flowering plants of the genus *Phyllospadix*.

### TCDD Equivalentents (TEQ)

The sum of the concentrations of chlorinated dibenzodioxins (2,3,7,8-CDDs) and chlorinated dibenzofurans (2,3,7,8-CDFs) multiplied by their respective toxicity equivalency factor (TEF), as shown in the table below.

| Congener                             | TEF   | Minimum Level (pg/L) |
|--------------------------------------|-------|----------------------|
| <u>chlorinated dibenzo-p-dioxins</u> |       |                      |
| 2,3,7,8-tetraCDD                     | 1.0   | 5                    |
| 1,2,3,7,8-pentaCDD                   | 0.5   | 25                   |
| 1,2,3,4,7,8-hexaCDD                  | 0.1   | 25                   |
| 1,2,3,6,7,8-hexaCDD                  | 0.1   | 25                   |
| 1,2,3,7,8,9-hexaCDD                  | 0.1   | 25                   |
| 1,2,3,4,6,7,8-heptaCDD               | 0.01  | 25                   |
| OctaCDD                              | 0.001 | 50                   |
| <u>chlorinated dibenzofurans</u>     |       |                      |
| 2,3,7,8-tetraCDF                     | 0.1   | 5                    |
| 1,2,3,7,8-pentaCDF                   | 0.05  | 25                   |
| 2,3,4,7,8-pentaCDF                   | 0.5   | 25                   |
| 1,2,3,4,7,8-hexaCDF                  | 0.1   | 25                   |
| 1,2,3,6,7,8-hexaCDF                  | 0.1   | 25                   |
| 1,2,3,7,8,9-hexaCDF                  | 0.1   | 25                   |
| 2,3,4,6,7,8-hexaCDF                  | 0.1   | 25                   |
| 1,2,3,4,6,7,8-heptaCDF               | 0.01  | 25                   |
| 1,2,3,4,7,8,9-heptaCDF               | 0.01  | 25                   |
| OctaCDF                              | 0.001 | 50                   |

### Test of Significant Toxicity (TST)

A statistical approach used to analyze toxicity test data. The TST incorporates a restated null hypothesis, Welch's t-test, and the biological effect thresholds for chronic and acute toxicity.

### Toxicity Identification Evaluation (TIE)

Set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.

**Toxicity Reduction Evaluation (TRE)**

A study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A TIE may be required as part of the TRE, if appropriate.

**Trash**

Means all improperly discarded solid material from any production, manufacturing, or processing operations including, but not limited to, products, product packaging, or containers constructed of plastic, steel, aluminum, glass, paper, or other synthetic or natural materials.

**Trash Provisions**

Are the water quality objective for Trash, as well as the prohibition of discharge set forth in Chapter III.I and implementation requirements set forth in Chapter III.L of the Ocean Plan.

**Waste**

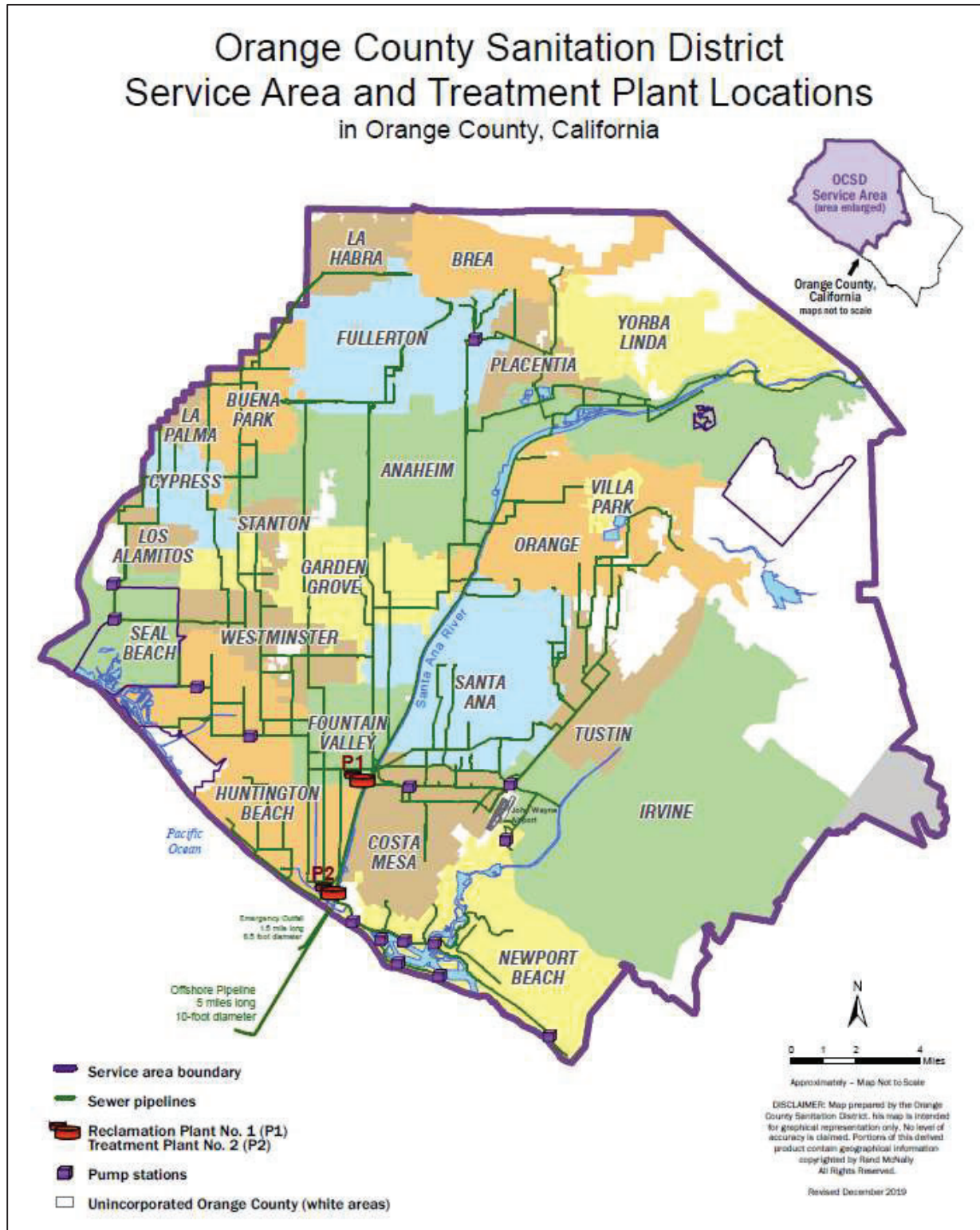
As used in the Ocean Plan, waste includes a discharger's total discharge, of whatever origin, i.e., gross, not net, discharge.

**Water Reclamation**

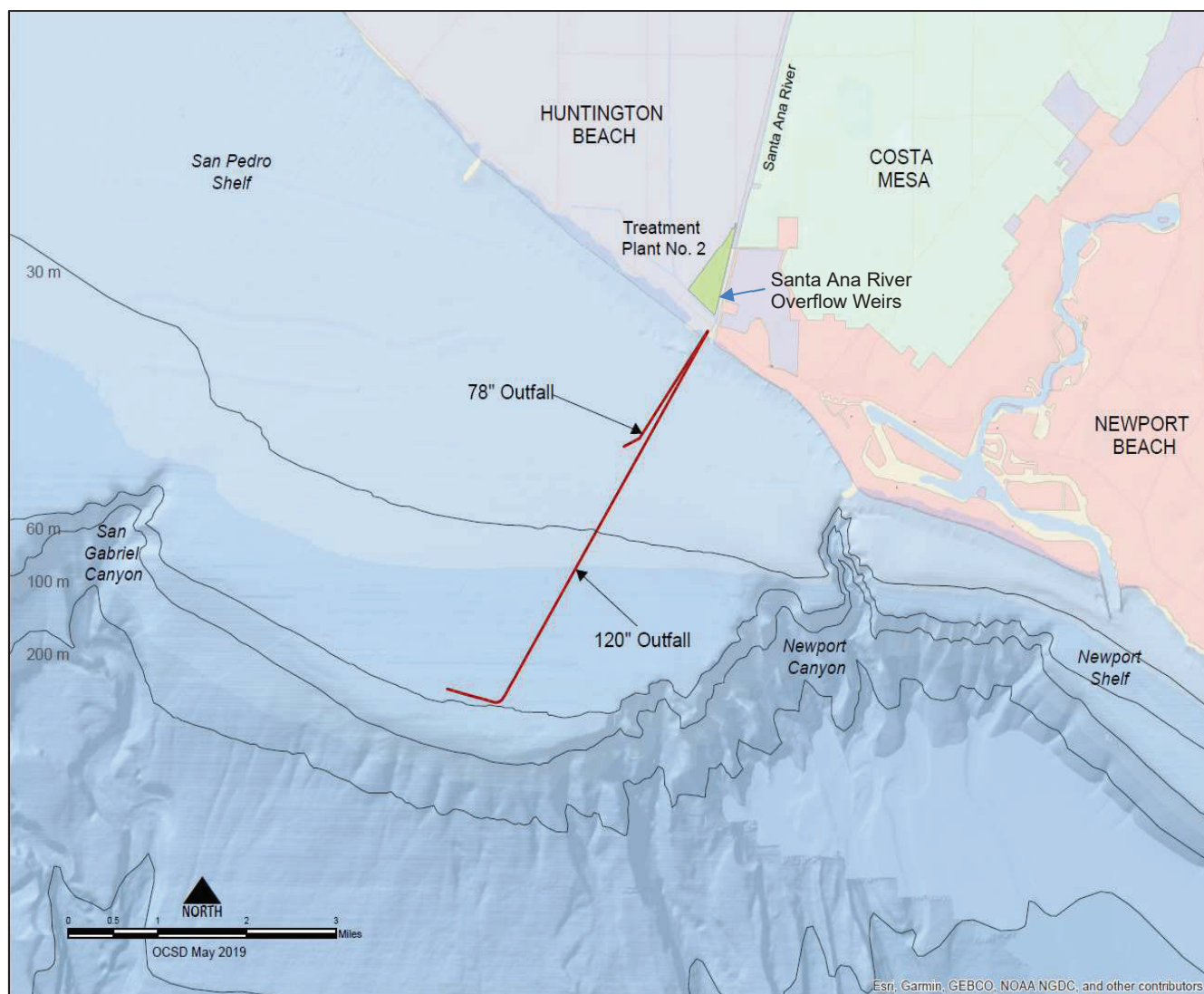
The treatment of wastewater to render it suitable for reuse, the transportation of treated wastewater to the place of use, and the actual use of treated wastewater for a direct beneficial use or controlled use that would not otherwise occur.

## ATTACHMENT B – MAPS

### Attachment B1 – Map of Service Area and Treatment Plant Locations

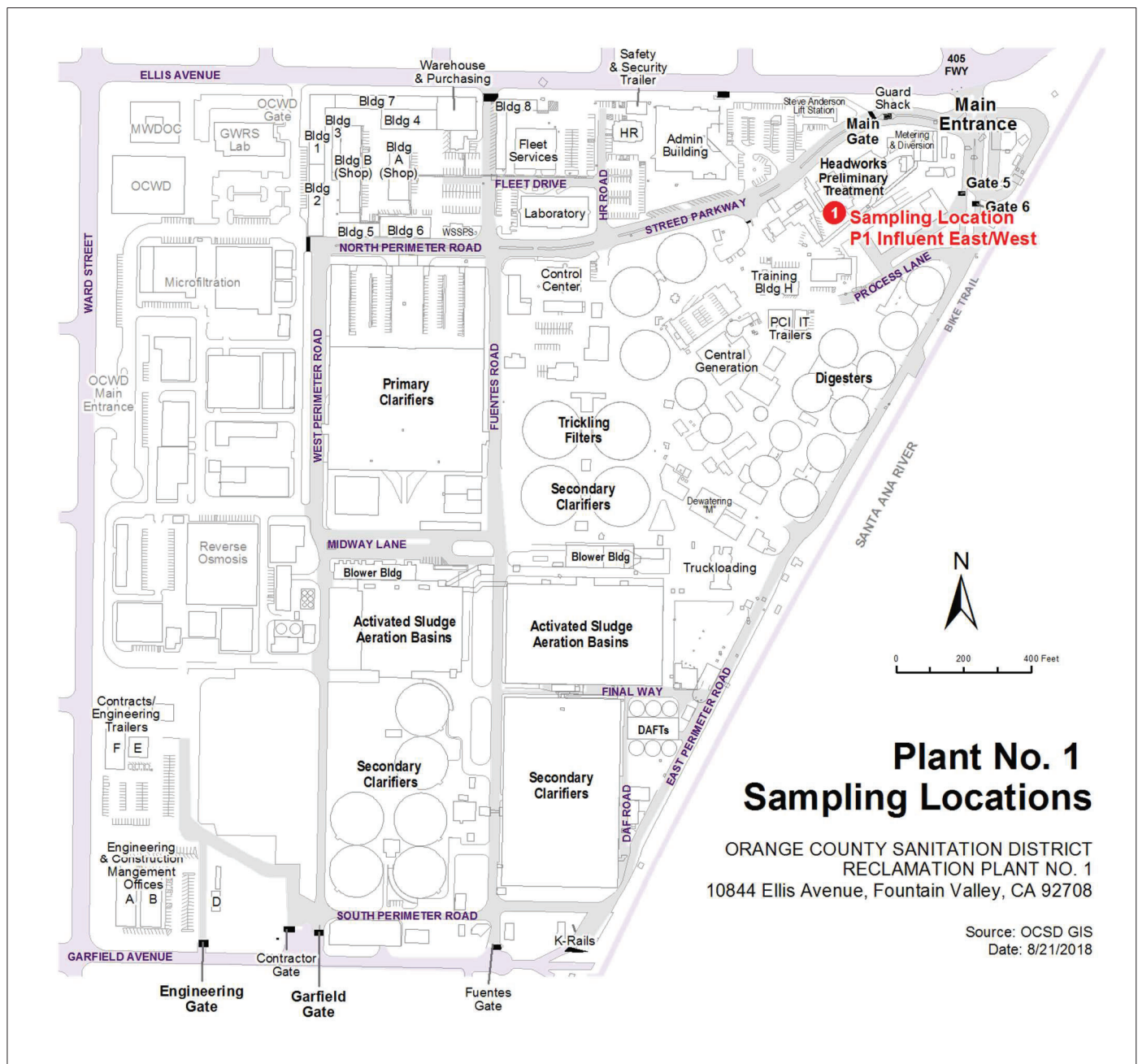


## Attachment B2 – Map of Outfall Location

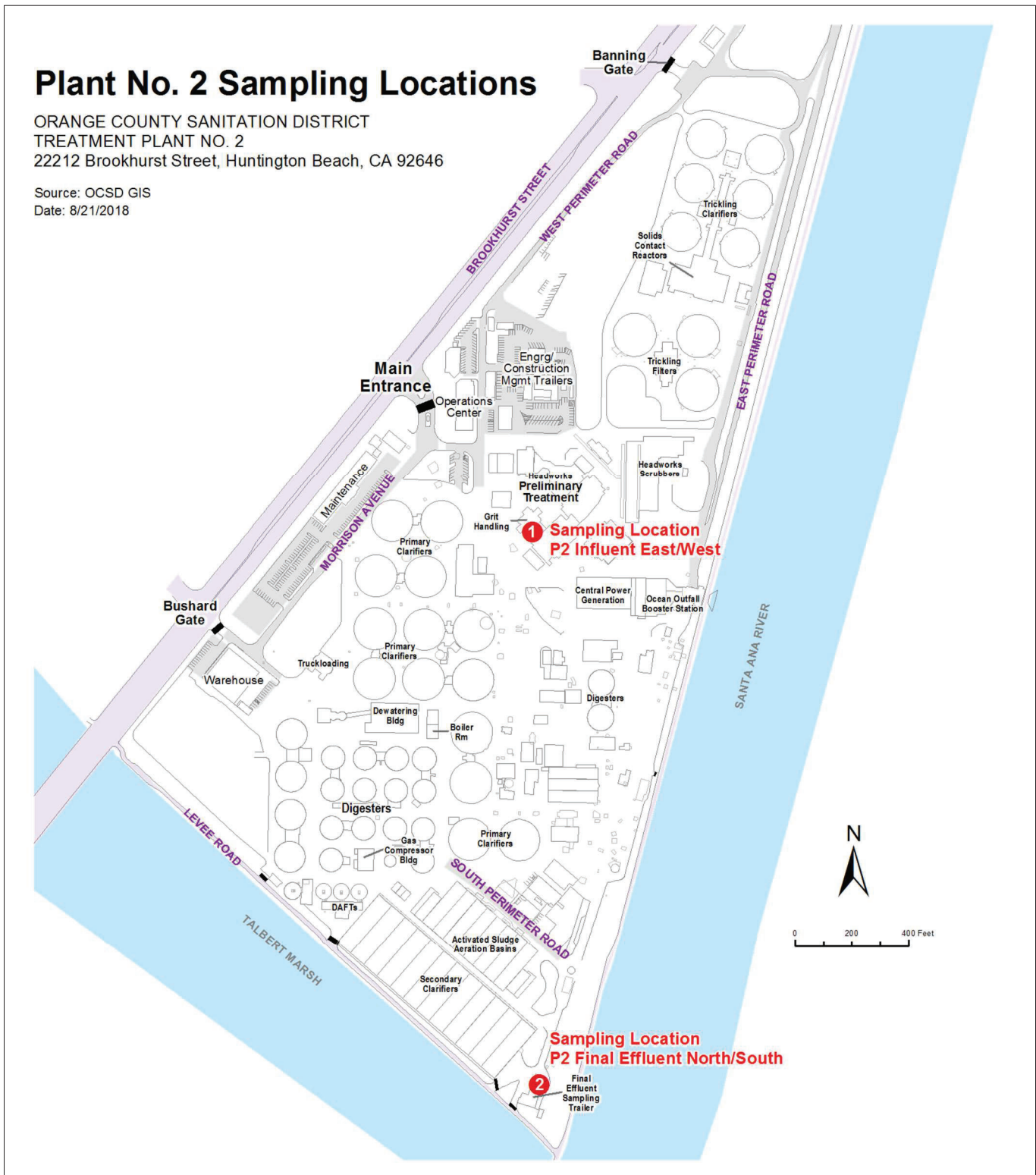




## Attachment B3 – Site Layout of Plant No. 1 and Sampling Locations



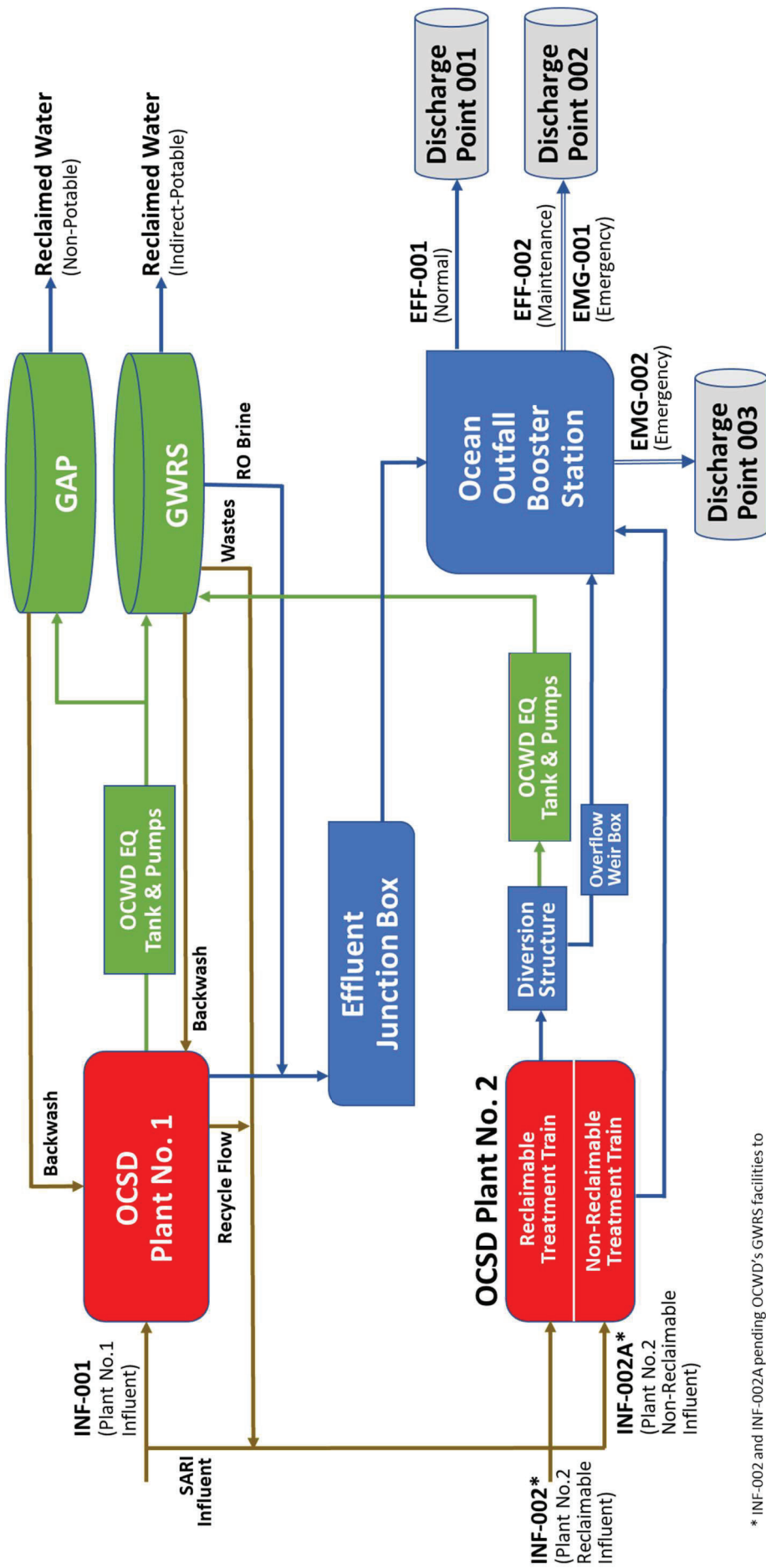
## Attachment B4 – Site Layout of Plant No. 2 and Sampling Locations





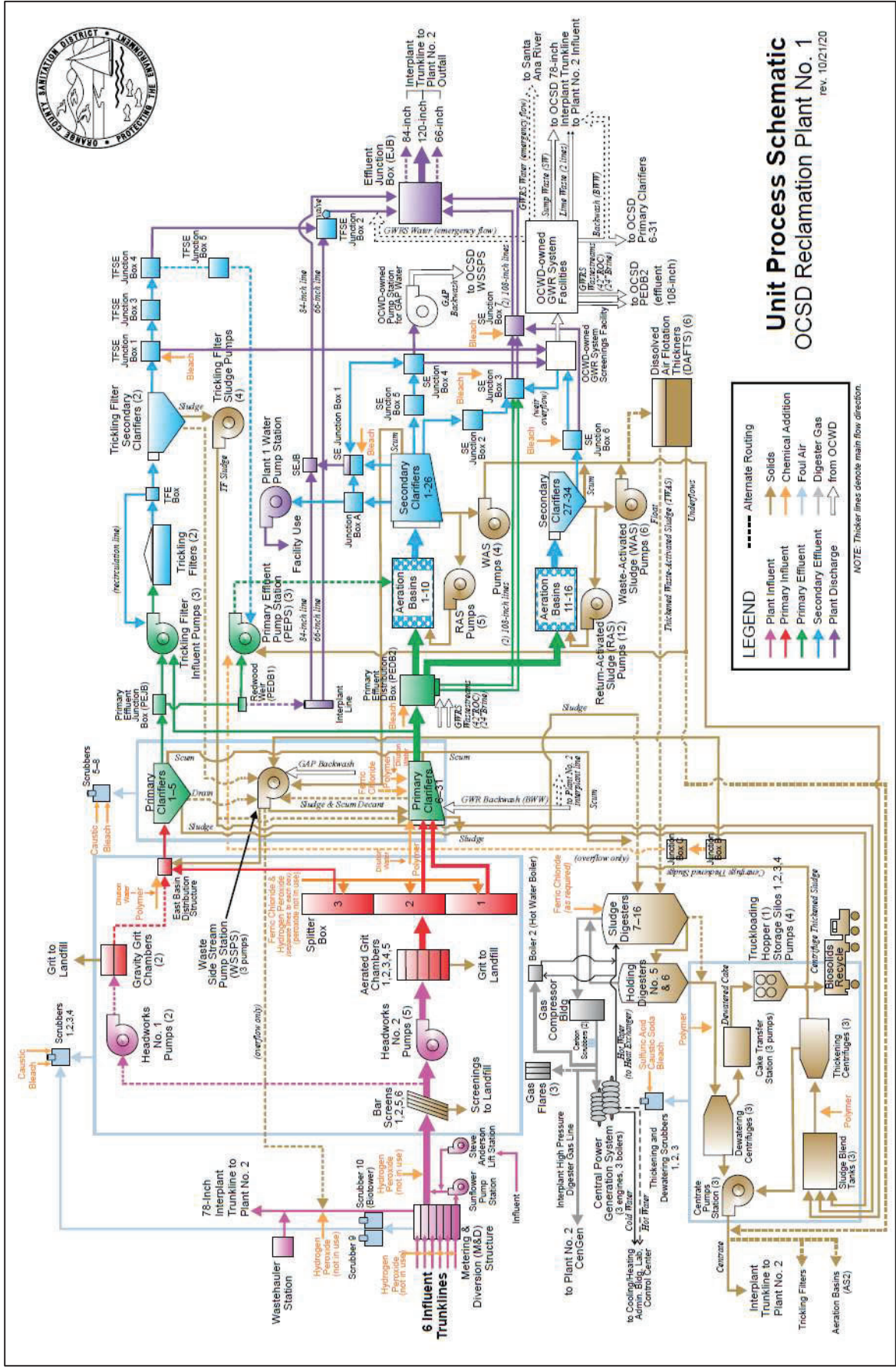
## ATTACHMENT C – FLOW SCHEMATICS

Figure C-1. Simplified Flow Schematic and Monitoring Locations after GWRS Final Expansion in 2023



\* INF-002 and INF-002A pending OCWD's GWRS facilities to receive secondary treated wastewater from Treatment Plant No. 2 (i.e., completion of the GWRS Final Expansion Project)

Figure C-2. Process Schematic for OC San Reclamation Plant No. 1





**Figure C-3. Process Schematic for OC San Treatment Plant No. 2**

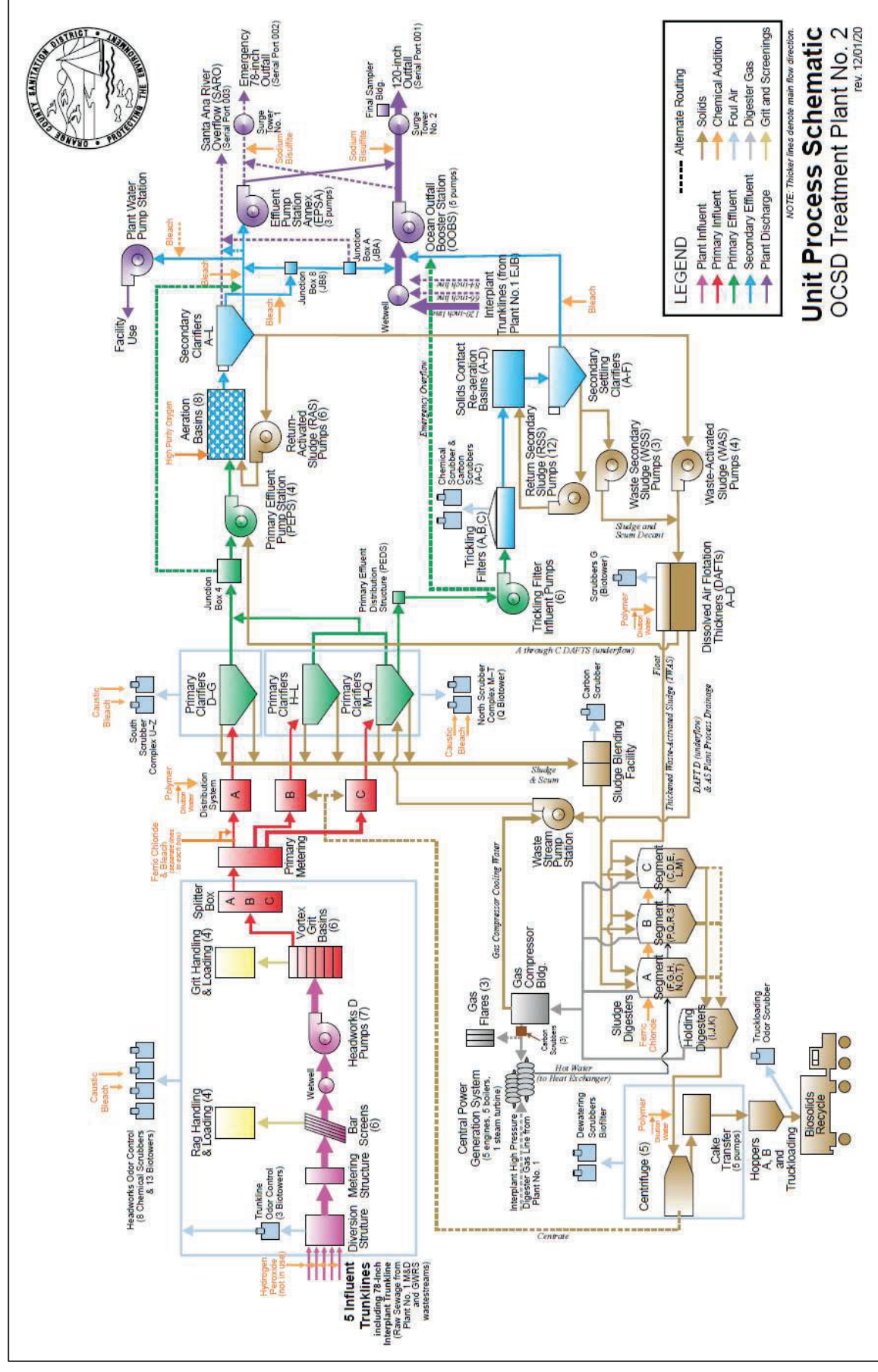
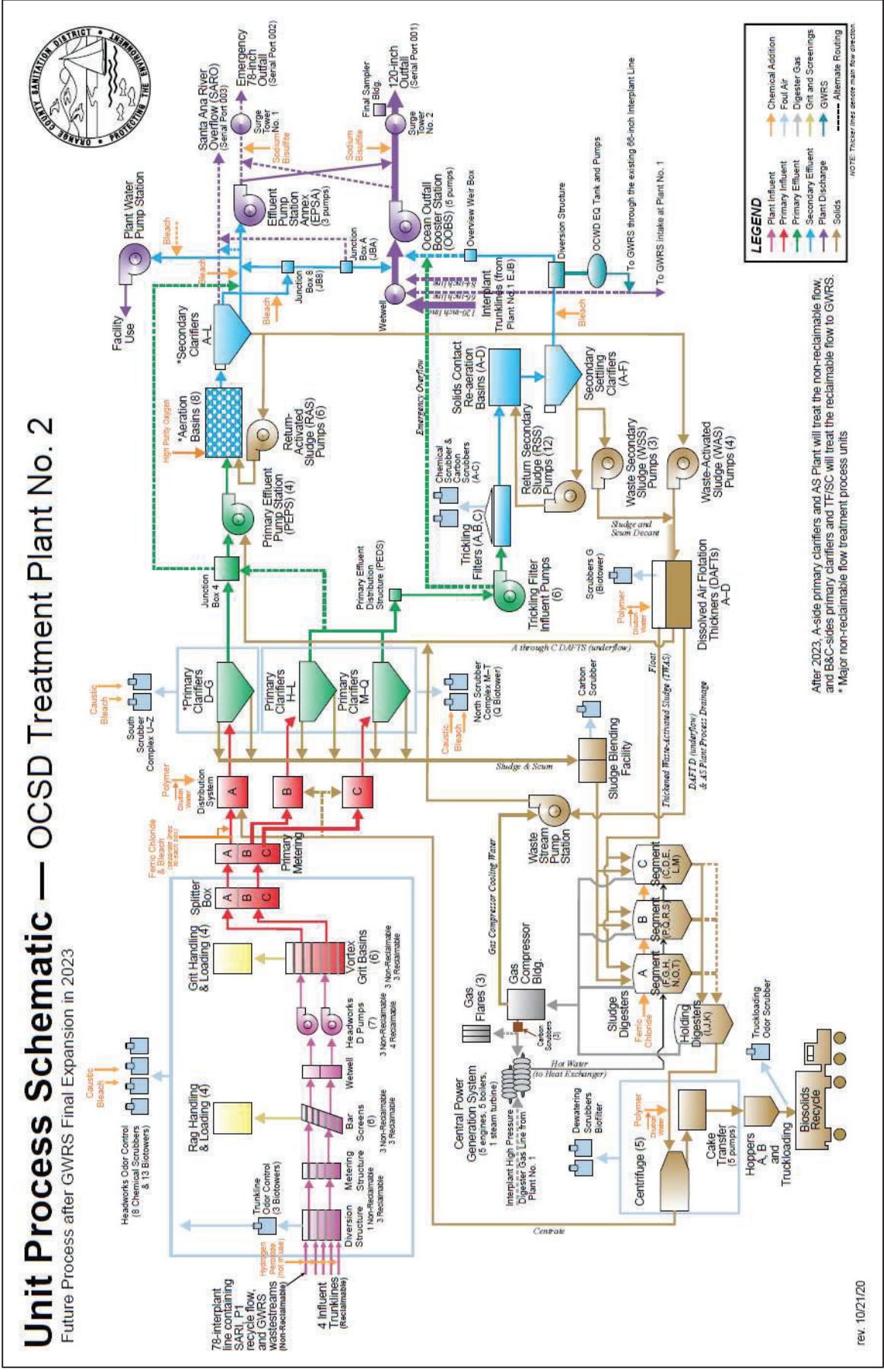


Figure C-4. Future Process Schematic for OC San Treatment Plant No. 2 after GWRS Final Expansion in 2023







## **ATTACHMENT D – STANDARD PROVISIONS**

### **I. STANDARD PROVISIONS – PERMIT COMPLIANCE**

#### **A. Duty to Comply**

1. The Discharger must comply with all conditions, terms, and requirements of this Order/Permit. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and California Water Code (CWC) and is grounds for enforcement action; for Order/Permit termination, revocation and reissuance, or modification; or denial of an Order/Permit renewal application. (40 CFR § 122.41(a); CWC 13261, 13263, 13265, 13268, 13000, 13001, 13304, 13350, 13385.)
2. The Discharger shall comply with effluent standards or prohibitions established under CWA section 307(a) for toxic pollutants and with standards for sewage sludge use and disposal established under CWA section 405(d) within the time provided in the regulations that establish these standards or prohibitions or standards of sewage sludge use or disposal, even if the Order/Permit has not yet been modified to incorporate the requirement. (40 CFR § 122.41(a)(1).)
3. The CWA provides that any person who violates section 301, 302, 306, 307, 308, 318 or 405 of the Act, or any permit condition or limitation implementing any such sections in a permit issued under section 402, or any requirement imposed in a pretreatment program approved under sections 402(a)(3) or 402(b)(8) of the Act, is subject to a civil penalty not to exceed \$25,000 per day for each violation, as adjusted annually for inflation pursuant to the Federal Civil Penalties Inflation Adjustment Act Improvements Act of 2015, and as currently set forth in 40 CFR § 19.4. The Clean Water Act provides that any person who negligently violates sections 301, 302, 306, 307, 308, 318, or 405 of the Act, or any condition or limitation implementing any of such sections in a permit issued under section 402 of the Act, or any requirement imposed in a pretreatment program approved under section 402(a)(3) or 402(b)(8) of the Act, is subject to criminal penalties of \$2,500 to \$25,000 per day of violation, or imprisonment of not more than 1 year, or both. In the case of a second or subsequent conviction for a negligent violation, a person shall be subject to criminal penalties of not more than \$50,000 per day of violation, or by imprisonment of not more than 2 years, or both. Any person who knowingly violates such sections, or such conditions or limitations is subject to criminal penalties of \$5,000 to \$50,000 per day of violation, or imprisonment for not more than 3 years, or both. In the case of a second or subsequent conviction for a knowing violation, a person shall be subject to criminal penalties of not more than \$100,000 per day of violation, or imprisonment of not more than 6 years, or both. Any person who knowingly violates section 301, 302, 303, 306, 307, 308, 318 or 405 of the Act, or any permit condition or limitation implementing any of such sections in a permit issued under section 402 of the Act, and who knows at that time that he thereby places another person in imminent danger of death or serious bodily injury, shall, upon conviction, be subject to a fine of not more than \$250,000 or imprisonment of not more than 15 years, or both. In the case of a second or



subsequent conviction for a knowing endangerment violation, a person shall be subject to a fine of not more than \$500,000 or by imprisonment of not more than 30 years, or both. An organization, as defined in section 309(c)(3)(B)(iii) of the CWA, shall, upon conviction of violating the imminent danger provision, be subject to a fine of not more than \$1,000,000 and can be fined up to \$2,000,000 for second or subsequent convictions. (40 CFR § 122.41(a)(2).)

4. Any person may be assessed an administrative penalty by the Administrator for violating section 301, 302, 306, 307, 308, 318 or 405 of this Act, or any permit condition or limitation implementing any of such sections in a permit issued under section 402 of this Act. Administrative penalties for Class I violations are not to exceed \$10,000 per violation, with the maximum amount of any Class I penalty assessed not to exceed \$25,000. Penalties for Class II violations are not to exceed \$10,000 per day for each day during which the violation continues, with the maximum amount of any Class II penalty not to exceed \$125,000. The civil and administrative penalty amounts are adjusted annually for inflation pursuant to the Federal Civil Penalties Inflation Adjustment Act Improvements Act of 2015, and the current penalty amounts are set forth in 40 CFR § 19.4.

#### **B. Need to Halt or Reduce Activity Not a Defense**

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order/Permit. (40 CFR § 122.41(c).)

#### **C. Duty to Mitigate**

The Discharger shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order/Permit which has a reasonable likelihood of adversely affecting human health or the environment. (40 CFR § 122.41(d).)

#### **D. Proper Operation and Maintenance**

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order/Permit. Proper operation and maintenance also include adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of the Order/Permit. (40 CFR § 122.41(e).)

#### **E. Property Rights**

1. This Order/Permit does not convey any property rights of any sort or any exclusive privilege. (40 CFR § 122.41(g).)
2. The issuance of this Order/Permit does not authorize any injury to persons or property or invasion of other private rights, or any infringement of State or local law or regulations. (40 CFR § 122.5(c).)

## **F. Inspection and Entry**

The Discharger shall allow the Santa Ana Water Board, State Water Board, USEPA, and/or an authorized representative (including an authorized contractor acting as their representative), upon presentation of credentials and other documents as may be required by law, to (33 USC 1318(a)(4)(b); 40 CFR § 122.41(i); CWC 13267, 13383):

1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order/Permit (33 USC 1318(a)(4)(b)(i); 40 CFR § 122.41(i)(1); CWC 13267, 13383);
2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order/Permit (33 USC 1318(a)(4)(b)(ii); 40 CFR § 122.41(i)(2); CWC 13267, 13383);
3. Inspect (and photograph) at reasonable times any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order/Permit (33 USC 1318(a)(4)(b)(ii); 40 CFR § 122.41(i)(3); CWC 13267, 13383); and
4. Sample or monitor, at reasonable times, for the purposes of assuring Order/Permit compliance or as otherwise authorized by the CWA or the California Water Code, any substances or parameters at any location. (33 USC 1318(a)(4)(b); 40 CFR § 122.41(i)(4); CWC 13267, 13383.)

## **G. Bypass**

1. Definitions
  - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 CFR § 122.41(m)(1)(i).)
  - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 CFR § 122.41(m)(1)(ii).)
2. Bypass not exceeding limitations. The Discharger may allow any bypass to occur which does not cause effluent limitations to be exceeded, but only if it is also for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5. (40 CFR § 122.41(m)(2).)
3. Prohibition of bypass. Bypass is prohibited, and the Santa Ana Water Board/USEPA may take enforcement action against a Discharger for bypass, unless (40 CFR § 122.41(m)(4)(i)):
  - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 CFR § 122.41(m)(4)(i)(A));

- b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 CFR § 122.41(m)(4)(i)(B)); and
  - c. The Discharger submitted notices as required under Standard Provisions – Permit Compliance I.G.5. (40 CFR § 122.41(m)(4)(i)(C).)
- 4. The Santa Ana Water Board/USEPA may approve an anticipated bypass, after considering its adverse effects, if the Santa Ana Water Board/USEPA determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance I.G.3. (40 CFR § 122.41(m)(4)(ii).)
- 5. Notice
  - a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit prior notice, if possible, at least ten days before the date of the bypass. The notice shall be sent to the Santa Ana Water Board and USEPA. As of December 21, 2025 all notices submitted in compliance with this section must be submitted electronically by the Discharger to the Santa Ana Water Board and USEPA or initial recipient, as defined in 40 CFR § 127.2(b), in compliance with this section and 40 CFR § 3 (including, in all cases, subpart D to part 3), 122.22 and part 127. Part 127 is not intended to undo existing requirements for electronic reporting. Prior to this date, and independent of part 127, the Discharger may be required to report electronically if specified by a particular Order/Permit or if required to do so by State law. (40 CFR § 122.41(m)(3)(i).)
  - b. Unanticipated bypass. The Discharger shall submit a notice of an unanticipated bypass as required in Attachment D. Standard Provisions – Reporting V.E (24-hour notice). As of December 21, 2025 all notices submitted in compliance with this section must be submitted electronically by the Discharger to the Santa Ana Water Board and USEPA or initial recipient, as defined in 40 CFR § 127.2(b), in compliance with this section and 40 CFR § 3 (including, in all cases, subpart D to part 3), 122.22 and part 127. Part 127 is not intended to undo existing requirements for electronic reporting. Prior to this date, and independent of part 127, the Discharger may be required to report electronically if specified by a particular Order/Permit or if required to do so by State law. (40 CFR § 122.41(m)(3)(ii).)

#### **H. Upset**

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed

treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 CFR § 122.41(n)(1).)

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 CFR § 122.41(n)(2).)
2. Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs, or other relevant evidence that (40 CFR § 122.41(n)(3)):
  - a. An upset occurred and that the Discharger can identify the cause(s) of the upset (40 CFR § 122.41(n)(3)(i));
  - b. The permitted facility was at the time being properly operated (40 CFR § 122.41(n)(3)(ii)); and
  - c. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b (24-hour notice) (40 CFR § 122.41(n)(3)(iii)).
  - d. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C. (40 CFR § 122.41(n)(3)(iv).)
3. Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof. (40 CFR § 122.41(n)(4).)

## **II. STANDARD PROVISIONS – PERMIT ACTION**

### **A. General**

This Order/Permit may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for Order/Permit modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order/Permit condition. (40 CFR § 122.41(f).)

### **B. Duty to Reapply**

If the Discharger wishes to continue an activity regulated by this Order/Permit after the expiration date of this Order/Permit, the Discharger must apply for and obtain a new Order/Permit. (40 CFR § 122.41(b).)

### **C. Transfers**

This Order/Permit is not transferable to any person except after notice to the Santa Ana Water Board/USEPA. The Santa Ana Water Board/USEPA may require modification or revocation and reissuance of the Order/Permit to change the name of the Discharger and incorporate such other requirements as may be necessary under the CWA and CWC. (40 CFR § 122.41(l)(3), 122.61.)

### **III. STANDARD PROVISIONS – MONITORING**

- A.** Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 CFR § 122.41(j)(1).)
- B.** Monitoring must be conducted according to test procedures approved under 40 CFR § 136 for the analyses of pollutants unless another method is required under 40 CFR § 1, subchapter N or O. (40 CFR § 122.41(j)(4).)
- C.** Monitoring for quantitative data shall be conducted in accordance with sufficiently sensitive analytical methods approved under 40 CFR § 136 or required under 40 CFR § 1, subchapter N or O. For the purposes of this requirement, a method approved under 40 CFR § 136 or required under 40 CFR § 1, subchapter N or O is “sufficiently sensitive” when:
  - 1. The method minimum level (ML) is at or below the level of the most stringent effluent limitation established in the Order/Permit for the measured pollutant or pollutant parameter, and either the method ML is at or below the level of the most stringent applicable water quality criterion for the measured pollutant or pollutant parameter or the method ML is above the applicable water quality criterion but the amount of the pollutant or pollutant parameter in the facility’s discharge is high enough that the method detects and quantifies the level of the pollutant or pollutant parameter in the discharge; or
  - 2. The method has the lowest ML of the analytical methods approved under 40 CFR § 136 or required under 40 CFR § 1, subchapter N or O for the measured pollutant or pollutant parameter.

In the case of pollutants or pollutant parameters for which there are no approved methods under 40 CFR § 136 or otherwise required under 40 CFR § 1, subchapter N or O, monitoring must be conducted according to a test procedure specified in this Order for such pollutants or pollutant parameters. (40 C.F.R. §§ 122.21(e)(3), 122.41(j)(4), 122.44(i)(1)(iv).)

### **IV. STANDARD PROVISIONS – RECORDS**

- A.** Except for records of monitoring information required by this Order/Permit related to the Discharger’s sewage sludge use and disposal activities, which shall be retained for a period of at least five (5) years (or longer as required by 40 CFR § 503), the Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order/Permit, and records of all data used to complete the application for this Order/Permit, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Santa Ana Water Board/USEPA at any time. (40 CFR § 122.41(j)(2).)
- B.** Records of monitoring information shall include:
  - 1. The date, exact place, and time of sampling or measurements (40 CFR § 122.41(j)(3)(i));



2. The individual(s) who performed the sampling or measurements (40 CFR § 122.41(j)(3)(ii));
  3. The date(s) analyses were performed (40 CFR § 122.41(j)(3)(iii));
  4. The individual(s) who performed the analyses (40 CFR § 122.41(j)(3)(iv));
  5. The analytical techniques or methods used (40 CFR § 122.41(j)(3)(v)); and
  6. The results of such analyses. (40 CFR § 122.41(j)(3)(vi).)
- C. Claims of confidentiality for the following information will be denied (40 C.F.R. § 122.7(b)):
1. The name and address of any permit applicant or Discharger. (40 C.F.R. § 122.7(b)(1)); and
  2. Permit applications (and attachments), permits and effluent data. (40 CFR § 122.7(b)(2).)

## **V. STANDARD PROVISIONS – REPORTING**

### **A. Duty to Provide Information**

The Discharger shall furnish to the Santa Ana Water Board, State Water Board, and/or USEPA within a reasonable time, any information which the Regional Water Board, State Water Board, and/or USEPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order/Permit or to determine compliance with this Order/Permit. Upon request, the Discharger shall also furnish to the Santa Ana Water Board, State Water Board, and/or USEPA copies of records required to be kept by this Order/Permit. (40 CFR § 122.41(h); CWC 13267, 13383.)

### **B. Signatory and Certification Requirements**

1. All applications, reports, or information submitted to the Santa Ana Water Board, State Water Board, and/or USEPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, V.B.5, and V.B.6 below. (40 CFR § 122.41(k)(1).)
2. The CWA provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this Order/Permit, including monitoring reports or reports of compliance or non-compliance shall, upon conviction, be punished by a fine of not more than \$10,000 per violation, or by imprisonment for not more than 6 months per violation, or by both. (40 CFR § 122.41(k)(2).)
3. For a municipality, State, federal, or other public agency. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of USEPA. (40 CFR § 122.22(a)(3).)

4. All reports required by this Order/Permit and other information requested by the Santa Ana Water Board, State Water Board, and/or USEPA shall be signed by a person described in Standard Provisions – Reporting V.B.3, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
  - a. The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.3 above (40 CFR § 122.22(b)(1).);
  - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company, (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 CFR § 122.22(b)(2).); and
  - c. The written authorization is submitted to the Santa Ana Water Board, State Water Board, and USEPA (40 CFR § 122.22(b)(3).)
5. Changes to authorization. If an authorization under Standard Provisions – Reporting V.B.4 is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting V.B.4 must be submitted to the Santa Ana Water Board, State Water Board, and USEPA prior to or together with any reports, information, or applications to be signed by an authorized representative. (40 CFR § 122.22(c).)
6. Certification. Any person signing a document under Standard Provisions – Reporting V.B.3 or V.B.4 shall make the following certification:

“I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.” (40 CFR § 122.22(d).)
7. Electronic reporting. If documents described in Standard Provisions – Reporting V.B.1, V.B.3, or V.B.4 are submitted electronically by or on behalf of the NPDES-regulated facility, any person providing the electronic signature for such documents shall meet all relevant requirements of Standard Provisions – Reporting V.B, and shall ensure that all of the relevant requirements of 40 CFR § 3 (including, in all cases, subpart D to part 3) (Cross-Media Electronic Reporting) and 40 CFR § 127 (NPDES Electronic Reporting Requirements) are met for that submission. (40 CFR § 122.22(e).)

### **C. Monitoring Reports**

1. Monitoring results shall be reported at the intervals specified elsewhere in this Order/Permit. (40 CFR § 122.41 (l)(4).)
2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) or forms provided or specified by the Santa Ana Water Board, State Water Board, and/or USEPA for reporting results of monitoring of sludge use or disposal practices. As of December 21, 2016 all reports and forms submitted in compliance with this section must be submitted electronically by the Discharger to the initial recipient defined in Standard Provisions – Reporting V.J and comply with this section and 40 CFR § 3, 40 CFR § 122.22, and 40 CFR § 127. (40 CFR § 122.41(l)(4)(i).)
3. If the Discharger monitors any pollutant more frequently than required by this Order/Permit using test procedures approved under 40 CFR § 136, or another method required for an industry-specific waste stream under 40 CFR § 1, subchapter N or O, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Santa Ana Water Board, State Water Board, and/or USEPA. (40 CFR § 122.41(l)(4)(ii).)
4. Calculations for all limitations which require averaging of measurements shall utilize an arithmetic mean unless otherwise specified in this Order/Permit. (40 CFR § 122.41(l)(4)(iii).)

### **D. Compliance Schedules**

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order/Permit shall be submitted no later than 14 days following each schedule date. (40 CFR § 122.41(l)(5).)

### **E. Twenty-Four Hour Reporting**

1. The Discharger shall report any noncompliance which may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A report shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The report shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance.

For noncompliance events related to combined sewer overflows, sanitary sewer overflows or bypass events, these reports must include the data described above (with the exception of time of discovery) as well as the type of event (combined sewer overflows, sanitary sewer overflows or bypass events), type of overflow structure (e.g., manhole, combined sewer overflows), discharge volumes untreated by the treatment works treating domestic sewage, types of human

health and environmental impacts of the sewer overflow event, and whether the noncompliance was related to wet weather.

As of December 21, 2025, all reports related to combined sewer overflows, sanitary sewer overflows or bypass events must be submitted to the Santa Ana Water Board/USEPA and must be submitted electronically by the Discharger to the initial recipient defined in Standard Provisions – Reporting V.J. The reports shall comply with 40 CFR § 3, 40 CFR § 122.22, and 40 CFR § 127. The Santa Ana Water Board/USEPA may also require the Discharger to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows or bypass events under this section. (40 CFR § 122.41(l)(6)(i).)

2. The following shall be included as information which must be reported within 24 hours:
  - a. Any unanticipated bypass which exceeds any effluent limitation in this Order/Permit. (40 CFR § 122.41(l)(6)(ii)(A).)
  - b. Any upset which exceeds any effluent limitation in the Order/Permit. (40 CFR § 122.41(l)(6)(ii)(B).)
  - c. Violation of a maximum daily discharge limitation for any of the pollutants listed by the Santa Ana Water Board/USEPA in the Order/Permit to be reporting within 24 hours. (See 122.44(g).) (40 CFR § 122.41(l)(6)(ii)(C).)
3. The Santa Ana Water Board/USEPA may waive the above required written report on a case-by-case basis if an oral report has been received within 24 hours. (40 CFR § 122.41(l)(6)(iii).)

#### **F. Planned Changes**

The Discharger shall give notice to the Santa Ana Water Board/USEPA as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required only when (40 CFR § 122.41(l)(1)):

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in 40 CFR § 122.29(b) (40 CFR § 122.41(l)(1)(i)); or
2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants which are subject neither to effluent limitations in the Order/Permit, nor to notification requirements under 40 CFR § 122.42(a)(1) (see Additional Provisions—Notification Levels VII.A). (40 CFR § 122.41(l)(1)(ii).)
3. The alteration or addition results in a significant change in the Discharger's sludge use or disposal practices, and such alteration, addition, or change may justify the application of Order/Permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during the Order/Permit application process or not reported pursuant to an approved land application plan. (40 CFR § 122.41(l)(1)(iii).)

**G. Anticipated Noncompliance**

The Discharger shall give advance notice to the Santa Ana Water Board/USEPA of any planned changes in the permitted facility or activity that may result in noncompliance with Order/Permit requirements. (40 CFR § 122.41(l)(2).)

**H. Other Noncompliance**

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E, at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E. For noncompliance events related to combined sewer overflows, sanitary sewer overflows or bypass events, these reports shall contain the information described in Standard Provision – Reporting V.E and the applicable required data in appendix A to 40 CFR § 127. As of December 21, 2025, all reports related to combined sewer overflows, sanitary sewer overflows or bypass events submitted in compliance with this section must be submitted electronically by the Discharger to the Santa Ana Water Board/USEPA or initial recipient, as defined in 40 CFR § 127.2(b), in compliance with this section and 40 CFR § 3 (including, in all cases, subpart D to 3), 122.22, and 40 CFR § 127. (40 CFR § 122.41(l)(7).)

**I. Other Information**

Where the Discharger becomes aware that it failed to submit any relevant facts in an order/permit application, or submitted incorrect information in an order/permit application or in any report to the Santa Ana Water Board, State Water Board, or USEPA, it shall promptly submit such facts or information. (40 CFR § 122.41(l)(8).)

**J. Identification of the Initial Recipient for NPDES Electronic Reporting Data**

The owner, operator, or the duly authorized representative of an NPDES-regulated entity is required to electronically submit the required NPDES information (as specified in appendix A to 40 CFR § 127) to the appropriate initial recipient defined in 40 CFR § 127.2(b). USEPA will identify and publish the list of initial recipients on its website and in the Federal Register, by state and by NPDES data group (see 40 CFR § 127.2(c)). USEPA will update and maintain this listing. (40 CFR § 122.41(l)(9).)

**VI. STANDARD PROVISIONS – ENFORCEMENT**

- A.** The Regional Water Board and/or USEPA is authorized to enforce the terms of this permit under several provisions of the CWC, including, but not limited to, sections 13268, 13385, 13386, and 13387.

**VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS**

**A. Publicly-Owned Treatment Works (POTWs)**

All POTWs shall provide adequate notice to the Santa Ana Water Board/USEPA of the following (40 CFR § 122.42(b)):



1. Any new introduction of pollutants into the POTW from an indirect discharger that would be subject to CWA sections 301 or 306 if it were directly discharging those pollutants (40 CFR § 122.42(b)(1)); and
2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of issuance of the Order/Permit. (40 CFR § 122.42(b)(2).)
3. For the purposes of this paragraph, adequate notice shall include information on the quality and quantity of effluent introduced into the POTW, and any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 CFR § 122.42(b)(3))

## ATTACHMENT E – MONITORING AND REPORTING PROGRAM

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## ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

Section 308 of the federal Clean Water Act (CWA) and 40 CFR § 122.41(h), (j)-(l), 122.44(i) and 122.48 require that all NPDES permits specify monitoring and reporting requirements. California Water Code (CWC) sections 13267 and 13383 also authorize the Santa Ana Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. This MRP establishes monitoring, reporting, and recordkeeping requirements that implement federal and California laws and/or regulations. The Discharger shall comply with this MRP of this Order/Permit.

### I. GENERAL MONITORING PROVISIONS

- A. The Santa Ana Water Board and USEPA, working with other groups, have developed a comprehensive basis for effluent and receiving water monitoring appropriate to large publicly-owned treatment works (POTWs) discharging to waters of the Southern California Bight. This effort has culminated in the publication by the Southern California Coastal Water Research Project (SCCWRP) of the Model Monitoring Program guidance document (Schiff, K.C., J.S. Brown and S.B. Weisberg. 2001. *Model Monitoring Program for Large Ocean Dischargers in Southern California*. SCCWRP Tech. Rep. #357. Southern California Coastal Water Research Project, Westminster, CA. 101 pp.). This guidance provides the principles, framework and recommended design for effluent and receiving water monitoring elements that have guided development of the monitoring program described below.
- B. This conceptual framework along with the California Ocean Plan has three components that comprise a range of spatial and temporal scales: (1) core monitoring; (2) regional monitoring; and (3) strategic process studies.
  1. **Core Monitoring.** Core monitoring is local in nature and focuses on monitoring trends in quality and effects of the point source discharge. This includes discharge monitoring, as well as some aspects of receiving water monitoring. Core monitoring results for the effluent shall be submitted on monthly Discharge Monitoring Reports/State Monitoring Reports (DMR/SMR) and summarized in the annual receiving water monitoring report if needed. Core monitoring results for receiving water, including annotated QA/QC findings, shall be described and summarized in the annual receiving water monitoring report, due by March 15th of each year, for the previous fiscal year (July 1 through June 30). The annual receiving water monitoring report shall include the specified parameters for each station along with more detailed statistical comparisons, including analyses to elucidate spatial and temporal trends in the data, and in relation to the wastewater plume. Statistical methods shall include, but are not limited to, various multivariate techniques such as cluster analysis, ordination, and regression. The applicability and choice of statistical methods shall be explained in the report.
  2. **Regional Monitoring.** Regional monitoring is focused on questions best answered by a region-wide approach that incorporates coordinated survey design and sampling techniques. Key components of regional monitoring include elements to address pollutant mass emission estimates, public health concerns, monitoring trends in natural resources, assessment of regional impacts from all contaminant

sources, and beneficial use protection. The final designs of regional monitoring programs are developed by means of steering and technical committees comprised of participating agencies. For each component of regional monitoring, this Order/Permit specifies the required degree and nature of participation by Orange County Sanitation District (OC San), based upon its past participation in regional monitoring programs. The degree and nature of OC San's participation in regional monitoring programs shall be briefly described and summarized in the annual receiving water monitoring report. Each year, as part of the annual receiving monitoring report, the Discharger shall provide an informational report summarizing to date its contributing activities towards coordinated implementation of regional monitoring programs.

Although participation in regional monitoring programs is required under this Order/Permit, revisions to Attachment E, at the direction of the Santa Ana Water Board and USEPA, may be necessary to accomplish the goals of regional monitoring.

3. **Strategic Process Studies.** Strategic process studies are focused on refined questions regarding specific effects or development of monitoring techniques and are anticipated to be of short duration and/or small scale, although multi-year studies may be needed. Questions regarding discharge or receiving water quality, discharge impacts, ocean processes in the area of the discharge, or development of techniques for monitoring the same, arising out of the results of core, regional monitoring, or other relevant studies shall be pursued through these studies. These studies are by nature ad hoc and, typically, cannot be anticipated in advance of the five-year permit cycle. Monitoring efforts, status of in-progress studies, and summary results for completed strategic process studies shall be briefly described and summarized in the annual receiving water monitoring report.

In the spring, beginning in 2022 and continuing every-other year during the term of this Order/Permit, the Discharger, Santa Ana Water Board, and USEPA shall consult to determine the need for strategic process studies. By October 1st, the Discharger shall submit proposals to the Santa Ana Water Board and USEPA for the following fiscal year's (July 1 through June 30) monitoring effort, or a letter explaining why no special studies are proposed. Final scopes of work, including reporting schedules, shall be presented by the Discharger at a spring meeting with Santa Ana Water Board and USEPA to obtain Santa Ana Water Board and USEPA approval. Upon approval, the Discharger shall implement its strategic process studies.

- C. Every five years SCCWRP coordinates receiving water regional monitoring within the Southern California Bight and compiles monitoring data collected by the dischargers and other participating entities. The sixth regional monitoring program (Bight '18) occurred primarily during the summer of 2018. The next (seventh) regional monitoring program (Bight '23) is expected to take place during 2023. While participation in regional monitoring programs is required under this Order/Permit, revisions to the Discharger's monitoring program at the direction of the Santa Ana Water Board and



USEPA may be necessary to accomplish the goals of regional monitoring or to allow the performance of special studies to investigate regional or site-specific water issues of concern. These revisions may include a reduction or increase in the number of parameters to be monitored, the frequency of monitoring, or the number and size of samples to be collected, which are applicable to receiving water monitoring program only. Such changes shall be authorized by the Santa Ana Water Board Executive Officer and USEPA Director upon written notification to the Discharger.

Permittee participation in regional monitoring programs is required as a condition of this Order/Permit. The Discharger shall complete collection and analysis of samples in accordance with the schedule established by the Steering Committee directing the Bight-wide regional monitoring surveys. The level of participation shall be similar to that provided by the Discharger in previous regional surveys conducted in 1994, 1998, 2003, 2008, 2013, and 2018.

- D. All plant samples shall be representative of the waste discharge under conditions of peak load. Results of quarterly, semiannual, and annual analyses shall be reported by the due date specified in Table E-16 of the MRP. Should there be instances when monitoring could not be performed during these specified months, the Discharger must notify the Santa Ana Water Board and USEPA, state the reason why monitoring could not be conducted, and obtain approval from the Santa Ana Water Board and USEPA for an alternate schedule.
- E. Pollutants shall be analyzed using the analytical methods described in 40 CFR § 136; or where no methods are specified for a given pollutant, by methods approved by the Santa Ana Water Board, the State Water Board, and/or USEPA. If more than one analytical test method is listed for a given parameter, the Discharger must select from the listed methods and corresponding Minimum Level. USEPA published regulations for the Sufficiently Sensitive Methods Rule (SSM Rule) which became effective September 18, 2015. For the purposes of monitoring and reporting under the NPDES program, when more than one test procedure is approved under 40 CFR § 136 for the analysis of a pollutant or pollutant parameter, the test procedure must be sufficiently sensitive as defined at 40 CFR § 122.21(e)(3) and 122.44(i)(1)(iv). A USEPA-approved analytical method is sufficiently sensitive where:
  - 1. The ML is at or below both the level of the applicable water quality criterion/objective and the permit limitation or performance goal for the measured pollutant or pollutant parameter; or
  - 2. In permit applications, the ML is above the applicable water quality criterion/objective, but the amount of the pollutant or pollutant parameter in a facility's discharge is high enough that the method detects and quantifies the level of the pollutant or pollutant parameter in the discharge; or
  - 3. The method has the lowest ML of the USEPA-approved analytical methods where none of the USEPA-approved analytical methods for a pollutant can achieve the MLs necessary to assess the need for effluent limitations or to monitor compliance with a permit limitation.

The MLs in Ocean Plan Appendix II remain applicable. However, there may be situations when analytical methods are published with MLs that are more sensitive than the MLs for analytical methods listed in the Ocean Plan. For instance, USEPA Method 1631E for mercury is not currently listed in Ocean Plan Table II, but it is published with an ML of 0.5 ng/L that makes it a sufficiently sensitive analytical method.

- F. In conformance with federal regulations 40 CFR § 122.45(c), analyses to determine compliance with the effluent limitations for metals shall be conducted using the total recoverable method. For chromium (III) and (VI), the Discharger may, at its option, meet both the chromium (III) and the chromium (VI) limitations by analyzing for total recoverable chromium.
- G. Water/wastewater samples must be analyzed within allowable holding time limits as specified in 40 CFR § 136.3. All quality assurance/quality control (QA/QC) analyses must be run in the same preparation and analytical batches in which samples are actually analyzed. The Discharger shall retain the QA/QC documentation in its files and make available for inspection and/or submit this documentation when requested by the Santa Ana Water Board and/or USEPA. Proper chain of custody procedures must be followed, and a copy of this documentation shall be submitted with the monthly report.
- H. If the Discharger samples and performs analyses (other than for process/operational control, startup, research, or equipment testing) on any influent, effluent, or receiving water constituent more frequently than required by this Order/Permit using approved analytical methods, the results of those analyses shall be included in the monitoring report. These results shall be reflected in the calculation of the average (or median) used in demonstrating compliance with limitations set forth in this Order/Permit.
- I. All monitoring instruments and devices used by the Discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy. In the event that continuous monitoring equipment is out of service for greater than a 24-hour period, the Discharger shall obtain a representative grab sample each day the equipment is out of service. The Discharger shall correct the cause(s) of failure of the continuous monitoring equipment as soon as practicable. In its monitoring report, the Discharger shall specify the period(s) during which the equipment was out of service and, if the problem has not been corrected, shall identify the steps which the Discharger is taking or proposes to take to bring the equipment back into service and the schedule for these actions.
- J. For every item of monitoring data where the requirements are not met, the monitoring report shall include a statement discussing the reasons for noncompliance, the actions undertaken or proposed that will bring the discharge into full compliance with requirements at the earliest time and an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Santa Ana Water Board and USEPA by letter when compliance with the time schedule has been achieved.
- K. **Laboratory Certification.** Laboratories analyzing monitoring samples shall be certified by the State Water Board, Environmental Laboratory Accreditation Program (ELAP), in accordance with CWC section 13176, and must include QA/QC data in their reports.

- L. The Discharger shall have and implement an acceptable written QA plan for laboratory analyses. The Discharger shall ensure that the results of the Discharge Monitoring Report-Quality Assurance (DMR-QA) Study or the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board the following address:

State Water Resources Control Board  
 Quality Assurance Program Officer  
 Office of Information Management and Analysis  
 1001 I Street, Sacramento, CA 95814

## II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order/Permit:

**Table E-1. Monitoring Station Locations**

| Monitoring Location Type                                                                     | Monitoring Location Name | Monitoring Location Description                                                                                                                                                                                    | Latitude      | Longitude      |
|----------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------|
| <b>Influent Monitoring Stations (n=2)</b>                                                    |                          |                                                                                                                                                                                                                    |               |                |
| Reclamation Plant No. 1 Influent                                                             | INF-001                  | Reclamation Plant No. 1 sampling stations shall be located at each point of inflow to the treatment plant and upstream of any in-plant return flows, where representative samples of the influent can be obtained. | 33° 41.588' N | 117° 56.294' W |
| Treatment Plant No. 2 Influent (Before OCWD receives reclaimed water from Plant No.2)        | INF-002                  | Treatment Plant No. 2 sampling stations shall be located at each point of inflow to the treatment plant and upstream of any in-plant return flows, where representative samples of the influent can be obtained.   | 33° 38.342' N | 117° 57.462' W |
| Treatment Plant No. 2 Influent (Pending until OCWD receives reclaimed water from Plant No.2) | INF-002                  | INF-002 sampling stations shall be located at each point of reclaimable inflow to the Treatment Plant No.2, where representative samples of the reclaimable influent can be obtained.                              | 33° 38.342' N | 117° 57.462' W |

| Monitoring Location Type                                                                                  | Monitoring Location Name | Monitoring Location Description                                                                                                                                                                                   | Latitude      | Longitude      |
|-----------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------|
|                                                                                                           | INF-002A                 | INF-002A sampling station shall be located at a point of non-reclaimable influent from the 78-inch interplant trunkline containing SARI influent, Reclamation Plant No.1 recycle flow, and GWRS wastestream.      | 33° 38.317' N | 117° 57.453' W |
| <b>Effluent Monitoring Stations (n=2)</b>                                                                 |                          |                                                                                                                                                                                                                   |               |                |
| Effluent discharged to Discharge Point 001 (during normal operation)                                      | EFF-001                  | Sampling station shall be located downstream of any in-plant return flows, but before entering 120-inch outfall, where representative effluent samples can be obtained.                                           | 33° 38.012' N | 117° 57.452' W |
| Effluent discharged to Discharge Point 002 (during essential maintenance or capital improvement projects) | EFF-002                  | Sampling station can be same as EFF-001, but before entering 78-inch outfall, where representative samples of the disinfected effluent discharge can be obtained.                                                 | 33° 38.012' N | 117° 57.452' W |
| <b>Emergency Discharge Monitoring Stations (n=2)</b>                                                      |                          |                                                                                                                                                                                                                   |               |                |
| Emergency discharges to Discharge Point 002 (during an emergency)                                         | EMG-001                  | Sampling station shall be located downstream of any in-plant return flows, but before entering the emergency 78-inch outfall, where representative samples of the disinfected effluent discharge can be obtained. | 33° 38.012' N | 117° 57.452' W |
| Santa Ana River Overflow to Discharge Point 003 (during an extreme emergency)                             | EMG-002                  | Sampling station shall be located before entering the Santa Ana River overflow weirs, where representative samples of the disinfected effluent discharge can be obtained.                                         | 33° 38.297' N | 117° 57.356' W |

**Table E-2. Receiving Water Core and Regional Monitoring Station Locations**

| Station Description                                                        | Station Location Name | Latitude      | Longitude      | Depth (m) | CTD Sampling Depths (1 m intervals) | Discrete Sampling Depths (m) |
|----------------------------------------------------------------------------|-----------------------|---------------|----------------|-----------|-------------------------------------|------------------------------|
| <b>Receiving Water Core Monitoring Stations</b>                            |                       |               |                |           |                                     |                              |
| <b>Monthly Water Quality Monitoring Stations (n=28)</b>                    |                       |               |                |           |                                     |                              |
| * = Station sampled for ammonia (NH <sub>3</sub> -N) and nitrate nitrogen. |                       |               |                |           |                                     |                              |
| *                                                                          | 2103                  | 33° 35.089' N | 117° 56.678' W | 110       | 1 - 75 m                            | 1, 10, 20, 30, 40, 50, 60    |
| *                                                                          | 2104                  | 33° 34.199' N | 117° 57.414' W | 143       | 1 - 75 m                            | 1, 10, 20, 30, 40, 50, 60    |
| *                                                                          | 2105                  | 33° 33.309' N | 117° 58.150' W | 280       | 1 - 75 m                            | 1, 10, 20, 30, 40, 50, 60    |
| *                                                                          | 2106                  | 33° 32.420' N | 117° 58.885' W | 309       | 1 - 75 m                            | 1, 10, 20, 30, 40, 50, 60    |
| *                                                                          | 2183                  | 33° 35.701' N | 117° 57.744' W | 36        | 1 - 2 m above bottom                | 1, 10, 20, 30, 34            |
| *                                                                          | 2184                  | 33° 34.811' N | 117° 58.480' W | 51        | 1 - 2 m above bottom                | 1, 10, 20, 30, 40, 49        |
| *                                                                          | 2185                  | 33° 33.922' N | 117° 59.215' W | 114       | 1 - 75 m                            | 1, 10, 20, 30, 40, 50, 60    |
| *                                                                          | 2186                  | 33° 33.032' N | 117° 59.951' W | 247       | 1 - 75 m                            | 1, 10, 20, 30, 40, 50, 60    |
| *                                                                          | 2203                  | 33° 36.313' N | 117° 58.810' W | 25        | 1 - 2 m above bottom                | 1, 10, 20, 23                |
| *                                                                          | 2204                  | 33° 35.423' N | 117° 59.546' W | 39        | 1 - 2 m above bottom                | 1, 10, 20, 30, 37            |
| ZID boundary; *                                                            | 2205                  | 33° 34.534' N | 118° 00.282' W | 57        | 1 - 2 m above bottom                | 1, 10, 20, 30, 40, 50, 55    |
| *                                                                          | 2206                  | 33° 33.644' N | 118° 01.018' W | 185       | 1 - 75 m                            | 1, 10, 20, 30, 40, 50, 60    |
| *                                                                          | 2223                  | 33° 36.924' N | 117° 59.871' W | 22        | 1 - 2 m above bottom                | 1, 10, 20                    |
| *                                                                          | 2224                  | 33° 36.035' N | 118° 00.608' W | 31        | 1 - 2 m above bottom                | 1, 10, 20, 29                |



| Station Description | Station Location Name | Latitude      | Longitude      | Depth (m) | CTD Sampling Depths (1 m intervals) | Discrete Sampling Depths (m) |
|---------------------|-----------------------|---------------|----------------|-----------|-------------------------------------|------------------------------|
| *                   | 2225                  | 33° 35.146' N | 118° 01.346' W | 47        | 1 - 2 m above bottom                | 1, 10, 20, 30, 40, 45        |
| *                   | 2226                  | 33° 34.257' N | 118° 02.083' W | 135       | 1 - 75 m                            | 1, 10, 20, 30, 40, 50, 60    |
| *                   | 2303                  | 33° 37.537' N | 118° 00.936' W | 21        | 1 - 2 m above bottom                | 1, 10, 19                    |
| *                   | 2304                  | 33° 36.649' N | 118° 01.674' W | 29        | 1 - 2 m above bottom                | 1, 10, 20, 27                |
| *                   | 2305                  | 33° 35.760' N | 118° 02.412' W | 38        | 1 - 2 m above bottom                | 1, 10, 20, 30, 36            |
| *                   | 2306                  | 33° 34.871' N | 118° 03.149' W | 114       | 1 - 75 m                            | 1, 10, 20, 30, 40, 50, 60    |
| --                  | 2351                  | 33° 38.151' N | 118° 02.001' W | 21        | 1 - 2 m above bottom                | None                         |
| --                  | 2352                  | 33° 37.262' N | 118° 02.739' W | 29        | 1 - 2 m above bottom                | None                         |
| --                  | 2353                  | 33° 36.373' N | 118° 03.477' W | 37        | 1 - 2 m above bottom                | None                         |
| --                  | 2354                  | 33° 35.484' N | 118° 04.214' W | 123       | 1 - 75 m                            | None                         |
| --                  | 2403                  | 33° 38.765' N | 118° 03.072' W | 21        | 1 - 2 m above bottom                | None                         |
| --                  | 2404                  | 33° 37.875' N | 118° 03.808' W | 30        | 1 - 2 m above bottom                | None                         |
| --                  | 2405                  | 33° 36.986' N | 118° 04.544' W | 37        | 1 - 2 m above bottom                | None                         |
| --                  | 2406                  | 33° 36.096' N | 118° 05.280' W | 60        | 1 - 2 m above bottom                | None                         |

**Quarterly REC-1 Water Quality Monitoring Stations (Offshore Zone) (n=8)**

Quarterly REC-1 stations are monitored 5 days over a 30-day period in spring, summer, fall, and winter for geometric mean calculation ("spring" means April, May, or June; "summer" means July, August, or September; "fall" means October, November, or December; and "winter" means January, February, or March).

|    |      |               |                |     |          |                           |
|----|------|---------------|----------------|-----|----------|---------------------------|
| -- | 2103 | 33° 35.089' N | 117° 56.678' W | 110 | 1 - 75 m | 1, 10, 20, 30, 40, 50, 60 |
|----|------|---------------|----------------|-----|----------|---------------------------|

| Station Description | Station Location Name | Latitude      | Longitude      | Depth (m) | CTD Sampling Depths (1 m intervals) | Discrete Sampling Depths (m) |
|---------------------|-----------------------|---------------|----------------|-----------|-------------------------------------|------------------------------|
| --                  | 2104                  | 33° 34.199' N | 117° 57.414' W | 143       | 1 - 75 m                            | 1, 10, 20, 30, 40, 50, 60    |
| --                  | 2183                  | 33° 35.701' N | 117° 57.744' W | 36        | 1 - 2 m above bottom                | 1, 10, 20, 30, 34            |
| --                  | 2203                  | 33° 36.313' N | 117° 58.810' W | 25        | 1 - 2 m above bottom                | 1, 10, 20, 23                |
| --                  | 2223                  | 33° 36.924' N | 117° 59.871' W | 22        | 1 - 2 m above bottom                | 1, 10, 20                    |
| --                  | 2303                  | 33° 37.537' N | 118° 00.936' W | 21        | 1 - 2 m above bottom                | 1, 10, 19                    |
| --                  | 2351                  | 33° 38.151' N | 118° 02.001' W | 21        | 1 - 2 m above bottom                | 1, 10, 19                    |
| --                  | 2403                  | 33° 38.765' N | 118° 03.072' W | 21        | 1 - 2 m above bottom                | 1, 10, 19                    |

#### Quarterly Benthic Monitoring Stations (n=11)

Quarterly benthic stations are monitored for infauna and sediment geochemistry (except pesticides) every quarter in spring, summer, fall, and winter ("spring" means April, May, or June; "summer" means July, August, or September; "fall" means October, November, or December; and "winter" means January, February, or March) as well as sampled annually in summer for pesticides and annually for whole sediment toxicity.

|              |    |               |                |    |    |    |
|--------------|----|---------------|----------------|----|----|----|
| ZID Boundary | 0  | 33° 34.573' N | 118° 00.598' W | 56 | -- | -- |
| --           | 1  | 33° 34.657' N | 118° 00.968' W | 56 | -- | -- |
| ZID Boundary | 4  | 33° 34.498' N | 117° 59.761' W | 56 | -- | -- |
| --           | 9  | 33° 34.363' N | 117° 59.510' W | 59 | -- | -- |
| --           | 73 | 33° 34.596' N | 118° 00.709' W | 55 | -- | -- |
| ZID Boundary | 76 | 33° 34.459' N | 118° 00.297' W | 58 | -- | -- |
| --           | 77 | 33° 34.373' N | 117° 59.730' W | 60 | -- | -- |
| --           | 84 | 33° 34.648' N | 118° 00.543' W | 54 | -- | -- |
| --           | 85 | 33° 34.532' N | 118° 00.679' W | 57 | -- | -- |

| Station Description                                                                                                                               | Station Location Name | Latitude      | Longitude      | Depth (m) | CTD Sampling Depths (1 m intervals) | Discrete Sampling Depths (m) |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------|----------------|-----------|-------------------------------------|------------------------------|
| Farfield Reference                                                                                                                                | CON                   | 33° 36.037' N | 118° 05.387' W | 59        | --                                  | --                           |
| ZID Boundary                                                                                                                                      | ZB                    | 33° 34.545' N | 118° 00.274' W | 56        | --                                  | --                           |
| <b>Annual Benthic Monitoring Stations (n=11)</b>                                                                                                  |                       |               |                |           |                                     |                              |
| Annual benthic stations are monitored only in the “summer” (July, August, or September) for infauna and sediment geochemistry.                    |                       |               |                |           |                                     |                              |
| --                                                                                                                                                | 3                     | 33° 34.434' N | 118° 00.660' W | 60        | --                                  | --                           |
| --                                                                                                                                                | 5                     | 33° 34.749' N | 118° 01.612' W | 59        | --                                  | --                           |
| --                                                                                                                                                | 10                    | 33° 34.902' N | 118° 02.081' W | 62        | --                                  | --                           |
| --                                                                                                                                                | 12                    | 33° 34.385' N | 117° 59.054' W | 58        | --                                  | --                           |
| --                                                                                                                                                | 13                    | 33° 35.307' N | 118° 02.944' W | 59        | --                                  | --                           |
| --                                                                                                                                                | 37                    | 33° 34.832' N | 117° 57.369' W | 56        | --                                  | --                           |
| --                                                                                                                                                | 74                    | 33° 34.616' N | 118° 00.230' W | 57        | --                                  | --                           |
| --                                                                                                                                                | 75                    | 33° 34.559' N | 117° 59.974' W | 60        | --                                  | --                           |
| --                                                                                                                                                | 78                    | 33° 34.329' N | 118° 00.035' W | 63        | --                                  | --                           |
| --                                                                                                                                                | 86                    | 33° 34.400' N | 118° 00.380' W | 57        | --                                  | --                           |
| --                                                                                                                                                | 87                    | 33° 34.780' N | 118° 00.842' W | 60        | --                                  | --                           |
| <b>1/5-year Benthic Monitoring Stations (n=35)</b>                                                                                                |                       |               |                |           |                                     |                              |
| 1/5-year benthic stations are monitored once every five years in the “summer” (July, August, or September) for infauna and sediment geochemistry. |                       |               |                |           |                                     |                              |
| --                                                                                                                                                | 7                     | 33° 35.325' N | 118° 00.367' W | 41        | --                                  | --                           |
| --                                                                                                                                                | 8                     | 33° 35.164' N | 117° 59.555' W | 44        | --                                  | --                           |
| --                                                                                                                                                | 17                    | 33° 33.961' N | 118° 00.187' W | 91        | --                                  | --                           |
| --                                                                                                                                                | 18                    | 33° 34.064' N | 118° 00.750' W | 91        | --                                  | --                           |
| --                                                                                                                                                | 20                    | 33° 34.599' N | 118° 02.229' W | 100       | --                                  | --                           |
| --                                                                                                                                                | 21                    | 33° 35.313' N | 118° 01.891' W | 44        | --                                  | --                           |
| --                                                                                                                                                | 22                    | 33° 35.204' N | 117° 59.028' W | 45        | --                                  | --                           |
| --                                                                                                                                                | 23                    | 33° 33.968' N | 117° 59.147' W | 100       | --                                  | --                           |

| Station Description | Station Location Name | Latitude      | Longitude      | Depth (m) | CTD Sampling Depths (1 m intervals) | Discrete Sampling Depths (m) |
|---------------------|-----------------------|---------------|----------------|-----------|-------------------------------------|------------------------------|
| --                  | 24                    | 33° 33.563' N | 118° 01.140' W | 200       | --                                  | --                           |
| --                  | 25                    | 33° 33.924' N | 118° 02.176' W | 200       | --                                  | --                           |
| --                  | 27                    | 33° 33.326' N | 117° 59.708' W | 200       | --                                  | --                           |
| --                  | 29                    | 33° 35.033' N | 118° 03.113' W | 100       | --                                  | --                           |
| --                  | 30                    | 33° 35.493' N | 118° 02.899' W | 46        | --                                  | --                           |
| --                  | 33                    | 33° 34.349' N | 117° 57.866' W | 100       | --                                  | --                           |
| --                  | 36                    | 33° 35.308' N | 117° 57.495' W | 45        | --                                  | --                           |
| --                  | 38                    | 33° 34.634' N | 117° 57.317' W | 100       | --                                  | --                           |
| --                  | 39                    | 33° 33.283' N | 117° 58.531' W | 200       | --                                  | --                           |
| --                  | 40                    | 33° 32.496' N | 117° 59.775' W | 303       | --                                  | --                           |
| --                  | 41                    | 33° 32.690' N | 118° 01.149' W | 303       | --                                  | --                           |
| --                  | 42                    | 33° 33.098' N | 118° 02.598' W | 303       | --                                  | --                           |
| --                  | 44                    | 33° 34.586' N | 118° 05.422' W | 241       | --                                  | --                           |
| --                  | 55                    | 33° 36.739' N | 118° 05.413' W | 40        | --                                  | --                           |
| --                  | 56                    | 33° 35.665' N | 118° 05.417' W | 100       | --                                  | --                           |
| --                  | 57                    | 33° 34.970' N | 118° 05.418' W | 200       | --                                  | --                           |
| --                  | 58                    | 33° 33.365' N | 118° 05.347' W | 300       | --                                  | --                           |
| --                  | 59                    | 33° 36.070' N | 118° 03.701' W | 40        | --                                  | --                           |
| --                  | 60                    | 33° 35.532' N | 118° 04.017' W | 100       | --                                  | --                           |
| --                  | 61                    | 33° 35.011' N | 118° 04.326' W | 200       | --                                  | --                           |
| --                  | 62                    | 33° 34.069' N | 118° 04.568' W | 300       | --                                  | --                           |
| --                  | 63                    | 33° 34.173' N | 118° 03.407' W | 200       | --                                  | --                           |
| --                  | 64                    | 33° 33.484' N | 118° 03.663' W | 300       | --                                  | --                           |
| --                  | 65                    | 33° 33.859' N | 117° 57.230' W | 200       | --                                  | --                           |
| --                  | 83                    | 33° 34.239' N | 118° 01.414' W | 100       | --                                  | --                           |
| --                  | C4                    | 33° 35.056' N | 117° 55.833' W | 187       | --                                  | --                           |
| --                  | C5                    | 33° 33.920' N | 117° 55.620' W | 296       | --                                  | --                           |

| Station Description                                                                                                                                                                                                                                                                                                                                                               | Station Location Name | Latitude      | Longitude      | Depth (m) | CTD Sampling Depths (1 m intervals) | Discrete Sampling Depths (m) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------|----------------|-----------|-------------------------------------|------------------------------|
| <b>Semi-annual Trawl Fish and Epibenthic Macroinvertebrate Monitoring Stations (n=6)</b><br>Semi-annual trawl stations are monitored in summer (July, August, or September) and winter (January, February, or March).<br>* = Station is also sampled <u>annually</u> in one of the four quarters for demersal fish liver tissue chemistry and demersal fish liver histopathology. |                       |               |                |           |                                     |                              |
| Outfall; *                                                                                                                                                                                                                                                                                                                                                                        | T1                    | 33° 34.641' N | 118° 00.567' W | 55        | --                                  | --                           |
| Farfield reference; *                                                                                                                                                                                                                                                                                                                                                             | T11                   | 33° 36.055' N | 118° 05.199' W | 60        | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T12                   | 33° 34.868' N | 118° 01.670' W | 57        | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T17                   | 33° 35.309' N | 118° 02.987' W | 60        | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T22                   | 33° 34.326' N | 117° 59.856' W | 60        | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T23                   | 33° 34.336' N | 117° 59.051' W | 58        | --                                  | --                           |
| <b>Annual Trawl Fish and Epibenthic Macroinvertebrate Monitoring Stations (n=8)</b><br>Annual trawl stations are monitored in summer (July, August, or September).                                                                                                                                                                                                                |                       |               |                |           |                                     |                              |
| --                                                                                                                                                                                                                                                                                                                                                                                | T2                    | 33° 35.688' N | 117° 59.561' W | 35        | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T6                    | 33° 35.946' N | 118° 02.785' W | 36        | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T10                   | 33° 33.771' N | 118° 00.250' W | 137       | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T14                   | 33° 34.672' N | 118° 03.200' W | 137       | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T18                   | 33° 36.960' N | 118° 05.273' W | 36        | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T19                   | 33° 35.394' N | 118° 05.424' W | 137       | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T24                   | 33° 35.648' N | 118° 01.274' W | 36        | --                                  | --                           |
| --                                                                                                                                                                                                                                                                                                                                                                                | T25                   | 33° 34.245' N | 118° 01.967' W | 137       | --                                  | --                           |
| <b>Annual Rig Fish Monitoring Zones (n=2)</b><br>Annual rig fishing stations are monitored in summer (July, August, or September).<br>* All station positions and depths shall be determined prior to the first sampling.                                                                                                                                                         |                       |               |                |           |                                     |                              |



| Station Description                                                                                                       | Station Location Name | Latitude                                                                                                                                                                                                                                                                        | Longitude      | Depth (m) | CTD Sampling Depths (1 m intervals) | Discrete Sampling Depths (m) |
|---------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|-------------------------------------|------------------------------|
| Zone 1 (outfall)                                                                                                          | RF1                   | Inshore of the 60 m depth contour bounded by coordinates:<br>33° 36.272' N / 117° 57.264' W,<br>33° 37.522' N / 117° 59.374' W along the 15 m contour;<br>33° 34.698' N / 118° 01.713' W along the 80 m contour; and<br>33° 33.475' N / 117° 59.583' W along the 180 m contour. |                | *         | --                                  | --                           |
| Zone 3 (farfield reference)                                                                                               | RF3                   | Offshore of Huntington Beach along 60 m depth contour bounded by the coordinates:<br>33° 35.885' N / 118° 08.013' W;<br>33° 35.407' N / 118° 07.408' W;<br>33° 34.213' N / 118° 08.628' W;<br>33° 34.830' N / 118° 09.065' W.                                                   |                | *         | --                                  | --                           |
| Receiving Water Regional Monitoring Stations                                                                              |                       |                                                                                                                                                                                                                                                                                 |                |           |                                     |                              |
| Quarterly Southern California Bight Regional Water Quality Monitoring Stations (n=60)                                     |                       |                                                                                                                                                                                                                                                                                 |                |           |                                     |                              |
| * = Core water quality monitoring station sampled during Southern California Bight Regional Water Quality surveys (n=16). |                       |                                                                                                                                                                                                                                                                                 |                |           |                                     |                              |
| --                                                                                                                        | 1701                  | 33° 29.878' N                                                                                                                                                                                                                                                                   | 117° 44.721' W | 10        | 1 - 2 m above bottom                | --                           |
| --                                                                                                                        | 1702                  | 33° 29.180' N                                                                                                                                                                                                                                                                   | 117° 45.120' W | 40        | 1 - 2 m above bottom                | --                           |
| --                                                                                                                        | 1703                  | 33° 28.472' N                                                                                                                                                                                                                                                                   | 117° 45.524' W | 60        | 1 - 2 m above bottom                | --                           |
| --                                                                                                                        | 1704                  | 33° 28.071' N                                                                                                                                                                                                                                                                   | 117° 45.752' W | 100       | 1 - 100 m                           | --                           |
| --                                                                                                                        | 1705                  | 33° 27.434' N                                                                                                                                                                                                                                                                   | 117° 46.115' W | 400       | 1 - 100 m                           | --                           |
| --                                                                                                                        | 1706                  | 33° 26.455' N                                                                                                                                                                                                                                                                   | 117° 46.679' W | 600       | 1 - 100 m                           | --                           |
| --                                                                                                                        | 1801                  | 33° 32.027' N                                                                                                                                                                                                                                                                   | 117° 46.910' W | 10        | 1 - 2 m above bottom                | --                           |
| --                                                                                                                        | 1802                  | 33° 31.591' N                                                                                                                                                                                                                                                                   | 117° 47.158' W | 40        | 1 - 2 m above bottom                | --                           |

| Station Description | Station Location Name | Latitude      | Longitude      | Depth (m) | CTD Sampling Depths (1 m intervals) | Discrete Sampling Depths (m) |
|---------------------|-----------------------|---------------|----------------|-----------|-------------------------------------|------------------------------|
| --                  | 1803                  | 33° 31.154' N | 117° 47.406' W | 60        | 1 - 2 m above bottom                | --                           |
| --                  | 1804                  | 33° 30.612' N | 117° 47.710' W | 100       | 1 - 100 m                           | --                           |
| --                  | 1805                  | 33° 29.782' N | 117° 48.277' W | 500       | 1 - 100 m                           | --                           |
| --                  | 1806                  | 33° 28.618' N | 117° 48.847' W | 600       | 1 - 100 m                           | --                           |
| --                  | 1901                  | 33° 33.682' N | 117° 49.654' W | 10        | 1 - 2 m above bottom                | --                           |
| --                  | 1902                  | 33° 33.165' N | 117° 49.944' W | 60        | 1 - 2 m above bottom                | --                           |
| --                  | 1903                  | 33° 32.762' N | 117° 50.182' W | 100       | 1 - 75 m                            | --                           |
| --                  | 1904                  | 33° 31.787' N | 117° 50.734' W | 405       | 1 - 75 m                            | --                           |
| --                  | 1905                  | 33° 30.810' N | 117° 51.285' W | 510       | 1 - 75 m                            | --                           |
| --                  | 1906                  | 33° 29.829' N | 117° 51.842' W | 550       | 1 - 75 m                            | --                           |
| --                  | 2001                  | 33° 35.335' N | 117° 52.692' W | 10        | 1 - 2 m above bottom                | --                           |
| --                  | 2002                  | 33° 34.755' N | 117° 53.028' W | 60        | 1 - 2 m above bottom                | --                           |
| --                  | 2003                  | 33° 34.565' N | 117° 53.144' W | 100       | 1 - 75 m                            | --                           |
| --                  | 2004                  | 33° 33.589' N | 117° 53.708' W | 345       | 1 - 75 m                            | --                           |
| --                  | 2005                  | 33° 32.613' N | 117° 54.063' W | 410       | 1 - 75 m                            | --                           |
| --                  | 2006                  | 33° 31.647' N | 117° 54.824' W | 470       | 1 - 75 m                            | --                           |
| --                  | 2041                  | 33° 35.969' N | 117° 54.567' W | 10        | 1 - 2 m above bottom                | --                           |
| --                  | 2042                  | 33° 35.413' N | 117° 54.930' W | 53        | 1 - 2 m above bottom                | --                           |
| --                  | 2043                  | 33° 34.908' N | 117° 55.265' W | 165       | 1 - 75 m                            | --                           |
| --                  | 2044                  | 33° 33.951' N | 117° 55.887' W | 300       | 1 - 75 m                            | --                           |
| --                  | 2045                  | 33° 33.013' N | 117° 56.500' W | 390       | 1 - 75 m                            | --                           |
| --                  | 2046                  | 33° 32.080' N | 117° 57.110' W | 432       | 1 - 75 m                            | --                           |
| --                  | 2101                  | 33° 36.183' N | 117° 55.749' W | 10        | 1 - 2 m above bottom                | --                           |

| Station Description | Station Location Name | Latitude      | Longitude      | Depth (m) | CTD Sampling Depths (1 m intervals) | Discrete Sampling Depths (m) |
|---------------------|-----------------------|---------------|----------------|-----------|-------------------------------------|------------------------------|
| --                  | 2102                  | 33° 35.631' N | 117° 56.206' W | 26        | 1 - 2 m above bottom                | --                           |
| *                   | 2103                  | 33° 35.089' N | 117° 56.678' W | 110       | 1 - 75 m                            | --                           |
| *                   | 2104                  | 33° 34.199' N | 117° 57.414' W | 143       | 1 - 75 m                            | --                           |
| *                   | 2105                  | 33° 33.309' N | 117° 58.150' W | 280       | 1 - 75 m                            | --                           |
| *                   | 2106                  | 33° 32.420' N | 117° 58.885' W | 309       | 1 - 75 m                            | --                           |
| --                  | 2201                  | 33° 37.493' N | 117° 57.831' W | 10        | 1 - 2 m above bottom                | --                           |
| --                  | 2202                  | 33° 36.901' N | 117° 58.314' W | 16        | 1 - 2 m above bottom                | --                           |
| *                   | 2203                  | 33° 36.313' N | 117° 58.810' W | 25        | 1 - 2 m above bottom                | --                           |
| *                   | 2204                  | 33° 35.423' N | 117° 59.546' W | 39        | 1 - 2 m above bottom                | --                           |
| *                   | 2205                  | 33° 34.534' N | 118° 00.282' W | 57        | 1 - 2 m above bottom                | --                           |
| *                   | 2206                  | 33° 33.644' N | 118° 01.018' W | 185       | 1 - 75 m                            | --                           |
| --                  | 2301                  | 33° 38.572' N | 118° 00.064' W | 10        | 1 - 2 m above bottom                | --                           |
| --                  | 2302                  | 33° 38.053' N | 118° 00.495' W | 15        | 1 - 2 m above bottom                | --                           |
| *                   | 2303                  | 33° 37.537' N | 118° 00.936' W | 21        | 1 - 2 m above bottom                | --                           |
| *                   | 2304                  | 33° 36.649' N | 118° 01.674' W | 29        | 1 - 2 m above bottom                | --                           |
| *                   | 2305                  | 33° 35.760' N | 118° 02.412' W | 38        | 1 - 2 m above bottom                | --                           |
| *                   | 2306                  | 33° 34.871' N | 118° 03.149' W | 114       | 1 - 75 m                            | --                           |
| --                  | 2401                  | 33° 39.920' N | 118° 02.103' W | 10        | 1 - 2 m above bottom                | --                           |
| --                  | 2402                  | 33° 39.342' N | 118° 02.593' W | 16        | 1 - 2 m above bottom                | --                           |

| Station Description | Station Location Name | Latitude      | Longitude      | Depth (m) | CTD Sampling Depths (1 m intervals) | Discrete Sampling Depths (m) |
|---------------------|-----------------------|---------------|----------------|-----------|-------------------------------------|------------------------------|
| *                   | 2403                  | 33° 38.765' N | 118° 03.072' W | 21        | 1 - 2 m above bottom                | --                           |
| *                   | 2404                  | 33° 37.875' N | 118° 03.808' W | 29        | 1 - 2 m above bottom                | --                           |
| *                   | 2405                  | 33° 36.986' N | 118° 04.544' W | 37        | 1 - 2 m above bottom                | --                           |
| *                   | 2406                  | 33° 36.096' N | 118° 05.280' W | 60        | 1 - 2 m above bottom                | --                           |
| --                  | 2451                  | 33° 41.475' N | 118° 03.944' W | 10        | 1 - 2 m above bottom                | --                           |
| --                  | 2452                  | 33° 40.739' N | 118° 04.584' W | 17        | 1 - 2 m above bottom                | --                           |
| --                  | 2453                  | 33° 39.987' N | 118° 05.204' W | 22        | 1 - 2 m above bottom                | --                           |
| --                  | 2454                  | 33° 39.098' N | 118° 05.946' W | 30        | 1 - 2 m above bottom                | --                           |
| --                  | 2455                  | 33° 38.210' N | 118° 06.675' W | 36        | 1 - 2 m above bottom                | --                           |
| --                  | 2456                  | 33° 37.318' N | 118° 07.411' W | 42        | 1 - 2 m above bottom                | --                           |

| Responsible Agency                                                                                                                                                                                                                                                                                                     | Station Location Name | Latitude      | Longitude      | Depth | Station Location            | Station Description                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------|----------------|-------|-----------------------------|---------------------------------------|
| <b>Orange County Regional Shoreline REC-1 Cooperative Monitoring Program Stations (Nearshore Zone) (n=36)</b><br>r = Regional OCHCA station. c = OC San station.<br>Orange County Regional Shoreline REC-1 stations are monitored at least once per week.<br>* = These stations are monitored at least twice per week. |                       |               |                |       |                             |                                       |
| OCSD-r                                                                                                                                                                                                                                                                                                                 | OSB03                 | 33° 44.355' N | 118° 06.449' W | surf  | Seal Beach/<br>Sunset Beach | Projection of<br>8 <sup>th</sup> St.  |
| OCSD-r                                                                                                                                                                                                                                                                                                                 | OSB05                 | 33° 44.296' N | 118° 06.378' W | surf  | Seal Beach/<br>Sunset Beach | 100 yards<br>south of Pier            |
| OCSD-r                                                                                                                                                                                                                                                                                                                 | OSB04                 | 33° 44.209' N | 118° 06.121' W | surf  | Seal Beach/<br>Sunset Beach | Projection of<br>14 <sup>th</sup> St. |

| Responsible Agency | Station Location Name | Latitude      | Longitude      | Depth | Station Location                   | Station Description                     |
|--------------------|-----------------------|---------------|----------------|-------|------------------------------------|-----------------------------------------|
| OCSD-r             | OSB01                 | 33° 43.603' N | 118° 05.041' W | surf  | Seal Beach/<br>Sunset Beach        | Projection of<br>Seaway                 |
| OCSD-r             | OSUB1                 | 33° 42.986' N | 118° 04.341' W | surf  | Seal Beach/<br>Sunset Beach        | Projection of<br>Broadway               |
| OCSD-c             | 39N                   | 33° 42.114' N | 118° 03.321' W | surf  | Bolsa Chica/<br>Huntington Beach   | Bolsa Chica<br>Beach                    |
| OCSD-c             | 33N                   | 33° 41.281' N | 118° 02.495' W | surf  | Bolsa Chica/<br>Huntington Beach   | Projection of<br>Bolsa Chica<br>Reserve |
| OCSD-r             | BCO-1                 | 33° 40.994' N | 118° 02.138' W | surf  | Bolsa Chica/<br>Huntington Beach   | Bolsa Chica<br>Wetlands<br>Channel      |
| OCSD-c             | 27N                   | 33° 40.587' N | 118° 01.712' W | surf  | Bolsa<br>Chica/Huntington<br>Beach | Bluffs at Sea<br>Pointe (Dog<br>Beach)  |
| OCSD-r             | HB1                   | 33° 40.065' N | 118° 01.937' W | surf  | Bolsa Chica/<br>Huntington Beach   | PCH &<br>Goldenwest                     |
| OCSD-r             | HB2                   | 33° 40.022' N | 118° 01.937' W | surf  | Bolsa Chica/<br>Huntington Beach   | PCH & 22 <sup>nd</sup><br>St.           |
| OCSD-r             | HB3                   | 33° 39.952' N | 118° 00.933' W | surf  | Bolsa Chica/<br>Huntington Beach   | PCH & 20 <sup>th</sup><br>St.           |
| OCSD-c             | 21N                   | 33° 39.843' N | 118° 00.785' W | surf  | Bolsa Chica/<br>Huntington Beach   | Projection of<br>17 <sup>th</sup> St.   |
| OCSD-r             | HB4                   | 33° 39.680' N | 118° 00.613' W | surf  | Bolsa Chica/<br>Huntington Beach   | PCH & 13 <sup>th</sup><br>St.           |
| OCSD-r             | HB5                   | 33° 39.414' N | 118° 00.310' W | surf  | Bolsa Chica/<br>Huntington Beach   | PCH & 6 <sup>th</sup> St.               |
| OCSD-c             | 15N                   | 33° 39.114' N | 117° 59.846' W | surf  | Bolsa Chica/<br>Huntington Beach   | Projection of<br>Jack's Snack<br>Bar    |
| OCSD-c             | 12N                   | 33° 38.854' N | 117° 59.413' W | surf  | Bolsa Chica/<br>Huntington Beach   | Projection of<br>Beach Blvd             |
| OCSD-c             | 9N*                   | 33° 38.565' N | 117° 58.924' W | surf  | Bolsa Chica/<br>Huntington Beach   | Projection of<br>Newland St.            |

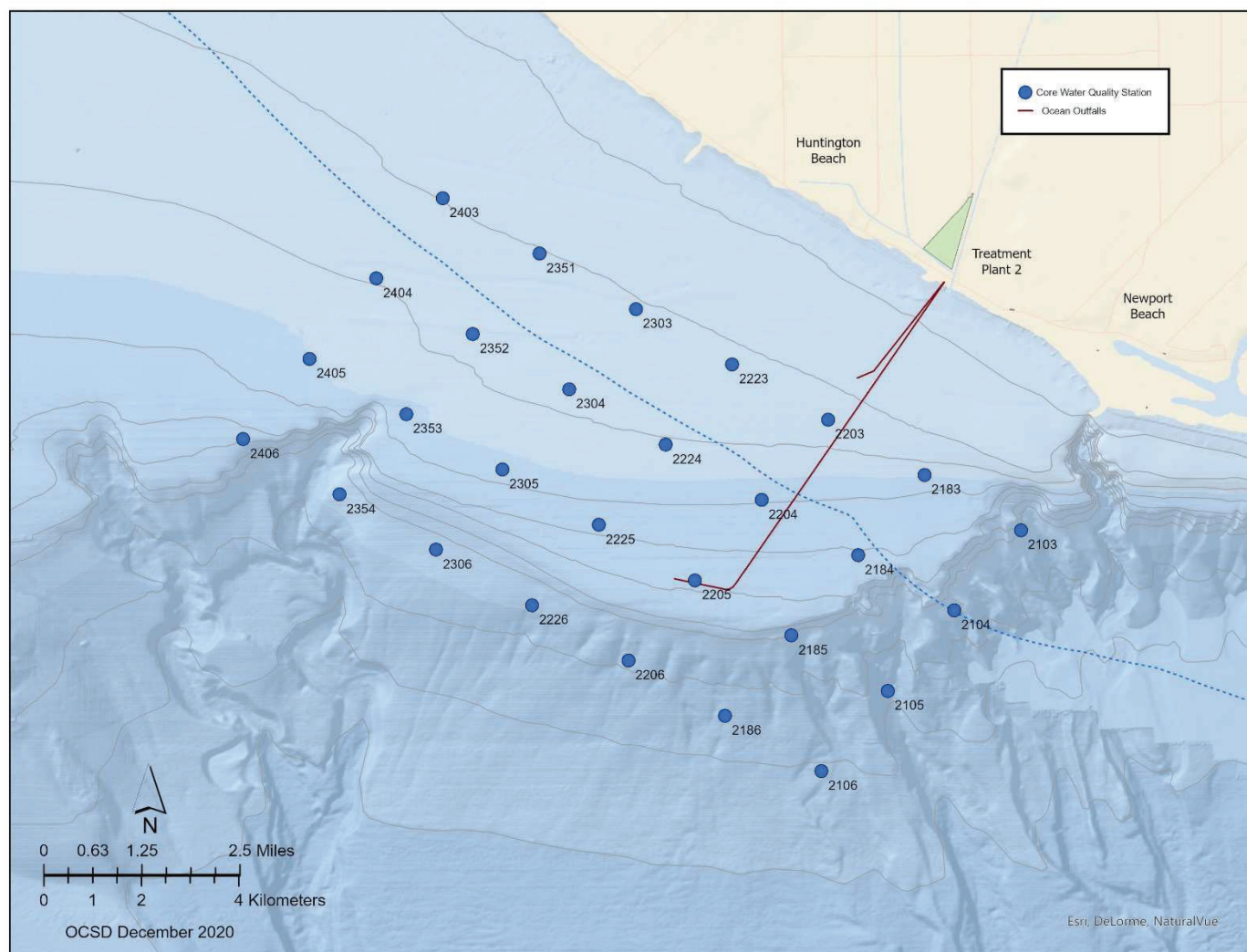


| Responsible Agency | Station Location Name | Latitude      | Longitude      | Depth | Station Location                 | Station Description                                     |
|--------------------|-----------------------|---------------|----------------|-------|----------------------------------|---------------------------------------------------------|
| OCSD-c             | 6N*                   | 33° 38.331' N | 117° 58.573' W | surf  | Bolsa Chica/<br>Huntington Beach | Projection of<br>Magnolia St.                           |
| OCSD-c             | 3N*                   | 33° 38.018' N | 117° 58.032' W | surf  | Bolsa Chica/<br>Huntington Beach | Projection of<br>Brookhurst St.                         |
| OCSD-c             | 0*                    | 33° 37.764' N | 117° 57.598' W | surf  | Bolsa Chica/<br>Huntington Beach | Santa Ana<br>River mouth                                |
| OCSD-r             | TM                    | 33° 37.994' N | 117° 57.645' W | surf  | Bolsa Chica/<br>Huntington Beach | PCH Bridge<br>at Talbert<br>Marsh                       |
| OCSD-r             | SAR-N                 | 33° 37.870' N | 117° 57.434' W | surf  | Bolsa Chica/<br>Huntington Beach | Santa Ana<br>River mouth                                |
| OCSD-c             | 3S                    | 33° 37.619' N | 117° 57.264' W | surf  | Newport Beach                    | Projection<br>Orange St.                                |
| OCSD-c             | 6S                    | 33° 37.337' N | 117° 56.704' W | surf  | Newport Beach                    | Projection<br>52 <sup>nd</sup> /53 <sup>rd</sup> St.    |
| OCSD-c             | 9S                    | 33° 37.033' N | 117° 56.283' W | surf  | Newport Beach                    | Projection<br>38 <sup>th</sup> St.                      |
| OCSD-c             | 15S                   | 33° 36.342' N | 117° 55.459' W | surf  | Newport Beach                    | Projection of<br>15 <sup>th</sup> /16 <sup>th</sup> St. |
| OCSD-c             | 21S                   | 33° 36.059' N | 117° 54.213' W | surf  | Newport Beach                    | Upcoast of<br>Balboa Pier                               |
| OCSD-c             | 27S                   | 33° 35.646' N | 117° 52.910' W | surf  | Newport Beach                    | The Wedge                                               |
| OCSD-c             | 29S                   | 33° 35.559' N | 117° 52.508' W | surf  | Newport Beach                    | Corona del<br>Mar State<br>Beach                        |
| OCSD-r             | BGC                   | 33° 35.384' N | 117° 52.117' W | surf  | Newport Beach                    | Little Corona<br>Beach                                  |
| OCSD-r             | PPC                   | 33° 34.490' N | 117° 50.512' W | surf  | Newport<br>Beach/Crystal<br>Cove | Pelican Point<br>Beach (reef)                           |
| OCSD-c             | 39S                   | 33° 34.700' N | 117° 51.946' W | surf  | Newport<br>Beach/Crystal<br>Cove | Pelican Point<br>(ramp)                                 |

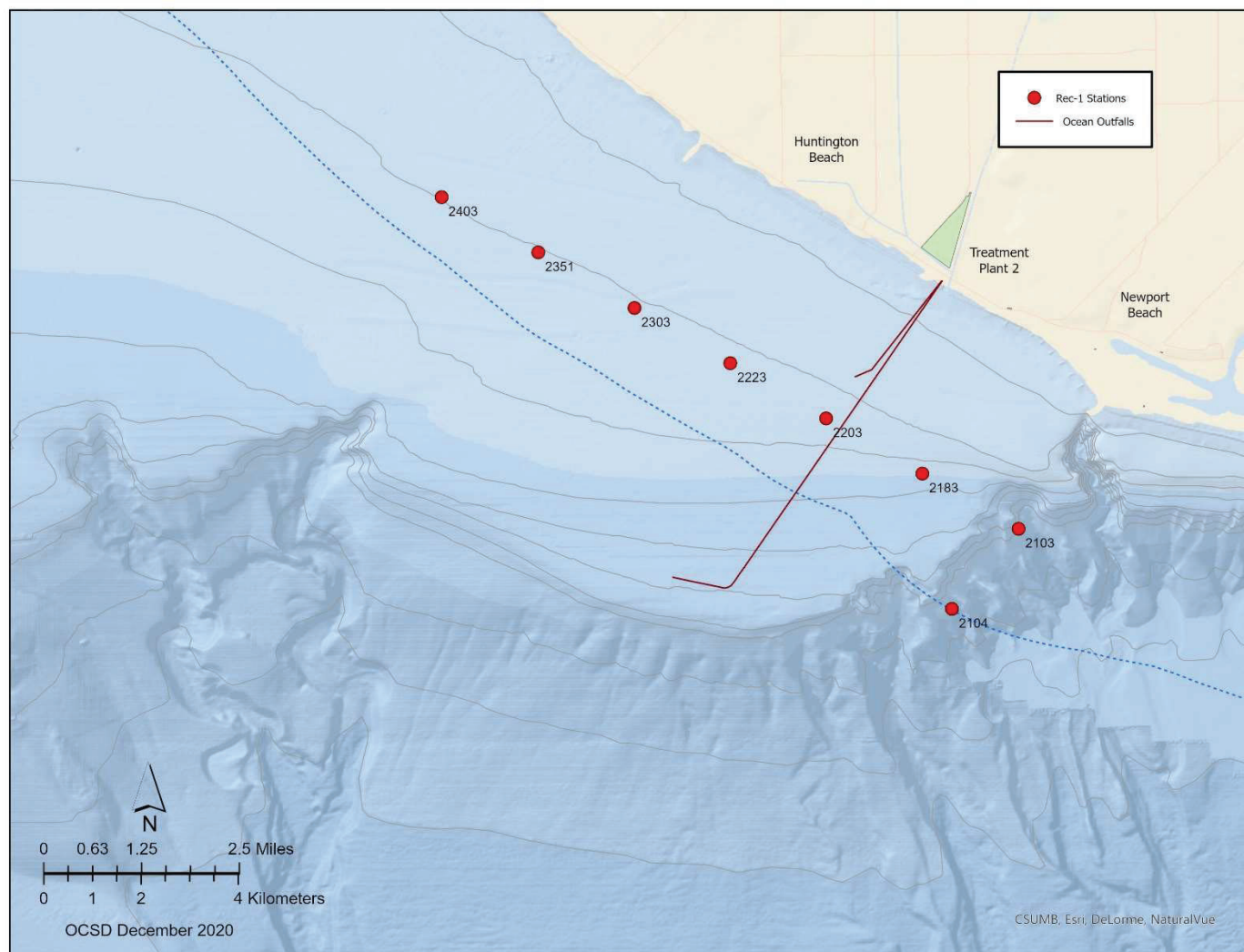
| Responsible Agency | Station Location Name | Latitude      | Longitude      | Depth | Station Location           | Station Description            |
|--------------------|-----------------------|---------------|----------------|-------|----------------------------|--------------------------------|
| OCSD-r             | WFC                   | 33° 34.887' N | 117° 51.342' W | surf  | Newport Beach/Crystal Cove | Pelican Hill Waterfall         |
| OCSD-r             | ONB39                 | 33° 34.450' N | 117° 50.449' W | surf  | Newport Beach/Crystal Cove | Crystal Cove - Los Trancos     |
| OCSD-r             | MDC                   | 33° 33.607' N | 117° 49.323' W | surf  | Newport Beach/Crystal Cove | Muddy Creek Beach (Reef Point) |

The North latitude and West longitude information in Table E-1 and E-2 are approximate for administrative purposes.

**Figure E-1. Monthly Water Quality Monitoring Stations (n=28)**

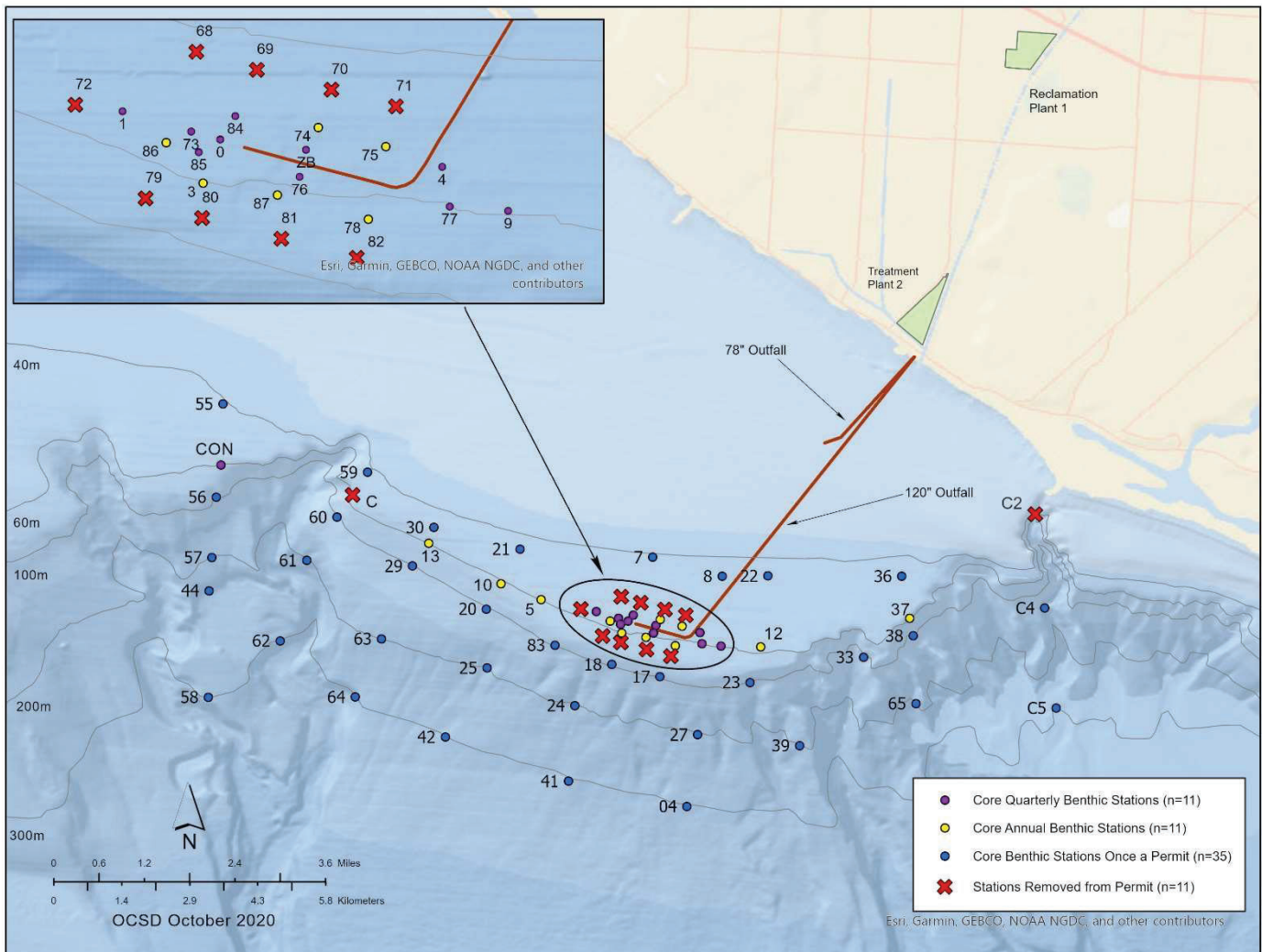


**Figure E-2. Quarterly REC-1 Water Quality Monitoring Stations (Offshore Zone) (n=8)**

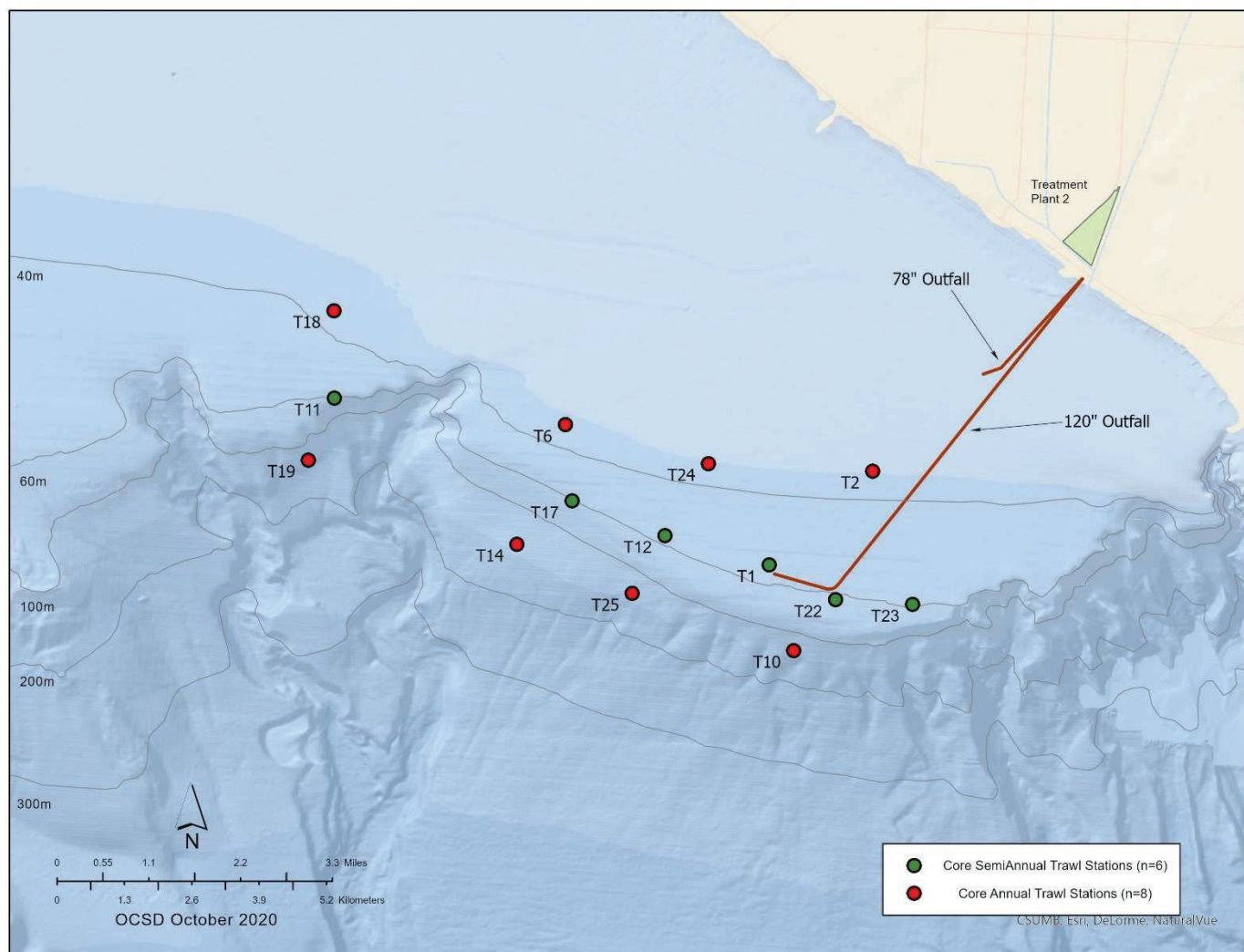




**Figure E-3. Benthic Monitoring Stations: Quarterly (n=11), Annual (n=11), and 1/5-year (n=35)**

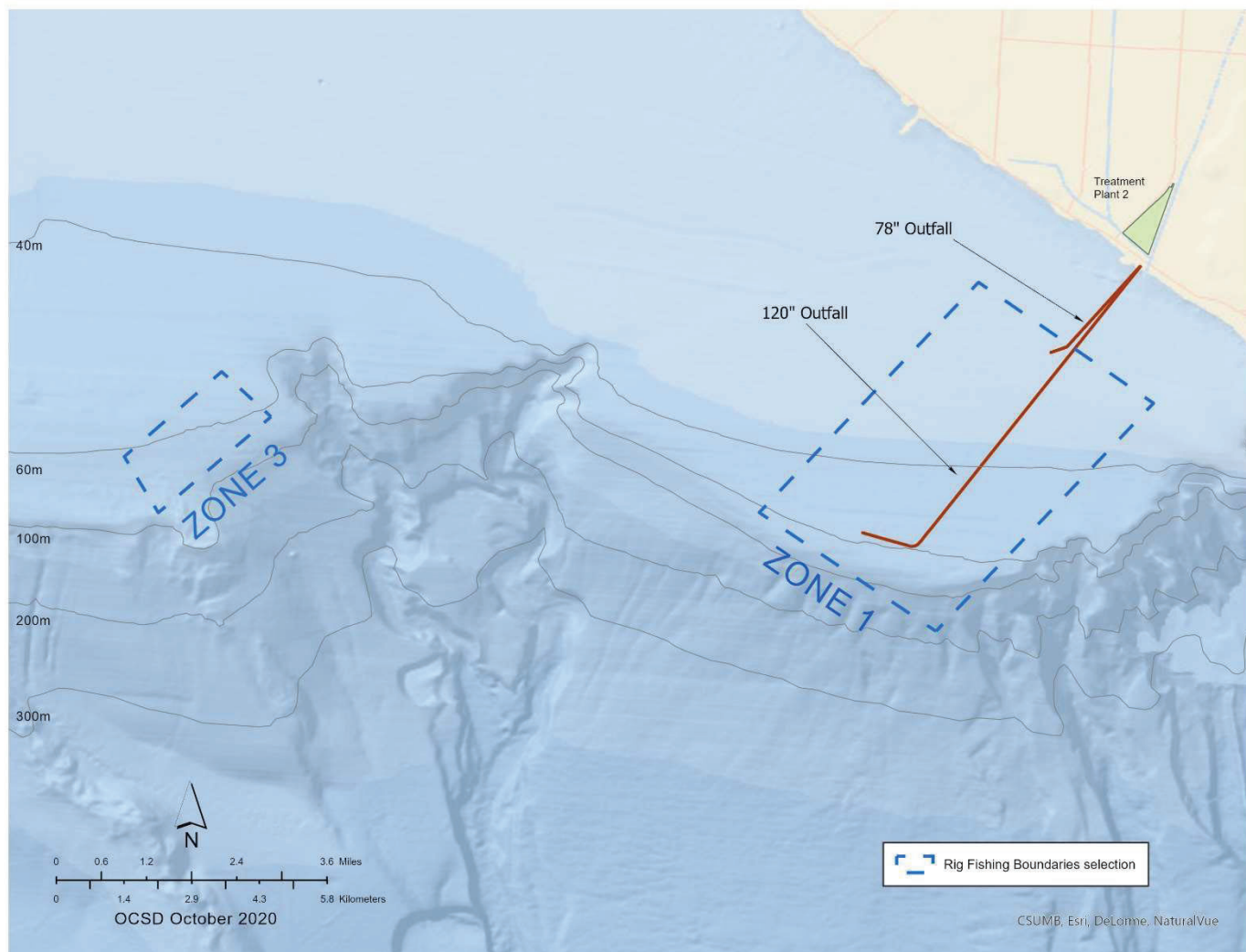


**Figure E-4. Semi-annual Trawl Fish Monitoring Stations (n=6) and Annual Trawl Fish Monitoring Stations (n=8)**

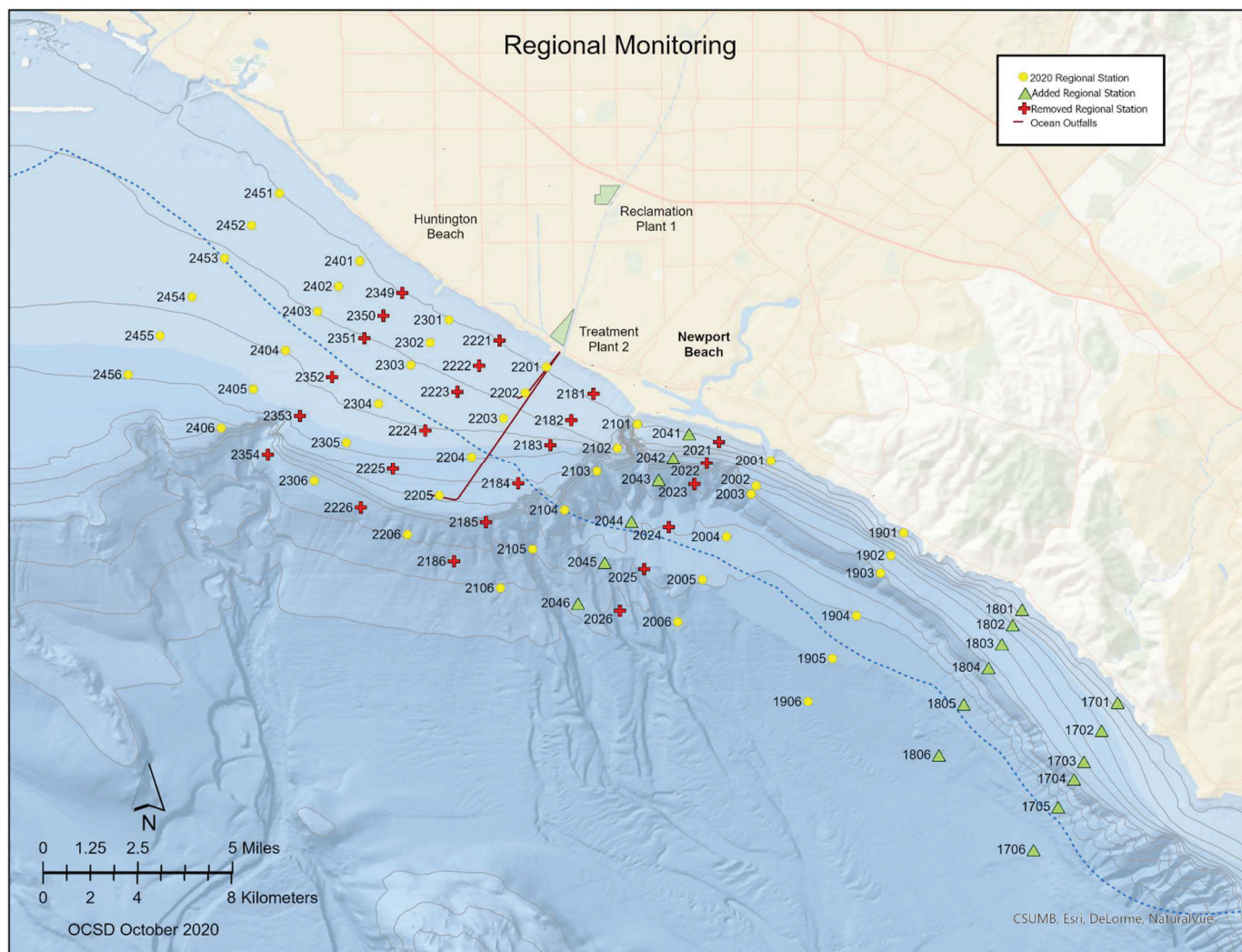




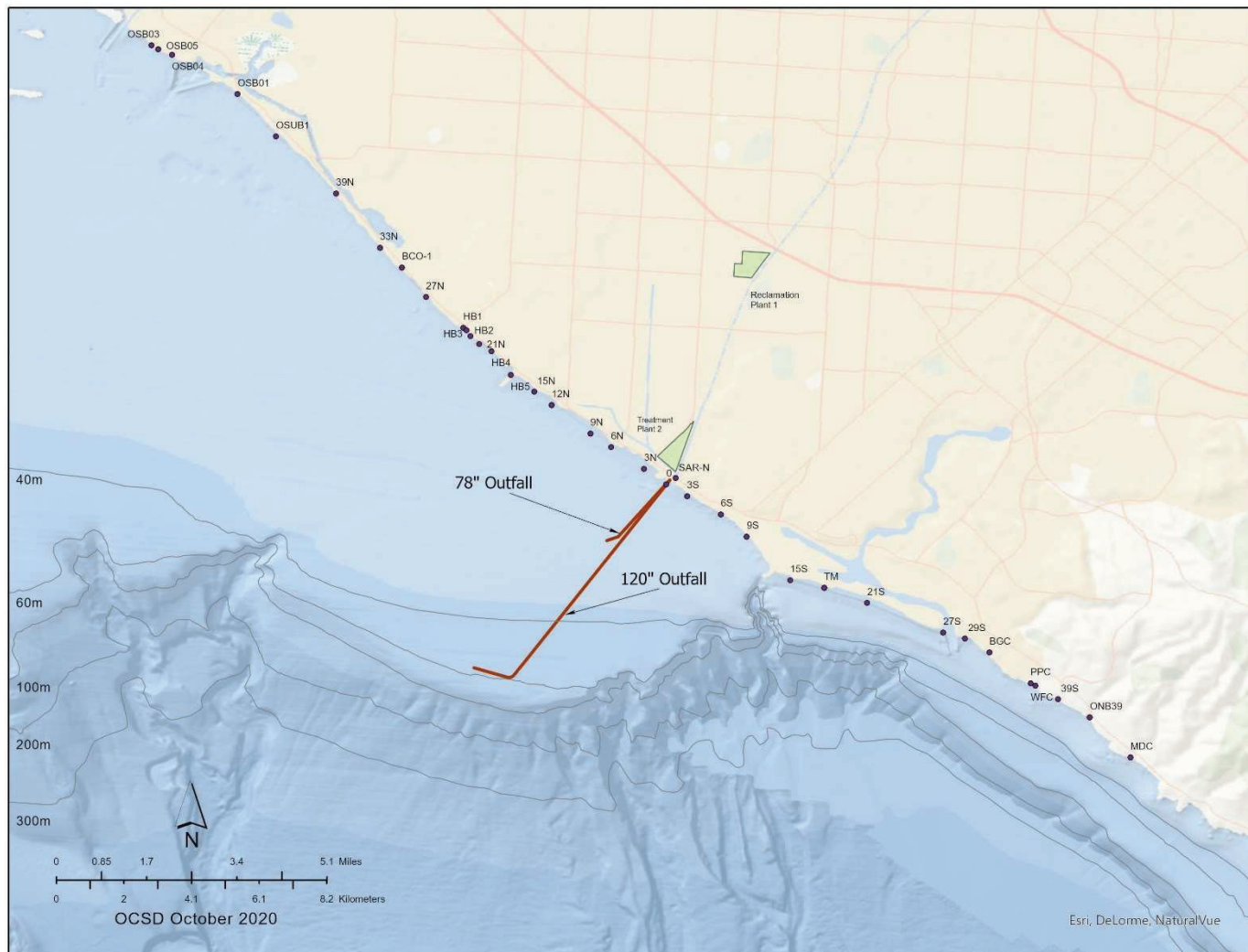
**Figure E-5. Annual Rig Fish Monitoring Zones (n=2)**



**Figure E-6. Quarterly Central Bight Water Quality Monitoring Stations (n=60)**



**Figure E-7. Orange County Shoreline REC-1 Cooperative Monitoring Stations  
(Nearshore Zone) (n=36)**



### III. INFLUENT MONITORING REQUIREMENTS

#### A. Monitoring Location – Influent Monitoring Stations (n=2)

The Discharger shall monitor influent to Reclamation Plant No. 1 at INF-001 and Treatment Plant No. 2 at INF-002 (see Table E-1), as follows.

**Table E-3. Influent Monitoring**

| Parameter                                                                                              | Units          | Sample Type        | Minimum Sampling Frequency | Required Analytical Method |
|--------------------------------------------------------------------------------------------------------|----------------|--------------------|----------------------------|----------------------------|
| <b>Miscellaneous</b>                                                                                   |                |                    |                            |                            |
| Flow rate                                                                                              | MGD            | recorder/totalizer | continuous                 | 1                          |
| Nitrite nitrogen                                                                                       | mg/L           | 24-hr composite    | 1/quarter                  | 1                          |
| Nitrate nitrogen                                                                                       | mg/L           | 24-hr composite    | 1/quarter                  | 1                          |
| Organic nitrogen                                                                                       | mg/L           | 24-hr composite    | 1/quarter                  | 1                          |
| Total phosphorous (as P)                                                                               | mg/L           | 24-hr composite    | 1/quarter                  | 1                          |
| <b>Secondary Treatment Standards and/or Ocean Plan Table 4 (formerly Table A) Effluent Limitations</b> |                |                    |                            |                            |
| Biochemical oxygen demand, 5-day @ 20°C (BOD <sub>5</sub> )                                            | mg/L           | 24-hr composite    | 1/week                     | 1                          |
| Carbonaceous biochemical oxygen demand, 5-day @ 20°C (CBOD <sub>5</sub> )                              | mg/L           | 24-hr composite    | 1/day                      | 1                          |
| Total suspended solids (TSS)                                                                           | mg/L           | 24-hr composite    | 1/day                      | 1                          |
| pH                                                                                                     | standard units | grab               | 1/day                      | 1                          |
| Grease and oil                                                                                         | mg/L           | grab               | 1/month                    | 1                          |
| Settleable solids                                                                                      | --             | --                 | --                         | --                         |
| Turbidity                                                                                              | --             | --                 | --                         | --                         |
| <b>Ocean Plan Table 3 (formerly Table B) for Protection of Marine Aquatic Life</b>                     |                |                    |                            |                            |
| Arsenic, total recoverable                                                                             | µg/L           | 24-hr composite    | 1/month                    | 1                          |
| Cadmium, total recoverable                                                                             | µg/L           | 24-hr composite    | 1/month                    | 1                          |
| Chromium (VI), total recoverable <sup>2</sup>                                                          | µg/L           | 24-hr composite    | 1/month                    | 1                          |



| Parameter                                                                                    | Units | Sample Type     | Minimum Sampling Frequency | Required Analytical Method |
|----------------------------------------------------------------------------------------------|-------|-----------------|----------------------------|----------------------------|
| Copper, total recoverable                                                                    | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Lead, total recoverable                                                                      | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Mercury, total recoverable                                                                   | ng/L  | 24-hr composite | 1/month                    | 1,3                        |
| Nickel, total recoverable                                                                    | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Selenium, total recoverable                                                                  | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Silver, total recoverable                                                                    | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Zinc, total recoverable                                                                      | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Cyanide                                                                                      | µg/L  | grab            | 1/month                    | 1,4                        |
| Total chlorine residual                                                                      | --    | --              | --                         | --                         |
| Ammonia (as N)                                                                               | mg/L  | 24-hr composite | 1/week                     | 1                          |
| Chlorinated phenols <sup>5</sup>                                                             | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Non-chlorinated phenols <sup>5</sup>                                                         | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Endosulfan <sup>5</sup>                                                                      | µg/L  | 24-hr composite | 2/year                     | 1                          |
| Endrin                                                                                       | µg/L  | 24-hr composite | 2/year                     | 1                          |
| HCH <sup>5</sup>                                                                             | µg/L  | 24-hr composite | 2/year                     | 1                          |
| Radioactivity                                                                                | pCi/L | 24-hr composite | 1/month                    | 1,6                        |
| <b>Ocean Plan Table 3 (formerly Table B) for Protection of Human Health – Noncarcinogens</b> |       |                 |                            |                            |
| Acrolein                                                                                     | µg/L  | grab            | 1/quarter                  | 1                          |
| Antimony, total recoverable                                                                  | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Bis(2-chloroethoxy)methane                                                                   | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Bis(2-chloroisopropyl)ether                                                                  | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Chlorobenzene                                                                                | µg/L  | grab            | 1/quarter                  | 1                          |
| Chromium (III), total recoverable <sup>2</sup>                                               | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Di-n-butyl phthalate                                                                         | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Dichlorobenzenes <sup>5</sup>                                                                | µg/L  | grab            | 1/quarter                  | 1                          |



| Parameter                                                                                 | Units | Sample Type     | Minimum Sampling Frequency | Required Analytical Method |
|-------------------------------------------------------------------------------------------|-------|-----------------|----------------------------|----------------------------|
| Diethyl phthalate                                                                         | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Dimethyl phthalate                                                                        | µg/L  | 24-hr composite | 1/month                    | 1                          |
| 4,6-dinitro-2-methylphenol                                                                | µg/L  | 24-hr composite | 1/month                    | 1                          |
| 2,4-dinitrophenol                                                                         | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Ethylbenzene                                                                              | µg/L  | grab            | 1/quarter                  | 1                          |
| Fluoranthene                                                                              | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Hexachlorocyclopentadiene                                                                 | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Nitrobenzene                                                                              | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Thallium, total recoverable                                                               | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Toluene                                                                                   | µg/L  | grab            | 1/quarter                  | 1                          |
| Tributyltin                                                                               | µg/L  | 24-hr composite | 1/quarter                  | 1,7                        |
| 1,1,1-trichloroethane                                                                     | µg/L  | grab            | 1/quarter                  | 1                          |
| <b>Ocean Plan Table 3 (formerly Table B) for Protection of Human Health – Carcinogens</b> |       |                 |                            |                            |
| Acrylonitrile                                                                             | µg/L  | grab            | 1/quarter                  | 1                          |
| Aldrin                                                                                    | µg/L  | 24-hr composite | 2/year                     | 1                          |
| Benzene                                                                                   | µg/L  | grab            | 1/quarter                  | 1                          |
| Benzidine                                                                                 | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Beryllium, total recoverable                                                              | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Bis(2-chloroethyl) ether                                                                  | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Bis(2-ethylhexyl) phthalate                                                               | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Carbon tetrachloride                                                                      | µg/L  | grab            | 1/quarter                  | 1                          |
| Chlordane <sup>5,8</sup>                                                                  | µg/L  | 24-hr composite | 2/year                     | 1                          |
| Chlorodibromomethane                                                                      | µg/L  | grab            | 1/quarter                  | 1                          |
| Chloroform                                                                                | µg/L  | grab            | 1/quarter                  | 1                          |
| DDT <sup>5</sup>                                                                          | µg/L  | 24-hr composite | 2/year                     | 1                          |
| 1,4-dichlorobenzene                                                                       | µg/L  | grab            | 1/quarter                  | 1                          |
| 3,3-dichlorobenzidine                                                                     | µg/L  | 24-hr composite | 1/month                    | 1                          |

| Parameter                        | Units | Sample Type     | Minimum Sampling Frequency | Required Analytical Method |
|----------------------------------|-------|-----------------|----------------------------|----------------------------|
| 1,2-dichloroethane               | µg/L  | grab            | 1/quarter                  | 1                          |
| 1,1-dichloroethylene             | µg/L  | grab            | 1/quarter                  | 1                          |
| Dichlorobromomethane             | µg/L  | grab            | 1/quarter                  | 1                          |
| Dichloromethane                  | µg/L  | grab            | 1/quarter                  | 1                          |
| 1,3-dichloropropene <sup>9</sup> | µg/L  | grab            | 1/quarter                  | 1                          |
| Dieldrin                         | µg/L  | 24-hr composite | 2/year                     | 1                          |
| 2,4-dinitrotoluene               | µg/L  | 24-hr composite | 1/month                    | 1                          |
| 1,2-diphenylhydrazine            | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Halomethanes <sup>5</sup>        | µg/L  | grab            | 1/quarter                  | 1                          |
| Heptachlor                       | µg/L  | 24-hr composite | 2/year                     | 1                          |
| Heptachlor epoxide               | µg/L  | 24-hr composite | 2/year                     | 1                          |
| Hexachlorobenzene                | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Hexachlorobutadiene              | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Hexachloroethane                 | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Isophorone                       | µg/L  | 24-hr composite | 1/month                    | 1                          |
| N-nitrosodimethylamine           | µg/L  | 24-hr composite | 1/month                    | 1                          |
| N-nitrosodi-N-propylamine        | µg/L  | 24-hr composite | 1/month                    | 1                          |
| N-nitrosodiphenylamine           | µg/L  | 24-hr composite | 1/month                    | 1                          |
| PAHs <sup>5</sup>                | µg/L  | 24-hr composite | 1/month                    | 1                          |
| PCBs <sup>5</sup>                | µg/L  | 24-hr composite | 2/year                     | 1                          |
| Individual PCB congeners         | --    | --              | --                         | --                         |
| TCDD equivalents <sup>5</sup>    | µg/L  | 24-hr composite | 1/quarter                  | 1,10                       |
| 1,1,2,2-tetrachloroethane        | µg/L  | grab            | 1/quarter                  | 1                          |
| Tetrachloroethylene              | µg/L  | grab            | 1/quarter                  | 1                          |
| Toxaphene                        | µg/L  | 24-hr composite | 2/year                     | 1                          |
| Trichloroethylene                | µg/L  | grab            | 1/quarter                  | 1                          |
| 1,1,2-trichloroethane            | µg/L  | grab            | 1/quarter                  | 1                          |

| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Units | Sample Type     | Minimum Sampling Frequency | Required Analytical Method |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------|----------------------------|----------------------------|
| 2,4,6-trichlorophenol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | µg/L  | 24-hr composite | 1/month                    | 1                          |
| Vinyl chloride                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | µg/L  | grab            | 1/quarter                  | 1                          |
| <p><sup>1</sup> As specified in 40 CFR § 136, or in this Order/Permit.</p> <p><sup>2</sup> For chromium (III) and (VI), the Discharger may, at its option, meet both the chromium (III) and the chromium (VI) limitations by analyzing for total recoverable chromium.</p> <p><sup>3</sup> Mercury, total recoverable: USEPA Method 1631E, with a quantitation level of 0.5 ng/L, shall be used to analyze total recoverable mercury in wastewater.</p> <p><sup>4</sup> Cyanide: If the Discharger can demonstrate to the satisfaction of the Santa Ana Water Board (subject to USEPA approval) that an analytical method is available to reliably distinguish between strongly and weakly complexed cyanide, effluent limitations for cyanide may be met (or performance goals may be evaluated) by the combined measurement of free cyanide, simple alkali metal cyanides, and weakly complexed organometallic cyanide complexes. In order for the analytical method to be acceptable, the recovery of free cyanide from metal complexes must be comparable to that achieved by the approved method in 40 CFR § 136.</p> <p><sup>5</sup> See Attachment A – Definitions.</p> <p><sup>6</sup> Radioactivity: The following methods shall be used: USEPA Method 900.0 or Standard Methods 7110B for gross alpha and gross beta; USEPA Method 903.0 or 903.1 for radium-226; USEPA Method 904.0 for radium-228; USEPA Method 906.0 for tritium; USEPA Method 905.0 for strontium-90; and USEPA Method 908.0, 908.1, or 200.8 for uranium. Analysis for combined radium-226 &amp; 228 shall be conducted only if gross alpha or gross beta results for the same sample exceed the stipulated criteria. If combined radium-226 &amp; 228 exceeds the stipulated criteria, then analyze for tritium, strontium-90, and uranium. Note that as of February 2021, the stipulated criteria for gross alpha, gross beta, and radium-226 &amp; 228 are 15 pCi/L, 50 pCi/L, and 5 pCi/L, respectively. These criteria are prospective, including future changes to any incorporated provisions of federal law, as the changes take effect.</p> <p><sup>7</sup> Tributyltin: SM 6710B or other improved methods approved by the Santa Ana Water Board and USEPA shall be used to analyze tributyltin in wastewater.</p> <p><sup>8</sup> Chlordane: The Discharger may temporarily suspend the monitoring requirements for alpha- and gamma-chlordene, if analytical standards for these compounds are not available. However, the Discharger is required to resume detection and quantification practices as soon as standards become available.</p> <p><sup>9</sup> 1,3-dichloropropene is the sum of cis- and trans-1,3-dichloropropene.</p> <p><sup>10</sup> TCDD equivalents: TCDD equivalents shall mean the sum of the concentrations of 2,3,7,8-CDDs and 2,3,7,8-CDFs multiplied by their respective toxicity equivalency factor (see Attachment A). For TCDD congeners, the Discharger shall use USEPA Method</p> |       |                 |                            |                            |

| Parameter                                                                                            | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Method |
|------------------------------------------------------------------------------------------------------|-------|-------------|----------------------------|----------------------------|
| 1613 for dioxins and furans and the minimum levels, as specified in Attachment A – TCDD Equivalents. |       |             |                            |                            |

#### IV. EFFLUENT MONITORING REQUIREMENTS

##### A. Effluent Monitoring Stations (n=2)

Upon discharge through Discharge Point 001 or Discharge Point 002 during periods of essential maintenance or capital improvement projects of the 120-inch outfall conducted under 40 CFR § 122.41(m)(2), the Discharger shall monitor effluent at EFF-001 or EFF-002 (see Table E-1), respectively, as follows.

**Table E-4. Effluent Monitoring**

| Parameter                                                                                              | Units                    | Sample Type         | Minimum Sampling Frequency | Required Analytical Test Method |
|--------------------------------------------------------------------------------------------------------|--------------------------|---------------------|----------------------------|---------------------------------|
| <b>Miscellaneous</b>                                                                                   |                          |                     |                            |                                 |
| Flow rate                                                                                              | MGD                      | recorder/ totalizer | continuous                 | 1                               |
| Fecal coliform density (Discharge Point 001)                                                           | MPN /100 mL <sup>2</sup> | grab                | 1/day                      | 1                               |
| Fecal coliform density (Discharge Point 002)                                                           | MPN /100 mL <sup>2</sup> | grab                | 3/day                      | 1                               |
| <i>Enterococcus</i> density (Discharge Point 001)                                                      | CFU /100 mL <sup>2</sup> | grab                | 1/day                      | 1                               |
| <i>Enterococcus</i> density (Discharge Point 002)                                                      | CFU /100 mL <sup>2</sup> | grab                | 3/day                      | 1                               |
| Nitrite nitrogen                                                                                       | mg/L                     | 24-hr composite     | 1/month                    | 1                               |
| Nitrate nitrogen                                                                                       | mg/L                     | 24-hr composite     | 1/month                    | 1                               |
| Organic nitrogen                                                                                       | mg/L                     | 24-hr composite     | 1/month                    | 1                               |
| Total nitrogen                                                                                         | lbs/year                 | calculated          | 1/year                     | --                              |
| Total phosphorous (as P)                                                                               | mg/L                     | 24-hr composite     | 1/month                    | 1                               |
| <b>Secondary Treatment Standards and/or Ocean Plan Table 4 (formerly Table A) Effluent Limitations</b> |                          |                     |                            |                                 |

| Parameter                                                                              | Units                          | Sample Type     | Minimum Sampling Frequency | Required Analytical Test Method |
|----------------------------------------------------------------------------------------|--------------------------------|-----------------|----------------------------|---------------------------------|
| Biochemical oxygen demand, 5-day @ 20°C (BOD <sub>5</sub> ) <sup>3</sup>               | mg/L                           | 24-hr composite | 1/week                     | 1                               |
| Carbonaceous biochemical oxygen demand, 5-day @ 20°C (CBOD <sub>5</sub> ) <sup>3</sup> | mg/L                           | 24-hr composite | 1/day                      | 1                               |
| Total suspended solids (TSS) <sup>3</sup>                                              | mg/L                           | 24-hr composite | 1/day                      | 1                               |
| pH                                                                                     | pH units                       | grab            | 1/day                      | 1                               |
| Grease and oil                                                                         | mg/L                           | grab            | 1/month                    | 1                               |
| Settleable solids                                                                      | ml/L                           | grab            | 1/day                      | 1                               |
| Turbidity                                                                              | NTU                            | 24-hr composite | 1/month                    | 1                               |
| <b>Ocean Plan Table 3 (formerly Table B) for Protection of Marine Aquatic Life</b>     |                                |                 |                            |                                 |
| Arsenic, total recoverable                                                             | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Cadmium, total recoverable                                                             | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Chromium (VI), total recoverable <sup>4</sup>                                          | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Copper, total recoverable                                                              | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Lead, total recoverable                                                                | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Mercury, total recoverable                                                             | ng/L                           | 24-hr composite | 1/month                    | 1,5                             |
| Nickel, total recoverable                                                              | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Selenium, total recoverable                                                            | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Silver, total recoverable                                                              | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Zinc, total recoverable                                                                | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Cyanide                                                                                | µg/L                           | grab            | 1/month                    | 1,6                             |
| Total chlorine residual                                                                | µg/L                           | grab            | 1/12 hours                 | 1,7                             |
| Ammonia (as N)                                                                         | mg/L                           | 24-hr composite | 1/week                     | 1                               |
| Acute toxicity, TST (Discharge Point 001)                                              | Pass "0" or Fail "1", % Effect | 24-hr composite | 1/quarter                  | 1                               |



| Parameter                                                                                    | Units                          | Sample Type     | Minimum Sampling Frequency | Required Analytical Test Method |
|----------------------------------------------------------------------------------------------|--------------------------------|-----------------|----------------------------|---------------------------------|
| Chronic toxicity, TST                                                                        | Pass "0" or Fail "1", % Effect | 24-hr composite | 1/month                    | 1                               |
| Chlorinated phenols <sup>8</sup>                                                             | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Non-chlorinated phenols <sup>8</sup>                                                         | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Endosulfan <sup>8</sup>                                                                      | µg/L                           | 24-hr composite | 2/year                     | 1                               |
| Endrin                                                                                       | µg/L                           | 24-hr composite | 2/year                     | 1                               |
| HCH <sup>8</sup>                                                                             | µg/L                           | 24-hr composite | 2/year                     | 1                               |
| Radioactivity                                                                                | pCi/L                          | 24-hr composite | 1/month                    | 1,9                             |
| <b>Ocean Plan Table 3 (formerly Table B) for Protection of Human Health – Noncarcinogens</b> |                                |                 |                            |                                 |
| Acrolein                                                                                     | µg/L                           | grab            | 1/quarter                  | 1                               |
| Antimony, total recoverable                                                                  | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Bis(2-chloroethoxy) methane                                                                  | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Bis(2-chloroisopropyl) ether                                                                 | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Chlorobenzene                                                                                | µg/L                           | grab            | 1/quarter                  | 1                               |
| Chromium (III), total recoverable <sup>4</sup>                                               | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Di-n-butyl phthalate                                                                         | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Dichlorobenzenes <sup>8</sup>                                                                | µg/L                           | grab            | 1/quarter                  | 1                               |
| Diethyl phthalate                                                                            | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Dimethyl phthalate                                                                           | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| 4,6-dinitro-2-methylphenol                                                                   | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| 2,4-dinitrophenol                                                                            | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Ethylbenzene                                                                                 | µg/L                           | grab            | 1/quarter                  | 1                               |
| Fluoranthene                                                                                 | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Hexachlorocyclopentadiene                                                                    | µg/L                           | 24-hr composite | 1/month                    | 1                               |
| Nitrobenzene                                                                                 | µg/L                           | 24-hr composite | 1/month                    | 1                               |

| Parameter                                                                                 | Units | Sample Type     | Minimum Sampling Frequency | Required Analytical Test Method |
|-------------------------------------------------------------------------------------------|-------|-----------------|----------------------------|---------------------------------|
| Thallium, total recoverable                                                               | µg/L  | 24-hr composite | 1/month                    | 1                               |
| Toluene                                                                                   | µg/L  | grab            | 1/quarter                  | 1                               |
| Tributyltin                                                                               | µg/L  | 24-hr composite | 1/quarter                  | 1,10                            |
| 1,1,1-trichloroethane                                                                     | µg/L  | grab            | 1/quarter                  | 1                               |
| <b>Ocean Plan Table 3 (formerly Table B) for Protection of Human Health – Carcinogens</b> |       |                 |                            |                                 |
| Acrylonitrile                                                                             | µg/L  | grab            | 1/quarter                  | 1                               |
| Aldrin                                                                                    | µg/L  | 24-hr composite | 2/year                     | 1                               |
| Benzene                                                                                   | µg/L  | grab            | 1/quarter                  | 1                               |
| Benzidine                                                                                 | µg/L  | 24-hr composite | 1/month                    | 1                               |
| Beryllium, total recoverable                                                              | µg/L  | 24-hr composite | 1/month                    | 1                               |
| Bis(2-chloroethyl) ether                                                                  | µg/L  | 24-hr composite | 1/month                    | 1                               |
| Bis(2-ethylhexyl) phthalate                                                               | µg/L  | 24-hr composite | 1/month                    | 1                               |
| Carbon tetrachloride                                                                      | µg/L  | grab            | 1/quarter                  | 1                               |
| Chlordane <sup>8,11</sup>                                                                 | µg/L  | 24-hr composite | 2/year                     | 1                               |
| Chlorodibromomethane                                                                      | µg/L  | grab            | 1/quarter                  | 1                               |
| Chloroform                                                                                | µg/L  | grab            | 1/quarter                  | 1                               |
| DDT <sup>8</sup>                                                                          | µg/L  | 24-hr composite | 2/year                     | 1                               |
| 1,4-dichlorobenzene                                                                       | µg/L  | grab            | 1/quarter                  | 1                               |
| 3,3-dichlorobenzidine                                                                     | µg/L  | 24-hr composite | 1/month                    | 1                               |
| 1,2-dichloroethane                                                                        | µg/L  | grab            | 1/quarter                  | 1                               |
| 1,1-dichloroethylene                                                                      | µg/L  | grab            | 1/quarter                  | 1                               |
| Dichlorobromomethane                                                                      | µg/L  | grab            | 1/quarter                  | 1                               |
| Dichloromethane                                                                           | µg/L  | grab            | 1/quarter                  | 1                               |
| 1,3-dichloropropene <sup>12</sup>                                                         | µg/L  | grab            | 1/quarter                  | 1                               |
| Dieldrin                                                                                  | µg/L  | 24-hr composite | 2/year                     | 1                               |
| 2,4-dinitrotoluene                                                                        | µg/L  | 24-hr composite | 1/month                    | 1                               |
| 1,2-diphenylhydrazine                                                                     | µg/L  | 24-hr composite | 1/month                    | 1                               |

| Parameter                     | Units | Sample Type     | Minimum Sampling Frequency | Required Analytical Test Method |
|-------------------------------|-------|-----------------|----------------------------|---------------------------------|
| Halomethanes <sup>8</sup>     | µg/L  | grab            | 1/quarter                  | 1                               |
| Heptachlor                    | µg/L  | 24-hr composite | 2/year                     | 1                               |
| Heptachlor epoxide            | µg/L  | 24-hr composite | 2/year                     | 1                               |
| Hexachlorobenzene             | µg/L  | 24-hr composite | 1/month                    | 1                               |
| Hexachlorobutadiene           | µg/L  | 24-hr composite | 1/month                    | 1                               |
| Hexachloroethane              | µg/L  | 24-hr composite | 1/month                    | 1                               |
| Isophorone                    | µg/L  | 24-hr composite | 1/month                    | 1                               |
| N-nitrosodimethylamine        | µg/L  | 24-hr composite | 1/month                    | 1                               |
| N-nitrosodi-N-propylamine     | µg/L  | 24-hr composite | 1/month                    | 1                               |
| N-nitrosodiphenylamine        | µg/L  | 24-hr composite | 1/month                    | 1                               |
| PAHs <sup>8</sup>             | µg/L  | 24-hr composite | 1/month                    | 1                               |
| PCBs <sup>8</sup>             | µg/L  | 24-hr composite | 2/year                     | 1                               |
| Individual PCB congeners      | µg/L  | 24-hr composite | 1/year                     | 1,13                            |
| TCDD equivalents <sup>8</sup> | µg/L  | 24-hr composite | 1/quarter                  | 1,14                            |
| 1,1,2,2-tetrachloroethane     | µg/L  | grab            | 1/quarter                  | 1                               |
| Tetrachloroethylene           | µg/L  | grab            | 1/quarter                  | 1                               |
| Toxaphene                     | µg/L  | 24-hr composite | 2/year                     | 1                               |
| Trichloroethylene             | µg/L  | grab            | 1/quarter                  | 1                               |
| 1,1,2-trichloroethane         | µg/L  | grab            | 1/quarter                  | 1                               |
| 2,4,6-trichlorophenol         | µg/L  | 24-hr composite | 1/month                    | 1                               |
| Vinyl chloride                | µg/L  | grab            | 1/quarter                  | 1                               |

<sup>1</sup> As specified in 40 CFR § 136, or in this Order/Permit.

<sup>2</sup> Results may be reported as either MPN/100 mL if the laboratory method used provides results in MPN/100 mL or CFU/100 mL if the laboratory method used provides results in CFU/100 mL.

<sup>3</sup> Percent removal shall be calculated based on mass where:

$$\% \text{ removal} = (\text{influent mass} - \text{effluent mass}) / \text{influent mass};$$

$$\text{influent mass (lbs/day)} = \text{influent flow (MGD)} \times \text{influent concentration (mg/L)} \times 8.34;$$

| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|----------------------------|---------------------------------|
| <p>effluent mass (lbs/day) = effluent flow (MGD) x effluent concentration (mg/L) x 8.34.</p> <p><sup>4</sup> For chromium (III) and (VI), the Discharger may, at its option, meet both the chromium (III) and the chromium (VI) limitations by analyzing for total recoverable chromium.</p> <p><sup>5</sup> Mercury, total recoverable: USEPA Method 1631E, with a quantitation level of 0.5 ng/L, shall be used to analyze total recoverable mercury in wastewater.</p> <p><sup>6</sup> Cyanide: If the Discharger can demonstrate to the satisfaction of the Santa Ana Water Board (subject to USEPA approval) that an analytical method is available to reliably distinguish between strongly and weakly complexed cyanide, effluent limitations for cyanide may be met (or performance goals may be evaluated) by the combined measurement of free cyanide, simple alkali metal cyanides, and weakly complexed organometallic cyanide complexes. In order for the analytical method to be acceptable, the recovery of free cyanide from metal complexes must be comparable to that achieved by the approved method in 40 CFR § 136.</p> <p><sup>7</sup> The minimum level (ML) for total chlorine residual in wastewater shall be less than or equal to 50 µg/L.</p> <p><sup>8</sup> See Attachment A – Definitions</p> <p><sup>9</sup> Radioactivity: The following methods shall be used: USEPA Method 900.0 or Standard Methods 7110B for gross alpha and gross beta; USEPA Method 903.0 or 903.1 for radium-226; USEPA Method 904.0 for radium-228; USEPA Method 906.0 for tritium; USEPA Method 905.0 for strontium-90; and USEPA Method 908.0, 908.1, or 200.8 for uranium. Analysis for combined radium-226 &amp; 228 shall be conducted only if gross alpha or gross beta results for the same sample exceed the stipulated criteria. If combined radium-226 &amp; 228 exceeds the stipulated criteria, then analyze for tritium, strontium-90, and uranium. Note that as of February 2021, the stipulated criteria for gross alpha, gross beta, and radium-226 &amp; 228 are 15 pCi/L, 50 pCi/L, and 5 pCi/L, respectively. These criteria are prospective, including future changes to any incorporated provisions of federal law, as the changes take effect.</p> <p><sup>10</sup> Tributyltin: SM 6710B or other improved methods approved by the Santa Ana Water Board and USEPA shall be used to analyze tributyltin in wastewater.</p> <p><sup>11</sup> Chlordane: The Discharger may temporarily suspend the monitoring requirements for alpha- and gamma-chlordene, if analytical standards for these compounds are not available. However, the Discharger is required to resume detection and quantification practices as soon as standards become available.</p> <p><sup>12</sup> 1,3-dichloropropene is the sum of cis- and trans-1,3-dichloropropene.</p> <p><sup>13</sup> PCBs: USEPA draft Method 1668c (and quantitation levels) shall be used to analyze PCB congeners in wastewater. To facilitate interpretation of sediment and fish tissue data, individual PCB congeners whose analytical characteristics resemble those of PCB-18, 28, 37, 44, 49, 52, 66, 70, 74, 77, 81, 87, 99, 101, 105, 110, 114, 118, 119, 123, 126, 128, 138,</p> |       |             |                            |                                 |

| Parameter                                                                                                                                                                                                                                                                                                                                                                | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|----------------------------|---------------------------------|
| 149, 151, 153/168, 156, 157, 158, 167, 169, 170, 177, 180, 183, 187, 189, 194, 201, and 206 shall be individually quantified and reported.                                                                                                                                                                                                                               |       |             |                            |                                 |
| <sup>14</sup> TCDD equivalents: TCDD equivalents shall mean the sum of the concentrations of 2,3,7,8-CDDs and 2,3,7,8-CDFs multiplied by their respective toxicity equivalency factor (see Attachment A). For TCDD congeners, the Discharger shall use USEPA Method 1613 for dioxins and furans and the minimum levels, as specified in Attachment A – TCDD Equivalents. |       |             |                            |                                 |

## B. Emergency Discharge Monitoring Stations (n=2)

Upon emergency discharge through Emergency Discharge Point 002, the Discharger shall monitor effluent at EMG-001 (see Table E-1). Upon emergency discharge through Emergency Discharge Point 003, the Discharger shall monitor effluent at EMG-002 (see Table E-1). At minimum, monitored parameters shall include bacteria indicator organisms (i.e., total coliform, fecal coliform, and enterococcus), parameters with secondary treatment and/or Ocean Plan Table 4 standards, and relevant pollutants of concern (e.g., Total chlorine residual and Ammonia (as N)) in Table E-4. During emergency discharge, the minimum sampling frequency shall be daily, until emergency discharge ceases.

## C. Mass Emission Benchmarks

Constituents that have been assigned Mass Emission Benchmarks are listed in the NPDES Order/Permit under Section V.B. The Mass Emission Benchmarks have been established for the discharge through Discharge Point 001 (120" outfall) and shall be reported in metric tons per year (MT/yr). The Discharger shall monitor and report annually the mass emission rate for all constituents that have mass emission benchmarks. For each constituent, the 12-month average mass emission rate, and the effluent concentrations and flows used to calculate that mass emission rate shall be reported in the annual pretreatment report and annual receiving water monitoring report (effluent chapter).

# V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

## A. Acute Whole Effluent Toxicity

### 1. Monitoring Frequency

The Discharger shall conduct acute toxicity tests on 24-hour composite effluent samples (Table E-4). Once each calendar year, at a different time of year from the previous years, the Discharger shall split a 24-hour composite effluent sample and concurrently conduct toxicity tests using a fish and an invertebrate species. The Discharger shall then continue to conduct routine quarterly toxicity testing using the single, most sensitive species.



Acute toxicity test samples shall be collected at the designated NPDES sampling station for the effluent. In order to better relate toxicity to other effluent characteristics, it is recommended that at least twice per year the Discharger process a split toxicity sample for analysis of all other monitored parameters specified by the effluent monitoring program.

2. Marine and Estuarine Species and Test Methods

Species and short-term test methods for estimating the acute toxicity of NPDES effluents are generally found in the fifth edition of *Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms* (EPA/821/R-02/012, 2002; Table IA, 40 CFR § 136).

For this Order/Permit, the Discharger shall conduct 96-hour static renewal toxicity tests with topsmelt, *Atherinops affinis*, representing a vertebrate species (Test Method 2006.0), and the mysid, *Americamysis bahia*, representing an invertebrate species (Test Method 2007.0).

If laboratory-held cultures of the topsmelt, *Atherinops affinis*, are not available for testing, then the Discharger shall conduct a 96-hour static renewal toxicity test with the inland silverside, *Menidia beryllina* (Test Method 2006.0).

3. Quality Assurance

- a. Quality assurance measures, instructions, and other recommendations and requirements are found in the test methods manuals previously referenced. Additional requirements are specified, below.
- b. For Discharge Point 001, an acute dilution allowance is authorized such that the critical acute instream waste concentration (IWC) is set at a percent effluent value lower than 100% effluent. The acute IWC for Discharge Point 001 is 5.56% effluent.
- c. Effluent dilution water and control water should be prepared and used as specified in the test methods manual for the test species. If the dilution water is different from test organism culture water, then a second control using culture water shall also be used. If the use of artificial sea salts is considered provisional in the test method, then artificial sea salts shall not be used to increase the salinity of the effluent sample prior to toxicity testing without written approval by the permitting authority.
- d. If organisms are not cultured in-house, then concurrent testing with a reference toxicant shall be conducted. If organisms are cultured in-house, then monthly reference toxicant testing is sufficient. Reference toxicant tests and effluent toxicity tests shall be conducted using the same test conditions (e.g., same test duration, etc.).
- e. If the effluent toxicity test during the reporting period for the month does not meet the Test Acceptability Criteria (TAC) in the WET method (EPA/600/R-95/136, 1995), then the permittee shall resample and retest within 14 days. The

results of this retest shall only replace that effluent toxicity test that did not meet TAC during the reporting period for the month.

- f. If the discharged effluent is disinfected using chlorine, then total chlorine residual shall not be removed from the effluent sample prior to toxicity testing.

## **B. Chronic Whole Effluent Toxicity**

### **1. Monitoring Frequency**

The Discharger shall conduct chronic toxicity tests on 24-hour composite effluent samples (Table E-4). Once each calendar year, at a different time of year from the previous years, the Discharger shall split a 24-hour composite effluent sample and concurrently conduct three toxicity tests using a fish, an invertebrate, and an algal species. The Discharger shall then continue to conduct routine monthly toxicity testing using the single, most sensitive species.

Chronic toxicity test samples shall be collected at the designated NPDES sampling station for the effluent. In order to better relate toxicity to other effluent characteristics, it is recommended that at least twice per year the Discharger process a split toxicity sample for analysis of all other monitored parameters specified by the effluent monitoring program.

### **2. Marine and Estuarine Species and Test Methods**

Species and short-term test methods for estimating the chronic toxicity of NPDES effluents are found in the first edition of *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995) and applicable water quality standards; also see 40 CFR § 122.41(j)(4) and 122.44(d)(1)(iv), and 40 CFR § 122.21(j)(5)(viii) for POTWs.

The Discharger shall conduct a static renewal toxicity test with the topsmelt, *Atherinops affinis* (Larval Survival and Growth Test Method 1006.0); a static non-renewal toxicity test with the giant kelp, *Macrocystis pyrifera* (Germination and Growth Test Method 1009.0); and a toxicity test with one of the following invertebrate species:

Static non-renewal toxicity test with the red abalone, *Haliotis rufescens* (Larval Shell Development Test Method);

Static non-renewal toxicity test with the purple sea urchin, *Strongylocentrotus purpuratus*, or the sand dollar, *Dendraster excentricus* (Fertilization Test Method 1008.0).

If laboratory-held cultures of the topsmelt, *Atherinops affinis*, are not available for testing, then the Discharger shall conduct a static renewal toxicity test with the inland silverside, *Menidia beryllina* (Larval Survival and Growth Test Method 1006.0) in the third edition of *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Marine and Estuarine Organisms* (EPA/821/R-02/014, 2002).

### 3. Quality Assurance

- a. Quality assurance measures, instructions, and other recommendations and requirements are found in the chronic test methods manuals previously referenced. Additional requirements are specified, below.
- b. For Discharge Point 001, a chronic dilution allowance is authorized such that the critical chronic instream waste concentration (IWC) is set at a percent effluent value lower than 100% effluent. The chronic IWC for Discharge Point 001 is 0.556% effluent.

For Discharge Point 002, a chronic dilution allowance is authorized such that the critical chronic instream waste concentration (IWC) is set at a percent effluent value lower than 100% effluent. The chronic IWC for Discharge Point 002 is 2.78% effluent.

- c. Effluent dilution water and control water should be prepared and used as specified in the test methods manual *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995) and/or *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Marine and Estuarine Organisms* (EPA/821/R-02/014, 2002). If the dilution water is different from test organism culture water, then a second control using culture water shall also be used. If the use of artificial sea salts is considered provisional in the test method, then artificial sea salts shall not be used to increase the salinity of the effluent sample prior to toxicity testing without written approval by the permitting authority.
- d. If organisms are not cultured in-house, then concurrent testing with a reference toxicant shall be conducted. If organisms are cultured in-house, then monthly reference toxicant testing is sufficient. Reference toxicant tests and effluent toxicity tests shall be conducted using the same test conditions (e.g., same test duration, etc.).
- e. If the effluent toxicity test during the reporting period for the month does not meet the Test Acceptability Criteria (TAC) in the WET method (EPA/600/R-95/136, 1995), then the permittee shall resample and retest within 14 days. The results of this retest shall only replace that effluent toxicity test that did not meet TAC during the reporting period for the month.
- f. If the discharged effluent is disinfected using chlorine, then total chlorine residual shall not be removed from the effluent sample prior to toxicity testing.
- g. pH drift during the toxicity test may contribute to artifactual toxicity when pH-dependent toxicants (e.g., ammonia, metals) are present in an effluent. To determine whether or not pH drift during the toxicity test is contributing to artifactual toxicity, the Discharger shall conduct three sets of parallel toxicity tests, in which the pH of one treatment is controlled at the pH of the effluent and the pH of the other treatment is not controlled, as described in Section 13.3.6 of the test methods manual, *Short-term Methods for Estimating the*

Chronic Toxicity of Effluents and Receiving Waters to Marine and Estuarine Organisms (EPA/821/R-02/014, 2002). Toxicity is confirmed to be artifactual and due to pH drift when no toxicity above the chronic WET permit limit is observed in the treatments controlled at the pH of the effluent. If toxicity is confirmed to be artifactual and due to pH drift, then, following written approval by the permitting authority, the Discharger may use the procedures outlined in Section 13.3.6 of the test methods manual to control sample pH during the toxicity test.

### **C. Analysis and Reporting of Acute and Chronic Toxicity Monitoring Results**

1. For Discharge Point 001, a full toxicity laboratory report for all acute and chronic toxicity testing shall be submitted as an attachment to the Self-Monitoring Report (SMR) for the month in which the toxicity tests are initiated. The laboratory report shall contain: all toxicity test results (raw data and statistical analyses) for each effluent and related reference toxicant tested; chain-of custody; the dates of sample collection and initiation of each toxicity test; control performance; all results for other effluent parameters monitored concurrently with the effluent toxicity tests via split samples; and schedule and progress reports on TRE/TIE studies.
2. For Discharge Point 002, a full toxicity laboratory report for all chronic toxicity testing shall be submitted as an attachment to the SMR for the month in which the toxicity tests are initiated. The laboratory report shall contain: all toxicity test results (raw data and statistical analyses) for each effluent and related reference toxicant tested; chain-of custody; the dates of sample collection and initiation of each toxicity test; control performance; all results for other effluent parameters monitored concurrently with the effluent toxicity tests via split samples; and schedule and progress reports on TRE/TIE studies.
3. The Discharger shall notify the Santa Ana Water Board and USEPA in writing within 14 days of exceedance of an acute or chronic WET permit limit. This notification shall describe actions the Discharger has taken or will take to investigate, identify, and correct the causes of toxicity; the status of actions required by this Order/Permit; and schedule for actions not yet completed; or reason(s) that no action has been taken.

## **VI. LAND DISCHARGE MONITORING REQUIREMENTS – NOT APPLICABLE**

## **VII. RECYCLING MONITORING REQUIREMENTS – NOT APPLICABLE**

## **VIII. RECEIVING WATER CORE MONITORING REQUIREMENTS**

### **A. Water Quality Monitoring**

1. Monitoring Locations – Monthly Water Quality Monitoring Stations (n=28)

The Discharger shall monitor the receiving water at the Monthly Water Quality Stations specified in Table E-2, as follows. Reference conditions shall be confirmed for each survey. Water column profiling protocols and analytical methods shall follow those described in Orange County Sanitation District – Ocean Monitoring Program, Quality Assurance and Project Plan (QAPP) (OCSD, MRP QAPP), and

Orange County Sanitation District – Laboratory Monitoring and Compliance Division, Laboratory Standard Operating Procedures (OCSD, Laboratory Quality Manual).

Compliance shall be evaluated based on statistical comparisons between water quality profiles in the reference and plume-affected zones. Appropriate reference stations and plume-impacted stations for each survey day shall be determined based on available current measurements and the presence or absence of typical plume “signals” (e.g., colored dissolved organic matter (CDOM), ammonia (NH<sub>3</sub>-N), and/or fecal indicator bacteria). Reference stations shall represent “natural” conditions, excluding the stations affected by the effluent plume.

**Table E-5. Water Quality Monitoring**

| Parameter                                 | Units                                         | Sample Type | Sample Depth                          | Minimum Sampling Frequency | Required Analytical Test Method |
|-------------------------------------------|-----------------------------------------------|-------------|---------------------------------------|----------------------------|---------------------------------|
| Surface Observations <sup>1</sup>         | --                                            | visual      | Surface                               | 1/month                    | --                              |
| Salinity (calculated)                     | psu                                           | profile     | CTD sampling depths in Table E-2      | 1/month                    | 2                               |
| Conductivity                              | S/m                                           | profile     |                                       | 1/month                    | 2                               |
| Temperature                               | °C                                            | profile     |                                       | 1/month                    | 2                               |
| Dissolved oxygen (DO)                     | mg/L                                          | profile     |                                       | 1/month                    | 2                               |
| Transmissivity                            | %                                             | profile     |                                       | 1/month                    | 2                               |
| Photosynthetically active radiation (PAR) | μEinsteins sec <sup>-1</sup> cm <sup>-2</sup> | profile     |                                       | 1/month                    | 2                               |
| Chlorophyll-a fluorescence                | μg/L                                          | profile     |                                       | 1/month                    | 2                               |
| pH                                        | standard unit                                 | profile     |                                       | 1/month                    | 2                               |
| Ammonia (NH <sub>3</sub> -N)              | mg/L                                          | grab        | Discrete sampling depths in Table E-2 | 1/month                    | 2                               |
| Nitrate nitrogen                          | mg/L                                          | grab        |                                       | 1/month                    | 2                               |

<sup>1</sup> Receiving water observations of any discoloration, turbidity, odor, trash (see Attachment A for the definition of trash), and unusual or abnormal amounts of floating or suspended matter in the water or on the beach, rocks, jetties, or beach structures, shall be made and recorded at stations. The character and extent of such matter shall be described. The dates, times, and depths of sampling and these observations shall also be reported. Recreational use at time of sampling, within a 100 meter radius of each sample location, shall also be recorded and submitted with results. In federal waters, the nature and extent of REC-1 activities shall be recorded and reported whenever a station is sampled. Recreational uses include, but are not limited to, swimming, wading, water-skiing, diving, surfing, and fishing.



| Parameter                                                                                    | Units | Sample Type | Sample Depth | Minimum Sampling Frequency | Required Analytical Test Method |
|----------------------------------------------------------------------------------------------|-------|-------------|--------------|----------------------------|---------------------------------|
| <sup>2</sup> As specified in 40 CFR § 136, or in the MRP QAPP and Laboratory Quality Manual. |       |             |              |                            |                                 |

## B. REC-1 Water Quality Monitoring (Offshore Zone)

Based on Chapter II.B.1 of the Ocean Plan, bacterial indicator standards shall be maintained throughout the water column in the Offshore Zone used for water contact sports, as determined by the Santa Ana Water Board or USEPA (i.e., waters designated as REC-1), to assure that the discharge does not pose a threat to water contact recreation.

1. Monitoring Locations – Quarterly REC-1 Water Quality Monitoring Stations (Offshore Zone) (n=8)

The Discharger shall monitor the receiving water at the Quarterly REC-1 Water Quality Monitoring Stations (Offshore Zone) specified in Table E-2, as follows. Water column profiling protocols and analytical methods shall follow those described in the MRP QAPP and Laboratory Quality Manual.

**Table E-6. REC-1 Water Quality Monitoring (Offshore Zone)**

| Parameter                                 | Units                                         | Sample Type | Sample Depth                     | Minimum Sampling Frequency                                   | Required Analytical Test Method |
|-------------------------------------------|-----------------------------------------------|-------------|----------------------------------|--------------------------------------------------------------|---------------------------------|
| Surface Observations <sup>1</sup>         | --                                            | visual      | Surface                          | 5 samples over a 30-day period/quarter<br><br>See Table E-2. | --                              |
| Salinity (calculated)                     | psu                                           | profile     | CTD sampling depths in Table E-2 |                                                              | 2                               |
| Conductivity                              | S/m                                           | profile     |                                  |                                                              | 2                               |
| Temperature                               | °C                                            | profile     |                                  |                                                              | 2                               |
| Dissolved oxygen (DO)                     | mg/L                                          | profile     |                                  |                                                              | 2                               |
| Transmissivity                            | %                                             | profile     |                                  |                                                              | 2                               |
| Photosynthetically active radiation (PAR) | μEinsteins sec <sup>-1</sup> cm <sup>-2</sup> | profile     |                                  |                                                              | 2                               |
| Chlorophyll-a fluorescence                | μg/L                                          | profile     |                                  |                                                              | 2                               |
| pH                                        | standard units                                | profile     |                                  |                                                              | 2                               |
| Ammonia (NH <sub>3</sub> -N)              | mg/L                                          | grab        | Discrete sampling                |                                                              | 2                               |
| Nitrate nitrogen                          | mg/L                                          | grab        |                                  |                                                              | 2                               |

| Parameter                                                       | Units                    | Sample Type | Sample Depth        | Minimum Sampling Frequency | Required Analytical Test Method |
|-----------------------------------------------------------------|--------------------------|-------------|---------------------|----------------------------|---------------------------------|
| Total coliform density                                          | MPN /100 mL <sup>3</sup> | grab        | depths in Table E-2 |                            | 2,4                             |
| <i>Enterococcus</i> density                                     | CFU /100 mL <sup>3</sup> | grab        |                     |                            | 2,4                             |
| Fecal coliform density                                          | MPN /100 mL <sup>3</sup> | calculated  |                     |                            | 2,4                             |
| <i>E. Coli</i> density<br>(converted to fecal coliform density) | MPN /100 mL <sup>3</sup> | grab        |                     |                            | 2,4                             |

- <sup>1</sup> Receiving water observations of any discoloration, turbidity, odor, trash (see Attachment A for the definition of trash), and unusual or abnormal amounts of floating or suspended matter in the water or on the beach, rocks, jetties, or beach structures, shall be made and recorded at stations. The character and extent of such matter shall be described. The dates, times, and depths of sampling and these observations shall also be reported. Recreational use at time of sampling, within a 100 meter radius of each sample location, shall also be recorded and submitted with results. In federal waters, the nature and extent of REC-1 activities shall be recorded and reported whenever a station is sampled. Recreational uses include, but are not limited to, swimming, wading, water-skiing, diving, surfing, and fishing.
- <sup>2</sup> As specified in 40 CFR § 136, or in the MRP QAPP and Laboratory Quality Manual.
- <sup>3</sup> Results may be reported as either MPN/100 mL if the laboratory method used provides results in MPN/100 mL or CFU/100 mL if the laboratory method used provides results in CFU/100 mL.
- <sup>4</sup> Total coliform and *E. coli* are analyzed using the Colilert-18 method and *Enterococcus* is analyzed using the Enterolert method. Values for *E. coli* are multiplied by 110% to determine fecal coliform values.

### C. Sediment Monitoring

1. Monitoring Locations – Quarterly Benthic Monitoring Stations (n=11), Annual Benthic Monitoring Stations (n=11), and 1/5-year Benthic Monitoring Stations (n=35)

**Sediment Chemistry.** Sediment samples collected for chemistry analyses shall be separate from those collected for benthic infauna community analyses or whole sediment toxicity testing. The Discharger shall monitor sediment chemistry at the Quarterly, Annual, and 1/5-year Benthic Monitoring Stations specified in Table E-2. Sampling protocols and analytical methods (and reporting limits) shall follow those described in the MRP QAPP and Laboratory Quality Manual. Sediment samples for chemistry analyses shall be collected from the top 2 cm of undisturbed surface material in a 0.1 m<sup>2</sup> Van Veen grab sample. Results of chemistry analyses shall be reported on a dry weight basis.

**Table E-7. Sediment Chemistry Monitoring**

| Parameter                    | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|------------------------------|-------|-------------|----------------------------|---------------------------------|
| Miscellaneous                |       |             |                            |                                 |
| Sediment grain size          | phi   | grab        | See Table E-2.             | 1                               |
| Total organic carbon         | %     | grab        |                            | 1                               |
| Dissolved sulfides           | mg/kg | grab        |                            | 1                               |
| Total nitrogen               | mg/kg | grab        |                            | 1                               |
| Total phosphorous            | mg/kg | grab        |                            | 1                               |
| Metals                       |       |             |                            |                                 |
| Aluminum, total recoverable  | mg/kg | grab        | See Table E-2.             | 1                               |
| Antimony, total recoverable  | mg/kg | grab        |                            | 1                               |
| Arsenic, total recoverable   | mg/kg | grab        |                            | 1                               |
| Barium, total recoverable    | mg/kg | grab        |                            | 1                               |
| Beryllium, total recoverable | mg/kg | grab        |                            | 1                               |
| Cadmium, total recoverable   | mg/kg | grab        |                            | 1                               |
| Chromium, total recoverable  | mg/kg | grab        |                            | 1                               |
| Copper, total recoverable    | mg/kg | grab        |                            | 1                               |
| Iron, total recoverable      | mg/kg | grab        |                            | 1                               |
| Lead, total recoverable      | mg/kg | grab        |                            | 1                               |
| Mercury, total recoverable   | mg/kg | grab        |                            | 1                               |
| Nickel, total recoverable    | mg/kg | grab        |                            | 1                               |
| Selenium, total recoverable  | mg/kg | grab        |                            | 1                               |
| Silver, total recoverable    | mg/kg | grab        |                            | 1                               |
| Zinc, total recoverable      | mg/kg | grab        |                            | 1                               |
| Pesticides <sup>2</sup>      |       |             |                            |                                 |
| 2,4'-DDT                     | µg/kg | grab        | See Table E-2.             | 1                               |
| 4,4'-DDT                     | µg/kg | grab        |                            | 1                               |
| 2,4'-DDD                     | µg/kg | grab        |                            | 1                               |
| 4,4'-DDD                     | µg/kg | grab        |                            | 1                               |

| Parameter                                                      | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|----------------------------------------------------------------|-------|-------------|----------------------------|---------------------------------|
| 2,4'-DDE                                                       | µg/kg | grab        |                            | 1                               |
| 4,4'-DDE                                                       | µg/kg | grab        |                            | 1                               |
| 4,4'-DDMU                                                      | µg/kg | grab        |                            | 1                               |
| Aldrin                                                         | µg/kg | grab        |                            | 1                               |
| Dieldrin                                                       | µg/kg | grab        |                            | 1                               |
| cis-Chlordane <sup>3</sup>                                     | µg/kg | grab        |                            | 1                               |
| trans-Chlordane <sup>3</sup>                                   | µg/kg | grab        |                            | 1                               |
| trans-Nonachlor                                                | µg/kg | grab        |                            | 1                               |
| Heptachlor                                                     | µg/kg | grab        |                            | 1                               |
| Heptachlor epoxide                                             | µg/kg | grab        |                            | 1                               |
| Endosulfan                                                     | µg/kg | grab        |                            | 1                               |
| Endrin                                                         | µg/kg | grab        |                            | 1                               |
| Hexachlorobenzene                                              | µg/kg | grab        |                            | 1                               |
| Lindane (gamma-BHC)                                            | µg/kg | grab        |                            | 1                               |
| Mirex                                                          | µg/kg | grab        |                            | 1                               |
| Polychlorinated Biphenyl (PCB) Congeners                       |       |             |                            |                                 |
| Individual PCB congeners <sup>4</sup>                          | µg/kg | grab        | See Table E-2.             | 1                               |
| Polycyclic Aromatic Hydrocarbons (PAHs) – Low Molecular Weight |       |             |                            |                                 |
| Acenaphthene                                                   | µg/kg | grab        | See Table E-2.             | 1                               |
| Acenaphthylene                                                 | µg/kg | grab        |                            | 1                               |
| Anthracene                                                     | µg/kg | grab        |                            | 1                               |
| Biphenyl                                                       | µg/kg | grab        |                            | 1                               |
| Fluorene                                                       | µg/kg | grab        |                            | 1                               |
| 2-Methylnapthalene                                             | µg/kg | grab        |                            | 1                               |
| 1-Methylphenanthrene                                           | µg/kg | grab        |                            | 1                               |
| Naphthalene                                                    | µg/kg | grab        |                            | 1                               |
| 1-Methylnapthalene                                             | µg/kg | grab        |                            | 1                               |
| 2,6-Dimethylnaphthalene                                        | µg/kg | grab        |                            | 1                               |

| Parameter                                                                                                                                                                                                                                                                                                                                                               | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|----------------------------|---------------------------------|
| 1,6,7-Trimethyl-naphthalene<br>(2,3,5-Trimethylnaphthalene)                                                                                                                                                                                                                                                                                                             | µg/kg | grab        |                            | 1                               |
| Phenanthrene                                                                                                                                                                                                                                                                                                                                                            | µg/kg | grab        |                            | 1                               |
| Polycyclic Aromatic Hydrocarbons (PAHs) – High Molecular Weight                                                                                                                                                                                                                                                                                                         |       |             |                            |                                 |
| Benz[a]anthracene                                                                                                                                                                                                                                                                                                                                                       | µg/kg | grab        | See Table E-2.             | 1                               |
| Benzo[a]pyrene                                                                                                                                                                                                                                                                                                                                                          | µg/kg | grab        |                            | 1                               |
| Benzo[b/j]fluoranthene                                                                                                                                                                                                                                                                                                                                                  | µg/kg | grab        |                            | 1                               |
| Benzo[e]pyrene                                                                                                                                                                                                                                                                                                                                                          | µg/kg | grab        |                            | 1                               |
| Benzo[g,h,i]perylene                                                                                                                                                                                                                                                                                                                                                    | µg/kg | grab        |                            | 1                               |
| Benzo[k]fluoranthene                                                                                                                                                                                                                                                                                                                                                    | µg/kg | grab        |                            | 1                               |
| Chrysene                                                                                                                                                                                                                                                                                                                                                                | µg/kg | grab        |                            | 1                               |
| Dibenz[a,h]anthracene                                                                                                                                                                                                                                                                                                                                                   | µg/kg | grab        |                            | 1                               |
| Fluoranthene                                                                                                                                                                                                                                                                                                                                                            | µg/kg | grab        |                            | 1                               |
| Indeno(1,2,3-c,d) pyrene                                                                                                                                                                                                                                                                                                                                                | µg/kg | grab        |                            | 1                               |
| Perylene                                                                                                                                                                                                                                                                                                                                                                | µg/kg | grab        |                            | 1                               |
| Pyrene                                                                                                                                                                                                                                                                                                                                                                  | µg/kg | grab        |                            | 1                               |
| <sup>1</sup> As specified in the MRP QAPP and Laboratory Quality Manual.                                                                                                                                                                                                                                                                                                |       |             |                            |                                 |
| <sup>2</sup> Pesticides: The Discharger shall monitor pesticides only in the summer (July, August, and September) at the quarterly and annual Benthic Monitoring Stations and once in five years (in summer) at the 1/5-year station specified in Table E-2.                                                                                                            |       |             |                            |                                 |
| <sup>3</sup> Chlordane: The Discharger may temporarily suspend the monitoring requirements for cis- and trans- chlordene, if analytical standards for these compounds are not available. However, the Discharger is required to resume detection and quantification practices as soon as standards become available.                                                    |       |             |                            |                                 |
| <sup>4</sup> Individual PCB congeners: Individual PCB congeners whose analytical characteristics resemble those of PCB-18, 28, 37, 44, 49, 52, 66, 70, 74, 77, 81, 87, 99, 101, 105, 110, 114, 118, 119, 123, 126, 128, 138, 149, 151, 153/168, 156, 157, 158, 167, 169, 170, 177, 180, 183, 187, 189, 194, 201, and 206 shall be individually quantified and reported. |       |             |                            |                                 |

2. Monitoring Locations – Quarterly Benthic Monitoring Stations (n=11), Annual Benthic Monitoring Stations (n=11), and 1/5-year Benthic Monitoring Stations (n=35)

**Benthic Infauna Community.** Sediment samples collected for benthic infauna community analyses shall be separate from those collected for chemistry analyses



or whole sediment toxicity testing. The Discharger shall monitor benthic infauna at the Quarterly, Annual, and 1/5-year Benthic Monitoring Stations specified in Table E-2. Sampling protocols, including treatment, storage, and analyses, shall follow those described in the MRP QAPP and Laboratory Quality Manual. Sediment samples for benthic infauna community analyses shall be washed and screened (1.0 mm mesh) from an entire 0.1 m<sup>2</sup> Van Veen grab sample and fixed and preserved for sorting. All retained organisms from the Quarterly, Annual, and 1/5-year stations shall be counted and identified to as low a taxon as possible.

**Table E-8. Benthic Infauna Community Monitoring**

| Parameter                                                     | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|---------------------------------------------------------------|-------|-------------|----------------------------|---------------------------------|
| Number of species, per grab sample                            | --    | grab        | See Table E-2.             | 1                               |
| Number of individuals per species, per grab sample            | --    | grab        |                            | 1                               |
| Benthic response index (BRI)                                  | --    | grab        |                            | 1                               |
| Shannon-Weiner's diversity index (H')                         | --    | grab        |                            | 1                               |
| Swartz's 75% dominance index (SDI)                            | --    | grab        |                            | 1                               |
| 1 As specified in the MRP QAPP and Laboratory Quality Manual. |       |             |                            |                                 |

### 3. Monitoring Locations – Quarterly Benthic Monitoring Stations (n=11)

**Whole Sediment Toxicity.** Sediment samples collected for whole sediment toxicity testing shall be separate from those collected for chemistry analyses or benthic infauna community analyses. The Discharger shall annually monitor whole sediment toxicity at the eleven (11) Quarterly Benthic Monitoring Stations specified in Table E-2. Sampling protocols and analyses shall follow those described in the MRP QAPP and Laboratory Quality Manual. Sediment samples for sediment toxicity testing shall be collected from the top 2 cm of undisturbed surface material in a 0.1 m<sup>2</sup> Van Veen grab sample.

If a station sample result is statistically significant using a standard t-test of no difference and the magnitude of difference compared to the control is greater than 20 %, the station shall be re-sampled and re-tested in the following quarter to determine if the observed toxicity is persistent. If toxicity is persistent (i.e., the second test also tests significantly toxic), then the Discharger shall report the toxicity test results to the Santa Ana Water Board and USEPA and investigate the causes and report the investigation results and mitigation efforts in the annual monitoring report.

**Table E-9. Whole Sediment Toxicity Monitoring**

| Parameter                                                                                                                                                                                                                               | Units     | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|----------------------------|---------------------------------|
| Whole sediment acute toxicity                                                                                                                                                                                                           | % of home | grab        | See Table E-2.             | <sup>1</sup>                    |
| <sup>1</sup> USEPA 10-day static amphipod ( <i>Eohaustorius estuarius</i> ) survival test: <i>Methods for Assessing the Toxicity of Sediment-Associated Contaminants with Estuarine and Marine Amphipods</i> (EPA/600/R- 94/025, 1994). |           |             |                            |                                 |

**D. Demersal Fish and Epibenthic Macroinvertebrate Monitoring**

1. Monitoring Locations – Semi-annual Trawl Fish and Epibenthic Macroinvertebrate Monitoring Stations (n=6) and Annual Trawl Fish and Epibenthic Macroinvertebrate Monitoring Stations (n=8)

**Demersal Fish and Epibenthic Macroinvertebrate Communities.** Trawl samples collected for demersal fish and epibenthic macroinvertebrate community structure analyses may be the same as those collected for demersal fish tissue chemistry analyses. The Discharger shall monitor demersal fish and epibenthic macroinvertebrates at the Semi-Annual and Annual Trawl Fish and Epibenthic Macroinvertebrate Monitoring Stations specified in Table E-2. Sampling and analysis protocols shall follow those described in the MRP QAPP and Laboratory Quality Manual. At each station, a single trawl sample shall be collected using one standard semi-balloon otter trawl towed for at least 10 minutes along the respective isobath. Samples shall be processed, with all demersal fish and epibenthic macroinvertebrates, identified to species, counted, measured (fish only), and weighed.

**Table E-10. Demersal Fish and Epibenthic Macroinvertebrate Community Monitoring**

| Parameter                                                 | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|-----------------------------------------------------------|-------|-------------|----------------------------|---------------------------------|
| Number of species, per trawl sample                       | --    | trawl       | See Table E-2.             | 1                               |
| Number of individuals per species, per trawl sample       | --    | trawl       |                            | 1                               |
| Wet weight of macroinvertebrate species, per trawl sample | kg    | trawl       |                            | 1                               |
| Wet weight of fish species, per trawl sample              | kg    | trawl       |                            | 1                               |

| Parameter                                      | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|------------------------------------------------|-------|-------------|----------------------------|---------------------------------|
| Size classes of fish species, per trawl sample | cm    | trawl       |                            | 1                               |
| Shannon-Weiner's diversity index (H')          | --    | trawl       |                            | 1                               |
| Swartz's 75% dominance index (SDI)             | --    | trawl       |                            | 1                               |
| Fish response index (FRI)                      | --    | trawl       |                            | 1                               |
| Abnormalities and disease symptoms             | --    | trawl       |                            | 1                               |

<sup>1</sup> As specified in the MRP QAPP and Laboratory Quality Manual.

## 2. Monitoring Locations – Semi-annual Trawl Fish Monitoring Stations (n=2)

**Demersal Fish Liver Tissue Chemistry.** The trawl samples collected for demersal fish liver tissue chemistry analyses may be the same as those collected for demersal fish and epibenthic macroinvertebrate community structure analyses. The Discharger shall annually monitor flatfish (e.g., Pacific Sanddab, Hornyhead Turbot, and English Sole) at the two Semi-annual Trawl Fish and Epibenthic Macroinvertebrate Monitoring Stations specified in Table E-2. Sampling and analysis protocols (including reporting limits) shall follow those described in the MRP QAPP and Laboratory Quality Manual. At each station, a single trawl sample shall be collected using one standard semi-balloon otter trawl towed for at least 10 minutes along the specified isobath. A reasonable level of effort (i.e., five (5) trawls per station) shall be used to collect a maximum twenty (20) individuals of flatfish between 15-20 cm standard length at each station. Target fish will be identified to species, counted, measured to the nearest millimeter, weighed, bagged, and transported on wet ice to the Discharger's laboratory for chemical analyses on liver tissue contaminants. Fish will be sorted into two composite samples per station, with a maximum of ten (10) individuals in each composite.

**Table E-11. Demersal Fish Liver Tissue Chemistry Monitoring**

| Parameter                                    | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|----------------------------------------------|-------|-------------|----------------------------|---------------------------------|
| Length of each fish sample, per trawl sample | cm    | trawl       | See Table E-2.             | 1                               |

| Parameter                                                                                                                                                                                 | Units    | Sample Type                                                                  | Minimum Sampling Frequency | Required Analytical Test Method |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------|----------------------------|---------------------------------|
| Wet weight of each fish sample, per trawl sample                                                                                                                                          | kg       | trawl                                                                        |                            | 1                               |
| Percent lipid                                                                                                                                                                             | %, wet g | composite of liver tissue, maximum of 20 individuals of flatfish per station |                            | 1                               |
| Arsenic                                                                                                                                                                                   | ng/wet g | composite of liver tissue, maximum of 20 individuals of flatfish per station |                            | 1                               |
| Mercury (methylmercury)                                                                                                                                                                   | ng/wet g | composite of liver tissue, maximum of 20 individuals of flatfish per station |                            | 1                               |
| Selenium                                                                                                                                                                                  | ng/wet g | composite of liver tissue, maximum of 20 individuals of flatfish per station |                            | 1                               |
| Sum of individual PCB congeners <sup>2</sup>                                                                                                                                              | ng/wet g | composite of liver tissue, maximum of 20 individuals of flatfish per station |                            | 1                               |
| Individual PCB congeners <sup>2</sup>                                                                                                                                                     | ng/wet g | composite of liver tissue, maximum of 20 individuals of flatfish per station |                            | 1                               |
| Sum of individual DDT derivatives <sup>3</sup>                                                                                                                                            | ng/wet g | composite of liver tissue, maximum of 20 individuals of flatfish per station |                            | 1                               |
| Sum of individual Chlordane derivatives <sup>4</sup>                                                                                                                                      | ng/wet g | composite of liver tissue, maximum of 20 individuals of flatfish per station |                            | 1                               |
| <sup>1</sup> As specified in the MRP QAPP and Laboratory Quality Manual.                                                                                                                  |          |                                                                              |                            |                                 |
| <sup>2</sup> Individual PCB congeners: Individual PCB congeners whose analytical characteristics resemble those of PCB-18, 28, 37, 44, 49, 52, 66, 70, 74, 77, 81, 87, 99, 101, 105, 110, |          |                                                                              |                            |                                 |

| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|----------------------------|---------------------------------|
| 114, 118, 119, 123, 126, 128, 138, 149, 151, 153/168, 156, 157, 158, 167, 169, 170, 177, 180, 183, 187, 189, 194, 201, and 206 shall be individually quantified and reported. The analytical report for individual PCB congeners shall be submitted electronically into the State Water Board's California Integrated Water Quality System (CIWQS) in a tabular format as an attachment.                                                                                   |       |             |                            |                                 |
| <sup>3</sup> Individual DDT derivatives: 2,4'- and 4,4'-isomers of DDT, DDE, and DDD, plus 4,4'-DDMU.                                                                                                                                                                                                                                                                                                                                                                      |       |             |                            |                                 |
| <sup>4</sup> Individual Chlordane derivatives: Cis- and trans-chlordane, cis- and trans-chlordene, heptachlor, heptachlor epoxide, cis- and trans-nonachlor, and oxychlordane. The Discharger may temporarily suspend the monitoring requirements for cis- and trans-chlordene, if analytical standards for these compounds are not available. However, the Discharger is required to resume detection and quantification practices as soon as standards become available. |       |             |                            |                                 |

3. Monitoring Locations – Semi-annual Trawl Fish and Epibenthic Macroinvertebrate Monitoring Stations (n=2)

**Demersal Fish Liver Histopathology.** Histopathological analyses shall be performed annually on liver tissues of flatfish from two Semi-annual Trawl Fish and Epibenthic Macroinvertebrate Monitoring Stations specified in Table E-2. A reasonable level of effort (i.e., five (5) trawls per location) shall be used to collect twenty (20) individuals of flatfish at each station. Fish species are the same as those targeted for liver tissue chemistry analyses (e.g., Pacific Sanddab, Hornyhead Turbot, and English Sole).

The increased frequency of histopathological analyses in this Order/Permit cycle (i.e., annual) is different from that of the previous Order/Permit in order to capture potential environmental effects associated with the Discharger's increased water reclamation efforts.

4. Monitoring Locations – Annual Rig Fishing Monitoring Zones (n=2)

**Sport Fish Muscle Chemistry.** The Discharger shall collect ten (10) sexually mature rockfish (e.g., Vermilion Rockfish, Copper Rockfish, and California Scorpionfish) at each Rig Fishing Monitoring Zone specified in Table E-2. Sampling and analysis protocols (including reporting limits) shall follow those described in the MRP QAPP and Laboratory Quality Manual. Target fish will be identified to species, counted, measured to the nearest millimeter, weighed, bagged, and transported on wet ice to the Discharger's laboratory for chemical analyses on muscle tissue contaminants. Fish will be sorted into two composite samples per zone, with a maximum of five (5) individuals in each composite.

**Table E-12. Sport Fish Muscle Chemistry Monitoring**

| Parameter                                            | Units    | Sample Type                                                                | Minimum Sampling Frequency | Required Analytical Test Method |
|------------------------------------------------------|----------|----------------------------------------------------------------------------|----------------------------|---------------------------------|
| Length of each fish sample                           | cm       | hook and line                                                              | See Table E-2.             | 1                               |
| Wet weight of each fish sample                       | kg       | hook and line                                                              |                            | 1                               |
| Percent lipid                                        | %, wet g | composite of muscle tissue, maximum of 10 individuals of rockfish per zone |                            | 1                               |
| Arsenic, total                                       | ng/wet g | composite of muscle tissue, maximum of 10 individuals of rockfish per zone |                            | 1                               |
| Mercury (methylmercury)                              | ng/wet g | composite of muscle tissue, maximum of 10 individuals of rockfish per zone |                            | 1                               |
| Selenium, total                                      | ng/wet g | composite of muscle tissue, maximum of 10 individuals of rockfish per zone |                            | 1                               |
| Sum of individual PCB congeners <sup>2</sup>         | ng/wet g | composite of muscle tissue, maximum of 10 individuals of rockfish per zone |                            | 1                               |
| Individual PCB congeners <sup>2</sup>                | ng/wet g | composite of muscle tissue, maximum of 10 individuals of rockfish per zone |                            | 1                               |
| Sum of individual DDT derivatives <sup>3</sup>       | ng/wet g | composite of muscle tissue, maximum of 10 individuals of rockfish per zone |                            | 1                               |
| Sum of individual Chlordane derivatives <sup>4</sup> | ng/wet g | composite of muscle tissue, maximum of 10 individuals of rockfish per zone |                            | 1                               |

<sup>1</sup> As specified in the MRP QAPP and Laboratory Quality Manual.

<sup>2</sup> Individual PCB congeners. Individual PCB congeners whose analytical characteristics resemble those of PCB-18, 28, 37, 44, 49, 52, 66, 70, 74, 77, 81, 87, 99, 101, 105, 110, 114, 118, 119, 123, 126, 128, 138, 149, 151, 153/168, 156, 157, 158, 167, 169, 170, 177, 180, 183, 187, 189, 194, 201, and 206 shall be individually quantified and reported. The analytical report



| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Units | Sample Type | Minimum Sampling Frequency | Required Analytical Test Method |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|----------------------------|---------------------------------|
| for individual PCB congeners shall be submitted electronically in the State Water Board's California Integrated Water Quality System (CIWQS) in a tabular format as an attachment.                                                                                                                                                                                                                                                                                         |       |             |                            |                                 |
| <sup>3</sup> Individual DDT derivatives. 2,4'- and 4,4'-isomers of DDT, DDE, and DDD, plus 4,4'-DDMU.                                                                                                                                                                                                                                                                                                                                                                      |       |             |                            |                                 |
| <sup>4</sup> Individual Chlordane derivatives. Cis- and trans-chlordane, cis- and trans-chlordene, heptachlor, heptachlor epoxide, cis- and trans-nonachlor, and oxychlordane. The Discharger may temporarily suspend the monitoring requirements for cis- and trans-chlordene, if analytical standards for these compounds are not available. However, the Discharger is required to resume detection and quantification practices as soon as standards become available. |       |             |                            |                                 |

## IX. RECEIVING WATER REGIONAL MONITORING REQUIREMENTS

Discharger participation in regional monitoring programs is a required condition of this Order/Permit. The Discharger shall participate in regional monitoring activities coordinated by the Southern California Coastal Water Research Project (SCCWRP), County of Orange Health Care Agency (HCA), the Southern California Coastal Ocean Observation System (SCCOOS), and other appropriate agencies once approved by the Santa Ana Water Board and USEPA.

The regional monitoring programs which must be conducted under this Order/Permit include:

### A. Southern California Bight Regional Monitoring Program

The Discharger shall continue to participate in Southern California Bight Regional Monitoring Program studies, including water quality, benthic infauna, sediment chemistry, fish communities, fish predator risk, and ocean acidification, which are coordinated through the Southern California Coastal Water Research Project (SCCWRP) (e.g., the 1994 Southern California Bight Pilot Project, Bight'98, Bight'03, Bight'08, Bight'13, and Bight'18).

Participation in the Southern California Bight Regional Monitoring Program shall include aspects of the program relevant to understanding regional trends and answering regional questions related to public health (i.e., bacterial contamination), water quality, sediment geochemistry, biological communities, and seafood safety (e.g., fish tissue contamination) in the receiving waters environment. For sediment chemistry and benthic infauna communities, this level of effort shall be similar to the 35 one-off (i.e., 1/5-year) summer samples. For trawls, participation shall be equivalent to the 8 annual summer samples. In both cases (i.e., the benthic and trawl programs), 22 quarterly/annual benthic stations and 6 semi-annual trawl stations identified in the monitoring program should continue to be sampled, even during regional monitoring program events, to assess compliance and trends near the discharge.

The Discharger shall complete collection, analysis, and reporting of samples in accordance with the schedules established by the next Bight regional program

development committee(s). Previous participation included method development, research, and monitoring activities involving microbiology, water quality data, marine sediments, fish/macrobenthic assemblages, fish tissue contamination, and harmful algal blooms related to point and nonpoint discharges to the marine environment. Levels of participation and areas of study are dependent upon the final study plans established by Bight regional program development committees, but will be equivalent to that provided by the Discharger in previous regional surveys conducted in 1994, 1998, 2003, 2008, 2013, and 2018. For Bight'18, this involved:

1. Participation in regional microbiological studies testing rapid methods and developing rapid methods for detection of fecal indicator bacteria in beach sands at 2 beach stations.
2. Collection and analysis of water quality samples for analyzing the relationship between nutrients discharged through POTW outfalls, upwelling, and harmful algal blooms. Note that this study was deferred to Bight'23 due to logistical sample deployment issues.
3. Collection and analysis of water quality samples for ocean acidification and hypoxia (OAH), including bongo net towing to collect pteropods, at 4 bongo net tow stations.
4. Collection and analysis of sediment grab samples for geochemistry and benthic infauna at 44 benthic stations .
5. Collection and analyses of fish and macroinvertebrate community structure at 21 trawl stations.
6. Collection and analysis of fish tissue from approximately two zones or the equivalent of 40 tissue samples for chemical contaminants (organics and metals), and analysis of selected biomarkers at a subset of these stations.

#### **B. Southern California Bight Regional Water Quality Program**

The Southern California Bight Regional Water Quality Program (SCBRWQP; previously known as the Central Bight Water Quality Cooperative Program or the Central Bight Regional Water Quality Monitoring Program) is a coordinated quarterly receiving water quality monitoring program conducted by OC San, the County Sanitation Districts of Los Angeles, the City of Los Angeles, the City of San Diego, and the City of Oxnard.

1. Monitoring Locations – Quarterly Southern California Bight Regional Water Quality Monitoring Stations (n=60)

The Discharger shall monitor the receiving water at the Quarterly Southern California Bight Regional Water Quality Monitoring Stations specified in Table E-2. The Discharger shall complete collection and analysis of samples. Results shall be reported in the annual receiving water monitoring report and uploaded, as available, to [www.sccoos.org](http://www.sccoos.org). The Discharger's level of participation shall be consistent with that provided in previous quarterly regional surveys.

**Table E-13. Southern California Bight Receiving Water Quality Monitoring**

| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                  | Units                                         | Sample Type | Sample Depth                     | Minimum Sampling Frequency | Required Analytical Test Method |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------|----------------------------------|----------------------------|---------------------------------|
| Surface Observations <sup>1</sup>                                                                                                                                                                                                                                                                                                                                                                                                          | --                                            | visual      | surface                          | 1/quarter                  | --                              |
| Salinity (calculated)                                                                                                                                                                                                                                                                                                                                                                                                                      | psu                                           | profile     | CTD sampling depths in Table E-2 | 1/quarter                  | 2                               |
| Conductivity                                                                                                                                                                                                                                                                                                                                                                                                                               | S/m                                           | profile     |                                  | 1/quarter                  | 2                               |
| Temperature                                                                                                                                                                                                                                                                                                                                                                                                                                | °C                                            | profile     |                                  | 1/quarter                  | 2                               |
| Dissolved oxygen (DO)                                                                                                                                                                                                                                                                                                                                                                                                                      | mg/L                                          | profile     |                                  | 1/quarter                  | 2                               |
| Transmissivity                                                                                                                                                                                                                                                                                                                                                                                                                             | %                                             | profile     |                                  | 1/quarter                  | 2                               |
| Photosynthetically active radiation (PAR)                                                                                                                                                                                                                                                                                                                                                                                                  | μEinsteins sec <sup>-1</sup> cm <sup>-2</sup> | profile     |                                  | 1/quarter                  | 2                               |
| Chlorophyll-a fluorescence                                                                                                                                                                                                                                                                                                                                                                                                                 | μg/L                                          | profile     |                                  | 1/quarter                  | 2                               |
| pH                                                                                                                                                                                                                                                                                                                                                                                                                                         | pH units                                      | profile     |                                  | 1/quarter                  | 2                               |
| <sup>1</sup> Wind direction and speed, weather, and sea and tidal condition shall be recorded, with the source(s) of the data documented. Observations of unusual water color, turbidity, odor, oil and grease, trash, or other physical evidence of waste discharge and trash in the water shall be noted on the log sheet prepared at the time of sample collection. These observations shall be recorded whenever a station is sampled. |                                               |             |                                  |                            |                                 |
| <sup>2</sup> As specified in 40 CFR § 136, or in the MRP QAPP and Laboratory Quality Manual.                                                                                                                                                                                                                                                                                                                                               |                                               |             |                                  |                            |                                 |

### C. Central Regional Kelp Survey

This regional survey is designed to determine if the extent of kelp beds in the Southern California Bight are changing over time and if the rate of change differs between kelp beds. A group of private and public agencies cooperatively monitors the health and standing crop of kelp beds using quarterly aerial imaging of kelp bed canopy cover within the central Bight. The data collected in this regional survey will be used to assess status and trends in kelp bed health and spatial extent. The regional nature of the survey will allow the status of kelp beds near to the discharge to be compared to regional trends.

The Discharger shall continue its current level of participation in the Central Region Kelp Survey Consortium (CRKSC) Monitoring Program to conduct regional kelp bed monitoring in Southern California coastal waters. The Discharger shall participate in the regional management and technical committees responsible for the development of the

survey design and the assessment of kelp bed resources in the Bight.

#### D. Orange County Regional Shoreline REC-1 Cooperative Monitoring Program

This regional program is a coordinated shoreline REC-1 water quality monitoring effort conducted by the Discharger, the Orange County Health Care Agency (OCHCA), the South Orange County Wastewater Authority, and the Orange County Public Works (OC Watersheds) in the Ocean Water Protection Program, along Orange County's coastal shoreline (from Seal Beach to San Clemente State Beach). OCHCA reviews collected bacteriological data to determine whether a station meets Ocean Water-Contact Sports Standard (i.e., Assembly Bill 411; AB411), and uses these results as the basis for health advisories, postings, or beach closures. This Order/Permit requires a minimum level of participation in microbiological regional monitoring.

##### 1. Monitoring Locations – Orange County Regional Shoreline REC-1 Cooperative Monitoring Stations (Nearshore Zone) (n=36)

The Discharger shall monitor the receiving water at least once per week at the Orange County Regional Shoreline REC-1 Cooperative Monitoring Stations (Nearshore Zone or Surfzone) specified in Table E-2. The Discharger shall complete collection, analysis, and reporting of the regional samples required under this Order/Permit. Results shall be reported in the annual receiving water monitoring report, except that microbiological results shall continue to be reported on a timely basis (approximately daily) to the OCHCA, Environmental Health and placed on the Internet each month.

**Table E-14. Orange County Regional Shoreline REC-1 Cooperative Monitoring**

| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Units                   | Sample Type | Minimum Sampling Frequency <sup>1</sup> | Required Analytical Test Method |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------|-----------------------------------------|---------------------------------|
| Surface Observations <sup>2</sup>                                                                                                                                                                                                                                                                                                                                                                                                                             | --                      | visual      | See Table E-2.                          | 3                               |
| Total coliform density                                                                                                                                                                                                                                                                                                                                                                                                                                        | CFU/100 mL <sup>4</sup> | grab        |                                         | 3,5                             |
| Fecal coliform density                                                                                                                                                                                                                                                                                                                                                                                                                                        | CFU/100 mL <sup>4</sup> | grab        |                                         | 3,5                             |
| <i>Enterococcus</i> density                                                                                                                                                                                                                                                                                                                                                                                                                                   | CFU/100 mL <sup>4</sup> | grab        |                                         | 3,5                             |
| <sup>1</sup> In the event of stormy weather which makes sampling hazardous at certain surfzone stations, collection of samples at such stations can be omitted, provided that such omissions do not occur more than 5 days in any calendar year or occur at consecutive sampling times. The observations shall still be recorded and reported to the Santa Ana Water Board and USEPA for these stations at the time the sample was attempted to be collected. |                         |             |                                         |                                 |
| <sup>2</sup> Wind direction and speed, weather, and sea and tidal condition shall be recorded, with the source(s) of the data documented. Observations of unusual water color, turbidity, odor, oil and grease, trash, or other physical evidence of waste discharge and trash in the water                                                                                                                                                                   |                         |             |                                         |                                 |

| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Units | Sample Type | Minimum Sampling Frequency <sup>1</sup> | Required Analytical Test Method |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|-----------------------------------------|---------------------------------|
| <p>shall be noted on the log sheet prepared at the time of sample collection. These observations shall be recorded whenever a station is sampled.</p> <p><sup>3</sup> As specified in 40 CFR § 136, or in the MRP QAPP and Laboratory Quality Manual.</p> <p><sup>4</sup> Results may be reported as either MPN/100 mL if the laboratory method used provides results in MPN/100 mL or CFU/100 mL if the laboratory method used provides results in CFU/100 mL.</p> <p><sup>5</sup> Test methods used for coliforms (total and fecal) shall be those presented in Table 1A of 40 CFR § 136, unless alternate methods have been approved in advance by USEPA pursuant to 40 CFR § 136, or improved methods have been determined by the Executive Officer and/or USEPA.</p> |       |             |                                         |                                 |

#### E. Ocean Acidification and Hypoxia (OAH) Mooring

The Discharger shall monitor ocean acidification and hypoxia at a single location. An automated data quality control system shall be established for telemetered data (i.e., real-time, online ocean acidification data) based on the Integrated Ocean Observation System (IOOS) protocols.

### X. STRATEGIC PROCESS STUDIES

The Discharger shall conduct Strategic Process Studies (SPS) to document the effectiveness of its source control and wastewater treatment operations in protecting the coastal ocean. SPS are designed to address unanswered questions raised by the Core monitoring program results and/or focus on issues of interest to the Discharger, the Santa Ana Water Board, and/or USEPA. SPS shall be proposed and be approved by the Santa Ana Water Board and USEPA to ensure appropriate focus and level of effort. Five studies below must be conducted under this Order/Permit using previously approved work plans as these studies have been approved during the term of the 2012 Order/Permit and are not yet completed:

- A. ROMS-BEC Ocean Outfall Modeling:** The Discharger works with SCCWRP and their collaborators to model and assess the spatial and temporal extent of its discharged effluent before and after the implementation of the GWRS Final Expansion.
- B. Microplastics Characterization:** This SPS characterizes the quantity and types of microplastics throughout the Discharger's treatment system. This study will also develop methods and analyses to help inform the transport, fate and impacts of microplastics through the Discharger's wastewater treatment process to the receiving environment.
- C. In-vitro Cell Bioassay Monitoring Assessment:** This study will provide a preliminary assessment of contaminants of emerging concern (CEC) in the receiving water using *in-vitro* cell bioassay techniques. Cell bioassays can be used as a screening tool to evaluate potential impacts resulting from changes in the effluent and receiving environment water quality associated with the GWRS Final Expansion.



- D. Sediment Linear Alkylbenzenes:** Linear Alkylbenzenes (LABs) were previously used to investigate whether other contaminants present in the sediment were associated with the effluent discharge. This SPS will provide updated data within the Discharger's monitoring area for evaluating future changes due to GWRS Final Expansion.
- E. Meiofauna Baseline:** The Discharger will investigate the impacts of increased RO concentrate flows from the GWRS Final Expansion on marine biota in the receiving water using meiofauna (animals ranging from 63 to 500 µm in size) which are known to be more sensitive to anthropogenic impacts than macrofauna. This SPS will characterize the meiofauna communities in the receiving environment and evaluate the suitability of using meiofauna for a before-after control-impact study of the GWRS Final Expansion.

## XI. OTHER MONITORING REQUIREMENTS

### A. Contaminants of Emerging Concern (CEC) Monitoring Study

The Discharger shall continue to investigate CEC in the discharge and/or receiving waters, following its approved CEC study workplan. Within six (6) months of the effective date of this Order/Permit, the Discharger shall submit for Executive Officer/Director approval an updated CEC study workplan and submit the special study final report with the application for permit reissuance. The workplan shall include, but is not limited to:

1. Updates of CEC identification for discharge monitoring (e.g., CEC concentrations and mass loadings, sample type, minimum sampling frequency, and analytical test method considering sensitivity, accuracy, availability, and cost). The parameters in Table E-15 may be re-evaluated and modified by the Executive officer/Director. The workplan may also propose surrogate or indicator CEC that may better characterized discharged CEC.

**Table E-15. CEC Discharge Monitoring**

| Parameter             | CAS #   | Units       | Sample Type     | Minimum Sampling Frequency | Required Analytical Test Method |
|-----------------------|---------|-------------|-----------------|----------------------------|---------------------------------|
| <b>Hormones</b>       |         |             |                 |                            |                                 |
| 17α-Ethynyl estradiol | 57-63-6 | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| 17α-Estradiol         | 57-91-0 | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| 17β-Estradiol         | 50-28-2 | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Estriol               | 50-27-1 | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Estrone               | 53-16-7 | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Progesterone          | 57-83-0 | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |



| Parameter                                                 | CAS #         | Units       | Sample Type     | Minimum Sampling Frequency | Required Analytical Test Method |
|-----------------------------------------------------------|---------------|-------------|-----------------|----------------------------|---------------------------------|
| Testosterone                                              | 58-22-0       | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| <b>Industrial Endocrine Disrupting Compounds (IEDCs)</b>  |               |             |                 |                            |                                 |
| Bisphenol A                                               | 80-05-7       | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| 4-para-Nonylphenol                                        | 84852-15-3    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Nonylphenol diethoxylate                                  | Not available | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Nonylphenol monoethoxylate                                | 68412-54-4    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Octylphenol                                               | 27193-28-8    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| 4-n-Octylphenol diethoxylate                              | 51437-90-2    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Octylphenol monoethoxylate                                | 5143-89-9     | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| <b>Pharmaceuticals and Personal Care Products (PPCPs)</b> |               |             |                 |                            |                                 |
| Acetaminophen                                             | 298-46-4      | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Caffeine                                                  | 58-08-2       | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Carbamazepine                                             | 298-46-4      | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| DEET<br>(N,N-Diethyl-m-toluamide)                         | 134-62-3      | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Diclofenac                                                | 15307-86-5    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Erythromycin                                              | 114-07-8      | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Fluoxetine hydrochloride                                  | 56296-78-7    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Galaxolide (HHCB)                                         | 1222-05-5     | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Gemfibrozil                                               | 25812-30-0    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Ibuprofen                                                 | 15687-27-1    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Oxybenzone                                                | 131-57-7      | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Primidone                                                 | 125-33-7      | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Sulfamethoxazole                                          | 723-46-6      | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Triclosan                                                 | 3380-34-5     | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |

| Parameter                                                  | CAS #         | Units       | Sample Type     | Minimum Sampling Frequency | Required Analytical Test Method |
|------------------------------------------------------------|---------------|-------------|-----------------|----------------------------|---------------------------------|
| <b>Pesticides and Insecticides</b>                         |               |             |                 |                            |                                 |
| Fipronil                                                   | 120068-37-3   | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Fipronil Sulfone                                           | 120068-36-2   | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Bifenthrin                                                 | 82657-04-3    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Total Permethrin                                           | 52645-53-1    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Chlorpyrifos                                               | 2921-88-2     | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| Diazinon                                                   | 333-41-5      | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| <b>Flame Retardants</b>                                    |               |             |                 |                            |                                 |
| BDE 47                                                     | Not available | ng/L & lb/d | 24-hr composite | 1/year                     | 1,2                             |
| BDE 99                                                     | Not available | ng/L & lb/d | 24-hr composite | 1/year                     | 1,2                             |
| BDE 100                                                    | Not available | ng/L & lb/d | 24-hr composite | 1/year                     | 1,2                             |
| BDE 183                                                    | Not available | ng/L & lb/d | 24-hr composite | 1/year                     | 1,2                             |
| TDCPP; tris(1,3-dichloro-2-propyl)phosphate                | 13674-87-8    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| TCEP; tris(2-chloroethyl)phosphate                         | 115-96-8      | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| TCPP; tris(1-chloro-2-propyl)phosphate                     | 13674-84-5    | ng/L & lb/d | 24-hr composite | 1/year                     | 1                               |
| <b>PFAS (per-and polyfluoroalkyl substances) compounds</b> |               |             |                 |                            |                                 |
| PFDA (Perfluorodecanoic acid)                              | 335-76-2      | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| PFDoA (Perfluorododecanoic acid)                           | 307-55-1      | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| PFHxA (Perfluorohexanoic acid)                             | 307-24-4      | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| PFHpA (Perfluoroheptanoic acid)                            | 375-85-9      | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| PFNA (Perfluorononanoic acid)                              | 375-95-1      | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |

| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CAS #      | Units       | Sample Type     | Minimum Sampling Frequency | Required Analytical Test Method |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|-----------------|----------------------------|---------------------------------|
| PFOA (Perfluorooctanoic acid)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 335-67-1   | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| PFTeDA (Perfluorotetradecanoic acid)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 376-06-7   | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| PFTrDA (Perfluorotridecanoic acid)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 72629-94-8 | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| PFUnDA (Perfluoroundecanoic acid)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2058-94-8  | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| PFBS (Perfluorobutane sulfonic acid)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 375-73-5   | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| PFHxS (Perfluorohexane sulfonic acid)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 355-46-4   | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| PFOS (Perfluorooctane sulfonic acid)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1763-23-1  | ng/L & lb/d | 24-hr composite | 1/year                     | 3                               |
| <sup>1</sup> USEPA Methods 1694 Modified or 1698 Modified; ASTM Methods D7065 or D7485; or other methodologies utilized by the U.S. Geological Survey, California Department of Public Health, or other federal or State agencies.<br><sup>2</sup> USEPA Methods 1614A Modified using GC/MS SIM or other methodologies utilized by the U.S. Geological Survey, California Department of Public Health, or other federal or State agencies.<br><sup>3</sup> USEPA Method 537.1 Modified with isotope dilution using LC/MS/MS or other methodologies to meet the required reporting limit of 50 ng/L. |            |             |                 |                            |                                 |

2. Characterization of CEC monitoring data for the final effluent and receiving water including sediments, using previously collected monitoring data and monitoring data collected for this study. At minimum, this characterization shall include: identification of all monitored CECs, monitoring dates, frequency and duration; QA/QC information; reporting limits, minimum levels, and minimum detection limits achieved for each methodology; and an analysis of trends.
3. A preliminary workplan for collaborative studies: (1) CECs identification in reclaimed wastewater with the Orange County Water District, and (2) methods development for measuring CECs in marine sediments with the Los Angeles County Sanitation Districts.

## B. Outfall and Diffuser System Inspection

The Discharger shall externally inspect each ocean outfall (i.e., 120" and 78" outfalls) every 2.5 years to ensure that the outfall structures are in serviceable condition and that they can continue to be operated safely. Inspections shall include general observations including any collection of debris within the diversion structure surge chamber, and photographic/videographic records of the external outfall pipes and adjacent ballasting system. The inspections may be conducted by remotely operated vehicle, diver, or manned submarine. Within 60 days of the completion of the outfall and diffuser system inspection, a summary report of the inspection findings, including videographic and/or photographic images shall be submitted to the Santa Ana Water Board and USEPA.

## **XII. REPORTING REQUIREMENTS**

### **A. General Monitoring and Reporting Requirements**

1. The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.
2. If there is no discharge during any reporting period, the report shall so state.
3. Each monitoring report shall contain a separate section titled "Summary of Non-compliance" which discusses the compliance record and the corrective actions taken or planned that may be needed to bring the discharge into full compliance with waste discharge requirements. This section shall clearly list all non-compliance with discharge requirements as well as all excursions of effluent limitations.
4. The Discharger shall inform the Santa Ana Water Board and USEPA well in advance of any proposed construction or maintenance activity, or modification to the POTW that could potentially affect compliance with applicable requirements.
5. The date and time of sampling (as appropriate) shall be reported with the analytical values determined.
6. The laboratory conducting analyses shall be certified by the State Water Resources Control Board, Environmental Laboratory Accreditation Program (ELAP), in accordance with CWC section 13176, and must include quality assurance/quality control (QA/QC) data in their reports.
7. The Discharger shall have and implement an acceptable written NPDES monitoring and reporting program QAPP for all monitoring and analyses required under this Order/Permit. For each target analyte, required quality control samples shall be analyzed as described in the laboratory Quality Manual. A summary report of the QA/QC findings for all monitoring and analysis for the previous fiscal year (July 1 through June 30) shall be provided in the annual receiving water monitoring report.
8. QA/QC data associated with a sample must be reported when requested by the Santa Ana Water Board or USEPA. The Santa Ana Water Board and USEPA will reject laboratory data if quality control data is unavailable or unacceptable.

9. When requested by the Santa Ana Water Board or USEPA, the Discharger shall participate in the NPDES DMR QA performance study.
10. Dischargers are to instruct their laboratories to establish calibration standards so that the ML (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve. In accordance with Ocean Plan Chapter III.C.5.b, the Discharger's laboratory may employ a calibration standard lower than the ML in Appendix II of the Ocean Plan.
11. Upon request by the Discharger, the Santa Ana Water Board, in consultation with the State Water Board's Quality Assurance Program and/or USEPA, may establish an ML that is not contained in Appendix II of the 2019 Ocean Plan, to be included in the Discharger's NPDES permit, in any of the following situations:
  - a. When the pollutant under consideration is not included in Appendix II;
  - b. When the Discharger agrees to use a test method that is more sensitive than those specified in 40 CFR § 136 (most recent revision);
  - c. When the Discharger agrees to use an ML lower than those listed in Appendix II;
  - d. When the Discharger demonstrates that the calibration standard matrix is sufficiently different from that used to establish the ML in Appendix II and proposes an appropriate ML for their matrix; or
  - e. When the Discharger uses a method whose quantification practices are not consistent with the definition of an ML. In such cases, the Discharger, Santa Ana Water Board, State Water Board and USEPA shall agree on a lowest quantifiable limit, and that limit will substitute for the ML for reporting and compliance determination purposes.
12. Records and reports of receiving water monitoring surveys conducted to meet receiving water monitoring requirements shall include, at a minimum, the following information:
  - a. A description of climatic and receiving water characteristics at the time of sampling (weather observations, unusual or abnormal amounts of floating debris, trash, discoloration, odor, oil and grease, wind speed and direction, swell or wave action, sea and tidal condition, time of sampling or measurements, etc.).
  - b. The date, exact place and description of sampling stations, including differences unique to each station (e.g., date, time, station location, depth, and sample type).
  - c. A list of the individuals participating in field collection of samples or data and description of the sample collection and preservation procedures used in the various surveys.

- d. A description of the specific method used for laboratory analysis, the date(s) the analyses were performed and the individuals participating in these analyses.
  - e. An in-depth discussion of the results of the surveys. All tabulations and computations shall be explained.
13. The results of all monitoring data for this Order/Permit shall be reported in a tabulated format, acceptable by the Santa Ana Water Board and USEPA, that allows direct comparison with the limitations and conditions of this Order/Permit and clearly illustrates whether the facility is operating in compliance with this Order/Permit.

All receiving water monitoring raw data files should be stored onsite and must be submitted to the Santa Ana Water Board and USEPA upon request.

#### **B. Self-Monitoring Reports (SMRs)**

1. The Discharger shall electronically submit SMRs using the State Water Board's California Integrated Water Quality System (CIWQS) Program website at [http://www.waterboards.ca.gov/water\\_issues/programs/ciwqs/](http://www.waterboards.ca.gov/water_issues/programs/ciwqs/). The CIWQS website will provide additional information for SMR submittal in the event there will be a planned service interruption for electronic submittal.
2. The Discharger shall report in the SMR the results for all monitoring specified in this Order/Permit. The Discharger shall submit monthly SMRs including the results of all required monitoring using USEPA-approved test methods or other test methods specified in this Order. SMRs are to include all new monitoring results obtained since the last SMR was submitted. If the Discharger monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.
3. Monitoring periods and reporting for all required monitoring shall be completed according to the schedule in Table E-16. Monitoring and reporting due dates are specified elsewhere in this Order/Permit for: biosolids program annual report; pretreatment program annual reports; pollutant minimization program annual report; spill preventative and contingency plans update report; storm water management plan update; Initial Investigation TRE Workplan update; dilution model update; ocean outfall condition assessment reports; NPDES monitoring data availability via Internet update report; annual receiving water monitoring report (including a chapter for annotated QA/QC findings, chapters summarizing monitored activities for regional monitoring and strategic process studies, and a chapter for the annual mass emission); and outfall and diffuser systems inspection report.



**Table E-16. Monitoring Periods and Reporting Schedule**

| <b>Sampling Frequency</b> | <b>Monitoring Period Begins On...</b>                                                            | <b>Monitoring Period</b>                                                                                              | <b>SMR/DMR Due Date</b>                                                                                 |
|---------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Continuous                | Effective date                                                                                   | All                                                                                                                   | Submit with monthly report                                                                              |
| Hourly                    | Effective date                                                                                   | Hourly                                                                                                                | Submit with monthly report                                                                              |
| Daily (1/day)             | Effective date                                                                                   | Midnight through 11:59 pm, or any 24-hr period that reasonably represents a calendar day for sampling purposes        | Submit with monthly report                                                                              |
| Weekly (1/week)           | Sunday following effective date, or on effective date if Sunday                                  | Sunday through Saturday                                                                                               | Submit with monthly report                                                                              |
| Monthly (1/month)         | First day of calendar month following effective date, or on effective date if first day of month | 1 <sup>st</sup> day of calendar month through last day of calendar month                                              | 15 <sup>th</sup> day of 2 <sup>nd</sup> month following the monitoring period, submit as monthly report |
| Quarterly (1/quarter)     | Closest of January 1, April 1, July 1, or October 1 following (or on) effective date             | January 1 through March 31<br>April 1 through June 30<br>July 1 through September 30<br>October 1 through December 31 | May 15, August 15, November 15, February 15, submit with monthly report                                 |
| Semi-annually (2/year)    | Closest of January 1 or July 1 following (or on) effective date                                  | January 1 through June 30<br>July 1 through December 31                                                               | August 15, February 15, submit with monthly report                                                      |
| Annually (1/year)         | January 1 following (or on) effective date                                                       | January 1 through December 31                                                                                         | February 15, submit with monthly report                                                                 |
| Other                     | On date specified in Order/Permit                                                                | As specified in Order/Permit                                                                                          | Submit with monthly report                                                                              |

4. **Reporting Protocols.** The Discharger shall report with each sample result the applicable reported Minimum Level (reported ML, also known as the Reporting Level, or RL) and the current Method Detection Limit (MDL), as determined by the procedure in 40 CFR § 136.

The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- a. Sample results greater than or equal to the reported ML shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
  - b. Sample results less than the reported ML, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported. For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ. The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ( $\pm$  a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.
  - c. Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.
  - d. Dischargers are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.
5. **Compliance Determination.** Compliance with effluent limitations for reportable pollutants shall be determined using sample reporting protocols defined above and Attachment A of this Order/Permit. For purposes of reporting and administrative enforcement by the Santa Ana Water Board, State Water Board, and USEPA, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the reportable pollutant in the monitoring sample is greater than the corresponding effluent limitation and greater than or equal to the reported Minimum Level (ML).
6. **Multiple Sample Data.** When determining compliance with a measure of central tendency (arithmetic mean, geometric mean, median, etc.) of multiple sample analyses and the data set contains one or more reported determinations of DNQ or ND, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:
- a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any).
  - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
7. The Discharger shall submit SMRs in accordance with the following requirements:

- a. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Discharger is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Discharger shall electronically submit the data in a tabular format as an attachment (e.g., annual receiving water monitoring report).
- b. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the waste discharge requirements; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.
- c. SMRs must be signed and certified as required by the Standard Provisions (Attachment D).

SMRs submitted to the State that are not DMRs shall also be submitted to USEPA at the following address, unless another USEPA address is specified in this Order/Permit. Signed and certified written reports and receiving water monitoring data may be submitted on a CD-ROM, with a signed cover letter.

U.S. EPA, Region 9  
ATTN: NPDES Data Team (ENF 2-3)  
Data Solution Section ECAD  
75 Hawthorne Street  
San Francisco, CA 94105-3901

### C. Discharge Monitoring Reports (DMRs)

1. DMRs are USEPA reporting requirements. The Discharger shall electronically certify and submit DMRs together with SMRs using Electronic Self-Monitoring Reports module eSMR 2.5 or any upgraded version. Electronic DMR submittal shall be in addition to electronic SMR submittal. Information about electronic DMR submittal is available at the DMR website at:  
<[http://www.waterboards.ca.gov/water\\_issues/programs/discharge\\_monitoring](http://www.waterboards.ca.gov/water_issues/programs/discharge_monitoring)>.

### D. Other Reports

1. **Pretreatment Report.** By October 31<sup>st</sup> of each year, the Discharger shall submit annual pretreatment reports to the Santa Ana Water Board, the State Water Board, and USEPA, describing the Discharger's pretreatment activities over the previous OCSD fiscal year (July 1 through June 30). The annual reports shall contain, but not be limited to, the information required in the attached *Pretreatment Reporting Requirements* (Attachment H), or an approved revised version thereof. If the Discharger is not in compliance with any conditions or requirements of this Order/Permit, the Discharger shall include the reasons for noncompliance and shall state how and when the Discharger will comply with such conditions and

requirements. Prior to the completion of the GWRS Final Expansion Project, the Discharger shall conduct annual technical review of local limits under 40 CFR § 403.5(c)(1) and submit the results as part of the annual pretreatment report. Within two (2) years of the completion of the GWRS Final Expansion Project, the Discharger shall provide a written technical evaluation of the need to revise local limits under 40 CFR § 403.5(c)(1).

2. **Biosolids Report.** By February 19th of each year, the Discharger shall submit an annual biosolids report into USEPA's CDX electronic reporting system, with an electronic copy to the Santa Ana Water Board by email at [santaana@waterboards.ca.gov](mailto:santaana@waterboards.ca.gov), for the period covering the previous calendar year (January 1 through December 31). The annual reports shall contain, but not be limited to, the information required in the attached *Biosolids Reporting Requirements* (Attachment G), or an approved revised version thereof. If the Discharger is not in compliance with any conditions or requirements of this Order/Permit, the Discharger shall include the reasons for noncompliance and shall state how and when the Discharger will comply with such conditions and requirements.
3. **Receiving Water Monitoring Report.** By March 15th of each year, the Discharger shall submit an annual receiving water monitoring report for monitored activities during the previous fiscal year (July 1 through June 30), to the Santa Ana Water Board and USEPA. The report shall contain both tabular and graphical summaries of receiving water monitoring data and discuss the compliance record and actions undertaken or proposed which will bring the discharge into full compliance with receiving water monitoring requirements at the earliest time, and an estimate of the date when the Discharger will be in compliance. Receiving water core monitoring results, including annotated QA/QC findings, shall be described and summarized. Separate report chapters for receiving water regional monitoring and strategic process studies shall provide summaries of monitored activities during the previous fiscal year. A separate report chapter for effluent monitoring shall also provide summaries of effluent monitoring including the annual mass emission during the previous fiscal year.
4. **Strategic Process Study Proposal.** In the spring, beginning in 2022 and continuing every-other year during the term of this Order/Permit, the Discharger, Santa Ana Water Board, and USEPA shall consult to determine the need for strategic process studies. By October 1st, the Discharger shall submit proposals to the Santa Ana Water Board and USEPA for the following fiscal year's (July 1 through June 30) monitoring effort, or a letter explaining why no special studies are proposed. Upon approval by the Santa Ana Water Board and USEPA, the Discharger shall implement its strategic process studies. Final scopes of work, including reporting schedules, shall be presented by the Discharger at a spring meeting with Santa Ana Water Board and USEPA staffs to obtain Santa Ana Water Board and USEPA approval.

5. **Other Workplans and Reports.** The Discharger shall report the results of any special studies (e.g., TRE work plan, dilution model update and ocean outfall condition assessment), Pollutant Minimization Program (PMP), updated Storm Water Management Plan, Spill Preventive and Contingency Plan (SPCP), Asset Management Plan (AMP), and Monitoring Data Accessibility Plan required by Special Provisions – Section VII.C of this Order/Permit. The Discharger shall also submit the workplans and report the results of special monitoring studies (e.g., CEC monitoring study and outfall and diffuser system inspection) required by Other Monitoring Requirements – Section XI of Attachment E. The Discharger shall submit workplans and/or reports with the first monthly SMR scheduled to be submitted on or immediately following the report due date in compliance with SMR reporting requirements described in subsection XII.B above.

## ATTACHMENT F – FACT SHEET

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## ATTACHMENT F – FACT SHEET

As described in section II.B of this Order/Permit, the California Regional Water Quality Control Board, Santa Ana Region (hereinafter Santa Ana Water Board) and USEPA incorporates this Fact Sheet as findings of the Santa Ana Water Board and USEPA supporting the issuance of this Order/Permit. This Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order/Permit.

This Order/Permit has been prepared under a standardized format to accommodate a broad range of discharge requirements for Dischargers in California. Only those sections or subsections of this Order/Permit that are specifically identified as “not applicable” have been determined not to apply to this Discharger. Sections or subsections of this Order/Permit not specifically identified as “not applicable” are fully applicable to this Discharger.

### I. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

**Table F-1. Facility Information**

|                                                     |                                                                                                                                                       |                                                       |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>WDID</b>                                         | 8 300110002                                                                                                                                           |                                                       |
| <b>Discharger</b>                                   | Orange County Sanitation District                                                                                                                     |                                                       |
| <b>Name of Facility (and POTW)</b>                  | Reclamation Plant No. 1 (Fountain Valley), collection system and outfalls<br>Treatment Plant No. 2 (Huntington Beach), collection system and outfalls |                                                       |
| <b>Facility Address</b>                             | Reclamation Plant No. 1                                                                                                                               | Treatment Plant No. 2                                 |
|                                                     | 10844 Ellis Avenue<br>Fountain Valley, CA 92708                                                                                                       | 22212 Brookhurst Street<br>Huntington Beach, CA 92646 |
|                                                     | Orange County                                                                                                                                         |                                                       |
| <b>Facility Contact, Title and Phone</b>            | James Herberg, General Manager<br>(714) 962-2411                                                                                                      |                                                       |
| <b>Authorized Person to Sign and Submit Reports</b> | James Herberg, General Manager<br>(714) 962-2411                                                                                                      |                                                       |
| <b>Mailing Address</b>                              | 10844 Ellis Avenue, Fountain Valley, CA 92708-7018                                                                                                    |                                                       |
| <b>Billing Address</b>                              | Same                                                                                                                                                  |                                                       |
| <b>Type of Facility</b>                             | Publicly Owned Treatment Works (POTW)                                                                                                                 |                                                       |
| <b>Major or Minor Facility</b>                      | Major                                                                                                                                                 |                                                       |
| <b>Threat to Water Quality</b>                      | 1                                                                                                                                                     |                                                       |
| <b>Complexity</b>                                   | A                                                                                                                                                     |                                                       |

|                                |                                                                                              |                            |
|--------------------------------|----------------------------------------------------------------------------------------------|----------------------------|
| <b>Pretreatment Program</b>    | Yes                                                                                          |                            |
| <b>Recycling Requirements</b>  | State Water Board Order WQ 2016-0068-DDW                                                     |                            |
| <b>Facility Permitted Flow</b> | 332 MGD (dry weather)                                                                        |                            |
| <b>Facility Design Flow</b>    | Reclamation Plant No. 1                                                                      | Treatment Plant No. 2      |
|                                | 182 MGD (dry weather)                                                                        | 150 MGD (dry weather)      |
|                                | 345 MGD (peak wet weather)                                                                   | 317 MGD (peak wet weather) |
| <b>Watershed</b>               | Santa Ana River Watershed                                                                    |                            |
| <b>Receiving Water</b>         | Pacific Ocean (Discharge Point Nos.001 and 002)<br>Santa Ana River (Discharge Point No. 003) |                            |
| <b>Receiving Water Type</b>    | Ocean and inland (tidal prism) waters                                                        |                            |

- A. Orange County Sanitation District (hereinafter OC San, Permittee or Discharger) is the owner and operator of Reclamation Plant No. 1 in Fountain Valley and Treatment Plant No. 2 in Huntington Beach (hereinafter treatment facilities), a Publicly-Owned Treatment Work (POTW).

For the purposes of this Order/Permit, references to the “discharger” or “permittee” in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

- B. The Facility discharges wastewater to the Pacific Ocean, a water of the United States. The Discharger was previously regulated by Order No. R8-2012-0035 and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0110604 adopted on June 15, 2012 and expired on July 19, 2017. The Discharger filed a report of waste discharge and submitted an application for reissuance of its waste discharge requirements (WDRs) and NPDES permit on January 3, 2017, and the previous order was administratively extended by operation of law. Attachment B provides a map of the area around the Facility. Attachment C provides a flow schematic of the Facility.
- C. Regulations at 40 CFR § 122.46 limit the duration of NPDES permits to a fixed term not to exceed five years. Accordingly, Table 3 of this Order/Permit limits the duration of the discharge authorization. However, pursuant to California Code of Regulations, title 23, section 2235.4, the terms and conditions of an expired permit are automatically continued pending reissuance of the permit if the Discharger complies with all federal NPDES requirements for continuation of expired permits. (See also 40 CFR § 122.6(d)).

## II. FACILITY DESCRIPTION

### A. Description of Wastewater and Biosolids Treatment and Controls

The Discharger’s POTW serves approximately 480 square miles of central and northern Orange County, California. Currently, the POTW consists of approximately

388 miles of trunk sewer systems and force mains, 15 offsite pump stations, two wastewater treatment plants, and three outfalls. In addition to the domestic, commercial, and industrial wastewater received from this service area, through interagency agreements, influent flows are contributed by the Santa Ana Watershed Project Authority (SAWPA) including Stringfellow Superfund Site (EPA ID #CAT080012826), Irvine Ranch Water District, and the Sanitation District of Los Angeles County. During dry weather conditions, the Discharger also accepts small volumes (average daily amount up to 10 MGD) of dry weather urban runoff from within the service area. During 2018-2019, the Discharger's wastewater treatment plants received and processed influent volumes averaging 191 MGD.

Raw wastewater entering either Reclamation Plant No. 1 or Treatment Plant No. 2 passes through metering and diversion structures and continues to headworks facilities for preliminary treatment. Note that influent flow from SAWPA is tributary to Reclamation Plant No. 1 via the Santa Ana River Interceptor (SARI). Because the California Department of Health does not permit SARI flow for the Groundwater Replenishment System (GWRS) reclamation, all SARI flow is diverted to Treatment Plant No. 2 through the Interplant Interceptor for treatment and ocean discharge. Preliminary treatment includes grit and screenings removal. Caustic soda shock, ferrous chloride, calcium nitrate, magnesium hydroxide and polymer can be added to reduce corrosion and odor and provide enhanced primary treatment. The wastewater then receives primary treatment in primary sedimentation facilities. The primary effluent is routed to either trickling filters or activated sludge facilities for secondary treatment. Secondary treated trickling filter and activated sludge effluents from Reclamation Plant No. 1 are delivered to the Orange County Water District (OCWD) for further tertiary treatment and reclamation. The OC San discharges secondary treated effluent from Treatment Plant No. 2 through Discharge Point 001 (primary 120" ocean outfall) to the Pacific Ocean, a water of the United States, except under emergency circumstances. In 2023, when the Final Expansion of the GWRS is complete, Treatment Plant No. 2 effluents will also be reclaimed at the GWRS. SARI flow will be separated from other domestic wastewater at Plant No. 2, where it will be secondary treated before being discharged to the ocean. The use of continuous disinfection prior to discharge through Discharge Point No. 001 (120" outfall) has been ceased since May 2015. Effluent disinfection with chlorine bleach followed by de-chlorination with sodium bisulfite occurs at the other two discharge points 002 and 003. Dewatered biosolids are shipped offsite to land application, composting, thermal conversion, or landfill facilities. Attachment B provides a map of the area around the treatment facilities. Attachment C provides flow schematics of the full secondary treatment facilities.

OCWD receives two streams of OC San secondary treated wastewater, one for the Green Acres Project (GAP, out of service in winter) and the second one for the GWRS. Both GAP and GWRS are major regional wastewater reclamation projects owned and operated by OCWD and located adjacent to Reclamation Plant No. 1. Advanced treatment for the GWRS includes microfiltration, reverse osmosis, and advance oxidation (ultraviolet/peroxide). The Discharger provides approximately 100 % of secondary treated effluent from Plant No.1 to GWRS for advanced treatment and

reuse. In Fiscal Year 2018-19, the average effluent flow to OCWD was 117 MGD. The GWRS Final Expansion has been under construction since November 2019, which supports the Discharger's strategic goal of maximizing water recycling up to 175 MGD at both plants, and OCWD's goal to produce up to 130 MGD of purified recycled water by 2023. GAP can divert up to 7.5 MGD of the Discharger's secondary treated wastewater for tertiary treatment, disinfection, and reuse for landscaped irrigation and industrial uses. In Fiscal Year 2018-19 summer months, average flow to GAP was 4 MGD. The GAP and part of the GWRS generates waste streams (e.g., microfiltration backwash, sump waste, and lime waste) are recycled back into the OC San's primary treatment facility, except for RO reject (or concentrate) which are returned to OC San, blended with other secondary treated waste streams and discharged via ocean outfalls.

In 2019, average daily total influent flow to both plants was 192 MGD, but because of water recycling efforts, the average daily ocean discharge was 100 MGD. Based on 2017 OCSD Facilities Master Plan, the average daily influent flow projected for 2025 is 188~206 MGD during the dry weather season and 478~523 MGD during the peak wet weather season. Note that lower and upper bound flows are based on the current trend and a conservative approach, respectively. The projected 2025 average daily ocean discharges are 55~73 MGD for dry weather and 344~389 MGD for peak wet weather, of which RO concentrate flow is 23 MGD.

## **B. Discharge Points and Receiving Waters**

Secondary treated effluents from Plant No. 1 and 2 are discharged to the Pacific Ocean through the Discharger's ocean outfall system. There are two booster pump stations (i.e., the Ocean Outfall Booster Station (OOPS) and the Effluent Pump Station Annex (EPSA)) that pump secondary effluent to two ocean outfalls: Discharge Point 001 via the 120" pipelines and Discharge Point 002 via the 78" pipelines. In addition to these two ocean outfalls, there are two overflow weirs at Plant No. 2 that discharge into the Santa Ana River.

- 1. Discharge Point 001 (120" outfall).** Discharge Point 001 (120" outfall) is the primary discharge point (activated ocean outfall, with a rated capacity of 480 MGD) to the Pacific Ocean terminating in a diffuser with 503 ports, at a depth of about 197 feet (60 meters) below mean sea level. Discharge Point 001 (120" outfall) is approximately 4.5 miles (3.9 nautical miles) offshore in federal waters as the territorial waters of the State end 3 nautical miles from shore. The Ocean Plan (Appendix 1, Ocean Waters) states, "If a discharge outside the territorial waters of the State could affect the quality of the waters of the State, the discharge may be regulated to assure no violation of the Ocean Plan will occur in ocean waters." This Order/Permit contains discharge prohibitions, effluent limitations, receiving water limitations, and other provisions to ensure that discharges from Discharge Point 001 do not affect State waters. The Ocean Plan minimum probable initial dilution ( $D_m = 180$ ) for this outfall is 181:1.
- 2. Discharge Point 002 (78" outfall).** Discharge Point 002 (78" outfall) is an emergency discharge point (deactivated ocean outfall, with a rated capacity of 230



MGD) to the Pacific Ocean terminating in a 130 ports diffuser, approximately 1.5 mile offshore from the mouth of the Santa Ana River, at a depth of about 65 feet (20 meters) below mean sea level. This outfall is used only during an emergency or planned essential maintenance or capital improvement projects to assure efficient operation of the 120" outfall under 40 CFR § 122.41(m)(2) since the 120" outfall was put into service in 1971. The Ocean Plan minimum probable initial dilution ( $D_m = 36$ ) for this outfall is 37:1.

- 3. Discharge Point 003 (Santa Ana River Overflow Weirs).** Discharge Point 003 (i.e., the Santa Ana River Overflow Weirs) is comprised of two extreme emergency discharge points (overflow) to the tidal prism in the Santa Ana River, which eventually flows to the Pacific Ocean. One of the overflow points is located at the termination structure upstream of the EPSA facility and other one is at the OOBs facility. The structure consists of a 50' long overflow weir with two 72" pipes. The total capacity is approximately 605 MGD (i.e., 130 MGD from EPSA and 475 MGD from OOBs), but the actual capacity depends on the water level at the river. This outfall has never discharged over the past 10 years; however, it remains available for extreme emergency uses (e.g., tsunami, earthquake, flood, and acts of war or terrorism).

**Table F-2. Discharge Location**

| Discharge Point                         | Effluent Description                                       | Discharge Point Latitude | Discharge Point Longitude | Receiving Water                       |
|-----------------------------------------|------------------------------------------------------------|--------------------------|---------------------------|---------------------------------------|
| 001<br>(120" outfall)                   | Secondary treated effluent and reverse osmosis concentrate | 33° 34' 36" N            | 118° 00' 36" W            | Pacific Ocean<br>(4.5 miles offshore) |
| 002<br>(78" outfall)                    | Secondary treated effluent and reverse osmosis concentrate | 33° 36' 56" N            | 117° 58' 13" W            | Pacific Ocean<br>(1.5 miles offshore) |
| 003<br>(Santa Ana River overflow weirs) | Secondary treated effluent and reverse osmosis concentrate | 33° 38' 06" N            | 117° 57' 20" W            | Santa Ana River<br>Tidal Prism        |

**C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data**

Effluent limitations contained in the existing Order/Permit for discharges from Discharge Point 001 (Monitoring Location EFF-001) and representative monitoring data for the period from May 2015 to December 2019 are as follows:

**Table F-3. Historic Technology Based Effluent Limitations and Monitoring Data at EFF-001**

| Parameter                                                                                 | Units             | Order No. R8-2012-0035<br>NPDES Permit No. CA0110604<br>Effluent Limitation |                   |                          | Discharge Monitoring Data<br>(May 2015 – December 2019) |                                           |                                                  |
|-------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------|-------------------|--------------------------|---------------------------------------------------------|-------------------------------------------|--------------------------------------------------|
|                                                                                           |                   | Average<br>Monthly                                                          | Average<br>Weekly | Instantaneous<br>Maximum | Highest<br>Average<br>Monthly<br>Discharge              | Highest<br>Average<br>Weekly<br>Discharge | Highest<br>Instantaneous<br>Maximum<br>Discharge |
| Secondary Treatment Standards and/or Ocean Plan Table 4 Effluent Limitations              |                   |                                                                             |                   |                          |                                                         |                                           |                                                  |
| Carbonaceous<br>Biochemical<br>Oxygen<br>Demand, 5-<br>day @ 20°C<br>(CBOD <sub>5</sub> ) | mg/L              | 25                                                                          | 40                | --                       | 6.5                                                     | 8.8                                       | --                                               |
|                                                                                           | lbs/day           | 57,129                                                                      | 91,406            | --                       | 6,072                                                   | 8,580                                     | --                                               |
| Total<br>Suspended<br>Solids (TSS)                                                        | mg/L              | 30                                                                          | 45                | --                       | 7.4                                                     | 9.6                                       | --                                               |
|                                                                                           | lbs/day           | 68,555                                                                      | 102,832           | --                       | 7,167                                                   | 11,451                                    | --                                               |
| pH                                                                                        | standard<br>units | 6.0 to 9.0<br>(instantaneous minimum/maximum)                               |                   |                          | 7.06 to 8.41                                            |                                           |                                                  |
| Oil and<br>grease                                                                         | mg/L              | 25                                                                          | 40                | 75                       | 1.39                                                    | 1.39                                      | 1.39                                             |
|                                                                                           | lbs/day           | 57,129                                                                      | 91,406            | 171,387                  | 1,106                                                   | 1,106                                     | --                                               |
| Settleable<br>Solids                                                                      | ml/L              | 1.0                                                                         | 1.5               | 3.0                      | --                                                      | --                                        | 2.5                                              |
| Turbidity                                                                                 | NTU               | 75                                                                          | 100               | 225                      | 6.73                                                    | 6.73                                      | 6.73                                             |

**Table F-4. Historic Water Quality Based Effluent Limitations and Monitoring Data  
at EFF-001**

| Parameter                                                                                | Units   | Order No. R8-2012-0035<br>NPDES Permit No. CA0110604<br>Effluent Limitation                                                                                                                                                                                                                              |                  |                          | Monitoring Data<br>(May 2015 – December 2019)            |                                          |                                                  |
|------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------|----------------------------------------------------------|------------------------------------------|--------------------------------------------------|
|                                                                                          |         | Average<br>Monthly<br>(or 6-Month<br>Median*)                                                                                                                                                                                                                                                            | Maximum<br>Daily | Instantaneous<br>Maximum | Highest<br>Average<br>Monthly<br>(or 6-Month<br>Median*) | Highest<br>Maximum<br>Daily<br>Discharge | Highest<br>Instantaneous<br>Maximum<br>Discharge |
| Ocean Plan Table 3 Water Quality Objectives for Protection of Marine Aquatic Life        |         |                                                                                                                                                                                                                                                                                                          |                  |                          |                                                          |                                          |                                                  |
| Total chlorine residual                                                                  | mg/L    | 0.36*                                                                                                                                                                                                                                                                                                    | 1.45             | 10.86                    | 0.09*                                                    | 0.3                                      | 1.2                                              |
|                                                                                          | lbs/day | 823*                                                                                                                                                                                                                                                                                                     | 3,313            | 24,817                   | 68*                                                      | 359                                      | --                                               |
| Acute toxicity, TST                                                                      | P or F  | --                                                                                                                                                                                                                                                                                                       | P                | --                       | Pass                                                     |                                          |                                                  |
| Chronic toxicity, TST                                                                    | P or F  | --                                                                                                                                                                                                                                                                                                       | P                | --                       | Pass                                                     |                                          |                                                  |
| <u>Radioactivity</u><br>Gross Alpha<br>Gross Beta<br>Radium-226 & 228                    | pCi/L   | Not to exceed limits specified in Title 17, Division 1, Chapter 5, Subchapter 4, Group 3, Article 3, section 30253 of the California Code of Regulations. Reference to section 30253 is prospective, including future changes to any incorporated provisions of federal law, as the changes take effect. |                  |                          | 36<br>70<br>1.14                                         | 36<br>70<br>1.14                         | --                                               |
| Ocean Plan Table 3 Water Quality Objectives for Protection of Human Health - Carcinogens |         |                                                                                                                                                                                                                                                                                                          |                  |                          |                                                          |                                          |                                                  |
| Benzidine                                                                                | µg/L    | 0.01249                                                                                                                                                                                                                                                                                                  | --               | --                       | ND                                                       | --                                       | --                                               |
|                                                                                          | lbs/day | 0.0285                                                                                                                                                                                                                                                                                                   | --               | --                       | ND                                                       | --                                       | --                                               |
| Hexachloro-benzene                                                                       | µg/L    | 0.0380                                                                                                                                                                                                                                                                                                   | --               | --                       | ND                                                       | --                                       | --                                               |
|                                                                                          | lbs/day | 0.0868                                                                                                                                                                                                                                                                                                   | --               | --                       | ND                                                       | --                                       | --                                               |
| PCBs <sup>1</sup>                                                                        | µg/L    | 0.0034                                                                                                                                                                                                                                                                                                   | --               | --                       | ND                                                       | --                                       | --                                               |
|                                                                                          | lbs/day | 0.0078                                                                                                                                                                                                                                                                                                   | --               | --                       | ND                                                       | --                                       | --                                               |
| TCDD equivalents <sup>1</sup>                                                            | pg/L    | 0.706                                                                                                                                                                                                                                                                                                    | --               | --                       | ND                                                       | --                                       | --                                               |
|                                                                                          | lbs/day | 0.000001613                                                                                                                                                                                                                                                                                              | --               | --                       | ND                                                       | --                                       | --                                               |

| Parameter                                                          | Units   | Order No. R8-2012-0035<br>NPDES Permit No. CA0110604<br>Effluent Limitation |                  |                          | Monitoring Data<br>(May 2015 – December 2019)            |                                          |                                                  |
|--------------------------------------------------------------------|---------|-----------------------------------------------------------------------------|------------------|--------------------------|----------------------------------------------------------|------------------------------------------|--------------------------------------------------|
|                                                                    |         | Average<br>Monthly<br>(or 6-Month<br>Median*)                               | Maximum<br>Daily | Instantaneous<br>Maximum | Highest<br>Average<br>Monthly<br>(or 6-Month<br>Median*) | Highest<br>Maximum<br>Daily<br>Discharge | Highest<br>Instantaneous<br>Maximum<br>Discharge |
| Toxaphene                                                          | µg/L    | 0.03801                                                                     | --               | --                       | ND                                                       | --                                       | --                                               |
|                                                                    | lbs/day | 0.0869                                                                      | --               | --                       | ND                                                       | --                                       | --                                               |
| <sup>1</sup> See Attachment A for definitions.<br>ND: Not detected |         |                                                                             |                  |                          |                                                          |                                          |                                                  |

#### D. Compliance Summary

##### Completion of Secondary Treatment Projects

Since the construction of the facilities that was required to achieve secondary treatment standards were not completed at the time the previous permit was issued, OC San entered into a consent decree mandating OC San to meet secondary treatment limits by December 31, 2012. After extensive process improvement and Capital Improvement Projects (CIP) including completion of the following:

- Trickling filter project at Reclamation Plant No. 1, completed in Spring 2006;
- Secondary treatment rehabilitation project at Treatment Plant No. 2, completed in Spring 2008;
- New trickling filters and solid contact basins at Treatment Plant No. 2, completed in Summer 2011. The project added 60 MGD of secondary treatment capacity; and
- New activated sludge system at Reclamation Plant No. 1, completed in Fall 2012. The project added 60 MGD of secondary treatment capacity.

The Discharger has officially declared the POTW at full secondary treatment on December 28, 2012 and continue operating at and meeting full secondary requirements.

##### Cessation of Continuous Disinfection Practices at Discharge Point 001 (120" outfall)

The disinfection program initiated in 2002 was based on using sodium hypochlorite (bleach) for disinfection and sodium bisulfite for dechlorination. Beginning in 2006, the Discharger started to note negative changes in the structure and health of benthic communities within the zone of initial dilution (ZID) around the diffuser to the 120-inch

outfall (the primary discharge point). To determine the causes of these changes to the marine environment, the Discharger conducted detailed studies divided into two phases: Phase I (2009-2010) for an analysis of historical data to identify any potential causes and Phase II (2011-2012) for investigation of operational changes and process improvements that occurred at the treatment facilities and the facility's dry-weather urban runoff diversion program. Based on the results of these investigations, the discharge of chlorinated effluent prior to 2012 resulted in a decline in benthic communities near the ZID. The benthic environment has been improved since 2012 due to reduction of the use of chlorine after achieving full secondary treatment.

After long-term beach water quality studies and oceanographic data evaluation by an Independent Advisory Panel (IAP) of experts hosted by the National Water Research Institute (NWRI), it was concluded that continuing to chlorinate full secondary treated effluent provides little to no public health benefit to local beaches. The Discharger has also achieved continuous compliance with the secondary treatment requirements of the Clean Water Act. On March 17, 2015, based on the observations and recommendations of the IAP and with public support, OC San received approval from the Santa Ana Regional Water Quality Control Board and USEPA to cease disinfection at Discharge Point 001 (120" outfall). OC San has ceased continuous chlorination/dechlorination since May 2015. The disinfection system would remain available for later use, if determined necessary. Use of disinfection remains at the other two discharge points (Discharge Points 002 and 003) to protect human health.

#### **Planned 6-Week Diversion to Discharge Point 002 (78" outfall)**

From September 11, 2012 through October 4, 2012, the Discharger temporarily diverted disinfected effluent from Discharge Point 001 (120" outfall) to Discharge Point 002 (78" outfall) as part of an infrastructure rehabilitation project (refer to as Capital Improvement Project J-112) to inspect, assess, and rehabilitate the Outfall Land Section and Ocean Outfall Booster Pump Station Piping. To minimize impacts to public health, the treated wastewater received enhanced disinfection so that state bacterial water contact standards were met prior to discharge and subsequent dilution. The Discharger conducted the J-112 environmental monitoring program including modeling to track the discharged plume, measure the effectiveness of the enhanced disinfection program, and determine environmental impacts to the receiving water during the diversion to the nearshore outfall. The lessons learned report and a more comprehensive report (i.e., 2012 OCSD Outfall Diversion Summary Report) were submitted to the Santa Ana Water Board and USEPA on March 25, 2014, which summarize the sampling methodology during the diversion and provide recommendations for improving monitoring during any future diversions to Discharge Point 002 (78" outfall).

#### **Unintentional Bypass of primary effluent from Plant No. 2 to Discharge Point 001 (120" outfall)**

The unintentional bypass of primary effluent into Discharge Point 001 (120" outfall) occurred on July 5, 2020 from 4:28 am to 5:02 am, a duration of 34 minutes, due to a power loss after an Edison transformer failed. The power loss affected equipment

throughout Treatment Plant No. 2 including the trickling filter influent pumps. The trickling filter weir overflowed with primary treated wastewater, approximately 0.94 million gallons of which discharged directly to the 120" outfall.

Another unanticipated bypass of primary effluent into the 120" outfall occurred at the same location due to failure of two of three lead pumps serving the trickling filters pump station (TFPS) during a wet weather event on December 28, 2020 at 7:09 pm for approximately 9 minutes. The estimated flow over the TFPS weir is total 78,500 gallons.

After discovering each incident, the Discharger notified the Santa Ana Water Board and USEPA and investigated the incident to determine the cause and duration of the incident, the volume discharged, and to evaluate and implement appropriate corrective actions to prevent this incident from reoccurring. As part of corrective actions, the TFPS lead pumps were disassembled, and their internal fans replaced. As a key sustainable action, OC San is currently in construction installing an upgraded power monitoring/power management control system to improve the power system response speed to prevent short term power loss event that can lock out equipment. This project is scheduled to be complete by 2023.

## **E. Planned Changes**

### **GWRS Final Expansion Project**

The GWRS is a regional water reclamation program that was initiated jointly by OC San and OCWD and began operation in January 2008. The GWRS Initial Expansion was completed in 2015, which allowed OCWD to take up to 135 MGD of secondary effluent from Plant No. 1 to produce 100 MGD of GWRS product water. Currently, the Discharger sends approximately 100% of secondary effluent from Plant No. 1 to GWRS to provide a drought-proof water supply for Orange County. Also, it delivers up to 4 MGD of secondary effluent from Plant No. 1 to GAP during the summer months to provide recycled water for landscaped irrigation and industrial uses.

The layout of the OCWD facilities has been designed to provide GWRS with an ultimate production capacity of 130 MGD. The GWRS ultimate production capacity of 130 MGD is referred to as the GWRS Final Expansion project. Based on the SP-173 Effluent Reuse Study for the evaluation of reclamation of Plant No. 2 effluent, the following projects are identified by the implementation plan for the GWRS Final Expansion and anticipated to be completed by 2023.

- Modify Plant 2 Headworks Facility to create a reclaimable and non-reclaimable treatment train. Reclaimable flows will go through existing preliminary, primary, and secondary (trickling filter solids contact) treatment before being pumped to GWRS.
- Build a Plant 2 effluent pump station that will convey approximately 50-85 MGD of secondary effluent from Plant No. 2 trickling filter solids contact process to the GWRS Facility.



- Build a 6 MG equalization storage tank at Plant No. 2 to store secondary effluent during peak day flows and be returned to the system during low flow periods to maintain a constant flow to the GWRS.
- Relocate the existing plant water pump station to a location more suitable for receiving the trickling filter solids contact treatment processes. After the GWRS Final Expansion, the activated sludge treatment process will treat only SARI and side-stream flows.
- Convert the unused 66-inch gravity reinforced concrete pipeline into a pressure pipeline using a trenchless pipe repair method, to convey the effluent pump station discharge.

This expansion would require a total flow from OC San plants of approximately 175 MGD to produce 130 MGD of indirect potable reuse water. The Final Expansion of the GWRS will enable the Discharger to recycle the majority of the wastewater generated in its service area and treated at its two wastewater treatment plants.

#### **Organic Waste Co-Digestion Project**

Construction and operation of an organic waste co-digestion facility is anticipated to be completed in 2022. An organic waste receiving facility will be constructed to receive, store, and feed approximately 150 tons of pre-processed food waste slurry to the digester complex at Plant No. 2 (by Digester M) to generate additional digester gas per recommendations of the Biosolids Master Plan. This facility will be replaced with a permanent receiving station following completion of a proposed program to replace existing digesters at Plant No. 2. Food waste will be fed into OC San's digesters, creating more gas and electricity, while producing a few extra truckloads of solids per week.

### **III. APPLICABLE PLANS, POLICIES, AND REGULATIONS**

The requirements contained in this Order are based on the requirements and authorities described in this section.

#### **A. Legal Authorities**

This Order serves as WDRs pursuant to article 4, chapter 4, division 7 of the California Water Code (CWC) (commencing with section 13260). This Order/Permit is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the USEPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). This Order shall serve as an NPDES permit for point source discharges from this POTW to waters of the United States. The Order/Permit are consolidated pursuant to 40 CFR. § 124.4(c)(2). Although Discharge Point 001 is beyond the limit of State-regulated ocean waters, effluent plume migration into State waters warrants joint regulation of the discharge by USEPA and the Santa Ana Water Board.

## B. California Environmental Quality Act (CEQA)

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of Chapter 3 of CEQA, (commencing with section 21100) of Division 13 of the Public Resources Code.

## C. State and Federal Laws, Regulations, Policies, and Plans

1. **Water Quality Control Plan.** The Santa Ana Regional Water Quality Control Board adopted a Water Quality Control Plan for the Santa Ana River Basin (hereinafter Basin Plan) in 1995, that designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for the Offshore Zone and Nearshore Zone of the Pacific Ocean and the tidal prism of the Santa Ana River. In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63 which established State policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. Requirements in this Order/Permit implement the Basin Plan. Basin Plan beneficial uses applicable to the Pacific Ocean and/or the Santa Ana River tidal prism are as follows:

**Table F-5. Basin Plan Beneficial Uses**

| Discharge Point                               | Receiving Water Name                                  | Beneficial Use(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001<br>(120" outfall)                         | Pacific Ocean,<br>Beyond the limit<br>of State waters | <b>Offshore Zone:</b> Industrial service supply; navigation; water contact recreation; non-contact water recreation; commercial and sportfishing; wildlife habitat; rare, threatened, or endangered species; spawning, reproduction, and development; and marine habitat.                                                                                                                                                                                           |
| 002<br>(78" outfall)                          | Pacific Ocean,<br>Offshore Zone                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 003<br>(Santa Ana<br>River overflow<br>weirs) | Santa Ana River<br>Tidal Prism,<br>Reach 1            | <b>Nearshore Zone:</b> Industrial service supply; navigation; water contact recreation; non-contact water recreation; commercial and sportfishing; wildlife habitat; rare, threatened, or endangered species; spawning, reproduction, and development; marine habitat; and shellfish harvesting.<br><br><b>Santa Ana River, Reach 1:</b> Water contact recreation (access prohibited); non-contact water recreation; warm freshwater habitat; and wildlife habitat. |

2. **California Ocean Plan.** The State Water Board adopted the Water Quality Control Plan for Ocean Waters of California, California Ocean Plan (hereinafter Ocean Plan) in 1972 and amended it in 1978, 1983, 1988, 1990, 1997, 2000, 2005, 2009, 2012, 2015, 2018, and 2019. The Ocean Plan is applicable, in its entirety, to point source discharges to the ocean. The Ocean Plan identifies

beneficial uses of ocean waters of the state to be protected as summarized below:

**Table F-6. Ocean Plan Beneficial Uses**

| Discharge Point | Receiving Water | Beneficial Uses                                                                                                                                                                                                                                                                                                                          |
|-----------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001 and 002     | Pacific Ocean   | Industrial water supply; water contact and non-contact recreation, including aesthetic enjoyment; navigation; commercial and sport fishing; mariculture; preservation and enhancement of designated Areas of Special Biological Significance (ASBS); rare and endangered species; marine habitat; fish spawning and shellfish harvesting |

In order to protect the beneficial uses, the Ocean Plan establishes water quality objectives and a program of implementation. Requirements of this Order/Permit implement the Ocean Plan.

3. **California Thermal Plan.** The State Water Board adopted the Water Quality Control Plan for Control of Temperature in the Coastal and Interstate Waters and Enclosed Bays and Estuaries of California (hereinafter Thermal Plan) on January 7, 1971, and amended this plan on September 18, 1975. This plan contains temperature objectives for coastal and inland surface waters. Requirements of this Order/Permit implement the Thermal Plan.
4. **Antidegradation Policy.** Federal regulation 40 CFR § 131.12 requires that the state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution 68-16 ("Statement of Policy with Respect to Maintaining High Quality of Waters in California"). Resolution 68-16 is deemed to incorporate the federal antidegradation policy where the federal policy applies under federal law. Resolution 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Santa Ana Regional Water Quality Control Board's Basin Plan implements, and incorporates by reference, both the state and federal antidegradation policies. The permitted discharge must be consistent with the antidegradation provision of section 131.12 and State Water Board Resolution 68-16.
5. **Anti-Backsliding Requirements.** Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 CFR § 122.44(l) restrict backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued Order/Permit must be as stringent as those in the previous Order/Permit, with some exceptions in which limitations may be relaxed.
6. **Alaska Rule.** USEPA has revised its regulation that specifies when new and revised State and tribal water quality standards become effective for CWA purposes. (40 CFR § 131.21; 65 Fed. Reg. 24641 (April 27, 2000).) Under the revised regulation (also known as the Alaska Rule), new and revised standards

submitted to USEPA after May 30, 2000 must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect and submitted to USEPA by May 30, 2000, may be used for CWA purposes, whether or not approved by USEPA.

7. **Section 403(c) of the Clean Water Act (CWA):** Discharges to marine waters are subject to Section 403 of the CWA, which sets forth criteria to prevent unreasonable degradation of the marine environment and authorized imposition of any additional effluent limits necessary to protect the marine environment. Pursuant to 40 CFR § 125.122, unreasonable degradation of the marine environment is evaluated based on ten factors or based on the application of a state's water quality standards. Specifically, 40 CFR § 125.122(b) states that discharges in compliance with State water quality standards "shall be presumed not to cause unreasonable degradation of the marine environment, for any specific pollutants or conditions specified in the variance or the standard." USEPA and the Santa Ana Water Board are applying the Basin Plan and the Ocean Plan as specified in section III.C.1 and 2 of this Fact Sheet, except for evaluating acute/chronic toxicity for Discharge Points 001 and 002 using the TST statistical approach. USEPA has reviewed the previous studies to examine the comparison of toxicity test results using the TST and NOEC statistical approaches and has determined that use of the TST statistical approach is consistent with the Ocean Plan and CWA § 403(c) in that it provides protection of the designated beneficial uses of ocean waters. TST statistical approach is also used in other NPDES permits for large publicly owned treatment works, including City of Los Angeles. Given the available dilution (i.e. 180), the receiving water monitoring requirements, the Discharger's analysis of the ocean discharge criteria as part of its application (See Appendix E.1 Ocean Discharge Criteria Response to CWA 403(c)), and USEPA's additional 403(c) analysis, USEPA makes a determination that the discharges authorized in this permit will not cause unreasonable degradation of the marine environment.
8. **Federal Permit Renewal Contingency.** The renewal of the Discharger's federal permit by USEPA is contingent upon determination by the U.S. Fish and Wildlife Service (USFWS) and NOAA National Marine Fisheries Service (NMFS) that the proposed discharge is consistent with the: (1) federal Endangered Species Act (ESA); (2) the Magnuson-Stevens Fishery Conservation and Management Act (MSA); and (3) the Santa Ana Water Board's certification/concurrence that the discharge will comply with applicable State water quality standards.

USEPA's reissuance of NPDES No. CA0110604 to OC San is subject to requirements of the ESA and MSA. USEPA may decide that changes to this permit are warranted based on the results of the completed consultation and a reopener provision to this effect has been included in the Order/Permit. Issuance of this NPDES permit which incorporates both federal requirements and State waste discharge requirements will serve as the State's concurrence that the discharge complied with State water quality standards. The California Coastal Commission has indicated that it is not necessary to obtain a consistency

certification pursuant to the Coastal Zone Management Act for the issuance of a federal NPDES permit containing secondary treatment standards.

9. **Endangered Species Act Requirements.** This Order/Permit does not authorize any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (Fish and Game Code, §§ 2050 to 2097) or the Federal Endangered Species Act (16 U.S.C.A. §§ 1531 to 1544). In July 2020, USEPA requested updated information related to: (1) essential fish habitat and managed and associated species, and (2) threatened and endangered species and their designated critical habitats, in the vicinity of the OC San's outfalls from NMFS. Subsequently, EPA and NMFS exchanged information about the facility, discharge characteristics, action area, OC San's monitoring program, and past consultations in Southern California. USEPA determined that the draft permit would have no effect on listed bird species; may affect but is not likely to adversely affect listed fish and abalone species; and is likely to adversely affect listed sea turtle, seal, and whale species. These effect determinations are explained in the biological evaluation. USEPA is consulting with the United States Fish and Wildlife Service and the National Marine Fisheries Service. This Order/Permit requires compliance with effluent limits, receiving water limits, and other requirements to protect beneficial uses, including protecting threatened and endangered species. The Discharger is responsible for meeting all requirements of the applicable Endangered Species Act.

**D. Impaired Water Bodies on the CWA section 303(d) List**

On April 6, 2018, USEPA issued a final decision regarding California's 2014- 2016 CWA Section 303(d) List of Impaired Waters. The list (hereinafter 303(d) list) identifies water bodies where water quality standards are not expected to be met after implementation of technology-based effluent limits by point sources. Huntington Beach State Park is included on the 303(d) list for PCBs (polychlorinated biphenyls). The offshore and nearshore zones off Huntington Beach State Park are the immediately affected receiving waters of discharges from the Facility. A total maximum daily load (TMDL) for PCBs is required but has not been established yet. As such, effluent limitations for PCBs have been established for the Facility until applicable waste load allocations are assigned in a TMDL. A TMDL to address the impairment is not currently scheduled for development.

**E. Other Plans, Policies and Regulations**

1. **Stringency of Requirements for Individual Pollutants.** This Order/Permit contains technology-based effluent limitations and WQBELs for individual pollutants. Technology-based effluent limitations consist of restrictions on carbonaceous biochemical oxygen demand (5-day), total suspended solids, and pH which implement the minimum, applicable federal technology-based requirements for POTWs. Also, effluent limitations consisting of restrictions on oil and grease, settleable solids, and turbidity are necessary to implement State treatment standards in Ocean Plan Table 4. Discharge Point 001 WQBELs



consisting of restrictions on total chlorine residual, acute toxicity, chronic toxicity, radioactivity, benzidine, hexachlorobenzene, PCBs, TCDD equivalents, and toxaphene more stringent than federal technology-based limitations are necessary to meet State water quality standards in the Ocean Plan Table 3. Discharge Point 002 WQBELs consisting of restrictions on fecal coliform density, *Enterococcus* density, total chlorine residual, ammonia (as nitrogen), chronic toxicity, radioactivity, and TCDD equivalents more stringent than federal technology-based limitations are necessary to meet State water quality standards in the Ocean Plan Table 3. All effluent limitations are discussed in Attachment F-Fact Sheet. Collectively, the restrictions on individual pollutants in this Order/Permit are no more stringent than required by the CWA.

2. **Federal Secondary Treatment Regulations.** The OC San achieved full secondary treatment on December 28, 2012 and continues discharging secondary treated flow. 40 CFR § 133 establishes the minimum levels of effluent quality to be achieved by secondary treatment. These limitations, established by USEPA, are incorporated into this Order/Permit, except where more stringent limitations are required by other applicable plans, policies, or regulations or to prevent backsliding.
3. **Industrial Storm Water Requirements.** The Discharger is not required to be covered under the statewide NPDES general permit for storm water discharges associated with industrial activities (NPDES General Permit No. CAS000001) because all stormwater at the plants is collected by various drains and conveyed downstream of both plants' headworks for treatment. No storm water is diverted untreated to the outfalls. The Discharger has provided certification to the Santa Ana Water Board and USEPA that industrial storm water is managed by internal drainage systems at Reclamation Plant No. 1 and Treatment Plant No. 2, where storm water is captured, treated, and discharged with the treated wastewater regulated under this Order/Permit. This Order/Permit requires the Discharger to submit and implement a Storm Water Management Plan to address onsite storm water quality.
4. **Sanitary Sewer Overflows.** The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order 2006-0003-DWQ (General Order) on May 2, 2006, which contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows (SSOs). The Monitoring and Reporting Program for the General Order was amended through the State Water Board's Order WQ 2013-0058-EXEC on August 6, 2013. This General Order requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the General Order. The General Order also requires agencies to develop sanitary sewer management plans and report all SSOs, among other requirements and prohibitions. The Discharger's collection system is part of the POTW that is subject to this Order/Permit. The Discharger must comply with both the General Order and this Order/Permit.



- 5. Pretreatment Requirements.** In compliance with 40 CFR § 403, OC San has developed an approved Pretreatment Program for the POTW that it owns and operates. This Order/Permit includes the Discharger's approved Pretreatment Program and requires the Discharger to continue implementation and control of the Program throughout the service area of its POTW, including contributing jurisdictions.

The POTW, as Control Authority, may exercise its authority over the entire service area directly, as provided by State law, or may elect to enter into contracts or other multi-jurisdictional agreements with contributing jurisdictions. In case the POTW elects to enter into inter-jurisdictional agreements, the POTW must ensure that discharges received from entities outside of its political boundaries are regulated to the same extent as are discharges within its political boundaries. OC San applies one set of local limits to all discharges to its treatment facilities.
- 6. Sewage Sludge/Biosolids Requirements.** Section 405 of the CWA and implementing regulations at 40 CFR § 503 require that producers of sewage sludge/biosolids meet certain reporting, handling, and use or disposal requirements. The State has not been delegated the authority to implement this program; therefore, USEPA is the implementing agency. This Order/Permit contains sewage sludge/biosolids requirements pursuant to 40 CFR § 503 that are applicable to the Discharger.
- 7. Standard and Special Provisions.** Standard Provisions, which apply to all NPDES permits in accordance with 40 CFR § 122.41, and additional conditions applicable to POTWs in accordance with 40 CFR § 122.42, are provided in Attachment D. The Santa Ana Water Board and USEPA have also included in this Order/Permit Special Provisions applicable to the Discharger. The rationale for the Special Provisions contained in this Order/Permit is provided in this Fact Sheet.

#### **IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS**

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in the Code of Federal Regulations: 40 CFR § 122.44(a) requires that permits include applicable technology-based limitations and standards; and 40 CFR § 122.44(d) requires that permits include water quality-based effluent limitations to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water.

##### **A. Discharge Prohibitions**

Discharge Prohibitions in this Order/Permit are based on Ocean Plan discharge prohibitions, Basin Plan discharge prohibitions, and discharge prohibitions in the existing Order/Permit.

## 1. California Ocean Plan Discharge Prohibitions

- a. **Prohibition III.B, III.C, III.D, III.E, and III.F:** This permit/order implements discharge prohibitions that are applicable under sections III.I.1.a, III.I.2.a, III.I.3.a, III.I.4.a, and III.I.6 of the 2019 California Ocean Plan, respectively.
  - b. **Prohibition III.G (Discharge at Discharge Point 001 without minimum initial dilution of at least 181:1):** This prohibition is necessary to ensure that the assumption used to derive the dilution credits established through this Order/Permit remains substantially the same so that the effluent limitations are protective of water quality. This Order considered a dilution credit of 180 ( $D_m = 180$ ) based on the Discharger's *GWRS Final Expansion Final Implementation Plan, Project No. SP-173, Effluent Reuse Study (October 2016)* to conduct the reasonable potential analysis described in Section IV.C.3 of this Fact Sheet. Moreover, the instream waste concentration (IWC) for the chronic toxicity effluent limitation is based on this dilution credit. When future discharge flows drop below 32 MGD, as a result of GWRS final expansion project, the outfall will likely need to be retrofitted to ensure adequate dilution and to prevent seawater intrusion and potential diffuser plugging. The initial dilution of 181:1 assumes that all 503 outfall ports are operational.
- ## 2. Santa Ana River Basin Plan Discharge Prohibitions
- a. **Prohibition III.H, III.I, and III.J:** This permit/order implements discharge prohibitions that are applicable under Chapter 5, Waste Discharge Prohibitions A, B.1, and B.2 of the 2019 Basin Plan, respectively.
- ## 3. Prohibition III.K (Discharge from any locations other than Discharge Point 001):
- This prohibition clarifies that any discharges other than those to Discharge Point 001 are unauthorized, except those to Discharge Point Nos. 002 and 003 as explicitly authorized during essential maintenance or an emergency.

## B. Technology-Based Effluent Limitations (TBELs)

### 1. Scope and Authority

Section 301(b) of the CWA and implementing USEPA permit regulations at 40 CFR § 122.44 require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order/Permit must meet minimum federal technology-based effluent limitations (TBELs) based on secondary treatment standards for POTWs at 40 CFR § 133.

The CWA requires that technology-based effluent limitations be established based on several levels of controls:

- a. Best practicable treatment control technology (BPT) represents the average of the best existing performance by well-operated facilities within an industrial

category or subcategory. BPT standards apply to toxic, conventional, and non-conventional pollutants.

- b. Best available technology economically achievable (BAT) represents the best existing performance of treatment technologies that are economically achievable within an industrial point source category. BAT standards apply to toxic and non-conventional pollutants.
- c. Best conventional pollutant control technology (BCT) represents the control from existing industrial point sources of conventional pollutants including BOD, TSS, fecal coliform, pH, and oil and grease. The BCT standard is established after considering a two-part reasonableness test. The first test compares the relationship between the costs of attaining a reduction in effluent discharge and the resulting benefits. The second test examines the cost and level of reduction of pollutants from the discharge from publicly-owned treatment works to the cost and level of reduction of such pollutants from a class or category of industrial sources. Effluent limitations must be reasonable under both tests.
- d. New source performance standards (NSPS) represent the best available demonstrated control technology standards. The intent of NSPS guidelines is to set limitations that represent state-of-the-art treatment technology for new sources.

The CWA requires USEPA to develop effluent limitations, guidelines and standards (ELGs) representing application of BPT, BAT, BCT, and NSPS. Section 402(a)(1) of the CWA and 40 CFR § 125.3 authorize the use of best professional judgment (BPJ) to derive technology-based effluent limitations on a case-by-case basis where ELGs are not available for certain industrial categories and/or pollutants of concern. Where BPJ is used, the Santa Ana Water Board and USEPA must consider specific factors outlined in 40 CFR § 125.3.

Regulations promulgated in 40 CFR § 125.3(a)(1) require technology-based effluent limitations for municipal Dischargers to be placed in NPDES permits based on Secondary Treatment Standards or Equivalent to Secondary Treatment Standards.

The Federal Water Pollution Control Act Amendments of 1972 (PL 92-500) established the minimum performance requirements for POTWs [defined in section 304(d)(1)]. Section 301(b)(1)(B) of that Act requires that such treatment works must, as a minimum, meet effluent limitations based on secondary treatment as defined by the USEPA Administrator.

Based on this statutory requirement, USEPA developed secondary treatment regulations, which are specified in 40 CFR § 133. These technology-based regulations apply to all municipal wastewater treatment plants and identify the minimum level of effluent quality attainable by secondary treatment in terms of biochemical oxygen demand (BOD<sub>5</sub>), total suspended solids (TSS), and pH.

**Table F-7. Federal Secondary Treatment Standards**

| Parameter                             | Units              | 30-day Average         | 7-day Average |
|---------------------------------------|--------------------|------------------------|---------------|
| BOD <sub>5</sub>                      | mg/L               | 30                     | 45            |
|                                       | influent % removal | not less than 85       | --            |
| CBOD <sub>5</sub><br>(if substituted) | mg/L               | 25                     | 40            |
|                                       | influent % removal | not less than 85       | --            |
| TSS                                   | mg/L               | 30                     | 45            |
|                                       | influent % removal | not less than 85       | --            |
| pH                                    | standard units     | 6.0 – 9.0 at all times |               |

On May 13, 2004, the Discharger requested the inclusion of effluent limitations for five-day carbonaceous biochemical oxygen demand (CBOD<sub>5</sub>), as allowed by 40 CFR § 133.102(a)(4), for the period following completion of expanded secondary treatment facilities. In the 2004 Order/Permit, CBOD<sub>5</sub> effluent limitations were applied to the final effluent during partial or full nitrification at the Discharger's secondary treatment facilities, where effluent nitrification is used to reduce ammonia toxicity associated with wastewater treatment and RO concentrate flow from the Groundwater Replenishment System. As nitrifying bacteria use oxygen to degrade nitrogenous compounds otherwise not significantly removed in the secondary treatment process, higher oxygen demand values for the final effluent result. Consequently, the use of CBOD<sub>5</sub> effluent limits have been found to ensure that federal secondary treatment standards for POTWs are achieved while allowing the Discharger to use the treatment process of nitrification to reduce ammonia toxicity in the discharged effluent and comply with Ocean Plan requirements for acute and chronic toxicity. CBOD<sub>5</sub> effluent limitations have been carried over to this Order/Permit and are applied in lieu of BOD<sub>5</sub> effluent limitations.

The effluent limitations consisting of restrictions on grease and oil, settleable solids, and turbidity are necessary to implement State treatment standards in Ocean Plan Table 4, which are summarized as follows.

**Table F-8. Ocean Plan Table 4 Effluent Limitations**

| Parameter        | Units              | 30-day Average   | 7-day Average | Instantaneous Maximum |
|------------------|--------------------|------------------|---------------|-----------------------|
| Grease and Oil   | mg/L               | 25               | 40            | 75                    |
| Suspended Solids | mg/L               | --               | --            | --                    |
|                  | influent % removal | not less than 75 | --            | --                    |

| Parameter         | Units          | 30-day Average         | 7-day Average | Instantaneous Maximum |
|-------------------|----------------|------------------------|---------------|-----------------------|
| Settleable Solids | ml/L           | 1.0                    | 1.5           | 3.0                   |
| Turbidity         | NTU            | 75                     | 100           | 225                   |
| pH                | standard units | 6.0 – 9.0 at all times |               |                       |

Compliance with mass emission effluent limitations for TBELs must be met end-of-pipe after Reclamation Plant No. 1 and Treatment Plant No. 2 effluents have commingled. The 2012 Order/Permit established mass emission effluent limitations based on the average daily influent flow projected for 2017 (i.e., 274 MGD). For Discharge Point 001, this Order/Permit establishes mass emission effluent limitations for TBELs based on the average daily upper bound influent flow of 206 MGD projected for 2025, taken from the Discharger's 2017 Master Plan. For Discharge Point 002, mass emission effluent limitations for TBELs continue to be based on the Discharger's conservative estimate for hydraulic flow capacity of the 78-inch outfall of 230 MGD.

## 2. Applicable Technology-Based Effluent Limitations

A summary of the applicable technology-based effluent limitations is shown in Tables F-14 and F-15, in section IV.E of this Fact Sheet.

## C. Water Quality-Based Effluent Limitations (WQBELs)

### 1. Scope and Authority

CWA Section 301(b) and 40 CFR § 122.44(d) require that NPDES permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards and State requirements.

Section 122.44(d)(1)(i) of 40 CFR requires that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, water quality-based effluent limitations (WQBELs) must be established using: (1) USEPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi). USEPA has applied CWA section 403(c) and 40 CFR § 125, Subpart M, following 40 CFR § 122.

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan, and achieve applicable water quality objectives and



criteria that are contained in other state plans and policies, or any applicable water quality criteria contained in the Ocean Plan.

## **2. Applicable Beneficial Uses and Water Quality Criteria and Objectives**

The Basin Plan and Ocean Plan designate beneficial uses, establish water quality objectives, and contain implementation programs and policies to achieve those objectives for all waters of the State. Applicable beneficial uses designated within the Basin Plan and Ocean Plan are listed in Tables F-5 and F-6, in section III.C of this fact sheet. The Basin Plan contains Water Quality Objectives for bacteria for water bodies designated for water contact recreation. The Ocean Plan establishes applicable water quality objectives for the protection of marine aquatic life and human health. The Ocean Plan also includes water quality objectives for bacterial characteristics, physical characteristics, chemical characteristics, biological characteristics, and radioactivity. The water quality objectives from the Ocean Plan and Basin Plan were incorporated into this Order/Permit as either final effluent limitations (based on reasonable potential) or receiving water limitations.

## **3. Determining the Need for WQBELs**

### **a. RPA Methodology**

The 2012 Order/Permit (Order No. R8-2012-0035) contains WQBELs for pollutant parameters based on Table B of the 2009 Ocean Plan. For this Order/Permit, the need for WQBELs based on water quality objectives in Table 3 of the 2019 Ocean Plan was statistically reevaluated in accordance with the Reasonable Potential Analysis (RPA) procedures contained in Appendix VI of the 2019 Ocean Plan.

The statistical RPA method accounts for the averaging period of the water quality objective, accounts for and captures the long-term variability of the pollutant in the effluent, accounts for limitations associated with sparse data sets, accounts for uncertainty associated with censored data sets, and assumes a lognormal distribution of the facility-specific effluent data. The program calculates the upper confidence bound (UCB) of an effluent population percentile after complete mixing. In the evaluation employed in this Order/Permit, the UCB is calculated as the one-sided, upper 95 percent confidence bound for the 95<sup>th</sup> percentile of the effluent distribution after complete mixing. The calculated UCB<sub>95/95</sub> is then compared to the appropriate water quality objective to determine the potential for an exceedance of that objective and the need for a WQBEL. For constituents that have an insufficient number of monitoring data or a substantial number of non-detected data with a reporting limit higher than the respective water quality objective, the RPA result is likely to be inconclusive.

According to the Ocean Plan RPA procedure, the RPA can yield three endpoints:



- Endpoint 1: An effluent limitation must be developed, and effluent monitoring for the pollutant is required;
- Endpoint 2: An effluent limitation is not required, but the Santa Ana Water Board and/or USEPA may require monitoring for the pollutant or for the whole effluent toxicity (WET) as appropriate; or
- Endpoint 3: The RPA is inconclusive, and monitoring for the pollutant or WET testing is required. An existing effluent limitation shall remain in the permit, otherwise a permit reopener clause shall be included to allow for subsequent modification of the permit to include an effluent limitation if the monitoring establishes that the discharge causes, has the reasonable potential to cause, or contributes to an excursion above the appropriate water quality objective.

**b. Minimum Initial Dilution or Dilution Credits**

The Discharger updated its dilution analysis as part of the planning process for the GWRS final expansion project (GWRS Final Expansion Final Implementation Plan, Project No. SP-173, Effluent Reuse Study, October 2016). The updated modeling included calibrating current conditions with the modeling completed in 2002 and 2008. The updated modeling continued to use NRFIELD in Visual Plumes, consistent with the past modeling.

The analysis included various scenarios for flow rates, including low effluent flows as the result of the GWRS final expansion project, densities (i.e. temperature and salinity changes based on low effluent flows), and critical stratification conditions (i.e. August to October 2014). The analysis shows that dilution is unlikely to be impacted by the increase in RO concentrate discharged because the total discharge flow will be decreased. As to be expected, model results were the most sensitive to discharge flow rate and stratification and ranged from 151:1 to 363:1 for the various scenarios. Based on the updated modeling, the Santa Ana Water Board and USEPA is carrying over the dilution ratio of 181:1, as this dilution ( $D_m = 180$ ) is a representative of reasonable worst-case scenario (i.e. high discharge flow rate under strong stratified conditions) and is consistent with antidegradation and anti-backsliding requirements. With exception of radioactivity, the dilution ratio of 181:1 ( $D_m = 180$ ) has been applied to the final effluent limitations for Discharge Point No. 001. The dilution ratio of 37:1 ( $D_m = 36$ ) also has retained for Discharge Point No. 002 in this Order/Permit. Based on the Ocean Plan, water quality objectives for radioactivity shall apply directly to the undiluted waste effluent (i.e.,  $D_m = 0$  for radioactivity).

**c. RPA for Pollutants in Table 3 of the Ocean Plan**

RPAs were conducted using the State Water Board's RPcalc 2.0 software tool and minimum probable initial dilution values for Discharge Point 001 (i.e.,  $D_m = 180$ ) and Discharge Point 002 (i.e.,  $D_m = 36$ ), except for radioactivity.

Effluent data provided in the Discharger's monitoring reports from May 2015 to December 2019, after the Discharger achieved full secondary treatment and ceased disinfection, were used in the analyses in order to be representative of current performance. The results of these RPAs are summarized in Table F-9 for Discharge Point 001 (120" outfall) and Table F-10 for Discharge Point 002 (78" outfall).

Based on the RPA for Discharge Point 001, Endpoint 1 is established for total chlorine residual. Consequently, WQBELs for total chlorine residual are included in the Order/Permit and periodic effluent monitoring is also required. The Discharger ceased continuous effluent disinfection using chlorination/dechlorination at Discharge Point 001 (120" outfall). However, sodium hypochlorite has been used as oxidant chemical to treat foul air (i.e., H<sub>2</sub>S control) in scrubbers at primary treatment process and to control the bulking from filamentous organisms in the return activated sludge (RAS), which probably causes RP for total chlorine residual. Endpoint 2 is established for the majority of detected Table 3 pollutants. Consequently, WQBELs for these pollutants are not included in the Order/Permit, but periodic effluent monitoring for all Endpoint 2 pollutants is required. Endpoint 3 with inconclusive results was reported for benzidine, hexachlorobenzene, PCBs, TCDD equivalents, and toxaphene. Consequently, existing WQBELs for these five pollutants are retained in this Order/Permit and periodic effluent monitoring is also required. This Order/Permit does not include effluent limitations for other pollutants displaying Endpoint 3; instead, monitoring requirements for those pollutants were included.

Based on the RPA for Discharge Point 002, Endpoint 1 is established for ammonia (as nitrogen), and total chlorine residual. Consequently, WQBELs for ammonia and total chlorine residual are included in the Order/Permit and periodic effluent monitoring is also required. Endpoint 2 is established for the majority of detected Table 3 pollutants. Consequently, WQBELs for these pollutants are not included in the Order/Permit, but periodic effluent monitoring for all Endpoint 2 pollutants is required. Endpoint 3 with inconclusive results was reported for TCDD equivalents and thus, existing WQBELs and quarterly effluent monitoring for TCDD equivalents are retained in this Order/Permit. This Order/Permit does not include effluent limitations for other pollutants displaying Endpoint 3; instead, monitoring requirements for those pollutants were included.

Step 13 of the RPA procedures in the Ocean Plan authorizes RPA based on best professional judgment (BPJ). Information may include the facility type, discharge type, solids loading analysis, lack of dilution, history of compliance problems, potential toxic impact of discharge, fish tissue residue data, water quality and beneficial uses of the receiving water, CWA 303(d) listing for the pollutant, the presence of endangered or threatened species or critical habitat, and other information. Based on Step 13 of the Ocean Plan (i.e., BPJ), reasonable potential for acute and chronic toxicity has been established

for Discharge Point 001 and chronic toxicity for Discharge Point 002 because toxicity levels at POTWs can exceed applicable water quality standards, and numerical water quality objectives/criteria for the majority of chemical pollutants have not been established. In addition, reasonable potential for radioactivity has been established for Discharge Points 001 and 002 because POTWs accept waste from various facilities, including hospitals that could potentially discharge radioactivity into the sewershed, and the Discharger's wastewater treatment plants are not typically designed to remove radioactivity from the effluent. The discharge also had detectable concentrations of radioactivity (i.e., maximum effluent concentration (MEC) for gross alpha of 36 pCi/L and MEC for gross beta of 70 pCi/L). Thus, WQBELs for acute/chronic toxicity and radioactivity at Discharge Point 001 and a WQBLEL for chronic toxicity and radioactivity at Discharge Point 002 are included in this Order/Permit.

**Table F-9. Reasonable Potential Analysis for Discharge Point No. 001 (120" Outfall)**

| Table 3 Pollutant                                       | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Concentration (µg/L) | Max Effluent Concentration After Mixing <sup>[1]</sup> (µg/L) | Projected 95 <sup>th</sup> Percentile (µg/L) | Result          |
|---------------------------------------------------------|---------------------------|----------------|--------------------|-----------------------------------|---------------------------------------------------------------|----------------------------------------------|-----------------|
| <b>Objectives for Protection of Marine Aquatic Life</b> |                           |                |                    |                                   |                                                               |                                              |                 |
| Ammonia (as N)                                          | 600                       | 1,699          | 0                  | 46,000                            | 254                                                           | 203                                          | Endpoint 2      |
| Arsenic                                                 | 8                         | 56             | 0                  | 6.91                              | 3.02                                                          | 3.02                                         | Endpoint 2      |
| Cadmium                                                 | 1                         | 56             | 53                 | 0.240                             | 0.0013                                                        | ---                                          | Endpoint 2      |
| Chlorinated Phenols                                     | 1                         | 55             | 54                 | 0.54                              | 0.0030                                                        | ---                                          | Endpoint 2      |
| Chromium (VI) <sup>[2]</sup>                            | 2                         | 56             | 6                  | 1.55 <sup>[2]</sup>               | 0.0086 <sup>[2]</sup>                                         | 0.0095 <sup>[2]</sup>                        | Endpoint 2      |
| Acute Toxicity                                          | PASS                      | 26             | 0                  | NA                                | PASS                                                          | NA                                           | Endpoint 1; BPJ |
| Chronic Toxicity                                        | PASS                      | 71             | 0                  | NA                                | PASS                                                          | NA                                           | Endpoint 1; BPJ |
| Copper                                                  | 3                         | 56             | 0                  | 25.2                              | 2.1282                                                        | 2.0901                                       | Endpoint 2      |
| Cyanide                                                 | 1                         | 52             | 1                  | 5.88                              | 0.0325                                                        | 0.0372                                       | Endpoint 2      |
| Endosulfan                                              | 0.009                     | 10             | 10                 | <0.0050                           | <2.8×10 <sup>-5</sup>                                         | ---                                          | Endpoint 3      |
| Endrin                                                  | 0.002                     | 10             | 10                 | <0.0028                           | <1.5×10 <sup>-5</sup>                                         | ---                                          | Endpoint 3      |
| HCH                                                     | 0.004                     | 10             | 10                 | <0.0050                           | <2.8×10 <sup>-5</sup>                                         | ---                                          | Endpoint 3      |
| Lead                                                    | 2                         | 56             | 47                 | 0.62                              | 0.0034                                                        | ---                                          | Endpoint 2      |
| Mercury                                                 | 0.04                      | 55             | 0                  | 0.0071                            | 5.4×10 <sup>-4</sup>                                          | 5.4×10 <sup>-4</sup>                         | Endpoint 2      |
| Nickel                                                  | 5                         | 56             | 0                  | 23.5                              | 0.1298                                                        | 0.1338                                       | Endpoint 2      |
| Non-chlorinated Phenols                                 | 30                        | 55             | 53                 | 1.56                              | 0.008715                                                      | ---                                          | Endpoint 2      |
| Radioactivity (Gross Alpha)                             | <sup>[3]</sup>            | 48             | 8                  | 36 pCi/L                          | 36 pCi/L                                                      | 54.4435 pCi/L                                | Endpoint 1; BPJ |
| Radioactivity (Gross Beta)                              | <sup>[3]</sup>            | 56             | 0                  | 70 pCi/L                          | 70 pCi/L                                                      | 76.3882 pCi/L                                | Endpoint 1; BPJ |
| Radioactivity (Radium, combined)                        | <sup>[3]</sup>            | 11             | 0                  | 1.14 pCi/L                        | 1.14 pCi/L                                                    | 7.5682 pCi/L                                 | Endpoint 1; BPJ |
| Selenium                                                | 15                        | 56             | 0                  | 21.9                              | 0.1210                                                        | 0.1509                                       | Endpoint 2      |

| Table 3 Pollutant                                                 |  | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Concentration (µg/L) | Max Effluent Concentration After Mixing <sup>[1]</sup> (µg/L) | Projected 95 <sup>th</sup> Percentile (µg/L) | Result         |
|-------------------------------------------------------------------|--|---------------------------|----------------|--------------------|-----------------------------------|---------------------------------------------------------------|----------------------------------------------|----------------|
| Silver                                                            |  | 0.7                       | 55             | 46                 | 0.16                              | 0.16                                                          | ---                                          | Endpoint 2     |
| Total Chlorine Residual <sup>[4]</sup>                            |  | 2                         | 5,100          | 358                | 1,200                             | 6.63                                                          | 0.7395                                       | Endpoint 1     |
| Zinc                                                              |  | 20                        | 56             | 0                  | 64.7                              | 8.31                                                          | 8.21                                         | Endpoint 2     |
| <b>Objectives for Protection of Human Health – Noncarcinogens</b> |  |                           |                |                    |                                   |                                                               |                                              |                |
| 1,1,1-Trichloroethane                                             |  | 540,000                   | 54             | 54                 | <1.43                             | <0.0079                                                       | ---                                          | Endpoint 3     |
| 2,4-Dinitrophenol                                                 |  | 4.0                       | 55             | 55                 | <3.5                              | <0.0193                                                       | ---                                          | Endpoint 3     |
| 2-Methyl-4,6-Dinitrophenol                                        |  | 220                       | 55             | 55                 | <1.7                              | <0.0094                                                       | ---                                          | Endpoint 3     |
| Acrolein                                                          |  | 220                       | 54             | 54                 | <4.48                             | <0.0248                                                       | ---                                          | Endpoint 3     |
| Antimony                                                          |  | 1,200                     | 56             | 0                  | 2.54                              | 0.0140                                                        | 0.0146                                       | Endpoint 2     |
| Bis(2-Chloroethoxy)Methane                                        |  | 4.4                       | 55             | 55                 | <1.1                              | <0.0061                                                       | ---                                          | Endpoint 3     |
| Bis(2-Chloroisopropyl)Ether                                       |  | 1,200                     | 55             | 55                 | <0.73                             | <0.0040                                                       | ---                                          | Endpoint 3     |
| Chlorobenzene                                                     |  | 570                       | 54             | 54                 | <0.9                              | <0.0050                                                       | ---                                          | Endpoint 3     |
| Chromium (III) <sup>[2]</sup>                                     |  | 190,000                   | 56             | 6                  | 1.55 <sup>[2]</sup>               | 0.0086 <sup>[2]</sup>                                         | 0.0095 <sup>[2]</sup>                        | Endpoint 2     |
| Dichlorobenzenes                                                  |  | 5,100                     | 55             | 55                 | <2                                | <0.0111                                                       | ---                                          | Endpoint 3     |
| Diethyl Phthalate                                                 |  | 33,000                    | 55             | 54                 | 0.76                              | 0.0042                                                        | ---                                          | Endpoint 2     |
| Dimethyl Phthalate                                                |  | 820,000                   | 55             | 55                 | <1.64                             | <0.0091                                                       | ---                                          | Endpoint 3     |
| Di-n-Butyl Phthalate                                              |  | 3,500                     | 55             | 54                 | 1.8                               | 0.0099                                                        | ---                                          | Endpoint 2     |
| Ethylbenzene                                                      |  | 4,100                     | 47             | 47                 | <1.8                              | <0.0099                                                       | ---                                          | Endpoint 3     |
| Fluoranthene                                                      |  | 15                        | 55             | 55                 | <0.49                             | <0.0027                                                       | ---                                          | Endpoint 3     |
| Hexachlorocyclopentadiene                                         |  | 58                        | 55             | 55                 | <1.6                              | <0.0088                                                       | ---                                          | Endpoint 3     |
| Nitrobenzene                                                      |  | 4.9                       | 55             | 54                 | 0.38                              | 0.0021                                                        | ---                                          | Endpoint 2     |
| Thallium                                                          |  | 2                         | 56             | 48                 | 0.20                              | 0.0011                                                        | ---                                          | Endpoint 2     |
| Toluene                                                           |  | 85,000                    | 54             | 50                 | 0.19                              | 0.0010                                                        | ---                                          | Endpoint 2     |
| Tributyltin                                                       |  | 0.0014                    | <sup>[5]</sup> | <sup>[5]</sup>     | <sup>[5]</sup>                    | <sup>[5]</sup>                                                | ---                                          | <sup>[5]</sup> |
| <b>Objectives for Protection of Human Health – Carcinogens</b>    |  |                           |                |                    |                                   |                                                               |                                              |                |
| 1,1,2,2-Tetrachloroethane                                         |  | 2.3                       | 54             | 54                 | <0.53                             | <0.0029                                                       | ---                                          | Endpoint 3     |

| Table 3 Pollutant          |  | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Concentration (µg/L) | Max Effluent Concentration After Mixing <sup>[1]</sup> (µg/L) | Projected 95 <sup>th</sup> Percentile (µg/L) | Result                             |
|----------------------------|--|---------------------------|----------------|--------------------|-----------------------------------|---------------------------------------------------------------|----------------------------------------------|------------------------------------|
| 1,1,2-Trichloroethane      |  | 9.4                       | 54             | 54                 | <0.1                              | <0.0006                                                       | ---                                          | Endpoint 3                         |
| 1,1-Dichloroethylene       |  | 0.9                       | 54             | 54                 | <1.44                             | <0.0005                                                       | ---                                          | Endpoint 3                         |
| 1,2-Dichloroethane         |  | 28                        | 54             | 54                 | <0.98                             | <0.0055                                                       | ---                                          | Endpoint 3                         |
| 1,2-Diphenylhydrazine      |  | 0.16                      | 54             | 54                 | <0.60                             | <0.0033                                                       | ---                                          | Endpoint 3                         |
| 1,3-Dichloropropylene      |  | 8.9                       | 54             | 54                 | <1.56                             | <0.0087                                                       | ---                                          | Endpoint 3                         |
| 1,4-Dichlorobenzene        |  | 18                        | 109            | 109                | <0.95                             | <0.0053                                                       | ---                                          | Endpoint 3                         |
| TCDD Equivalents           |  | $3.9 \times 10^{-9}$      | 18             | 18                 | $<7.0 \times 10^{-7}$             | $<3.9 \times 10^{-9}$                                         | ---                                          | Endpoint 3;<br>WQBEL<br>carry over |
| 2,4,6-Trichlorophenol      |  | 0.29                      | 54             | 53                 | 0.54                              | 0.0030                                                        | ---                                          | Endpoint 2                         |
| 2,4-Dinitrotoluene         |  | 2.6                       | 55             | 55                 | <2.2                              | <0.0123                                                       | ---                                          | Endpoint 3                         |
| 3,3'-Dichlorobenzidine     |  | 0.0081                    | 55             | 55                 | <2.58                             | <0.0144                                                       | ---                                          | Endpoint 3                         |
| Acrylonitrile              |  | 0.10                      | 54             | 54                 | <1.51                             | <0.0084                                                       | ---                                          | Endpoint 3                         |
| Aldrin                     |  | $2.2 \times 10^{-5}$      | 10             | 10                 | <0.01                             | <0.00006                                                      | ---                                          | Endpoint 3                         |
| Benzene                    |  | 5.9                       | 54             | 54                 | <1.37                             | <0.0077                                                       | ---                                          | Endpoint 3                         |
| Benzidine                  |  | $6.9 \times 10^{-5}$      | 54             | 54                 | <8.27                             | <0.0462                                                       | ---                                          | Endpoint 3;<br>WQBEL<br>carry over |
| Beryllium                  |  | 0.033                     | 55             | 55                 | <0.185                            | <0.0010                                                       | ---                                          | Endpoint 3                         |
| Bis(2-Chloroethyl)Ether    |  | 0.045                     | 55             | 55                 | <0.96                             | <0.0053                                                       | ---                                          | Endpoint 3                         |
| Bis(2-Ethylhexyl)Phthalate |  | 3.5                       | 55             | 50                 | 3.9                               | 0.0215                                                        | ---                                          | Endpoint 2                         |
| Carbon Tetrachloride       |  | 0.90                      | 54             | 54                 | <1.79                             | <0.01                                                         | ---                                          | Endpoint 3                         |
| Chlordane                  |  | $2.3 \times 10^{-5}$      | 10             | 10                 | <0.05                             | $<2.8 \times 10^{-4}$                                         | ---                                          | Endpoint 3                         |
| Chlorodibromomethane       |  | 8.6                       | 54             | 9                  | 4.86                              | 0.0269                                                        | 0.0235                                       | Endpoint 2                         |
| Chloroform                 |  | 130                       | 54             | 1                  | 16.6                              | 0.0917                                                        | 3.285                                        | Endpoint 2                         |
| DDT (total)                |  | 0.00017                   | 10             | 10                 | <0.0031                           | $<1.7 \times 10^{-5}$                                         | ---                                          | Endpoint 3                         |
| Dichlorobromomethane       |  | 6.2                       | 54             | 2                  | 8.98                              | 0.05                                                          | 0.0534                                       | Endpoint 2                         |
| Dichloromethane            |  | 450                       | 54             | 47                 | 8.9                               | 0.0492                                                        | ---                                          | Endpoint 2                         |



| Table 3 Pollutant         | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Concentration (µg/L) | Max Effluent Concentration After Mixing <sup>[1]</sup> (µg/L) | Projected 95 <sup>th</sup> Percentile (µg/L) | Result                             |
|---------------------------|---------------------------|----------------|--------------------|-----------------------------------|---------------------------------------------------------------|----------------------------------------------|------------------------------------|
| Dieldrin                  | 4.0×10 <sup>-5</sup>      | 10             | 10                 | <0.01                             | <5.587×10 <sup>-5</sup>                                       | ---                                          | Endpoint 3                         |
| Halomethanes              | 130                       | 54             | 49                 | 1.37                              | 0.0076                                                        | ---                                          | Endpoint 2                         |
| Heptachlor                | 5×10 <sup>-5</sup>        | 10             | 10                 | <0.01                             | <5.587×10 <sup>-5</sup>                                       | ---                                          | Endpoint 3                         |
| Heptachlor Epoxide        | 2×10 <sup>-5</sup>        | 10             | 10                 | <0.01                             | <5.587×10 <sup>-5</sup>                                       | ---                                          | Endpoint 3                         |
| Hexachlorobenzene         | 0.00021                   | 55             | 55                 | <0.77                             | <0.0043                                                       | ---                                          | Endpoint 3;<br>WQBEL<br>carry over |
| Hexachlorobutadiene       | 14                        | 55             | 55                 | <0.77                             | <0.0043                                                       | ---                                          | Endpoint 3                         |
| Hexachloroethane          | 2.5                       | 55             | 55                 | <0.63                             | <0.0035                                                       | ---                                          | Endpoint 3                         |
| Isophorone                | 730                       | 55             | 55                 | <0.77                             | <0.0043                                                       | ---                                          | Endpoint 3                         |
| N-Nitrosodimethylamine    | 7.3                       | 55             | 55                 | <2.7                              | <0.0151                                                       | ---                                          | Endpoint 3                         |
| N-Nitrosodi-n-Propylamine | 0.38                      | 55             | 55                 | <1.36                             | <0.0076                                                       | ---                                          | Endpoint 3                         |
| N-Nitrosodiphenylamine    | 2.5                       | 55             | 55                 | <1.47                             | <0.0082                                                       | ---                                          | Endpoint 3                         |
| PAHs (total)              | 0.0088                    | 55             | 55                 | <1.5                              | <0.0083                                                       | ---                                          | Endpoint 3                         |
| PCBs                      | 1.9×10 <sup>-5</sup>      | 10             | 10                 | <0.50                             | <0.0028                                                       | ---                                          | Endpoint 3;<br>WQBEL<br>carry over |
| Tetrachloroethylene       | 2.0                       | 54             | 52                 | 1.58                              | 0.0087                                                        | ---                                          | Endpoint 2                         |
| Toxaphene                 | 0.00021                   | 10             | 10                 | <1.2                              | <0.0067                                                       | ---                                          | Endpoint 3;<br>WQBEL<br>carry over |
| Trichloroethylene         | 27                        | 54             | 47                 | <1.49                             | <0.0083                                                       | ---                                          | Endpoint 3                         |
| Vinyl Chloride            | 36                        | 54             | 54                 | <1.73                             | <0.0097                                                       | ---                                          | Endpoint 3                         |

Abbreviations:

Max = maximum  
WQO = water quality objective  
µg/L = micrograms per liter  
pCi/L = pico-curies per liter  
NA = not applicable

Footnotes:

[1] Concentrations after mixing include a dilution factor of 180 ( $D_m$ ), with exception of radioactivity ( $D_m=0$ ), and applicable background seawater concentrations ( $C_s$ ) for arsenic, copper, mercury, silver, and zinc.

[2] The Discharger monitored for total chromium, in lieu of chromium (III) and chromium (VI). The maximum detected concentration of total chromium (1.55 µg/L) is less than the water quality objective for chromium (III) and chromium (VI) (i.e., 190,000 µg/L and 2 µg/L, respectively).

[3] Not to exceed limits specified in title 17, division 1, chapter 5, subchapter 4, group 3, article 3, section 30253 of the CCR.

[4] While the discharger ceased chlorine disinfection of the effluent in May 2015, chlorine is still used at the facility for odor control and in-house water use.

[5] The discharger did not monitor for Tributyltin.

**Table F-10. Reasonable Potential Analysis for Discharge Point No. 002 (78" Outfall)**

| Table 3 Pollutant                                       | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Concentration (µg/L) | Max Effluent Concentration After Mixing <sup>[1]</sup> (µg/L) | Projected 95 <sup>th</sup> Percentile (µg/L) | Result          |
|---------------------------------------------------------|---------------------------|----------------|--------------------|-----------------------------------|---------------------------------------------------------------|----------------------------------------------|-----------------|
| <b>Objectives for Protection of Marine Aquatic Life</b> |                           |                |                    |                                   |                                                               |                                              |                 |
| Ammonia (as nitrogen)                                   | 600                       | 1,699          | 0                  | 46,000                            | 1,243                                                         | 993                                          | Endpoint 1      |
| Arsenic                                                 | 8                         | 56             | 0                  | 6.91                              | 3.02                                                          | 3.1                                          | Endpoint 2      |
| Cadmium                                                 | 1                         | 56             | 53                 | 0.240                             | 0.0065                                                        | ---                                          | Endpoint 2      |
| Chlorinated Phenolics                                   | 1                         | 55             | 54                 | 0.54                              | 0.0146                                                        | ---                                          | Endpoint 2      |
| Chromium (VI) <sup>[2]</sup>                            | 2                         | 56             | 6                  | 1.55 <sup>[2]</sup>               | 0.0419 <sup>[2]</sup>                                         | 0.0464 <sup>[2]</sup>                        | Endpoint 2      |
| Chronic Toxicity                                        | 1.0 TU <sub>c</sub>       | 0              | 0                  | NA                                | NA                                                            | NA                                           | Endpoint 1; BPJ |

| Table 3 Pollutant                                                 |  | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Concentration (µg/L) | Max Effluent Concentration After Mixing <sup>[1]</sup> (µg/L) | Projected 95 <sup>th</sup> Percentile (µg/L) | Result          |
|-------------------------------------------------------------------|--|---------------------------|----------------|--------------------|-----------------------------------|---------------------------------------------------------------|----------------------------------------------|-----------------|
| Copper                                                            |  | 3                         | 56             | 0                  | 25.2                              | 2.627                                                         | 2.446                                        | Endpoint 2      |
| Cyanide                                                           |  | 1                         | 52             | 1                  | 5.88                              | 0.1589                                                        | 0.1819                                       | Endpoint 2      |
| Endosulfan (total)                                                |  | 0.009                     | 10             | 10                 | <0.0050                           | <0.0001                                                       | ---                                          | Endpoint 3      |
| Endrin                                                            |  | 0.002                     | 10             | 10                 | <0.0028                           | <7.6×10 <sup>-5</sup>                                         | ---                                          | Endpoint 3      |
| HCH                                                               |  | 0.004                     | 10             | 10                 | <0.0050                           | <0.0001                                                       | ---                                          | Endpoint 3      |
| Lead                                                              |  | 2                         | 56             | 47                 | 0.62                              | 0.0168                                                        | ---                                          | Endpoint 2      |
| Mercury                                                           |  | 0.04                      | 55             | 0                  | 0.0071                            | 0.0007                                                        | 0.0007                                       | Endpoint 2      |
| Nickel                                                            |  | 5                         | 56             | 0                  | 23.5                              | 0.6351                                                        | 0.6545                                       | Endpoint 2      |
| Non-chlorinated Phenolics                                         |  | 30                        | 55             | 53                 | 1.56                              | 0.0422                                                        | ---                                          | Endpoint 2      |
| Radioactivity (Gross Alpha)                                       |  | [3]                       | 48             | 8                  | 36 pCi/L                          | 36 pCi/L                                                      | 54.4435 pCi/L                                | Endpoint 1; BPJ |
| Radioactivity (Gross Beta)                                        |  | [3]                       | 56             | 0                  | 70 pCi/L                          | 70 pCi/L                                                      | 76.3882 pCi/L                                | Endpoint 1; BPJ |
| Radioactivity (Radium, combined)                                  |  | [3]                       | 11             | 0                  | 1.14 pCi/L                        | 1.14 pCi/L                                                    | 7.5682 pCi/L                                 | Endpoint 1; BPJ |
| Selenium                                                          |  | 15                        | 3461           | 3380               | 21.9                              | 0.5919                                                        | 0.7383                                       | Endpoint 2      |
| Silver                                                            |  | 0.7                       | 55             | 46                 | 0.16                              | 0.16                                                          | ---                                          | Endpoint 2      |
| Total Chlorine Residual <sup>[4]</sup>                            |  | 2                         | 5100           | 358                | 1,200                             | 32.4                                                          | 9.73                                         | Endpoint 1      |
| Zinc                                                              |  | 20                        | 56             | 0                  | 64.7                              | 9.53                                                          | 9.03                                         | Endpoint 2      |
| <b>Objectives for Protection of Human Health – Noncarcinogens</b> |  |                           |                |                    |                                   |                                                               |                                              |                 |
| 1,1,1-Trichloroethane                                             |  | 540,000                   | 54             | 54                 | <1.43                             | <0.0386                                                       | ---                                          | Endpoint 3      |
| 2,4-Dinitrophenol                                                 |  | 4.0                       | 55             | 55                 | <3.5                              | <0.0946                                                       | ---                                          | Endpoint 3      |
| 2-Methyl-4,6-Dinitrophenol                                        |  | 220                       | 55             | 55                 | <1.7                              | <0.0459                                                       | ---                                          | Endpoint 3      |
| Acrolein                                                          |  | 220                       | 54             | 54                 | <4.48                             | <0.01211                                                      | ---                                          | Endpoint 3      |
| Antimony                                                          |  | 1,200                     | 56             | 0                  | 2.54                              | 0.0686                                                        | 0.0712                                       | Endpoint 2      |
| Bis(2-Chloroethoxy)Methane                                        |  | 4.4                       | 55             | 55                 | <1.1                              | <0.0297                                                       | ---                                          | Endpoint 3      |

| Table 3 Pollutant                                              |                      | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Concentration (µg/L) | Max Effluent Concentration After Mixing <sup>[1]</sup> (µg/L) | Projected 95 <sup>th</sup> Percentile (µg/L) | Result                             |
|----------------------------------------------------------------|----------------------|---------------------------|----------------|--------------------|-----------------------------------|---------------------------------------------------------------|----------------------------------------------|------------------------------------|
| Bis(2-Chloroisopropyl)Ether                                    | 1,200                | 55                        | 55             | 55                 | <0.73                             | <0.0197                                                       | ---                                          | Endpoint 3                         |
| Chlorobenzene                                                  | 570                  | 54                        | 54             | 54                 | <0.9                              | <0.0243                                                       | ---                                          | Endpoint 3                         |
| Chromium (III) <sup>[2]</sup>                                  | 190,000              | 56                        | 56             | 6                  | 1.55 <sup>[2]</sup>               | 0.0419 <sup>[2]</sup>                                         | 0.0464 <sup>[2]</sup>                        | Endpoint 2                         |
| Dichlorobenzenes                                               | 5,100                | 55                        | 55             | 55                 | <2                                | <0.0556                                                       | ---                                          | Endpoint 3                         |
| Diethyl Phthalate                                              | 33,000               | 55                        | 55             | 54                 | 0.76                              | 0.0205                                                        | ---                                          | Endpoint 2                         |
| Dimethyl Phthalate                                             | 820,000              | 55                        | 55             | 55                 | <1.64                             | <0.0443                                                       | ---                                          | Endpoint 3                         |
| Di-n-Butyl Phthalate                                           | 3,500                | 55                        | 55             | 54                 | 1.8                               | 0.0486                                                        | ---                                          | Endpoint 2                         |
| Ethylbenzene                                                   | 4,100                | 47                        | 47             | 47                 | <1.8                              | <0.04866                                                      | ---                                          | Endpoint 3                         |
| Fluoranthene                                                   | 15                   | 55                        | 55             | 55                 | <0.49                             | <0.0132                                                       | ---                                          | Endpoint 3                         |
| Hexachlorocyclopentadiene                                      | 58                   | 55                        | 55             | 55                 | <1.6                              | <0.0432                                                       | ---                                          | Endpoint 3                         |
| Nitrobenzene                                                   | 4.9                  | 55                        | 55             | 54                 | 0.38                              | 0.0103                                                        | ---                                          | Endpoint 2                         |
| Thallium                                                       | 2                    | 56                        | 56             | 48                 | 0.20                              | 0.0054                                                        | ---                                          | Endpoint 2                         |
| Toluene                                                        | 85,000               | 54                        | 54             | 50                 | 0.19                              | 0.0051                                                        | ---                                          | Endpoint 2                         |
| Tributyltin                                                    | 0.0014               | <sup>[5]</sup>            | <sup>[5]</sup> | <sup>[5]</sup>     | <sup>[5]</sup>                    | <sup>[5]</sup>                                                | ---                                          | <sup>[5]</sup>                     |
| <b>Objectives for Protection of Human Health – Carcinogens</b> |                      |                           |                |                    |                                   |                                                               |                                              |                                    |
| 1,1,2,2-Tetrachloroethane                                      | 2.3                  | 54                        | 54             | 54                 | <0.53                             | <0.0143                                                       | ---                                          | Endpoint 3                         |
| 1,1,2-Trichloroethane                                          | 9.4                  | 54                        | 54             | 54                 | <0.1                              | <0.0027                                                       | ---                                          | Endpoint 3                         |
| 1,1-Dichloroethylene                                           | 0.9                  | 54                        | 54             | 54                 | <1.44                             | <0.0389                                                       | ---                                          | Endpoint 3                         |
| 1,2-Dichloroethane                                             | 28                   | 54                        | 54             | 54                 | <0.98                             | <0.0265                                                       | ---                                          | Endpoint 3                         |
| 1,2-Diphenylhydrazine                                          | 0.16                 | 54                        | 54             | 54                 | <0.60                             | <0.0162                                                       | ---                                          | Endpoint 3                         |
| 1,3-Dichloropropylene                                          | 8.9                  | 54                        | 54             | 54                 | <1.56                             | <0.0422                                                       | ---                                          | Endpoint 3                         |
| 1,4-Dichlorobenzene                                            | 18                   | 109                       | 109            | 109                | <0.95                             | <0.0257                                                       | ---                                          | Endpoint 3                         |
| TCDD Equivalents                                               | 3.9×10 <sup>-9</sup> | 18                        | 18             | 18                 | <7.0×10 <sup>-7</sup>             | <1.9×10 <sup>-8</sup>                                         | ----                                         | Endpoint 3;<br>WQBEL<br>carry over |
| 2,4,6-Trichlorophenol                                          | 0.29                 | 54                        | 54             | 53                 | 0.54                              | 0.0146                                                        | ---                                          | Endpoint 2                         |
| 2,4-Dinitrotoluene                                             | 2.6                  | 55                        | 55             | 55                 | <2.2                              | <0.0595                                                       | ---                                          | Endpoint 3                         |

| Table 3 Pollutant          | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Concentration (µg/L) | Max Effluent Concentration After Mixing <sup>[1]</sup> (µg/L) | Projected 95 <sup>th</sup> Percentile (µg/L) | Result     |
|----------------------------|---------------------------|----------------|--------------------|-----------------------------------|---------------------------------------------------------------|----------------------------------------------|------------|
| 3,3'-Dichlorobenzidine     | 0.0081                    | 55             | 55                 | <2.58                             | <0.0697                                                       | ---                                          | Endpoint 3 |
| Acrylonitrile              | 0.10                      | 54             | 54                 | <1.51                             | <0.0408                                                       | ---                                          | Endpoint 3 |
| Aldrin                     | 2.2×10 <sup>-5</sup>      | 10             | 10                 | <0.01                             | <0.0003                                                       | ---                                          | Endpoint 3 |
| Benzene                    | 5.9                       | 54             | 54                 | <1.37                             | <0.0370                                                       | ---                                          | Endpoint 3 |
| Benzidine                  | 6.9×10 <sup>-5</sup>      | 54             | 54                 | <8.27                             | <0.2235                                                       | ---                                          | Endpoint 3 |
| Beryllium                  | 0.033                     | 55             | 55                 | <0.185                            | <0.0051                                                       | ---                                          | Endpoint 3 |
| Bis(2-Chloroethyl)Ether    | 0.045                     | 55             | 55                 | <0.96                             | <0.0259                                                       | ---                                          | Endpoint 3 |
| Bis(2-Ethylhexyl)Phthalate | 3.5                       | 55             | 50                 | 3.9                               | 0.1054                                                        | ---                                          | Endpoint 2 |
| Carbon Tetrachloride       | 0.90                      | 54             | 54                 | <1.79                             | <0.484                                                        | ---                                          | Endpoint 3 |
| Chlordane                  | 2.3×10 <sup>-5</sup>      | 10             | 10                 | <0.05                             | <0.0014                                                       | ---                                          | Endpoint 3 |
| Chlorodibromomethane       | 8.6                       | 54             | 9                  | 4.86                              | 0.1314                                                        | 0.1152                                       | Endpoint 2 |
| Chloroform                 | 130                       | 54             | 1                  | 16.6                              | 0.4486                                                        | 2.467                                        | Endpoint 2 |
| DDT (total)                | 0.00017                   | 10             | 10                 | <0.0031                           | <0.0001                                                       | ---                                          | Endpoint 3 |
| Dichlorobromomethane       | 6.2                       | 54             | 2                  | 8.98                              | 0.2427                                                        | 0.2611                                       | Endpoint 2 |
| Dichloromethane            | 450                       | 54             | 47                 | 8.9                               | 0.2405                                                        | ---                                          | Endpoint 2 |
| Dieldrin                   | 4.0×10 <sup>-5</sup>      | 10             | 10                 | <0.01                             | <0.0003                                                       | ---                                          | Endpoint 3 |
| Halomethanes               | 130                       | 54             | 49                 | 1.37                              | 0.0370                                                        | ---                                          | Endpoint 2 |
| Heptachlor                 | 5×10 <sup>-5</sup>        | 10             | 10                 | <0.01                             | <0.0003                                                       | ---                                          | Endpoint 3 |
| Heptachlor Epoxide         | 2×10 <sup>-5</sup>        | 10             | 10                 | <0.01                             | <0.0003                                                       | ---                                          | Endpoint 3 |
| Hexachlorobenzene          | 0.00021                   | 55             | 55                 | <0.77                             | <0.0208                                                       | ---                                          | Endpoint 3 |
| Hexachlorobutadiene        | 14                        | 55             | 55                 | <0.77                             | <0.0208                                                       | ---                                          | Endpoint 3 |
| Hexachloroethane           | 2.5                       | 55             | 55                 | <0.63                             | <0.017                                                        | ---                                          | Endpoint 3 |
| Isophorone                 | 730                       | 55             | 55                 | <0.77                             | <0.0208                                                       | ---                                          | Endpoint 3 |
| N-Nitrosodimethylamine     | 7.3                       | 55             | 55                 | <2.7                              | <0.0730                                                       | ---                                          | Endpoint 3 |
| N-Nitrosodi-n-Propylamine  | 0.38                      | 55             | 55                 | <1.36                             | <0.0368                                                       | ---                                          | Endpoint 3 |
| N-Nitrosodiphenylamine     | 2.5                       | 55             | 55                 | <1.47                             | <0.0397                                                       | ---                                          | Endpoint 3 |
| PAHs (total)               | 0.0088                    | 55             | 55                 | <1.5                              | <0.0417                                                       | ---                                          | Endpoint 3 |

| Table 3 Pollutant   | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Concentration (µg/L) | Max Effluent Concentration After Mixing <sup>[1]</sup> (µg/L) | Projected 95 <sup>th</sup> Percentile (µg/L) | Result     |
|---------------------|---------------------------|----------------|--------------------|-----------------------------------|---------------------------------------------------------------|----------------------------------------------|------------|
| PCBs                | 1.9×10 <sup>-5</sup>      | 10             | 10                 | <0.05                             | <0.0014                                                       | ---                                          | Endpoint 3 |
| Tetrachloroethylene | 2.0                       | 54             | 52                 | 1.58                              | 0.0427                                                        | ---                                          | Endpoint 2 |
| Toxaphene           | 0.00021                   | 10             | 10                 | <1.2                              | <0.0324                                                       | ---                                          | Endpoint 3 |
| Trichloroethylene   | 27                        | 54             | 47                 | <1.49                             | <0.0403                                                       | ---                                          | Endpoint 3 |
| Vinyl Chloride      | 36                        | 54             | 54                 | <1.73                             | <0.0468                                                       | ---                                          | Endpoint 3 |

Abbreviations:

Max = maximum  
WQO = water quality objective  
µg/L = micrograms per liter  
pCi/L = pico-curies per liter  
NA = not applicable

Footnotes:

- <sup>[1]</sup> Concentrations after mixing include a dilution factor of 36 ( $D_m$ ), with exception of radioactivity ( $D_m=0$ ), and applicable background seawater concentrations ( $C_s$ ) for arsenic, copper, mercury, silver, and zinc.
- <sup>[2]</sup> Discharger monitored for total chromium, in lieu of chromium (III) and chromium (VI). The maximum detected concentration of total chromium (1.55 µg/L) is less than the water quality objective for chromium (III) and chromium (VI) (i.e., 190,000 µg/L and 2 µg/L, respectively).
- <sup>[3]</sup> Not to exceed limits specified in title 17, division 1, chapter 5, subchapter 4, group 3, article 3, section 30253 of the CCR.
- <sup>[4]</sup> While the discharger ceased chlorine disinfection of the effluent in May 2015, chlorine is still used at the facility for odor control and in-house water use.
- <sup>[5]</sup> The discharger did not monitor for Tributyltin.



#### 4. WQBEL Calculations

##### a. Concentration-based WQBEL Calculations

From the Table 3 water quality objectives in the 2019 Ocean Plan, concentration-based WQBELs are calculated according to the following equation for all pollutants, except for acute and chronic toxicity (if applicable):

$$C_e = C_o + D_m(C_o - C_s)$$

Where,

$C_e$  = the effluent limitation

$C_o$  = the water quality objective to be met at the completion of initial dilution

$C_s$  = background seawater concentration (see Table F-11 below)

$D_m$  = minimum probable initial dilution expressed as parts seawater per part wastewater (i.e., 180 for Discharge Point 001 and 36 for Discharge Point 002)

Background seawater concentration for all Table 3 pollutants is assumed to be zero ( $C_s = 0$ ), except for the following five pollutants:

**Table F-11. Pollutants with Background Seawater Concentrations ( $C_s$ )**

| Constituents | Background Seawater Concentration |
|--------------|-----------------------------------|
| Arsenic      | 3 µg/L                            |
| Copper       | 2 µg/L                            |
| Mercury      | 0.0005 µg/L                       |
| Silver       | 0.16 µg/L                         |
| Zinc         | 8 µg/L                            |

The calculation of WQBELs for copper, total chlorine residual, ammonia, and TCDDs are demonstrated below for Discharge Point 002, as examples:

**Table F-12. Example – Ocean Plan Table 3 Water Quality Objectives ( $C_o$ )**

| Constituents            | 6-Month Median | Daily Maximum | Instantaneous Maximum | 30 Day Average            |
|-------------------------|----------------|---------------|-----------------------|---------------------------|
| Copper                  | 3 µg/L         | 12 µg/L       | 30 µg/L               | --                        |
| Total Chlorine Residual | 2 µg/L         | 8 µg/L        | 60 µg/L               | --                        |
| Ammonia                 | 0.60 mg/L      | 2.4 mg/L      | 6 mg/L                | --                        |
| TCDD equivalents        | --             | --            | --                    | $3.9 \times 10^{-9}$ µg/L |

Using the equation,  $C_e = C_o + D_m(C_o - C_s)$ , effluent limitations are calculated as follows. All calculations are based on discharge through Discharge Point 002 and, therefore, a dilution ratio ( $D_m$ ) of 36 is applied.

**Example - Copper**

$$C_e = 3 + 36(3 - 2) = 39 \text{ } \mu\text{g/L (6-month Median)}$$

$$C_e = 12 + 36(12 - 2) = 372 \text{ } \mu\text{g/L (Maximum Daily)}$$

$$C_e = 30 + 36(30 - 2) = 1,038 \text{ } \mu\text{g/L (Instantaneous Maximum)}$$

**Total Chlorine Residual**

$$C_e = 2 + 36(2 - 0) = 74 \text{ } \mu\text{g/L (6-month Median)}$$

$$C_e = 8 + 36(8 - 0) = 296 \text{ } \mu\text{g/L (Maximum Daily)}$$

$$C_e = 60 + 36(60 - 0) = 2,220 \text{ } \mu\text{g/L (Instantaneous Maximum)}$$

**Ammonia**

$$C_e = 0.6 + 36(0.6 - 0) = 22.2 \text{ mg/L (6-month Median)}$$

$$C_e = 2.4 + 36(2.4 - 0) = 88.8 \text{ mg/L (Maximum Daily)}$$

$$C_e = 6 + 36(6 - 0) = 222 \text{ mg/L (Instantaneous Maximum)}$$

**TCDD equivalents**

$$C_e = 3.9 \times 10^{-9} + 36(3.9 \times 10^{-9} - 0) = 144.3 \times 10^{-9} \text{ } \mu\text{g/L} = 0.144 \text{ pg/L (Average Monthly)}$$

**b. Mass-based WQBEL Calculations**

The following equation from the 2019 Ocean Plan Chapter III.C.4.j is used to calculate all mass-based WQBELs.

$$\text{Mass (in lbs/day)} = 0.00834 \times C_e \times Q$$

where:

$C_e$  = the effluent concentration limitation ( $\mu\text{g/L}$ )

$Q$  = flow rate (MGD)

For example, in the case of PCBs where  $Q = 206$  MGD for Discharge Point 001, the average monthly mass-based WQBEL is:

$$C_e = 0.00834 \times 0.0034 \text{ } \mu\text{g/L} \times 206 \text{ MGD} = 0.0058 \text{ lbs/day}$$

For Discharge Point 001 (120" outfall), the previous Order/Permit established mass emission effluent limitations based on the average daily influent flow ( $Q$ ) of 274 MGD projected for 2017. This Order/Permit establishes mass emission effluent limitations based on the average daily upper bound influent flow of 206 MGD projected for 2025, which was taken from the Discharger's 2017 Master Plan. For Discharge Point

002 (78" outfall), consistent with the previous permit, mass emission effluent limitations are based on the Discharger's conservative estimate for the hydraulic flow capacity of the 78" outfall (Q) of 230 MGD. Since Discharge Point 002 is authorized for emergency situations, it is anticipated to use its maximum hydraulic flow capacity of 230 MGD during emergency discharges.

### C. Bacteria WQBEL Calculations

Bacteria effluent limitations for Discharge Point 002 remain in the Order/Permit to ensure protection of human health and receiving water quality for REC-1 beneficial uses and compliance with REC-1 water quality objectives for bacteria in the Pacific Ocean.

The Ocean Plan was amended in 2018 to revise the bacterial objectives for ocean waters used for water contact recreation, which includes removal of the previously established total coliform objectives and the revised enterococci objectives based on the EPA 2012 Recreational Criteria for marine waters. This Order/Permit has established bacteria effluent limitations at Discharge Point 002 (78" outfall) based on the amended water quality objectives in Table 1 and 2 of the 2019 Ocean Plan. Fecal coliform and enterococcus WQBELs for Discharge Point 002 are determined using the equation,  $C_e = C_o + D_m(C_o - C_s)$  and an Initial dilution of 37:1 ( $D_m=36$ ). The background conditions for fecal coliform and enterococcus are assumed to be zero.

**Table F-13. Bacteria Water Quality Objectives ( $C_o$ ) in the Ocean Plan**

| Indicator              | Magnitude                     | WQO             |
|------------------------|-------------------------------|-----------------|
| Fecal coliform density | 30-day geometric mean         | 200 /100 mL     |
|                        | Single sample maximum         | 400 /100 mL     |
| Enterococci            | 6-week rolling geometric mean | 30 CFU /100 mL  |
|                        | Statistical threshold value   | 110 CFU /100 mL |

#### **Fecal Coliform**

$$C_e = 200 + 36 (200 - 0) = 7,400 \text{ MPN/100 mL (30-day Geometric Mean)}$$

$$C_e = 400 + 36 (400 - 0) = 14,800 \text{ MPN/100 mL (Single Sample Maximum)}$$

#### **Enterococcus**

$$C_e = 30 + 36 (30 - 0) = 1,110 \text{ CFU/100 mL (6-Week Rolling Geometric Mean)}$$

$$C_e = 110 + 36 (110 - 0) = 4,070 \text{ CFU/100 mL (Statistical Threshold Value)}$$

Based on the implementing procedures described above, WQBELs have been calculated for bacteria and all Table 3 pollutants (excluding acute and chronic toxicity) from the 2019 Ocean Plan and incorporated into this Order/Permit when applicable. A summary of the applicable WQBELs are shown in Tables F-14 and F-15 in section IV.E of this Fact Sheet.

## 5. Whole Effluent Toxicity (WET)

Whole Effluent Toxicity (WET) testing protects receiving waters from the aggregate toxic effect of a mixture of pollutants in the effluent or pollutants that are not typically monitored. An acute toxicity test is conducted over a short time period and measures mortality. A chronic toxicity test is conducted over a short or a longer period of time and measures a sublethal endpoint such as reproduction or growth, and sometimes mortality. Because of the nature of industrial discharges and contaminants of emerging concern (CEC) used in households such as PPCPs, pesticides and PFAS which may be released into the POTW sewershed, toxic constituents in and of itself or in combination may be present in the effluent.

A total of 71 chronic and 26 acute WET tests was conducted on the effluent discharged to Discharge Point 001 (120" outfall) between May 2015 and December 2019. None of the testing results was reported as "Fail" for both acute and chronic toxicity. However, based on Step 13 of the RPA procedures in the Ocean Plan (i.e., best professional judgement), reasonable potential for acute and chronic toxicity has been established for Discharge Point 001 and chronic toxicity for Discharge Point 002 due to potentially toxic chemicals exceeding water quality standards and numerous chemicals having no specific water quality standard. Therefore, the protection of aquatic life through the evaluation of toxicity monitoring is necessary to examine the potential for the individual and/or combination of toxic chemicals that pass through the POTW.

The Ocean Plan addresses the application of chronic and acute toxicity requirements based on minimum probable initial dilutions ( $D_m$ ) for ocean discharges. Following the Ocean Plan, dischargers are required to conduct chronic toxicity monitoring for ocean discharges with  $D_m$  factors ranging from 99 to 349; also, Santa Ana Water Boards or USEPA may require acute toxicity monitoring in addition to chronic toxicity monitoring. Dischargers with  $D_m$  factors below 99 are required to conduct only chronic toxicity testing. Since the  $D_m$  for Discharge Point 001 is 180, both acute and chronic toxicity final effluent limitations have been assigned to Discharge Point 001. Since the  $D_m$  for Discharge Point 002 (i.e.,  $D_m = 36$ ) is less than 99, only chronic toxicity monitoring is specified for Discharge Point 002. No acute toxicity final effluent limitations have been assigned to Discharge Point 002 consistent with 40 CFR § 122.44(d)(1)(v), and because the chronic toxicity final effluent limitation is protective of both chronic and acute toxicity.

The Ocean Plan establishes a daily maximum chronic toxicity objective of  $1.0 \text{ TU}_c = 100/\text{NOEC}$  and a daily maximum acute toxicity objective of  $0.3 \text{ TU}_a = 100/\text{LC}_{50}$ . In 2010, USEPA published a guidance document titled *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, 2010), in which they recommend the following: "Permitting authorities should consider adding the Test of Significant Toxicity (TST) statistical approach to their implementation procedures for analyzing valid WET data for their current NPDES WET Program." The TST hypothesis testing

approach more reliably identifies toxicity—in relation to the chronic (equal to or greater than 0.25) and acute (equal to or greater than 0.20) mean responses of regulatory management decision—than the NOEC hypothesis-testing approach used in the Ocean Plan. The TST statistical approach is the improved approach for addressing statistical uncertainty when used in combination with USEPA's toxicity test methods and is implemented in federal permits issued by USEPA Region 9. Therefore, this Order/Permit includes final effluent limitations for both Discharge Point 001 (120" outfall) and Discharge Point 002 (78" outfall) using the TST hypothesis testing approach. This statistical approach is consistent with the Ocean Plan in that it provides maximum protection to the environment since it more reliably identifies acute and chronic toxicity than the NOEC hypothesis-testing approach (See Chapter III.F and Appendix I of the 2019 California Ocean Plan).

The discharge is subject to determination of "Pass" or "Fail" from acute and/or chronic toxicity tests using the TST statistical t-test approach described in the *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (USEPA 833-R-10-003, 2010), Appendix A, Figure A-1, Table A-1, and Appendix B, Table B-1.

The TST's null hypothesis ( $H_0$ ) for chronic toxicity is:

$H_0$ : Mean discharge In-stream Waste Concentration (IWC) response  $\leq 0.75 \times$  Mean control response.

The TST's null hypothesis ( $H_0$ ) for acute toxicity is:

$H_0$ : Mean discharge In-stream Waste Concentration (IWC) response  $\leq 0.80 \times$  Mean control response.

A test result that rejects this null hypothesis is reported as "Pass." A test result that does not reject this null hypothesis is reported as "Fail."

#### **WQBEL Calculation for Chronic Toxicity (Discharge Points 001 and 002)**

The chronic toxicity WQBEL for Discharge Point 001 is expressed as a null hypothesis ( $H_0$ ) and regulatory management decision (b value) of 0.75 for the chronic toxicity methods in Attachment E – Monitoring and Reporting Program. The chronic toxicity In-stream Waste Concentration (IWC) is  $100/D_m = 100/180 = 0.556\%$  for Discharge Point 001 (120" outfall) and  $IWC = 100/36 = 2.78\%$  for Discharge Point 002 (78" outfall).

The null hypothesis for chronic toxicity at Discharge Point 001 is:

$H_0$ : Mean response (0.556% effluent)  $\leq 0.75$  mean response (Control)

The null hypothesis for chronic toxicity at Discharge Point 002 is:

$H_0$ : Mean response (2.78% effluent)  $\leq 0.75$  mean response (Control)

#### **WQBEL Calculation for Acute Toxicity (Discharge Point 001)**

The acute toxicity WQBEL for Discharge Point 001 is expressed as a null hypothesis ( $H_0$ ) and regulatory management decision (b value) of 0.80 for the acute toxicity methods in Attachment E – Monitoring and Reporting Program. The acute toxicity In-stream Waste Concentration (in % effluent) for Discharge Point 001 is  $100/(0.1 \cdot D_m) = 100/(0.1 \cdot 180) = 5.56\%$ .

The null hypothesis for acute toxicity at Discharge Point 001 is:

$H_0$ : Mean response (5.56% effluent)  $\leq$  0.80 mean response (Control).

Results obtained from evaluating statistically the IWC compared to the control shall be analyzed using the TST hypothesis testing approach in Attachment E – Monitoring and Reporting Program. Compliance with these acute and chronic toxicity WQBELs is demonstrated by rejecting the null hypothesis and reporting “Pass” or “P”. The Maximum Daily Effluent Limitation (MDEL) for acute or chronic toxicity is exceeded and a violation will be flagged when WET test, analyzed using the TST statistical approach, results in “Fail”.

#### **D. Final Effluent Limitation Considerations**

##### **1. Anti-Backsliding Requirements**

Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 CFR § 122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. All effluent limitations and conditions in this Order/Permit are at least as stringent as those in the previous Order/Permit.

This Order/Permit implements different bacteria indicators and water quality objectives based on Chapter II.B.1 of the 2019 Ocean Plan (i.e., State Water Board Water-Contact Objectives for Bacteria). The California Ocean Plan was amended in 2018 to revise the bacterial objectives for ocean waters used for water contact recreation, which includes removal of the previously established total coliform objectives and the revised enterococci objectives based on the EPA 2012 Recreational Criteria for marine waters. This Order/Permit has established bacteria effluent limitations at Discharge Point 002 (78” outfall) based on the amended objectives in the 2019 Ocean Plan. The new effluent limitations are expressed using different averaging periods than the previous limits and are therefore not directly comparable.

##### **2. Antidegradation Policies**

Pursuant to 40 CFR 131.12, the state water quality standards must include an antidegradation policy consistent with the federal antidegradation policy. The State Water Board established California’s antidegradation policy in State Water Board Resolution No. 68-16. This resolution incorporates the federal antidegradation policy, where the federal policy applies under federal law. Resolution No. 68-16 requires that existing water quality be maintained unless it is demonstrated that any degradation is consistent with the maximum benefit to



the people of the State, will not unreasonably affect current or possible beneficial uses, and will not result in water quality less than prescribed in applicable policies.

A complete antidegradation analysis is required if the proposed activity results in a substantial increase in mass emissions of pollutants or if the activity results in significant impact to aquatic life. It is not necessary to do a complete antidegradation analysis if the reduction in water quality will be spatially localized or limited with respect to the waterbody. In such cases, a simple antidegradation analysis will suffice. The Santa Ana Water Board and USEPA believe that the simple antidegradation analysis presented herein is consistent with California and Federal antidegradation policies because effects from effluent discharges are spatially and temporally limited and not expected to have a significant impact on receiving water quality.

This Order/Permit includes both narrative and numeric final effluent limitations, receiving water limitations, performance goals, and mass emission benchmarks to maintain the chemical, physical, and biological characteristics, and to protect the beneficial uses, of the receiving water. These requirements ensure that all water quality objectives are being met outside the zone of initial dilution, thereby maintaining the beneficial uses. The Ocean Plan allows for minimal degradation within the zone of initial dilution as long as the water quality objectives are maintained just outside the zone of initial dilution. The minimal degradation permitted by the Ocean Plan is consistent with the antidegradation policy because it maintains maximum benefit to the people of the State, it will not unreasonably affect the present and anticipated beneficial uses, and it will not result in water quality less than that prescribed in the policies.

The final effluent limitations from the previous Order/Permit have been retained. Bacteria limits at Discharge Point 002 (78" outfall) were updated to implement new State water quality objectives for bacteria in the 2019 Ocean Plan. The new effluent limitations for fecal coliform and enterococcus are consistent with the State's antidegradation policy because the discharge is in compliance with new State Water Board water-contact objectives for bacteria in the Pacific Ocean. In addition, receiving water limitations for bacteria are included to further limit bacteria and to ensure that any increase in the bacteria levels due to cessation of disinfection at Discharge Point 001 (120" outfall) will be maintained below the applicable bacteria water quality standards. Full secondary treatment and these bacteria limitations will ensure that undisinfected effluent discharges will not adversely impact human health and receiving water quality for REC-1 beneficial uses.

The performance goal and mass emission benchmarks are additional incentives for the Discharger to maintain the current treatment quality since they set final effluent targets for the Discharger to meet based on current performance. Most mass emission benchmarks in this Order/Permit are more stringent due to improved performance by full secondary treatment and a basis of lower flow rate (i.e., the average daily influent flow of 206 MGD projected for 2025); however, the

mass emission benchmarks for some constituents (e.g., selenium) have increased due to increased concentrations in the influent and/or more sensitive analysis methods. Since the mass emission benchmarks are based on actual performance and do not exceed the water quality objectives for the receiving water, the increase of any mass emission benchmarks is not expected to result in additional degradation.

Compliance with all permit-specified concentration-based and mass emission-based limits, performance goals, mass emission benchmarks, and the secondary treatment requirements for final effluent discharges in the Order/Permit will result in the use of best practicable treatment or control of the discharge necessary to assure that a pollution or nuisance will not occur and that the highest water quality consistent with the maximum benefit to the people of the state will be maintained. Additionally, the secondary treatment facilities would support the Groundwater Replenishment System, reduce the effluent discharge volumes to the ocean, and reduce the need for emergency use of Discharge Point 002 (78" short outfall). Therefore, it is believed that these changes would represent socioeconomic and public benefits, and the Discharger meets the goals of the antidegradation policy. Based on these considerations, this Order/Permit is consistent with state and federal antidegradation requirements.

### **3. Stringency of Requirements for Individual Pollutants**

This Order contains both technology-based and water quality-based effluent limitations for individual pollutants. The technology-based effluent limitations consist of restrictions on biochemical oxygen demand (5-day), carbonaceous biochemical oxygen demand (5-day), total suspended solids, and pH which implement the minimum, applicable federal technology-based requirements for POTWs. Also, effluent limitations consisting of restrictions on oil and grease, settleable solids, and turbidity more stringent than federal technology-based limitations are necessary to implement State treatment standards in Ocean Plan Table 4.

Water quality-based effluent limitations (WQBELs) have been derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. The procedures for calculating the individual WQBELs are based on the Ocean Plan, which was approved by USEPA on February 14, 2006 and has since been further amended. All beneficial uses and water quality objectives contained in the Basin Plan were approved under state law and submitted to and approved by USEPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to USEPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 CFR § 131.21(c)(1).

Discharge Point 001 WQBELs consisting of restrictions on total chlorine residual, acute toxicity, chronic toxicity, radioactivity, benzidine, hexachlorobenzene,

PCBs, TCDD equivalents, and toxaphene more stringent than federal technology-based limitations are necessary to meet State water quality standards in the Ocean Plan. Discharge Point 002 WQBELs consisting of restrictions on ammonia (as N), total chlorine residual, fecal coliform density, *Enterococcus* density, chronic toxicity, radioactivity, and TCDD equivalents more stringent than federal technology-based limitations are necessary to meet State water quality standards in the Ocean Plan. All effluent limitations are discussed in Attachment F-Fact Sheet. Collectively, the restrictions on individual pollutants in this Order/Permit are no more stringent than required to implement the requirements of the CWA.

**E. Final Effluent Limitations for Discharge Points 001 and 002**

**Table F-14. Summary of Effluent Limitations – Discharge Point 001 (120" outfall)**

| Parameter                                                                                | Units                    | Effluent Limitations <sup>[1],[2]</sup>                      |                               |                                  |                                          |                                 | Basis                                                    |
|------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------|-------------------------------|----------------------------------|------------------------------------------|---------------------------------|----------------------------------------------------------|
|                                                                                          |                          | Average Monthly <sup>[3]</sup>                               | Average Weekly <sup>[3]</sup> | Maximum Daily <sup>[3],[4]</sup> | Instantaneous Maximum <sup>[3],[4]</sup> | Six-Month Median <sup>[3]</sup> |                                                          |
| Carbonaceous Biochemical Oxygen Demand, 5-day @ 20°C (CBOD <sub>5</sub> ) <sup>[5]</sup> | mg/L                     | 25                                                           | 40                            | –                                | –                                        | –                               | Existing/<br>Secondary treatment standard                |
|                                                                                          | lbs/day                  | 42,951                                                       | 68,722                        | –                                | –                                        | –                               |                                                          |
|                                                                                          | % removal <sup>[6]</sup> | ≥ 85                                                         | –                             | –                                | –                                        | –                               |                                                          |
| Total Suspended Solids (TSS)                                                             | mg/L                     | 30                                                           | 45                            | –                                | –                                        | –                               | Existing/<br>Secondary treatment standard                |
|                                                                                          | lbs/day                  | 51,541                                                       | 77,312                        | –                                | –                                        | –                               |                                                          |
|                                                                                          | % removal <sup>[6]</sup> | ≥ 85                                                         | –                             | –                                | –                                        | –                               |                                                          |
| pH                                                                                       | standard units           | 6.0 (instantaneous minimum)<br>– 9.0 (instantaneous maximum) |                               |                                  |                                          |                                 | Existing/<br>Secondary treatment standard/<br>Ocean plan |
| Oil and Grease                                                                           | mg/L                     | 25                                                           | 40                            | –                                | 75                                       | –                               | Existing/<br>Ocean Plan                                  |
|                                                                                          | lbs/day                  | 42,951                                                       | 68,722                        | –                                | 128,853                                  | –                               |                                                          |
| Settleable Solids                                                                        | ml/L                     | 1.0                                                          | 1.5                           | –                                | 3.0                                      | –                               | Existing/<br>Ocean Plan                                  |
| Turbidity                                                                                | NTU                      | 75                                                           | 100                           | –                                | 225                                      | –                               | Existing/<br>Ocean Plan                                  |
| Total chlorine residual                                                                  | mg/L                     | –                                                            | –                             | 1.45                             | 10.86                                    | 0.36                            | RP                                                       |
|                                                                                          | lbs/day                  | –                                                            | –                             | 2,491                            | 18,658                                   | 618                             |                                                          |

| Parameter                       | Units        | Effluent Limitations <sup>[1],[2]</sup>                                                                                                                                                                                                                                                                  |                               |                                  |                                          |                                 | Basis                        |
|---------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------|------------------------------------------|---------------------------------|------------------------------|
|                                 |              | Average Monthly <sup>[3]</sup>                                                                                                                                                                                                                                                                           | Average Weekly <sup>[3]</sup> | Maximum Daily <sup>[3],[4]</sup> | Instantaneous Maximum <sup>[3],[4]</sup> | Six-Month Median <sup>[3]</sup> |                              |
| Acute toxicity <sup>[7]</sup>   | Pass or Fail | —                                                                                                                                                                                                                                                                                                        | —                             | Pass                             | —                                        | —                               | RP/BPJ                       |
| Chronic toxicity <sup>[7]</sup> | Pass or Fail | —                                                                                                                                                                                                                                                                                                        | —                             | Pass                             | —                                        | —                               | RP/BPJ                       |
| Radioactivity                   | pCi/L        | Not to exceed limits specified in Title 17, Division 1, Chapter 5, Subchapter 4, Group 3, Article 3, Section 30253 of the California Code of Regulations. Reference to Section 30253 is prospective, including future changes to any incorporated provisions of federal law, as the changes take effect. |                               |                                  |                                          |                                 | RP/BPJ Ocean Plan            |
| Benzidine                       | µg/L         | 0.0125                                                                                                                                                                                                                                                                                                   | —                             | —                                | —                                        | —                               | Inconclusive/<br>Carry-over; |
|                                 | lbs/day      | 0.0215                                                                                                                                                                                                                                                                                                   | —                             | —                                | —                                        | —                               |                              |
| Hexachloro-benzene              | µg/L         | 0.0380                                                                                                                                                                                                                                                                                                   | —                             | —                                | —                                        | —                               | Inconclusive/<br>Carry-over; |
|                                 | lbs/day      | 0.0653                                                                                                                                                                                                                                                                                                   | —                             | —                                | —                                        | —                               |                              |
| Toxaphene                       | µg/L         | 0.0380                                                                                                                                                                                                                                                                                                   | —                             | —                                | —                                        | —                               | Inconclusive/<br>Carry-over; |
|                                 | lbs/day      | 0.0653                                                                                                                                                                                                                                                                                                   | —                             | —                                | —                                        | —                               |                              |
| PCBs <sup>[3]</sup>             | µg/L         | 0.0034                                                                                                                                                                                                                                                                                                   | —                             | —                                | —                                        | —                               | Inconclusive/<br>Carry-over; |
|                                 | lbs/day      | 0.0058                                                                                                                                                                                                                                                                                                   | —                             | —                                | —                                        | —                               |                              |
| TCDD Equivalents <sup>[3]</sup> | pg/L         | 0.7059                                                                                                                                                                                                                                                                                                   | —                             | —                                | —                                        | —                               | Inconclusive/<br>Carry-over  |
|                                 | lbs/day      | 0.0000012                                                                                                                                                                                                                                                                                                | —                             | —                                | —                                        | —                               |                              |

- <sup>[1]</sup> Mass emission effluent limitations (in lbs/day) are based on the average daily influent flow of 206 MGD projected for 2025, taken from the Discharger's 2017 Master Plan.
- <sup>[2]</sup> The minimum probable initial dilution used to calculate WQBELs for Ocean Plan Table 3 pollutants is 181:1 ( $D_m = 180$ ).
- <sup>[3]</sup> See Attachment A – Definitions.
- <sup>[4]</sup> The maximum daily effluent limitations shall apply to flow weighted 24-hour composite samples. The instantaneous maximum effluent limitations shall apply to grab samples.
- <sup>[5]</sup> In lieu of the parameter BOD<sub>5</sub> and the BOD<sub>5</sub> levels specified for effluent quality in 40 CFR § 133.102(a)(1), (2), and (3), the parameter CBOD<sub>5</sub> and the CBOD<sub>5</sub> levels specified for effluent quality in 40 CFR § 133.102(a)(4) are substituted and reported by the Discharger.
- <sup>[6]</sup> In each calendar month, the arithmetic mean of CBOD<sub>5</sub> and TSS, by mass, of effluent samples collected at Monitoring Location EFF-001 as described in the MRP, shall not exceed 15 percent of the arithmetic mean of CBOD<sub>5</sub> and TSS, by mass, of influent samples

collected at Monitoring Locations INF-001 and INF-002 as described in the MRP, at approximately the same times during the same periods.

- [7] The maximum daily effluent limitation shall be reported as “Pass” or “Fail”, and “% Effect”. See section V of Attachment E.

**Table F-15. Summary of Effluent Limitations – Discharge Point 002 (78” outfall)**

| Parameter                                                                               | Units                    | Effluent Limitations <sup>[1],[2]</sup>                      |                               |                                  |                                          |                                 | Basis                                              |
|-----------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------|-------------------------------|----------------------------------|------------------------------------------|---------------------------------|----------------------------------------------------|
|                                                                                         |                          | Average Monthly <sup>[3]</sup>                               | Average Weekly <sup>[3]</sup> | Maximum Daily <sup>[3],[4]</sup> | Instantaneous Maximum <sup>[3],[4]</sup> | Six-Month Median <sup>[3]</sup> |                                                    |
| Carbonaceous Biochemical Oxygen Demand 5-day @ 20°C (CBOD <sub>5</sub> ) <sup>[5]</sup> | mg/L                     | 25                                                           | 40                            | –                                | –                                        | –                               | Existing/ Secondary treatment standard             |
|                                                                                         | lbs/day                  | 47,955                                                       | 76,728                        | –                                | –                                        | –                               |                                                    |
|                                                                                         | % removal <sup>[6]</sup> | ≥ 85                                                         | –                             | –                                | –                                        | –                               |                                                    |
| Total Suspended Solids (TSS)                                                            | mg/L                     | 30                                                           | 45                            | –                                | –                                        | –                               | Existing/ Secondary treatment standard             |
|                                                                                         | lbs/day                  | 57,546                                                       | 86,319                        | –                                | –                                        | –                               |                                                    |
|                                                                                         | % removal <sup>[6]</sup> | ≥ 85                                                         | –                             | –                                | –                                        | –                               |                                                    |
| pH                                                                                      | standard units           | 6.0 (instantaneous minimum)<br>– 9.0 (instantaneous maximum) |                               |                                  |                                          |                                 | Existing/ Secondary treatment standard/ Ocean plan |
| Oil and Grease                                                                          | mg/L                     | 25                                                           | 40                            | –                                | 75                                       | –                               | Existing/ Carry-over; Ocean Plan                   |
|                                                                                         | lbs/day                  | 47,955                                                       | 76,728                        | –                                | 143,865                                  | –                               |                                                    |
| Settleable Solids                                                                       | ml/L                     | 1.0                                                          | 1.5                           | –                                | 3.0                                      | –                               | Existing/ Carry-over; Ocean Plan                   |
| Turbidity                                                                               | NTU                      | 75                                                           | 100                           | –                                | 225                                      | –                               | Existing/ Carry-over; Ocean Plan                   |
| Total chlorine residual                                                                 | mg/L                     | –                                                            | –                             | 0.296                            | 2.22                                     | 0.074                           | RP                                                 |
|                                                                                         | lbs/day                  | –                                                            | –                             | 568                              | 4,258                                    | 142                             |                                                    |
| Ammonia as N                                                                            | mg/L                     | –                                                            | –                             | 88.8                             | 222                                      | 22.2                            | RP                                                 |
|                                                                                         | lbs/day                  | –                                                            | –                             | 170,336                          | 425,840                                  | 42,584                          |                                                    |

| Parameter                       | Units                      | Effluent Limitations <sup>[1],[2]</sup>                                                                                                                                                                                                                                                                  |                               |                                  |                                          |                                 | Basis                     |
|---------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------|------------------------------------------|---------------------------------|---------------------------|
|                                 |                            | Average Monthly <sup>[3]</sup>                                                                                                                                                                                                                                                                           | Average Weekly <sup>[3]</sup> | Maximum Daily <sup>[3],[4]</sup> | Instantaneous Maximum <sup>[3],[4]</sup> | Six-Month Median <sup>[3]</sup> |                           |
| Fecal coliform density          | MPN /100 mL <sup>[7]</sup> | A 30-day geometric mean of fecal coliform density not to exceed 7,400 MPN/100 mL; and a single sample maximum not to exceed 14,800 MPN/100 mL.                                                                                                                                                           |                               |                                  |                                          |                                 | RP/BPJ Ocean Plan         |
| Enterococcus density            | CFU /100 mL <sup>[7]</sup> | A six-week rolling geometric mean of enterococcus not to exceed 1,110 CFU/100 mL; and a statistical threshold value (STV), corresponding to the 90 <sup>th</sup> percentile of all enterococcus samples collected in a calendar month, not to exceed of 4,070 CFU/100mL.                                 |                               |                                  |                                          |                                 | RP/BPJ Ocean Plan         |
| Chronic toxicity <sup>[8]</sup> | Pass or Fail               | –                                                                                                                                                                                                                                                                                                        | –                             | Pass                             | –                                        | –                               | RP/BPJ                    |
| Radioactivity                   | pCi/L                      | Not to exceed limits specified in Title 17, Division 1, Chapter 5, Subchapter 4, Group 3, Article 3, Section 30253 of the California Code of Regulations. Reference to Section 30253 is prospective, including future changes to any incorporated provisions of federal law, as the changes take effect. |                               |                                  |                                          |                                 | RP/BPJ Ocean Plan         |
| TCDD Equivalents <sup>[3]</sup> | pg/L                       | 0.14430                                                                                                                                                                                                                                                                                                  | –                             | –                                | –                                        | –                               | Inconclusive/ Carry-over; |
|                                 | lbs/day                    | 0.00000 028                                                                                                                                                                                                                                                                                              | –                             | –                                | –                                        | –                               |                           |

<sup>[1]</sup> Mass emission effluent limitations (in lbs/day) are based on the Discharger's conservative estimate for hydraulic flow capacity of the outfall of 230 MGD.

<sup>[2]</sup> The minimum probable initial dilution used to calculate WQBELs for Ocean Plan Table 3 pollutants and bacteria is 37:1 ( $D_m = 36$ ).

<sup>[3]</sup> See Attachment A – Definitions.

<sup>[4]</sup> The maximum daily effluent limitations shall apply to flow weighted 24-hour composite samples. The instantaneous maximum effluent limitations shall apply to grab samples.

<sup>[5]</sup> In lieu of the parameter BOD<sub>5</sub> and the BOD<sub>5</sub> levels specified for effluent quality in 40 CFR § 133.102(a)(1), (2), and (3), the parameter CBOD<sub>5</sub> and the CBOD<sub>5</sub> levels specified for effluent quality in 40 CFR § 133.102(a)(4) are substituted and reported by the Discharger.

<sup>[6]</sup> In each calendar month, the arithmetic mean of CBOD<sub>5</sub> and TSS, by mass, of effluent samples collected at Monitoring Location EFF-002 or EMG-001 as described in the MRP, shall not exceed 15 percent of the arithmetic mean of CBOD<sub>5</sub> and TSS, by mass, of influent samples collected at Monitoring Locations INF-001 and INF-002 as described in the MRP, at approximately the same times during the same periods.

<sup>[7]</sup> Results may be reported as either MPN/100 mL if the laboratory method used provides results in MPN/100 mL or CFU/100 mL if the laboratory method used provides results in CFU/100 mL.



- [8] The maximum daily effluent limitation shall be reported as “Pass” or “Fail”, and “% Effect”.  
See section V of Attachment E.

**F. Interim Effluent Limitations – Not Applicable**

**G. Land Discharge Specifications – Not Applicable**

**H. Recycling Specifications – Not Applicable**

**V. PERFORMANCE GOAL AND MASS EMISSION BENCHMARKS**

**A. Performance Goal**

Section III.F.1, of the 2019 Ocean Plan allows the Santa Ana Water Board to establish more restrictive water quality objectives and effluent limitations than those set forth in the Ocean Plan as necessary for the protection of the beneficial uses of ocean waters.

Pursuant to this provision, performance goals that are more stringent than those based on Ocean Plan objectives are prescribed in this Order/Permit. This approach is consistent with the antidegradation policy in that it requires the Discharger to maintain its treatment level and effluent quality, recognizing normal variations in treatment efficiency and sampling and analytical techniques. However, this approach does not address substantial changes in treatment plant operations that could significantly affect the quality of the treated effluent.

While performance goals were previously placed in many State-issued NPDES permits for POTWs, they have been discontinued for inland surface water discharges. For inland surface waters, the California Toxics Rule (40 CFR § 131.38) has resulted in effluent limitations as stringent as many performance goals. However, the Ocean Plan allows for significant dilution, and the continued use of performance goals serves to maintain existing treatment levels and effluent quality and supports State and federal antidegradation policies.

The performance goals are based upon the last 5-year actual performance of the OC San’s treatment plants and are specified only as an indication of the treatment efficiency of the Facility. The performance goals are not considered enforceable effluent limitations or standards for the regulation of discharge from the treatment facility. They are intended to minimize pollutant loading (primarily for toxics), while maintaining the incentive for future voluntary improvement of water quality whenever feasible, without the imposition of more stringent limits based on improved performance.

The Discharger shall maintain existing treatment levels and the effluent quality at or below the performance goal concentrations. Any two consecutive exceedances of the performance goals shall trigger an investigation into the cause of the exceedance. If the exceedance persists in three successive monitoring periods, the Discharger shall submit a written report to the Santa Ana Water Board and USEPA on the nature of the exceedance and the results of the investigation including the cause of the exceedance.

The Santa Ana Water Board and USEPA recognize that OC San is going through upgrades of treatment plant operations including the GWRS final expansion and

participating in collaborative studies and activities with regional partnerships. This Order/Permit may be reopened to modify any of the performance goals if the Discharger requests and has demonstrated that the change is warranted, including results of completion of the GWRS final expansion.

### **Procedures for the Determination of Performance Goals**

1. For constituents that have been routinely detected in the effluent (at least 20 percent detectable data), performance goals are based on the one-sided, upper 95 percent confidence bound for the 95<sup>th</sup> percentile of the effluent performance data ( $UCB_{95/95}$ ) from May 2015 through December 2019 using the RPA protocol contained in the 2019 Ocean Plan. Effluent data are assumed log normally distributed. Performance goals are calculated according to the equation  $C_{PG} = C_o + D_m(C_o - C_s)$  and setting  $C_o = UCB_{95/95}$ .
  - a. If the maximum detected effluent concentration (MEC) is greater than the calculated performance goal ( $C_{PG}$ ), then the calculated performance goal ( $C_{PG}$ ) is used as the performance goal; or
  - b. If the MEC is less than the calculated performance goal ( $C_{PG}$ ), then the MEC is used as the performance goal.
  - c. If the performance goal determined in part a or b above is greater than the Water Quality Objective (WQO) in the 2019 Ocean Plan after considering dilution, then the WQO is used as the performance goal.

For example, the performance goals for arsenic and chloroform at Discharge Point 001 (120" outfall) are calculated as follows:

#### **Arsenic**

$C_o = UCB_{95/95} = 3.02 \mu\text{g/L}$  ;  $D_m = 180$ ;  $C_s$  = background seawater concentration =  $3 \mu\text{g/L}$ ;  $MEC = 6.91 \mu\text{g/L}$ ;  $C_{PG}$  = Performance Goal =  $(3.02 \mu\text{g/L}) + 180 \cdot (3.02 \mu\text{g/L} - 3 \mu\text{g/L}) = 6.62 \mu\text{g/L}$ .

Since the MEC of  $6.91 \mu\text{g/L}$  is greater than the calculated PG of  $6.62 \mu\text{g/L}$ , the prescribed performance goal for arsenic is  $6.62 \mu\text{g/L}$ .

#### **Chloroform**

$C_o = UCB_{95/95} = 3.285 \mu\text{g/L}$  ;  $D_m = 180$ ;  $C_s$  = background seawater concentration =  $0 \mu\text{g/L}$ ;  $MEC = 16.6 \mu\text{g/L}$ ;  $C_{PG}$  = Performance Goal =  $(3.285 \mu\text{g/L}) + 180 \cdot (3.285 \mu\text{g/L} - 0 \mu\text{g/L}) = 594.6 \mu\text{g/L}$ .

Since the MEC of  $16.6 \mu\text{g/L}$  is less than the calculated PG of  $594.6 \mu\text{g/L}$ , the prescribed performance goal for chloroform is  $16.6 \mu\text{g/L}$ .

2. For constituents where monitoring data have consistently shown nondetectable levels (less than 20 percent detectable data), performance goals are set at the multiplying factor of 2.13 times the Minimum Levels (ML) used for analysis, which is based on statistical procedures outlined in USEPA manual, the *Technical Support Document for Water Quality-Based Toxics Control* (TSD, 1991). These performance goals are estimated assuming a coefficient of variation of 0.6 and the

95 percent confidence interval of the 95<sup>th</sup> percentile based on an assumed lognormal distribution of daily efficient values (see Table 5-2 of USEPA's TSD)

- a. If the MEC is greater than the calculated performance goal based on ML (i.e.,  $2.13 \times \text{ML}$ ), then the calculated performance goal (i.e.,  $2.13 \times \text{ML}$ ) is used as the performance goal; or
- d. If the MEC is less than the calculated performance goal based on ML (i.e.,  $2.13 \times \text{ML}$ ), then the MEC is used as the performance goal.
- e. If the performance goal determined in part a or b above is greater than the Water Quality Objective (WQO) in the 2019 Ocean Plan after considering dilution, then the WQO is used as the performance goal.

For example, the performance goals for halomethanes and heptachlor at Discharge Point 001 (120" outfall) are calculated as follows:

**Halomethanes (5 detected out of 54 samples)**

$\text{MEC} = 1.37 \mu\text{g/L}$ ;  $\text{ML} = 0.2 \mu\text{g/L}$ ;  $\text{C}_{\text{PG}} = \text{Performance Goal} = 2.13 \times 0.2 = 0.43 \mu\text{g/L}$ .

Since the MEC of  $1.37 \mu\text{g/L}$  is greater than the calculated PG of  $0.43 \mu\text{g/L}$ , the prescribed performance goal for halomethanes is  $0.43 \mu\text{g/L}$ .

**Heptachlor (all nondetected)**

$\text{MEC} = \text{NA}$  (all ND);  $\text{ML} = 0.01 \mu\text{g/L}$ ;  $\text{C}_{\text{PG}} = \text{Performance Goal} = 2.13 \times 0.01 = 0.012 \mu\text{g/L}$ .

$\text{C}_o = \text{WQO} = 0.00005 \mu\text{g/L}$ ;  $\text{D}_m = 180$ ;  $\text{C}_s = \text{background seawater concentration} = 0 \mu\text{g/L}$ ;  $\text{C}_{\text{PG}} = \text{Performance Goal} = (0.00005 \mu\text{g/L}) + 180 \times (0.00005 \mu\text{g/L} - 0 \mu\text{g/L}) = 0.009 \mu\text{g/L}$ .

Since the performance goal based on ML of  $0.012 \mu\text{g/L}$  is greater than the calculated WQO-based performance goal of  $0.009 \mu\text{g/L}$ , so the prescribed performance goal for heptachlor is  $0.009 \mu\text{g/L}$ .

- 3. For constituents with effluent limitations, if the performance goal derived from the steps, above, exceeds respective effluent limitation, then a performance goal is not prescribed for that constituent.

**B. Mass Emission Benchmarks**

To address the uncertainty due to potential increases in toxic pollutant loadings from the discharge to the marine environment during the five-year Order/Permit term, and to establish a framework for evaluating the need for an antidegradation analysis to determine compliance with State and federal antidegradation requirements at the time of Order/Permit reissuance, 12-month average mass emission benchmarks have been established for effluent discharged through Discharge Point 001 (120" outfall). These mass emission benchmarks are not enforceable water quality-based effluent limitations. They may be re-evaluated and revised during the five-year Order/Permit term.

The mass emission benchmarks (in metric tons per year; MT/yr) for the discharge were re-evaluated using the same procedures as described above in Section V.A of this Fact Sheet for the calculation of the Performance Goals. The concentration-based Performance Goals were calculated using effluent monitoring data from May 2015 through December 2019 and were converted to mass-based Benchmarks using the Discharger's annual average influent flow of 206 MGD projected for 2025. The following equation was used for the calculation of the Mass Emission Benchmarks:

Mass Emission Benchmark (MT/yr) = ( $C_{PG}$   $\mu\text{g/L}$ )  $\times$  ( $Q$  gal/day)  $\times$  (3.785 L/gal)  $\times$  (365 days/yr)  $\times$  (1 MT/ $10^{12}$   $\mu\text{g}$ )

where,

$C_{PG}$ : Final performance goal calculated from Section V.A of the fact sheet ( $\mu\text{g/L}$ )

$Q$ : Annual average influent flow projected for 2025 =  $206 \times 10^6$  gal/day

Most mass emission benchmarks in this Order/Permit are more stringent due to improved performance by completion of full secondary treatment facilities, the use of more sensitive analytical method with lower minimum levels, implementation of an extensive source control pretreatment program, and decreased discharge flowrate (i.e., the projected average daily influent flow reduced from 274 MGD to 206 MGD).

## **VI. RATIONALE FOR RECEIVING WATER LIMITATIONS**

### **A. Surface Water**

The Ocean Plan and Basin Plan contain numeric and narrative water quality standards applicable to surface waters of California. Water quality objectives include a policy to maintain the high-quality waters pursuant to federal regulations (40 CFR § 131.12) and State Water Board Resolution No. 68-16. Receiving water limitations in the Order/Permit are based on the water quality objectives contained in the Ocean Plan chapters II.B, II.C, II.D, II.E, and II.F.

The Ocean Plan was amended in 2018 to revise the bacterial water quality objectives for ocean waters used for water contact recreation, which includes removal of the previously established total coliform objectives and the revised enterococci objectives based on the EPA 2012 Recreational Criteria for marine waters. The previously established fecal coliform objective has been retained as recent epidemiological studies conducted at southern California beaches showed that fecal coliform may be a better indicator of gastrointestinal illness than enterococci during certain types of exposures and environmental conditions. This Order/Permit has established receiving water limitations for bacteria based on these amended objectives in the 2019 Ocean Plan. Bacteria receiving water limitations are necessary to protect human health and the water contact recreation uses of the receiving water.

The Santa Ana Water Board and USEPA has determined that bacterial indicator standards for water contact recreation are applied throughout the water column of the Nearshore Zone (i.e., designated REC-1). Based on Chapter II.B.1 of the Ocean Plan, bacterial indicator standards shall also be maintained throughout the water

column in the Offshore Zone used for water contact sports, as determined by the Santa Ana Water Board and/or USEPA (i.e., waters designated as REC-1), to assure that the discharge does not pose a threat to water contact recreation. Receiving water limitations for enterococcus density in ocean waters beyond the outer limit of the State of California territorial marine waters are based on CWA section 304(a) water quality criteria (i.e., Recreational Water Quality Criteria). These criteria must be achieved beyond the zone of initial dilution in areas where primary contact recreation, as defined in USEPA guidance, occurs. USEPA describes the “primary contact recreation” use as protective when the potential for ingestion of, or immersion in, water is likely. Activities usually include swimming, water-skiing, skin-diving, surfing, and other activities likely to result in immersion (Water Quality Standards Handbook, EPA-823-B-94-005a, 1994, p. 2-2). The nature and extent of primary contact recreational use in federal waters is recorded and reported during water quality monitoring conducted under the receiving water core monitoring program.

**B. Groundwater – Not Applicable**

**VII. RATIONALE FOR PROVISIONS**

**A. Standard Provisions**

Standard Provisions, which apply to all NPDES permits in accordance with 40 CFR § 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 CFR § 122.42, are provided in Attachment D of this Order/Permit.

40 CFR § 122.41(a)(1) and (b) through (n) establish conditions that apply to all State-issued NPDES permits. The Order/Permit also incorporates federal conditions that address enforcement authority specified in 40 CFR § 122.41(a)(2) and (3). These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order/Permit.

40 CFR § 123.25(a)(12) allows the state to omit or modify conditions to impose more stringent requirements. In accordance with 40 CFR § 123.25, this Order/Permit omits federal conditions that address enforcement authority specified in 40 CFR § 122.41(j)(5) and (k)(2) because the enforcement authority under the Water Code is more stringent. In lieu of these conditions, this Order/Permit incorporates by reference Water Code section 13387(e).

**B. Special Provisions**

**1. Reopener Provisions**

The reopener provisions are based on 40 CFR 122.44(c) and 40 CFR § 123.25. The Santa Ana Water Board and USEPA may reopen the Order/Permit to modify conditions and requirements for cause. Causes for modification can include, but are not limited to, the promulgation of new regulations, modification in sludge use or disposal practices, or adoption of new regulations by the State Water Board,



Santa Ana Water Board, or USEPA including revisions to the Ocean Plan and Basin Plan.

## **2. Southern California Bight Monitoring Exchange**

The MRP (Attachment E) may be modified by the Santa Ana Water Board and USEPA to enable the Discharger to participate in comprehensive regional monitoring activities conducted in the Southern California Bight during the term of this permit. The intent of regional monitoring activities is to maximize the efforts of all monitoring partners using a cost-effective monitoring design and to best utilize the pooled scientific resources of the region. During these coordinated monitoring efforts, the Discharger's sampling and analytical effort may be reallocated to provide a regional assessment of the impact of wastewater discharges to the Southern California Bight; however, certain core elements (i.e., monthly water quality monitoring, quarterly REC-1 water quality monitoring, quarterly benthic monitoring, semi-annual trawl fish monitoring, and weekly Orange County Regional Shoreline REC-1 cooperative monitoring) shall remain unchanged. Anticipated modifications to the monitoring program will be coordinated so as to provide a comprehensive picture of the ecological and statistical significance of monitoring results and to determine cumulative impacts of various pollutant sources. If predictable relationships among the biological, water quality and effluent monitoring variables can be demonstrated, it may be appropriate to decrease the Discharger's monitoring effort. Conversely, the monitoring program may be intensified if it appears that the objectives cannot be achieved through the Discharger's existing monitoring program. These changes will improve the overall effectiveness of monitoring in the Southern California Bight. Minor changes may be made without further public notice.

## **3. Special Studies, Technical Reports, and Additional Monitoring Requirements**

### **a. Toxicity Reduction Requirements**

This Order/Permit requires the Discharger to develop procedures to conduct Toxicity Identification and Reduction Evaluations. This provision is based on Chapter III.C.10 of the Ocean Plan. If the discharge consistently exceeds an effluent limitation for toxicity as specified in this Order/Permit, the Discharger shall conduct a Toxicity Reduction Evaluation (TRE) as detailed in section VII.C.3.a of the Order/Permit. The TRE will help the Discharger identify the possible source(s) of toxicity. Once the source(s) of toxicity is identified, the Discharger shall take all reasonable steps to reduce toxicity to the required level.

### **b. Dilution Model Update**

This Permit/Order requires the Discharger to complete an updated dilution analysis for all outfalls as part of its permit application for the next permit reissuance. The Discharger shall update its dilution modeling that is reflective of the future operations upon completion of the GWRS final expansion



project, which includes, but not limited to the increase in RO concentrate and density and lower discharge flow rate. The updated dilution analysis must continue to use near-field modeling to consider how the effluent plumes behave in the zone of initial dilution (ZID) and demonstrate the impacts of final GWRS expansion to dilution under a “reasonable worst-case scenario” (e.g., critical stratification condition).

**c. Ocean Outfall Condition Assessment and Scoping Study**

As the 120-inch outfall approaches the end of life in 2021, the Discharger is undertaking a condition assessment of the 120-inch ocean outfall (referred to as PS18-09). The project includes removal of accumulated biofouling (i.e., barnacle collar) surrounding the 504 diffuser ports, removal/replacement of manhole covers, the full-length underwater interior inspection of the 120-inch outfall pipe, 3-D scanning and modeling, and analysis of low flow impacts on the diffuser functionality. The first field activity for the 120” inch outfall assessment project (i.e., removal of the barnacles from the outfall ports) was started on July 20, 2020 after the receipt of the Army Corp. of Engineers’ permit. OC San also plans to conduct a similar study for Discharge Point 002 (78” outfall) using lessons learned from this 120” outfall condition assessment and scoping study.

Upon completion of each outfall condition assessment project, a summary report of the field work findings, including videographic and/or photographic record of the interior of the outfall, 3-D mapping, and any environmental impacts caused during the field works shall be submitted to the Santa Ana Water Board and USEPA. In addition, a separate analysis report of the impacts of low flows on diffuser hydraulics and plugging shall be provided.

**4. Best Management Practices and Pollution Prevention**

**a. Pollution Minimization Program**

This provision is based on the requirements of Chapter III.C.9 of the 2019 Ocean Plan. The goal of the Pollutant Minimization Program is to reduce all potential sources of a pollutant through pollutant minimization strategies, including pollution prevention measures, in order to maintain the effluent concentration at or below the effluent limitation.

**b. Storm Water Management Plan**

This Order/Permit requires the Discharger to update its Storm Water Management Plan to implement the 2015 trash amendments of the Ocean Plan.

**5. Construction, Operation, and Maintenance Specifications**

This provision is based on the requirements of 40 CFR §122.41(e) and the previous Order/Permit.

**a. Roster of Personnel**

The Discharger shall update and report annually a roster of personnel who supervises and operates the wastewater treatment plants.

**b. Operation and Maintenance Manual**

This Order/Permit requires the Discharger to update an Operation and Maintenance Manual prior to start of operations and specifies its periodic updates.

**c. Spill Preventive and Contingency Plan (SPCP)**

Since spills or overflows are a common event at the POTW, this Order/Permit requires the Discharger to review and update, if necessary, its SPCP whenever there is a change which materially affects the potential for spills or after each incident. The Discharger shall ensure that the up-to-date SPCP is readily available to the sewage system personnel at all times and that the sewage personnel are familiar with it.

**6. Special Provisions for Publicly-Owned Treatment Works (POTWs)**

**a. Biosolids**

To implement CWA section 405(d), on February 19, 1993, USEPA promulgated 40 CFR § 503 to regulate the use and disposal of municipal sewage sludge. This regulation was amended on September 3, 1999. The regulation requires that producers of sewage sludge meet certain reporting, handling, and disposal requirements. It is the responsibility of the Discharger to comply with said regulations that are enforceable by USEPA, because California has not been delegated the authority to implement this biosolid program.

**b. Pretreatment**

This Order/Permit contains pretreatment requirements consistent with applicable effluent limitations, national standards of performance, and toxic and performance effluent standards established pursuant to Sections 208(b), 301, 302, 303(d), 304, 306, 307, 403, 404, 405, and 501 of the CWA, and amendments thereto. This permit contains requirements for the implementation of an effective pretreatment program pursuant to Section 307 of the CWA; 40 CFR § 35 and 403; and/or Section 2233, Title 23, California Code of Regulations.

**c. Collection System**

The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order 2006-0003-DWQ (General Order) on May 2, 2006. The State Water Board amended the Monitoring and Reporting Program for the General Order through Order WQ 2013-0058-EXEC on August 6, 2013. The General Order requires public agencies that own or operate sanitary sewer systems with sewer lines one mile of pipe or greater to enroll for coverage and comply with the General Order. The General Order requires agencies to develop sanitary sewer management

plans and report all sanitary sewer overflows, among other requirements and prohibitions.

The General Order contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows that are more extensive, and therefore, more stringent than the requirements under federal standard provisions. The Discharger and public agencies that are discharging wastewater into the facility's collection system were required to obtain enrollment for regulation under the General Order by December 1, 2006.

d. **Resource Recovery from Anaerobically Digestible Material.**

Some POTWs choose to accept organic material such as food waste, fats, oils, and grease into their anaerobic digesters for co-digestion to increase production of methane and other biogases for energy production and to prevent such materials from being discharged into the collection system, which could cause sanitary sewer overflows. The California Department of Resources Recycling and Recovery has proposed an exemption from requiring Process Facility/Transfer Station permits where this activity is regulated under waste discharge requirements or NPDES permits. The proposed exemption is restricted to anaerobically digestible material that has been prescreened, slurried, and processed/conveyed in a closed system to be co-digested with regular POTW sludge. The proposed exemption requires that a POTW develop Standard Operating Procedures for the proper handling, processing, tracking, and management of the anaerobically digestible material before it is received by the POTW.

Standard Operating Procedures are required for POTWs that accept hauled food waste, fats, oil, and grease for injection into anaerobic digesters. The development and implementation of Standard Operating Procedures for management of these materials is intended to allow the California Department of Resources Recycling and Recovery to exempt this activity from separate and redundant permitting programs. If the POTW does not accept food waste, fats, oil, or grease for resource recovery purposes, it is not required to develop and implement Standard Operating Procedures.

e. **Ensuring Adequate Treatment Capacity**

The Discharger shall submit a written report to the Santa Ana Water Board and USEPA within 90 days after the monthly average influent flow rate equals or exceeds 75 percent of the secondary design capacity of the POTW to prevent the waste flow exceeding the capacity of the POTW.

f. **Asset Management**

This Order/Permit requires the Discharger to develop an asset management program to cover the POTW.

**7. Other Special Provisions**

a. **Monitoring Data Accessibility**

This Order/Permit requires the Discharge to make monitoring data accessible to the public via the internet.

**8. Compliance Schedules – Not Applicable**

**VIII. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS**

CWA section 308 and 40 CFR § 122.41(h), (j)-(l), 122.44(i), and 122.48 require that all NPDES permits specify monitoring and reporting requirements. California Water Code sections 13267 and 13383 also authorize the Santa Ana Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. The Monitoring and Reporting Program (MRP), Attachment E of this Order establishes monitoring, reporting, and recordkeeping requirements that implement federal and state requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for this facility.

The MRP is guided, in part, by the principals, framework, and recommended design for discharge and receiving water monitoring presented in the Ocean Plan and *Model Monitoring Program for Large Ocean Dischargers in Southern California* (SCCWRP Tech. Rep. #357. Southern California Coastal Water Research Project, Westminster, CA. 2002). The conceptual framework for the MRP has three components that comprise a range of spatial and temporal scales: (1) core monitoring, (2) regional monitoring, and (3) strategic process studies.

**Core Monitoring.** Core monitoring is local in nature and focuses on monitoring trends in quality and effects of the point source discharge. This includes effluent monitoring, as well as many aspects of receiving water monitoring.

**Regional Monitoring.** Regional monitoring is focused on questions best answered by a region-wide approach that incorporates coordinated survey design and sampling techniques. Key components of regional monitoring include elements to address pollutant mass emission estimates, public health concerns, monitoring trends in natural resources, assessment of regional impacts from all contaminant sources, and beneficial use protection. The final designs of regional monitoring programs are developed by means of steering and technical committees comprised of participating agencies and organizations. For each component of regional monitoring, the MRP specifies the required degree and nature of participation by the Discharger, based upon its past participation in regional monitoring programs.

**Strategic Process Studies.** Strategic process studies are focused on refined questions regarding specific effects or development of monitoring techniques and are anticipated to be of short duration and/or small scale, although multi-year studies also may be needed. Questions regarding discharge or receiving water quality, discharge impacts, ocean processes in the area of the discharge, or development of techniques for monitoring the same, arising out of the results of core or regional monitoring, may be pursued through strategic process studies. These studies are by nature ad hoc and, typically, cannot be typically anticipated in advance of the five-year permit cycle.

### **A. Influent Monitoring**

Influent monitoring is required to determine compliance with effluent limitations and permit conditions, assess the performance of treatment facilities and evaluate the effectiveness of pretreatment and nonindustrial source control programs.

Influent monitoring in this Order/Permit follows the influent monitoring requirements in the previous Order/Permit with minor changes. The sample type and monitoring frequencies for halomethanes and 1,4-Dichlorobenzene were changed to grab and 1/quarter, respectively, to be consistent with other volatiles constituents. The monitoring frequency for ammonia nitrogen has been changed to weekly for proper self-monitoring as scheduled in Table E-16. The monitoring requirements for tributyltin and TCDD equivalents have been included.

### **B. Effluent Monitoring**

Effluent monitoring is required to determine compliance with effluent limitations and permit conditions, and to identify operational problems and improve treatment facility performance. Effluent monitoring also provides information on discharge characteristics and flows for use in interpreting receiving water monitoring data.

Effluent monitoring requirements also addresses the three management questions for effluent monitoring in *Model Monitoring Program for Large Ocean Discharges in Southern California* (SCCWRP, 2002; Model Monitoring Program):

Is the effluent concentration of selected constituents below levels that will protect human health and aquatic life?

What is the mass of selected constituents that are discharged annually?

Is the effluent concentration or mass changing over time?

Noteworthy changes for this Order/Permit include removal of total coliform monitoring at Discharge Point 002 (78" outfall) due to new bacteria water quality objectives in the 2019 Ocean Plan, more frequent monitoring for nutrient parameters, addition of annual monitoring for total nitrogen and quarterly monitoring for tributyltin, changes in monitoring frequencies for ammonia nitrogen for proper self-monitoring as scheduled in Table E-16, and changes in the sample type and/or monitoring frequencies for halomethanes and 1,4-Dichlorobenzene to be consistent with other volatile constituents.

In addition, effluent monitoring stations have been defined separately, for the effluent discharged to Discharge Point 001 during normal operations as EFF-001 and to Discharge Point 002 during planned essential maintenance or capital improvement projects as EFF-002, in order to determine compliance with each permit limitation and requirement.

### **C. Whole Effluent Toxicity Testing Requirements**

The rationale for WET has been discussed extensively in Section IV.C.5. of this Fact Sheet. Noteworthy WET changes for this Order/Permit include WET testing for Discharge Point 002 (78" outfall) using the TST hypothesis testing approach instead of



NOEC statistical approach. This TST statistical approach is consistent with the Ocean Plan in that it provides maximum protection to the environment since it more reliably identifies chronic toxicity than the previous NOEC hypothesis-testing approach.

#### **D. Receiving Water Core Monitoring**

To evaluate potential environmental and human health impacts from its discharge of final effluent into the Pacific Ocean, the Discharger conducts extensive water quality, sediment quality, fish and invertebrate community, and fish health monitoring off the coastal cities of Newport Beach and Huntington Beach, California.

##### **1. Water Quality Monitoring**

Offshore water quality monitoring data are used to determine compliance with receiving water limitations and Ocean Plan objectives for physical and chemical parameters, and assist in the interpretation of biological data. Water quality data collected provide the information necessary to demonstrate compliance with the water quality standards.

Water quality monitoring requirements also addresses the two management questions for water quality monitoring in the Model Monitoring Program:

Are water column physical and chemical parameters within ranges that ensure protection of the ecosystem?

What is the fate of the discharge plume?

Monthly water quality monitoring will be carried out over a large grid of 28 stations centered on Discharge Point 001 (120" outfall). The station grid covers from the coastline of Huntington Beach to Newport Beach.

Noteworthy changes for this Order/Permit include the addition of monthly monitoring for nitrate nitrogen to further investigate nutrients driving ocean acidification and the relationship between nutrients discharged through Discharge Point 001 (120" outfall) and harmful algal blooms.

##### **2. REC-1 Water Quality Monitoring (Offshore Zone)**

This REC-1 water quality monitoring is designed to determine if Ocean Plan water quality objectives for physical and chemical parameters and bacteria are being met. Data collected at these REC-1 monitoring stations provide the means to determine whether water quality standard and bacteriological objectives for water contact are being met.

REC-1 monitoring requirements also addresses the two management questions for REC-1 monitoring in the Model Monitoring Program:

Does sewage effluent reach water contact zones?

Are densities of bacteria in water contact zones below levels that will ensure public safety?

Quarterly (5-days/30-day period), REC-1 monitoring will be carried out over a picket-line of the 8 stations centered inshore of Discharge Point 001 (120"



outfall), in State waters. The south-to-north station alignment is adjacent to the coastline of Huntington Beach and Newport Beach.

### 3. Sediment Monitoring

The purpose of sediment monitoring is to map the area of impact and detect spatial and temporal trends in sediment pollutants and benthic infauna in the area of the discharge, and to assess compliance with State water quality standards and federal criteria. The data collected are used for regular assessment of trends in sediment contamination and biological response along a fixed grid of sites within the influence of the discharge.

Sediment monitoring requirements also addresses the two management questions for sediment monitoring in the Model Monitoring Program:

Are sediments in the vicinity of the discharge impaired? If so, what is the spatial extent of impairment?

Are sediment conditions changing over time?

The previous benthic sampling grid consisted of 29 semi-annual monitoring stations and 39 annual monitoring stations. The previous semi-annual station array is based on a sediment mapping study conducted by the Southern California Coastal Water Research Project (SCCWRP) from 2009 to 2012. This study recommended the design to best capture changes in sediment quality resulting from the effluent discharge.

The investigation into changes in benthic community health as a result of the District's effluent disinfection with chlorine bleach (*Changes in Biological Communities near the OCSD Outfall*, OCSD Marine Monitoring, 2013) revealed that impacts to benthic communities were localized to stations at and near (< 1 km) the 120" outfall diffuser. Impacts were seen at Stations 0, 1, 3, 4, 9, 84, 85, 86, 87, and ZB. All other stations were unaffected even though the degree of impact was severe at near outfall stations. Based on these results and over three decades of monitoring data, a more effective and efficient sediment monitoring station design has been constructed in this Order/Permit as below:

- a. Quarterly sampling for sediment chemistry and benthic infauna at 11 stations immediately surrounding the 120" outfall diffuser along the 60-meter contour line, including 3 stations within the ZID boundaries. (Note that these quarterly monitoring stations were the first and the most severely impacted stations by effluent chlorination.)
- b. Annual whole sediment toxicity sampling at the 11 quarterly monitoring stations.
- c. Annual (summer) sampling for sediment chemistry and benthic infauna at 11 stations located near the outfall and extending northward and southward along the 60-meter contour line.

- d. Once per 5 year (1/5-year) sampling for sediment chemistry and benthic infauna at an additional 35 stations centered on Discharge Point 001 (120" outfall) from Huntington Beach to Newport Beach coastlines.

Stations 68, 69, 70, 71, 72, 79, 80, 81, and 82 were eliminated as they are redundant to the station design. Downcoast Station C2 and farfield upcoast Station C were also removed as the sediment quality and biological community at these stations have been consistently different from those of other 60 m stations including control station, considering outliers. Therefore, total number of stations is reduced from 68 stations to 57 stations. The reduction in taxonomic and sediment geochemical analyses would enable the Discharger to redirect its efforts toward SPS and special studies to evaluate potential environmental impacts related to increased wastewater reclamation efforts and investigate CEC identification and characterization in effluent and sediments and toward the regional monitoring programs including Bight 18.

With the new sediment monitoring design, the number of stations sampled annually in summer is reduced from 68 to 22, which can reduce the turn-around time for summer data from approximately 9 to 12 months to 3 months, especially for the 11 quarterly stations in all seasons. This will provide a rapid assessment of sediment conditions and a more rapid response to changes in sediment quality and/or benthic infauna community within the monitoring area.

Noteworthy changes for this Order/Permit also include the change in sampling frequency for 35 monitoring stations that previously sampled annually to one-off (i.e., once per 5 years) since these stations are either not directly or not significantly impacted by the effluent discharge. Three (3) stations (i.e., Stations 10, 13, and 37) at the 60-m contour line remains as annual monitoring stations. Sampling these 1/5-year monitoring stations allows long-term trend analysis to continue in these areas where change occurs much more slowly in response to changes in effluent quality than at the outfall diffuser.

#### **4. Demersal Fish and Epibenthic Macroinvertebrate Monitoring**

The purpose of demersal fish and epibenthic macroinvertebrate monitoring is to detect spatial and temporal trends in demersal fish and epibenthic community structure, demersal fish liver tissue chemistry and liver histopathology, and sport fish muscle tissue chemistry in the area of the discharge, and to assess compliance with State water quality standards and federal criteria. The demersal fish and epibenthic macroinvertebrate monitoring data collected are used for regular assessment of temporal trends in community structure along an array of sites within the influence of the discharge. The sport fish monitoring data collected are used to provide information necessary for the management of local seafood consumption advisories.

Demersal fish and epibenthic invertebrate monitoring requirements also address the four management questions for fish and epibenthic invertebrate monitoring and seafood safety monitoring in the Model Monitoring Program:

Is the health of fish populations and communities impaired? Are fish populations and communities changing over time? Is fish tissue contamination changing over time?

Are seafood tissue concentrations below levels that will ensure public safety?

Annual demersal fish and epibenthic community monitoring continues to be carried out over a grid of 14 stations upcoast of Discharge Point 001 (120" outfall); of these 14 stations, the 6 stations at the outfall depth (60 meters) are monitored semi-annually. The monitoring area covers approximately 27 square kilometers adjacent to the coastline of Huntington Beach and Newport Beach. Annual demersal fish tissue chemistry monitoring is carried out over two stations upcoast of Discharge Point 001 at the outfall depth. Annual sport fish muscle tissue chemistry monitoring is also conducted in summer at two zones. Noteworthy changes for this Order/Permit are listed below:

- a. **Target fish species.** For demersal fish and epibenthic community monitoring, the previous Order/Permit required at least 10 individuals of each target species (i.e., Hornyhead Turbot and English Sole) per station. This Order/Permit allows the Discharger to collect 20 individuals total of flat fish (e.g., Pacific Sanddab, Hornyhead Turbot and English Sole) per station. For sport fish monitoring, the Discharger can collect 10 individuals total of rockfish (e.g., Vermilion Rockfish, Copper Rockfish, and California Scorpionfish) at each rig fish monitoring zone. Kelp bass and Sandbass are removed from the target species list in this Order/Permit because none of these fishes was caught by the Discharger at any of the three rig fishing zones in the past three monitoring periods (2014 – 2016).
- b. **Demersal fish tissue chemistry monitoring - Target tissue.** The previous Order/Permit required sampling and analysis of both fish liver and muscle tissues in Hornyhead Turbot and English Sole for the assessment of predator wildlife risk by consuming these two target fish species. Noteworthy change in demersal fish tissue chemistry monitoring for this Order/Permit includes removal of muscle tissue and analysis of contaminants in fish liver tissue only. The concentration of contaminants in demersal fish muscle tissue were relatively negligible for two permit cycles compared with liver tissues, given that liver tissue has higher lipid content than muscle tissue and therefore can bioaccumulate higher concentrations of lipid-soluble contaminants such as PCBs and DDT. It stands to reason that when this demersal fish is consumed whole by a fish predator, it is likely that the predator will acquire considerably higher contaminant loads from liver tissue as compared to muscle tissue, rendering the muscle tissue analysis dispensable.
- c. **Chemical analysis changes – Composite sampling.** To monitor bioaccumulation of organic contaminants (i.e., PCBs, DDT, and Chlordane) and metals (i.e., Arsenic, Selenium, and Mercury) in either demersal fish liver tissue or sport fish muscle tissue, composite samples of target fish will be

analyzed instead of individual sample analysis. This composite tissue analysis can minimize the variation between contaminant measurements, increase the ability to detect target compounds that are present in low concentrations, and improve the representativeness of the contaminant measurements at each sampling site, which is also consistent with fish tissue analysis protocols of other POTW ocean monitoring programs within the Southern California Bight.

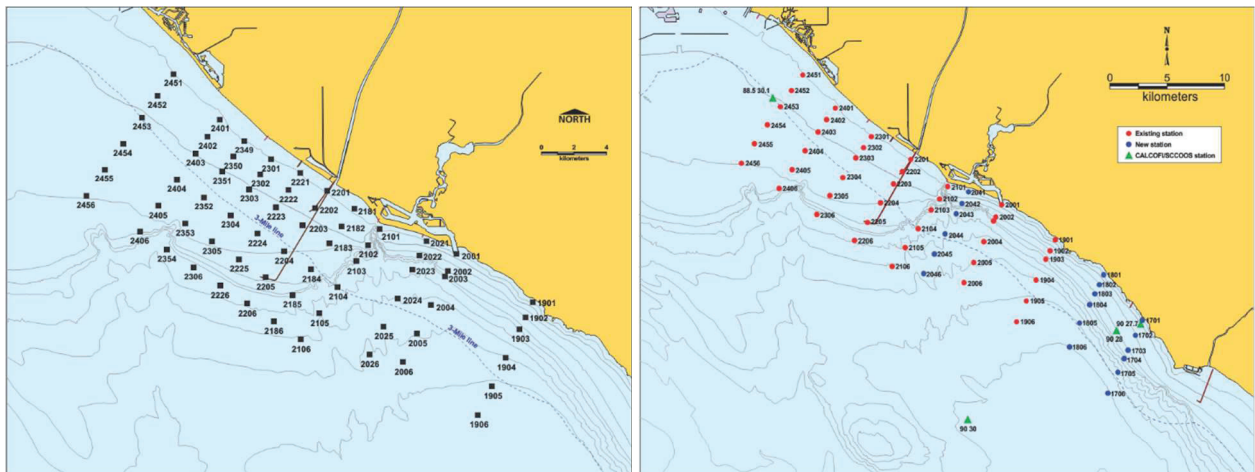
- d. Monitoring frequency of demersal fish liver histopathology.** Liver histopathology will be conducted on an annual basis instead of once every permit cycle to facilitate rapid detection of potential adverse effects on fish health by the anticipated changes in effluent quality and characteristics with GWRS final expansion (e.g., increased RO concentrate flow). Fish liver histopathology was used in the investigation of effluent chlorination effects and was found to be a useful biomarker of impact and the only significant effect on fish health.
- e. Sport fish monitoring - Farfield reference zone.** The Discharger was not able to catch the number of fish samples required by the previous Order/Permit at the farfield reference zone (Zone 2) in two consecutive years (2014 – 2015) after the issuance of the previous 2012 Order/Permit. After consulting with scientists from the California Department of Fish and Wildlife, the reference zone was relocated to an area near the Huntington Beach oil rigs (Zone 3) as this site only met the species and number of fish requirements among several candidate sites. In this Order/Permit, Zone 2 was replaced with Zone 3 as a farfield reference zone.

## **E. Receiving Water Regional Monitoring**

Discharger participation in regional monitoring programs continues to be a required condition of the Order/Permit. Regional monitoring programs which must be conducted under the Order/Permit include: Southern California Bight Regional Monitoring Program, Southern California Bight Regional Water Quality Program, Central Regional Kelp Survey, Orange County Regional Shoreline REC-1 Cooperative Monitoring Program, and Ocean Acidification and Hypoxia (OAH) Mooring. The Discharger currently participates in all five programs. For the regional monitoring program, collaboration with other water sectors (e.g., water supply, wastewater, groundwater, and stormwater) is encouraged to evaluate implementation of integrated water resource projects that helps achieve sustainable integrated water resources management.

Noteworthy changes for the Southern California Bight Regional Water Quality Program under this Order/Permit include the addition of four field sampling surveys per year, removal of stations transects 2021, 2181, 2221, and 2349 for more equal spacing of transects, and creation of transect lines 1701, 1801, and 2041 to extend sampling downcoast to Dana Point.

**Figure F-1. Previous and Current Southern California Bight Regional Water Quality Program Station Transects.**



Overall, the number of Southern California Bight Regional Water Quality Program stations were reduced from 66 to 60. The number of Orange County Regional Shoreline REC-1 Cooperative Monitoring Program stations has been changed from 38 to 36 due to removal of two Stations (i.e., OSB02 and ELMORO) at the request of the Orange County Health Care Agency.

## F. Strategic Process Studies

Discharger investigations conducted through strategic process studies continues to be a required condition of the Order/Permit. Strategic process studies which must be conducted under the Order/Permit include: ROMS-BEC Ocean Outfall Modeling, Microplastics Characterization, Contaminants of Emerging Concern Monitoring, Sediment Linear Alkylbenzenes, and Meiofauna Baseline. These studies have been approved by the Santa Ana Water Board and USEPA during the term of the 2012 Order/Permit but not yet completed by the Discharger. Additional studies will be proposed over the next permit cycle to ascertain if GWRS final expansion, which will result in lower outfall flows and higher concentration of the final effluent due to RO reject flow, will result in deleterious environmental impacts.

## G. Other Monitoring Requirements

### 1. Contaminants of Emerging Concern (CEC) Monitoring Study

The Discharger shall continue to investigate CECs identification and characterization in the final effluent, listed in Table E-15 of the MRP (Attachment E). The Discharger has annually screened for 15 pharmaceuticals and personal care products (PPCPs), 7 hormones, 7 industrial endocrine disrupting compounds (IEDCs), and 9 flame retardants in the final effluent in the previous permit. In addition, this Order/Permit requires the Discharger to conduct monitoring for new CECs of 6 pesticides and insecticides (i.e., Fipronil, Fipronil Sulfone, Bifenthrin, Total Permethrin, Chlorpyrifos, and Diazinon), 3 flame



retardants (i.e., TDCPP, TCEP, and TCPP) and 12 per- and polyfluoroalkyl substances (PFAS) compounds.

With regard to PFAS compounds, POTWs are potentially significant receivers of PFAS from various sources, including disposal of landfill leachate and firefighting foam, results in PFAS in the influent to POTWs. Typical POTW treatment systems are not designed to remove PFAS, and therefore, the discharge from POTWs, especially those with industrial inputs, are possible contributors of PFAS to the surface waters. Based on the State Water Board's PFAS analytical data obtained from PFAS Investigative Orders, 12 PFAS compounds (i.e., PFDA, PFDoA, PFNA, PFHxA, PFHpA, PFOA, PFTA, PFTTrDA, PFUnA, PFBS, PFHxS, and PFOS) are included in this Order/Permit to identify and understand PFAS in wastewater.

## **2. Outfall and Diffuser System Inspection**

This survey investigates the condition of the outfall structures to determine if the structures are in serviceable condition to ensure their continued safe operation. The inspection frequency has been increased to twice per permit term as the 120-inch outfall approaches the end of life in 2021. The data collected will be used for a periodic assessment of the integrity of the outfall pipes and diffuser system.

## **3. Discharge Monitoring Report-Quality Assurance (DMR-QA) Study Program**

Under the authority of section 308 of the CWA (33 U.S.C. § 1318), USEPA requires major and selected minor dischargers under the NPDES Program to participate in the annual DMR-QA Study Program. The DMR-QA Study evaluates the analytical ability of laboratories that routinely perform or support self-monitoring analyses required by NPDES permits. There are two options to satisfy the requirements of the DMR-QA Study Program: (1) The Discharger can obtain and analyze a DMR-QA sample as part of the DMR-QA Study; or (2) Per the waiver issued by USEPA to the State Water Board, the Discharger can submit the results of the most recent Water Pollution Performance Evaluation Study from its own laboratories or its contract laboratories. A Water Pollution Performance Evaluation Study is similar to the DMR-QA Study. Thus, it also evaluates a laboratory's ability to analyze wastewater samples to produce quality data that ensure the integrity of the NPDES Program. The Discharger shall ensure that the results of the DMR-QA Study or the results of the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board. The State Water Board's Quality Assurance Program Officer will send the DMR-QA Study results or the results of the most recent Water Pollution Performance Evaluation Study to USEPA's DMR-QA Coordinator and Quality Assurance Manager.

## **IX. PUBLIC PARTICIPATION**

The Santa Ana Water Board and USEPA have considered the issuance of WDRs and an NPDES permit for OC San. As a step in this process, the Santa Ana Water Board and



USEPA staff have developed a tentative WDRs and draft permit and have encouraged public participation in the reissuance process.

**A. Notification of Interested Parties**

The Santa Ana Water Board and USEPA have notified the Discharger and interested agencies and persons of their intent to reissue the Order/Permit for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Notification was provided through the posting of the Notice and tentative Order/Permit on the Santa Ana Water Board's and USEPA's websites.

The public had access to the agenda and any changes in dates and locations through the Santa Ana Water Board's website at <http://www.waterboards.ca.gov/santaana/> and USEPA's website at <https://www.epa.gov/npdes-permits/npdes-permits-epas-pacific-southwest-region-region-9>.

**B. Written Comments**

Interested persons were invited to submit written comments concerning the tentative WDRs and NPDES Order/Permit as provided through the notification process. Comments should be submitted either in person or by mail to the Executive Office at the Santa Ana Water Board at the address on the cover page of this Order/Permit, or by email to [Ryan.Harris@waterboards.ca.gov](mailto:Ryan.Harris@waterboards.ca.gov) or [Song.Julie@epa.gov](mailto:Song.Julie@epa.gov).

To be fully responded to by staff and considered by the Santa Ana Water Board and USEPA, the written comments were due by 5:00 p.m. on March 4, 2021.

**C. Public Hearing**

The Santa Ana Water Board held a public hearing on the tentative WDRs during its regular Board meeting on the following date and time and at the following location:

Date: June 18, 2021  
Time: 9 am  
Location: Meeting was remote, due to COVID-19 restrictions.

Interested persons were invited to attend. At the public hearing, the Santa Ana Water Board heard testimony, pertinent to the discharge, WDRs, and permit. For accuracy of the record, extensive testimony was requested in writing.

**D. Reconsideration of Waste Discharge Requirements**

Any aggrieved person by the adoption of the WDRs and Order/Permit may petition the State Water Board to review the decision of the Santa Ana Water Board and USEPA regarding the final WDRs and Order/Permit in accordance with Water Code section 13320 and California Code of Regulations, title 23, sections 2050 and following. The State Water Board must receive the petition by 5:00 p.m., within 30 calendar days of the date of adoption of this Order at the following address, except that if the thirtieth day following the date of this Order falls on a Saturday, Sunday, or state holiday, the petition must be received by the State Water Board by 5:00 p.m. on the next business day:

State Water Resources Control Board  
Office of Chief Counsel  
P.O. Box 100, 1001 I Street  
Sacramento, CA 95812-0100

Or by email at [waterqualitypetitions@waterboards.ca.gov](mailto:waterqualitypetitions@waterboards.ca.gov)

For instructions on how to file a petition for review, see the [Water Quality Petitions Website](https://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instructions.shtml):

([https://www.waterboards.ca.gov/public\\_notices/petitions/water\\_quality/wqpetition\\_instructions.shtml](https://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instructions.shtml))

#### **E. Federal NPDES Permit Appeals**

When a final NPDES permit is issued by USEPA, it will become effective 33 days following the date it is mailed to the Discharger, unless a request for review is filed. If a request for review is filed, only those permit conditions which are uncontested will go into effect pending disposition of the request for review. Requests for review must be filed within 33 days following the date the final permit is mailed and must meet the requirements of 40 CFR 124.19. All requests for review should be addressed to the Environmental Appeals Board (EAB) as follows. Requests sent through the U.S. Postal Service (except by Express Mail) must be addressed to the EAB's mailing address, which is:

U.S. Environmental Protection Agency  
Clerk of the Board  
Environmental Appeals Board (MC 1103B) Ariel Rios Building  
1200 Pennsylvania Avenue, N.W.  
Washington, D.C. 20460-0001

All filings delivered by hand or courier, including Federal Express, UPS, and U.S. Postal Express Mail, should be directed to the following address:

Environmental Appeals Board  
U.S. Environmental Protection Agency Colorado Building  
1341 G Street, N.W., Suite 600  
Washington, D.C. 20460

Those persons filing a request for review must have filed written comments on the draft federal permit. Otherwise, any such request for review may be filed only to the extent of changes from the draft to the final permit decision.

#### **F. Information and Copying**

The Report of Waste Discharge (ROWD), other supporting documents, and comments received are on file and may be inspected by appointment at the Santa Ana Water Board and USEPA addresses on the cover page of this Order/Permit, at any time between 9:00 a.m. and 3:00 p.m., Monday through Friday. Copying of documents may be arranged through either the Santa Ana Water Board by calling (951) 782-4130 or USEPA by calling (415) 972-3035.

#### **G. Register of Interested Persons**

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Santa Ana Water Board, reference this facility, and provide a name, address, and phone number.

**H. Additional Information**

Requests for additional information or questions regarding this Order/Permit should be directed to the Santa Ana Water Board or USEPA staffs.

Santa Ana Water Board: Ryan Harris at (951) 320-2008 or  
[Ryan.Harris@waterboards.ca.gov](mailto:Ryan.Harris@waterboards.ca.gov).

USEPA: Julie Song at (415) 972-3035 or [Song.Julie@epa.gov](mailto:Song.Julie@epa.gov).

## **ATTACHMENT G – BIOSOLIDS**

As described in section VII.C.6.a of this Order/Permit, the Santa Ana Water Board and USEPA incorporates these biosolids conditions as requirements of this Order/Permit.

### **I. GENERAL REQUIREMENTS**

- A. All biosolids generated by the Discharger shall be used or disposed of biosolids generated by the Discharger shall be used or disposed of in compliance with the applicable portions of 40 CFR § 257, 258 and 503, and the applicable State regulations. The Discharger is responsible for assuring that all biosolids produced at the facility are used or disposed of in accordance with these rules, whether the Discharger uses or disposes of the biosolids itself or transfers them to another party for further treatment and use or disposal. The Discharger is responsible for informing subsequent preparers, applicators, and disposers of the requirements that they must meet under these rules, and any monitoring requirements, including required frequencies of monitoring and maximum hold times for pathogen and indicator organism samples.
- B. Duty to mitigate. The Discharger shall take all reasonable steps to prevent or minimize any biosolids use or disposal which has a likelihood of adversely affecting human health or the environment.
- C. No biosolids shall be allowed to enter wetlands or other waters of the United States.
- D. Biosolids treatment, storage, and use or disposal shall not contaminate groundwater.
- E. Biosolids treatment, storage, and use or disposal shall be performed in a manner as to minimize nuisances such as objectionable odors or flies.
- F. The Discharger shall assure that haulers transporting biosolids off site for further treatment, storage, use, or disposal take all necessary measures to keep the biosolids contained. The Discharger shall maintain and have haulers adhere to a spill clean-up plan. Any spills shall be reported to USEPA and State agency in which the spill occurred. All trucks hauling biosolids that are not Class A, as defined at 40 CFR 503.32(a), shall be cleaned as necessary after loading and after unloading, so as to have no biosolids on the exterior of the truck or wheels.
- G. Trucks used to haul Class B biosolids shall not be used to haul animal feed or food on the return trip, unless approved by USEPA after a demonstration of the truck cleaning methods at the unloading site has been made.
- H. If biosolids are stored for over two years from the time they are generated by the Discharger or their contractor, the Discharger must submit a written notification to USEPA with the information in 40 CFR § 503.20(b), demonstrating the need for longer temporary storage.
- I. Any biosolids treatment, disposal, or storage site shall have facilities adequate to divert surface runoff from adjacent areas, to protect the site boundaries from erosion, and to prevent any conditions that would cause drainage from the materials in the site to

escape from the site. Adequate protection is defined as protection from at least a 100-year storm and from the highest tidal stage that may occur.

## **II. REQUIREMENTS FOR BIOSOLIDS LAND APPLICATION**

“Land application” is the placement of biosolids on the land for the specific purpose of growing a crop or other vegetation. Land application requirements are addressed in 40 CFR § 503 subpart B.

- A. A representative sample shall be collected and analyzed for the pollutants required under 40 CFR § 503.13 and for total nitrogen, organic nitrogen, and ammonia nitrogen, at the following frequency, based on the tonnage of biosolids produced per year (as expressed on a 100% solids basis). All results shall be reported on a 100% dry weight basis:

Less than 290 dry metric tons (dmt)/year: once/year.

290 to 1,500 dmt/year: once/quarter or 4 samples of accumulated biosolids

1,500 to 15,000 dmt/year: once/two months or 6 samples of accumulated biosolids.

Over 15,000 dmt/year: monthly or 12 samples of accumulated biosolids.

- B. The Discharger shall demonstrate that the biosolids meet Class A or Class B pathogen reduction levels by one of the methods listed in 40 CFR § 503.32.
- C. If Class B is demonstrated by testing fecal coliform, during each sampling event, 7 grab samples must be collected and analyzed, and the geometric mean of these samples calculated to determine the fecal coliform level for the sampling period.
- D. When using fecal coliforms to demonstrate Class A, in conjunction with operational parameters or in conjunction with testing of enteric viruses and helminth ova, four grab samples of fecal coliform shall be collected and analyzed each sampling period. Each of these samples must have levels of <1,000 MPN/gram, dry weight basis.
- E. If Class A or B pathogen requirements are met by monitoring pathogens and/or indicator organisms, samples must be collected in sterile containers, immediately cooled, and analysis started within the USEPA-specified holding times for these analyses: 8 hours for fecal coliform (24 hours for fecal coliform if the biosolids have been digested or composted), 24 hours for salmonella, 2 weeks for enteric viruses when frozen, 1 month for helminth ova when cooled to 4 degrees C).
- F. If pathogen reduction is demonstrated using a Process to Significantly/Further Reduce Pathogens, the Discharger shall maintain daily records of the operating parameters used to achieve this reduction.
- G. The Discharger shall track and keep records of the operational parameters used to achieve Vector Attraction Reduction (VAR) requirements in 40 CFR § 503.33(b). If VAR is met at the application site by incorporation or covering, the Discharger must obtain certification that these requirements have been met from the land applier or surface disposal site operator, and maintain these with their records.

### III. REQUIREMENTS FOR SURFACE DISPOSAL

“Surface disposal” is the placement of biosolids on the land in a sludge-only dedicated land disposal site or monofill for the purpose of disposal. Surface disposal requirements are addressed in 40 CFR § 503 subpart C.

- A. If the surface disposal site is unlined, a representative sample shall be collected and analyzed for the pollutants required under 40 CFR § 503.23, at the following frequency, based on the tonnage of biosolids produced per year (as expressed on a 100% solids basis). All results shall be reported on a 100% dry weight basis:
  - Less than 290 dry metric tons (dmt)/year: once/year.
  - 290 to 1,500 dmt/year: once/quarter or 4 samples of accumulated biosolids
  - 1,500 to 15,000 dmt/year: once/two months or 6 samples of accumulated biosolids.
  - Over 15,000 dmt/year: monthly or 12 samples of accumulated biosolids
- B. The Discharger shall demonstrate that the biosolids meet Class A or Class B pathogen reduction levels by one of the methods listed in 40 CFR § 503.32, or cover the site at the end of each operating day.
- C. If Class B is demonstrated by testing fecal coliform, during each sampling event, 7 grab samples must be collected and analyzed, and the geometric mean of these samples calculated to determine the fecal coliform level for the sampling period.
- D. If Class A or B pathogen requirements are met by monitoring pathogens and/or indicator organisms, samples must be collected in sterile containers, immediately cooled, and analysis started within the USEPA-specified holding times for these analyses: 8 hours for fecal coliform (24 hours for fecal coliform if the biosolids have been digested or composted), 24 hours for salmonella, 2 weeks for enteric viruses when frozen, 1 month for helminth ova when cooled to 4 degrees C).
- E. If pathogen reduction is demonstrated using a Process to Significantly/Further Reduce Pathogens, the Discharger shall maintain daily records of the operating parameters used to achieve this reduction.
- F. The Discharger shall track and keep records of the operational parameters used to achieve Vector Attraction Reduction (VAR) requirements in 40 CFR § 503.33(b). If VAR is met at the surface disposal site by incorporation or covering, the Discharger must obtain certification that these requirements have been met from the land applier or surface disposal site operator, and maintain these with their records.

### IV. REQUIREMENTS FOR DISPOSAL IN MUNICIPAL LANDFILL

“Disposal in a municipal landfill” is the placement of biosolids in a landfill subject to the requirements in 40 CFR § 258 where it is mixed with other materials being placed in the landfill, or used as alternative daily or final cover at the landfill.

- A. The Discharger shall ensure that the landfill used is in compliance with 40 CFR § 258 requirements and applicable State or Tribal requirements.



- B. If the biosolids are less than 15% solids, the Discharger shall run a paint filter test on an as-needed basis to demonstrate that the biosolids does not contain free liquids.

## **V. NOTIFICATION REQUIREMENTS**

The Discharger either directly or through contractual arrangements with their biosolids management contractors, shall comply with the following notification requirements.

- A. Notification of non-compliance: The Discharger shall notify USEPA and the applicable Santa Ana Water Board or State agency of any non-compliance within 24 hours by phone or e-mail if the non-compliance may seriously endanger public health or the environment. A written report shall also be submitted within 5 working days of knowing the non-compliance. For other instances of non-compliance, the Discharger shall notify USEPA and the Santa Ana Water Board of the non-compliance in writing within 5 working days of becoming aware of the non-compliance. The Discharger shall require their biosolids management contractors to notify USEPA and the Santa Ana Water Board of any non-compliance within the same time-frames.
- B. If biosolids are shipped to another State or to Tribal Lands, the Discharger shall send 30 days prior notice of the shipment to the USEPA and permitting authorities in the receiving State/Tribal authority.
- C. The Discharger shall notify USEPA and the Santa Ana Water Board at least 60 days prior to starting a new biosolids use or disposal practice.

## **VI. REPORTING REQUIREMENTS**

- A. The Discharger shall submit an annual biosolids report into USEPA's CDX electronic reporting system, with an electronic copy to the Santa Ana Water Board by email at [santaana@waterboards.ca.gov](mailto:santaana@waterboards.ca.gov), by February 19 of each year for the period covering the previous calendar year. The report shall include the tonnages of biosolids (reported in dry metric tons, 100% dry weight), that were land applied (without further treatment by another party), land applied after further treatment by another preparer, disposed in a sludge-only surface disposal site, sent to a landfill for alternative cover or fill, stored on site or off site, or used for another purpose. The report shall include the following attachments:
  - 1. Monitoring results from laboratories (results only, QA/QC pages not required). Copies of original lab reports must be available upon request and confirm the results are on a 100% dry weight basis. Lab reports for fecal coliforms must show the time the samples were collected and the time analysis was started.
  - 2. If operational parameters were used to demonstrate compliance with pathogen reduction and vector attraction reduction, the minimum mean of these parameters for each sampling period (i.e., minimum mean cell residence times (MCRTs) and minimum temperatures).
  - 3. If biosolids are stored on-site or off-site for more than 2 years, the information required in 40 CFR § 503.20(b) to demonstrate that the storage is temporary.

- B. If biosolids were land applied, the Discharger shall have the person applying the biosolids submit a pdf report to USEPA and State agency showing the name of each field; location, ownership, size in acres; the dates of applications, seedings, harvesting; the tonnage applied to field, in actual and dry weight; the calculated Plant Available Nitrogen; and copies of applier's certifications of management practices and site restrictions.

## ATTACHMENT H – PRETREATMENT

As described in section VII.C.6.b of this Order/Permit, the Santa Ana Water Board and USEPA incorporates these pretreatment conditions as requirements of this Order/Permit.

### I. GENERAL PRETREATMENT REQUIREMENTS

- A. The Discharger shall be responsible and liable for the performance of all Control Authority pretreatment requirements contained in 40 CFR § 403, including any subsequent regulatory revisions to Part 403. Where Part 403 or subsequent revision places mandatory actions upon the Discharger as Control Authority but does not specify a timetable for completion of the actions, the Discharger shall complete the required actions within one (1) year from the issuance date of this permit or the effective date of the Part 403 revisions, whichever comes later. For violations of pretreatment requirements, the Discharger shall be subject to enforcement actions, penalties, fines and other remedies by the USEPA or other appropriate parties, as provided in the Act. USEPA may initiate enforcement action against a nondomestic user for noncompliance with applicable standards and requirements as provided in the Act.
- B. Prior to the completion of GWRS Final Expansion project, the Discharger shall conduct annual Technical Review of local limits under 40 CFR § 403.5(c)(1) and submit the results as part of the annual pretreatment report. Within two (2) years of the completion of the GWRS Final Expansion project, the Discharger shall provide a written technical evaluation of the need to revise local limits under 40 CFR § 403.5(c)(1), as required in 40 CFR § 122.44(j)(2)(ii).
- C. The Discharger shall enforce the requirements promulgated under CWA sections 307(b), 307(c), 307(d) and 402(b) with timely, appropriate and effective enforcement actions. The Discharger shall cause all nondomestic users subject to federal categorical standards to achieve compliance no later than the date specified in those requirements or, in the case of a new nondomestic user, upon commencement of the discharge.
- D. The Discharger shall perform the pretreatment functions as required in 40 CFR § 403 including, but not limited to:
  1. Implement the necessary legal authorities as provided in 40 CFR § 403.8(f)(1);
  2. Enforce the pretreatment requirements under 40 CFR § 403.5 and 403.6;
  3. Implement the programmatic functions as provided in 40 CFR § 403.8(f)(2); and
  4. Provide the requisite funding and personnel to implement the pretreatment program as provided in 40 CFR § 403.8(f)(3).
- E. The Discharger shall submit annually a report to USEPA and the Santa Ana Water Board describing its pretreatment activities over the previous year. In the event the Discharger is not in compliance with any conditions or requirements of this Order/Permit, then the Discharger shall also include the reasons for noncompliance and state how and when the Discharger shall comply with such conditions and requirements. This annual report shall cover operations from July 1 through June 30

and is due on October 31st of each year. The report shall contain, but not be limited to, the following information:

1. A summary of analytical results from representative, flow proportioned, 24-hour composite sampling of the POTW's influent and effluent for those pollutants USEPA has identified under CWA section 307(a) which are known or suspected to be discharged by nondomestic users. This will consist of an annual full priority pollutant scan, with quarterly samples analyzed only for those pollutants detected in the full scan. The Discharger is not required to sample and analyze for asbestos. Sludge sampling and analysis are covered in the sludge section of this Order/Permit. The Discharger shall also provide any influent or effluent monitoring data for nonpriority pollutants which the Discharger believes may be causing or contributing to interference or pass through. Sampling and analysis shall be performed with the techniques prescribed in 40 CFR § 136.
2. A discussion of Upset, Interference or Pass Through incidents, if any, at the treatment plant which the Discharger knows or suspects were caused by nondomestic users of the POTW system. The discussion shall include the reasons why the incidents occurred, the corrective actions taken and, if known, the name and address of the nondomestic user(s) responsible. The discussion shall also include a review of the applicable pollutant limitations to determine whether any additional limitations, or changes to existing requirements, may be necessary to prevent pass through or interference.
3. An updated list of the Discharger's significant industrial users (SIUs) including their names and addresses, and a list of deletions, additions and SIU name changes keyed to the previously submitted list. The Discharger shall provide a brief explanation for each change. The list shall identify the SIUs subject to federal categorical standards by specifying which set(s) of standards are applicable to each SIU. The list shall also indicate which SIUs are subject to local limitations.
4. The Discharger shall characterize the compliance status of each SIU by providing a list or table which includes the following information: Name of the SIU; category, if subject to federal categorical standards; type of wastewater treatment or control processes in place; number of samples taken by the POTW during the year; number of samples taken by the SIU during the year; for an SIU subject to discharge requirements for total toxic organics, whether all required certifications were provided; a list of the standards violated during the year, where categorical standards and/or local limits violations are identified; whether the facility is in significant noncompliance (SNC) as defined at 40 CFR § 403.8(f)(2)(viii) at any time during the year; and a summary of enforcement or other actions taken during the year to return the SIU to compliance, where the type of action, final compliance date, and the amount of fines and penalties collected, if any, are described, including any proposed actions for bringing the SIU into compliance.
5. A brief description of any programs the POTW implements to reduce pollutants from nondomestic users that are not classified as SIUs.

6. A brief description of any significant changes in operating the pretreatment program which differ from the previous year including, but not limited to, changes concerning the program's administrative structure, local limits, monitoring program or monitoring frequencies, legal authority, enforcement policy, funding levels, or staffing levels;
7. A summary of the annual mass emission, and the effluent concentrations and flows used to calculate the annual mass emission (see section V.B of the Order/Permit);
8. A summary of the annual pretreatment budget, including the cost of pretreatment program functions and equipment purchases;
9. A summary of activities to involve and inform the public of the program including a copy of the newspaper notice, if any, required under 40 CFR § 403.8(f)(2)(viii); and
10. Results from annual Technical Review of local limits which is conducted under 40 CFR § 403.5(c)(1). See Section I.B of Attachment H.